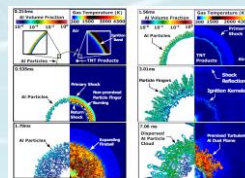
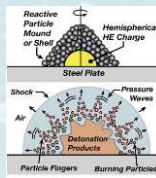
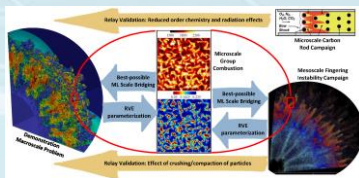


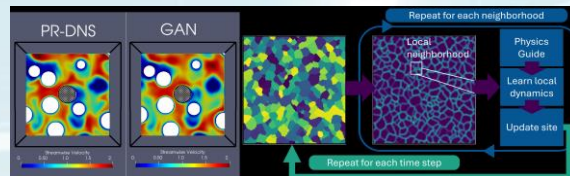
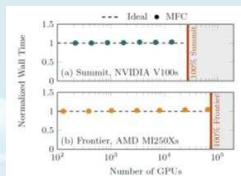
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Center for Multiscale Modeling of
Multiphase Combustion

TST Meeting April 1-2, 2026



MFC + Group Combustion



GPU + ML on Emerging Architecture



Center for Multiscale Modeling of Multiphase Combustion (CM³C)

Agenda for TST Site Visit

April 1 and 2, 2026

Emerson Alumni Hall

Wednesday April 1, 2026

- 8:15-9:00 **Full Breakfast Provided (Review team meets with students)**
- 9:00-10:10 T1 – CM³C Overview and Management (Bala, Jackson)
- 10:10-10:30 T2 – MFC for Particle Resolved DNS – BC Kim (20 min)
- 10:30-10:45 **Coffee break**
- 10:50-11:10 T3 – MFC for Euler-Lagrange Simulations – Jose Aguiar (20 min)
- 11:15-11:50 T4 – Extreme-scale Many-physics CFD – Spencer Bryngelson (30 min)
- 11:50-1:00 **Lunch Provided**
- 1:00-1:40 T5 – Reactive Flow Simulations for Group Combustion – Ryan Houim (30 min)
- 1:40-2:20 T6 – Experiments – Daniel Guildenbecher (30 min)
- 2:20-3:00 T7 – Rigorous EL Closure & Exascale Implementation – Thierry Daoud (30 min)
- 3:00-3:30 **Coffee break – Visiting team – Student Interactions**
- 3:30-4:10 T8 – Generative ML models - Erica Lindbeck (30 min)
- 4:10-4:50 T9 – Operator learning ML models – Cameron Lott (30 min)
- 5:00 **Day 1 Meeting Ends**
- 5:00-5:30 **TST caucus (if needed)**
- 5:30-7:30 **Social, followed by Buffet Dinner, at Hilton University of Florida (Visitors, Faculty, Students)**





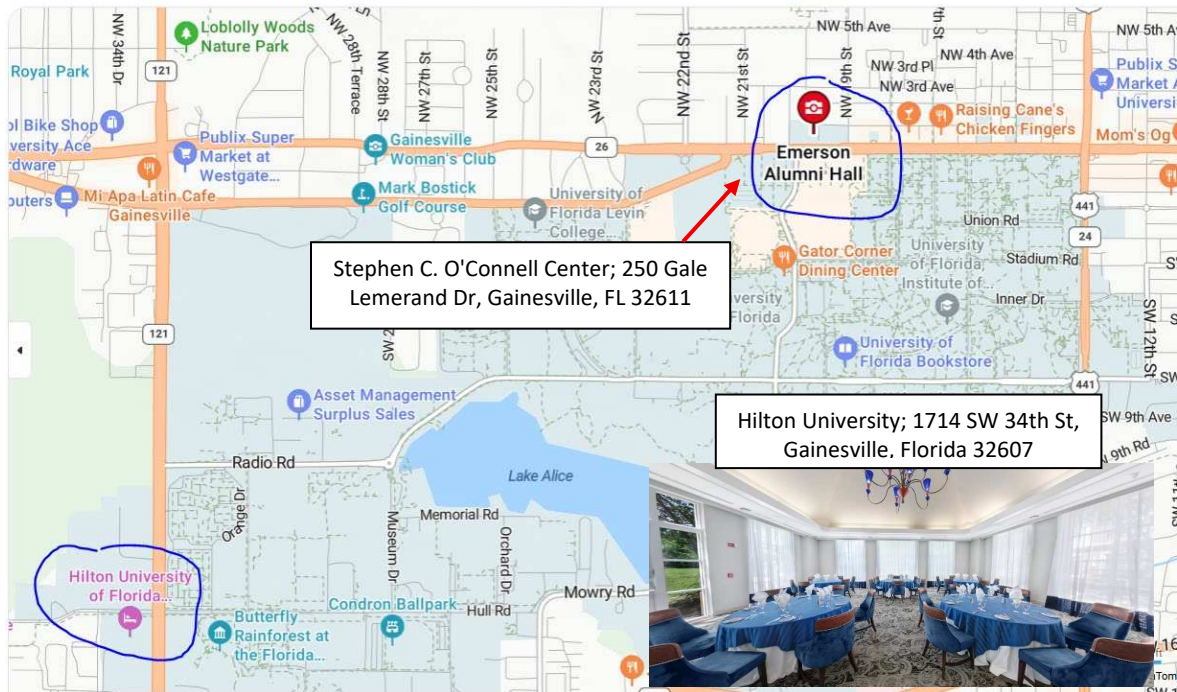
Center for Multiscale Modeling of Multiphase Combustion (CM³C)

Thursday April 2, 2026

- 8:15-9:00 **Continental Breakfast Provided (Review team meets with students)**
- 9:00-9:30 T10 – Immersed Boundary Development in MFC – Daniel Vickers (20 min)
- 9:30-9:50 T11 – Group Combustion Modeling – Davin Anderson 20 (min)
- 9:50-10:10 T12 – Carbon Rod Experiments – Alex Millauer-Hsu (20 min)
- 10:15-10:30 Short Break
- 10:30-11:00 T13 – Physics Informed NN and Uncertainty (Isaly Tappan/Juan Diaz) (30 min)
- 11:00-11:30 T14 – Validation, UQ, Uncertainty Budget – Nam Ho Kim (20 min)
- 11:30-12:00 **Final Discussions and Feedback**
- 12:00 **Box Lunch Provided + Student Interactions**

Meeting Location: Emerson Alumni Hall, 1938 W University Ave, Gainesville, FL 32603

Wednesday Social and Dinner: Hilton University, 1714 SW 34th St, Gainesville, FL 32607



Center for Multiscale Modeling of Multiphase Combustion (CM³C)

TST Review April 1-2, 2026, Attendee List

Faculty

S. Balachandar “Bala”	University of Florida	bala1s@ufl.edu
Spencer Bryngelson	Georgia Tech	shb@gatech.edu
James Fairbanks	University of Florida	fairbanksj@ufl.edu
Daniel Guildenbecher	Purdue University	dguilden@purdue.edu
Joel Harley	University of Florida	joel.harley@ufl.edu
Ryan Houim	Purdue University	rhouim@purdue.edu
Tom Jackson	University of Florida	tlj@ufl.edu
Nam-Ho Kim	University of Florida	nkim@ufl.edu
Alina Zare	University of Florida	azare@eng.ufl.edu
ST Thakur	University of Florida	sst@ufl.edu

Alliance Strategy Team (AST)

Tim Germann (Chair)	LANL	tgg@lanl.gov
Erik Draeger	LLNL	draeger1@llnl.gov
Sriram Swaminarayan	LANL	sriram@lanl.gov
Joshlynn Christie	NNSA HQ	joshlynn.christie@nnsa.doe.gov

Tri-Lab Sponsor Team (TST)

Flint Pierce (Chair)	SNL	fpierce@sandia.gov
Georges Akiki	LLNL	akiki1@llnl.gov
Bradford Durant	LANL	bdurant@lanl.gov
Christopher Malone	LANL	cmalone@lanl.gov
Carol Woodward	LLNL	woodward6@llnl.gov

Research Staff

Jose Aguilar	University of Florida	j.aguilar@ufl.edu
BC Kim	University of Florida	bck505@naver.com
Erica Lindbeck	University of Florida	elindbeck@ufl.edu
Colton Willhardt *	Purdue University	cwillhar@purdue.edu
Jacob Posey *	Purdue University	jwposey@purdue.edu

Administration Staff

Linda Curwick	University of Florida	lindacurwick@ufl.edu
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* Unable to attend



Center for Multiscale Modeling of Multiphase Combustion (CM³C)

Students in Attendance

Davin Anderson	University of Florida	andersondavin@ufl.edu
Thierry Daoud	University of Florida	t.daoud@ufl.edu
Joshua Gillis	University of Florida	joshuagillis@ufl.edu
K. Krishnaprasad	University of Florida	krishnap.kalivel@ufl.edu
Cameron Lott	University of Florida	lottc@ufl.edu
Fahad Nabid	University of Florida	fahadnabid@ufl.edu
Mike VandenBoom	University of Florida	m.vandenboom@ufl.edu
Matthew VandenBoom	University of Florida	mvandenboom@ufl.edu
Matthew Cuffaro	University of Florida	cuffaro.m@ufl.edu
Joana Bou Barcelo	University of Florida	jboubarcelo@ufl.edu
Luke Morris	University of Florida	luke.morris@ufl.edu
George Rauta	University of Florida	grauta@ufl.edu
Richard Samuelson	University of Florida	rsamuelson@ufl.edu
Isaly Tappan	University of Florida	i.tappan@ufl.edu
Zhihui Tian	University of Florida	zhihui.tian@ufl.edu
Khanh Tran	University of Florida	khanhtran@ufl.edu
Sam Gallic	University of Florida	samgallic@ufl.edu
Juan Diaz	University of Florida	jdiaz1@ufl.edu
Alex Millauer-Hsu	Purdue University	amillaue@purdue.edu
Daniel Vickers	Georgia Tech	dvickers6@gatech.edu

Students Unable to Attend

Avery Couture	University of Florida	avery.couture@ufl.edu
Dimitrios Adam	Georgia Tech	dadam7@gatech.edu
Benjamin Wilfong	Georgia Tech	bwilfong3@gatech.edu
Anand Radhakrishnan	Georgia Tech	aradhakr34@gatech.edu
Cohen Nunes	Purdue University	cnunes@purdue.edu
Amogh Natu	Purdue University	natua@purdue.edu
Rees Verleur	Purdue University	rverleur@purdue.edu
Gregory O'Brien	Purdue University	greg99@purdue.edu
Chrissie Spangler	Purdue University	spanglc@purdue.edu



CM³C

CM³C: Center for Multiscale Modeling of Multiphase Combustion

T1: Overview

Principal Investigator: S. Balachandar

Department of Mechanical and Aerospace Engineering



Our PSAAP-IV Team

CM³C Physics and Simulations



S. Balachandar
(UF)



Thomas Jackson
(UF)



Ryan Houim
(Purdue)

Exascale Computing



Spencer Bryngelson
(GTech)

Experiments



Daniel Guildenbecher
(Purdue)

V&V UQ and Workflow



Nam-Ho Kim
(UF)



James Fairbanks
(UF)

Machine Learning



Alina Zare
(UF)



Joel Harley
(UF)

Partners



Rodney Fox
(Iowa State)



Brian Taylor
(Eglin AFB)



Don Littrell
(Eglin AFB)



John Spencer
(NSWC Crane)

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Agenda to Follow – What to Watch For

- T2 – MFC for PR-DNS (**B.C. Kim**)
- T3 – MFC for EL Simulations (**Jose**)
- T4 – MFC at Exascale (Spencer)
- T5 - Reactive Flow Simulations (Ryan)
- T6 – Experimental plan (Dan)
- T7 – Rigorous EL Closure (**Thierry**)
- T8 – Generative ML (**Erica**)
- T9 – Operator Learning ML (**Cameron**)
- T10 – IB in MFC (**Daniel**)
- T11 – Group combustion modeling (**Davin**)
- T12 – Carbon rod experiments (**Alex**)
- T13 – UQ in ML and PINNs (**Isaly and Juan**)
- T14 – UQ for the center (Nam Ho)



T2. BC Kim, UF



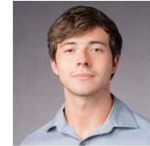
T3. Jose Aguilar, UF



T7. Thierry Daoud, UF



T8. Erica Lindbeck, UF



T9. Cameron Lott, UF

T10. Daniel Vickers
Georgia TechT11. Davin
Anderson, UFT12. Alex Millauer
Hsu, PurdueT13a. Juan Diaz
UFT13b. Isaly Tappan
UFCM³C

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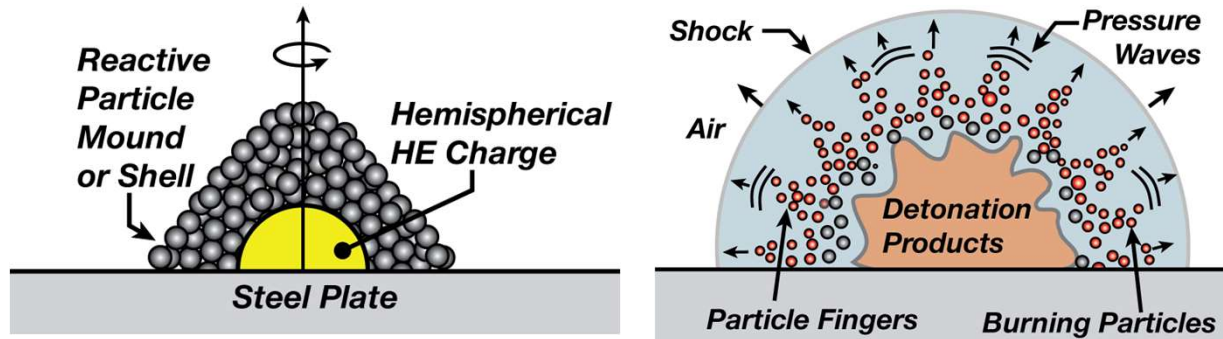
Outline

- Go over the challenges of our demonstration problem
- Revisit technical objectives and scientific goals
- Our closure modeling philosophy
- Multiscale modeling by interweaving simulations and ML modeling
- Experiments, validation, and uncertainty
- What we have accomplished so far
- What lies ahead

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Demonstration Problem



- Yearly integrated simulations starting from year-1
- Validation experiments & measurements at Purdue (and partners)

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CM³C – Intellectual Objectives

Center for Multiscale Modeling of Multiphase Combustion (CM³C)

- Radically advance the understanding of *Group Combustion Physics* built upon hierarchical physics-informed ML models with uncertainty quantification
- Develop a trace-based load-balancing strategy for multiphase simulations using the proven exascale code MFC (Multicomponent Flow Code)
- Develop workflow strategies for seamless integration of experiments, simulations, ML models, and UQ across scales
- Perform validation-quality demonstration, meso and microscale experiments for relay validation and uncertainty reduction
- Educate students and postdocs in exascale simulation science and place them at NNSA laboratories

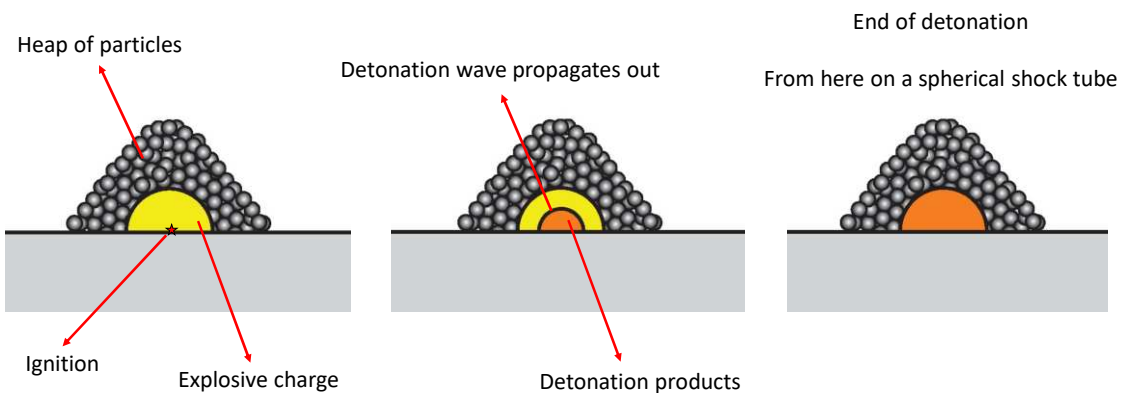
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Our Single Most Lasting Contribution After 5 Years

Revolutionary Modular Generator Superposition (MGS) framework that melds exascale simulations and modern ML tools for “best-possible” closures to be used with unprecedented confidence in engineering analysis of multiphase flows

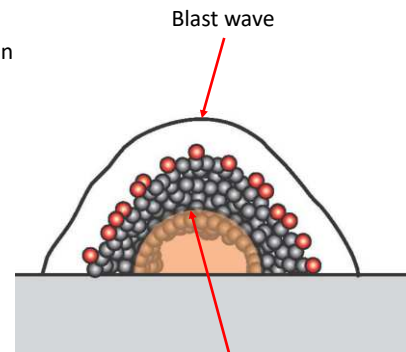
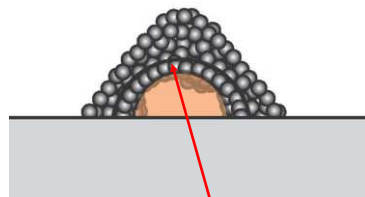
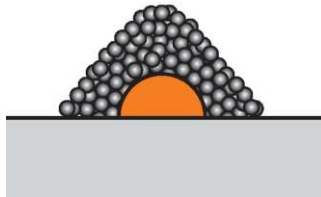
Demonstration Problem – Detonation Phase



Demonstration Problem – Compaction Phase

Three propagating waves

- Blast wave – compacts the bed
- Contact discontinuity – adds to compaction
- Expansion fan – expands the gas/particles



Blast wave

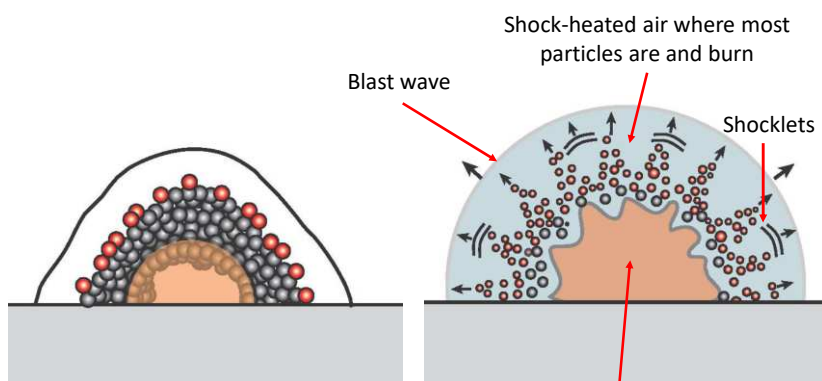
Air & blast-product contact interface

Only the inner most particle layer is strongly compacted

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Demonstration Problem – Dispersal Phase



Blast wave

Shock-heated air where most particles are and burn

Shocklets

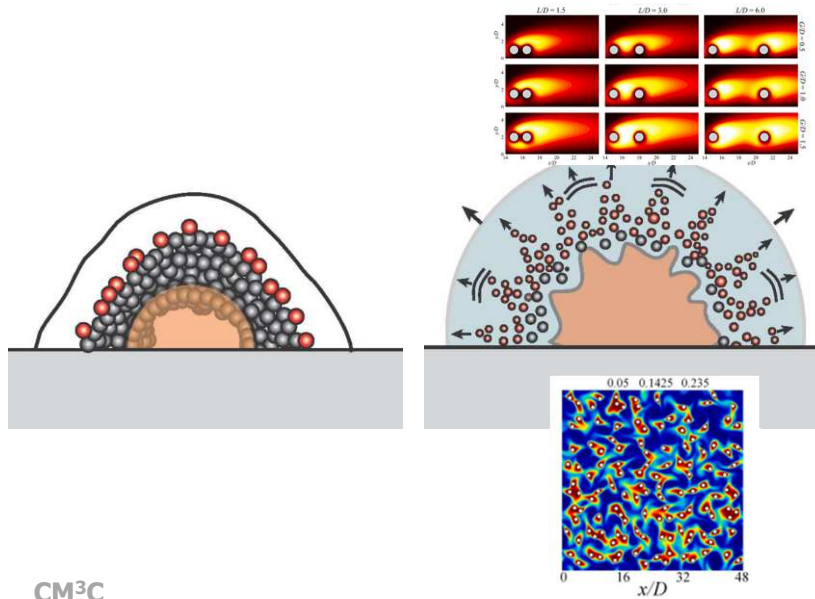
Blast-product undergoing secondary combustion

Few particles remain and burn here

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Importance of Group Ignition & Combustion

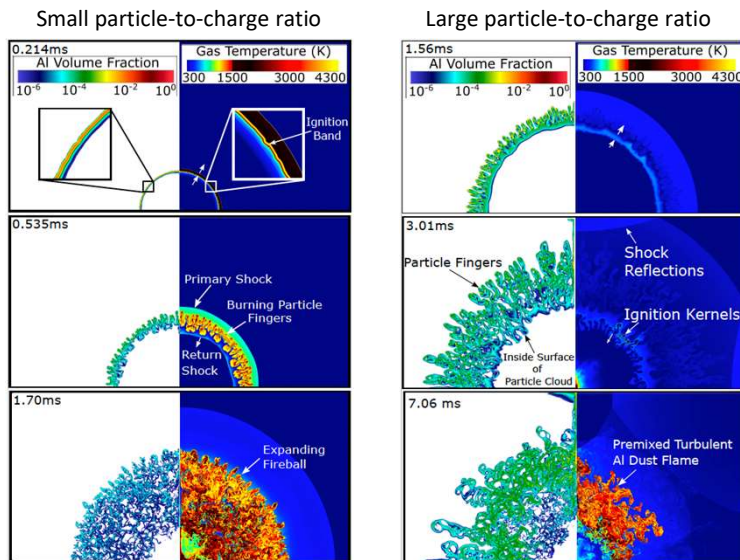


- Particles are non-uniformly distributed (fingering instability)
- It matters where the particles are (shock-heated vs det-prod)
- Ignition of one particle can trigger ignition of the cluster
- A burning particle increases temperature, but decreases oxidizer in the wake
- Burn rate of a cluster is vastly different from that of an isolated particle

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Global Influence – Diverse Large-scale Behaviors



- Coarse grained Euler-Euler simulations of Al particles
- Left: small particle-to-charge ratio
 - Prompt ignition
 - Non-premixed burning
- Right: large particle-to-charge ratio
 - Delayed ignition
 - Premixed burning



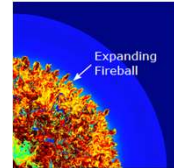
Ryan - See T5

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Technical Objectives & QoI

- Use exascale to predict (approach first-principle approach in accuracy)
 - Non-uniform distribution of explosively dispersing particles
 - Resulting spatio-temporal variation of burn-rate
 - Multiphase afterburn effect on blast dynamics
- Impact the predictive capability of engineering tools
 - With the “Best-possible” closure models



Quantities of Interest (QoI)

QoI1: Time-dependent blast location

QoI4: Temp and heat release rate

QoI2: Time-dep pressure/impulse at probes

QoI5: Number & amplitude of instabilities

QoI3: Volume fraction of reactive particles

QoI6: PDFs of pressure & volume fraction

Basic metrics

Higher-order metrics

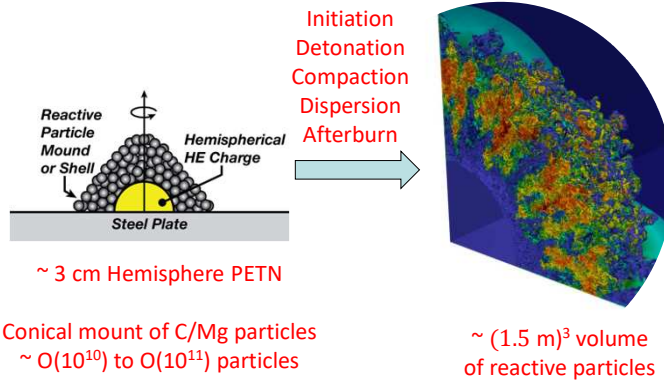
Science Questions

- How best to incorporate group combustion physics in terms of particle-gas-particle interactions $\Rightarrow$ *N-body combustion interaction*
- What is the role of shock propagation, diffraction, and focusing on group ignition and combustion $\Rightarrow$ *unsteady effects of N-body interaction*
- How clustering, fingering, and channeling instabilities alter and altered by group combustion $\Rightarrow$ *Two-way coupling between instabilities and group combustion*
- What is the role of early receptivity and subsequent growth of Rayleigh-Taylor (RT) and Richtmyer-Meshkov (RM) instabilities on group combustion

The fundamental understanding of group combustion and their translation into closure models will have broad impact, including their potential incorporation into NNSA codes (e.g., ALE3D, CTH, and RAGE)

Proposed Strategy & Rationale

Demonstration Macroscale Problem

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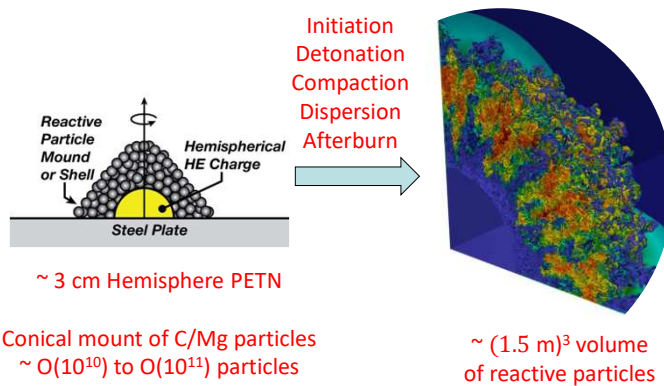
Modeling & Simulation Challenges

- Stiff chemical kinetics
 - Numerous reactions & species
 - Surface/gas-phase, premixed/non-premixed
- High-speed flow
 - Intense shocks and other discontinuities
 - Non-classical turbulence
 - Real gas EOS and properties
- Multiphase
 - Trillions of time-varying polydisperse particles
 - Direct particle-particle collisions
 - Droplet deformation and break-up

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What Can Be Simulated at Exascale

Demonstration Macroscale Problem

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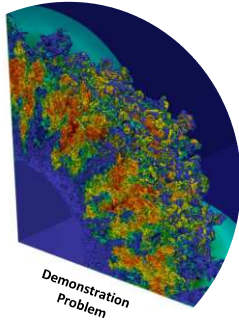
- Fully-resolved DNS
 - Full chemistry, particles, turbulence
- Euler-Lagrange (EL) LES
 - $O(10^{12})$ grid at $\sim 300 \mu\text{m}$ resolution
 - $O(10^{10})$ to $O(10^{11})$ particles
 - $O(10)$ to $O(100)$ chemical reactions
 - JWL or BKW EOS, variable properties
- Euler-Euler (EE) LES/URANS
 - $O(10^8)$ grid at 6 mm resolution
 - No particle tracking
 - ...



DNS: Direct numerical simulation; LES: Large eddy simulation; URANS: Unsteady Reynolds averaged Navier Stokes

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Two Technical Objectives



Develop “best-possible” EL closure models, so that exascale- LES is “statistically” as good as HF-PR

Develop “best possible” EE closure models to impact engineering simulations

- Fully-resolved DNS
 - Full chemistry, particles, turbulence
- Euler-Lagrange (EL) LES
 - $O(10^{12})$ grid at 300 μm resolution
 - $O(10^{10})$ to $O(10^{11})$ particles
 - $O(10)$ to $O(100)$ chemical reactions
 - JWL or BKW EOS, variable properties
- Euler-Euler (EE) LES/URANS
 - $O(10^8)$ grid at 6 mm resolution
 - No particle tracking
 - ...

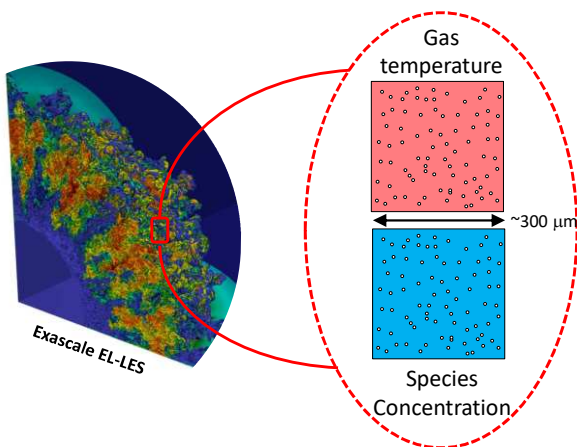
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DNS: Direct numerical simulation; LES: Large eddy simulation; URANS: Unsteady Reynolds averaged Navier Stokes

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Rigorously Derived EL Equations

Quantities computed within a RVE
Exascale EL-LES (coarse-grained)



Filtered LES equations

Particle quantities (Lagrangian)

Other LES parameters

$$\bar{\mathcal{F}}(\bar{\mathbf{u}}(\mathbf{x}, t), \mathbf{v}(j, t), F(j, t), \lambda) = 0$$

Fluid quantities (Eulerian)

Phase-coupling closure quantities

- Force, torque on each particle
- Evaporation/burn rate of each particle

Subgrid closure quantities

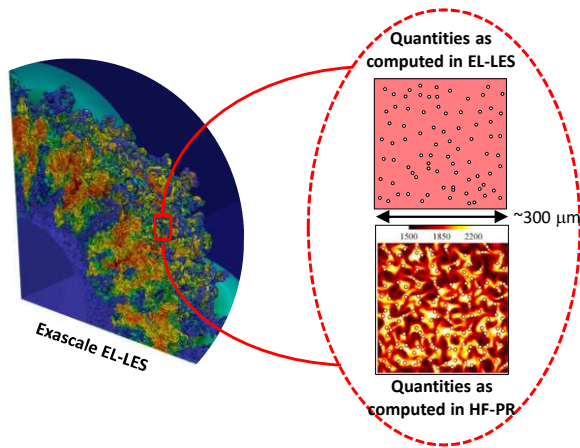
- Subgrid Reynolds stress
- Subgrid energy flux
- Subgrid kinetic energy flux

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RVE: Representative volume element = one $(\sim 300 \mu\text{m})^3$ grid cell

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Traditional Closure Models



EL closure quantities

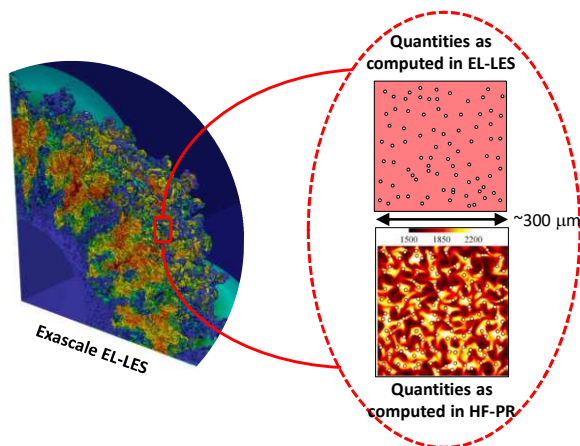
$$\bar{\mathcal{F}}(\bar{\mathbf{u}}(\mathbf{x}, t), \mathbf{v}(j, t), F(j, t), \lambda) = 0$$

- Phase-coupling (drag, burn rate) models
 - Based on an isolated particle in steady uniform flow
 - Only average influence of neighbors, assume uniform particle distribution
- Subgrid (Reynolds stress, flux) models
 - Assume Smagorinsky model
 - Ignore the effect of burning particles

CM³CRVE: Representative volume element = one ($\sim 300 \mu\text{m}$)³ grid cell

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Four Key Principles Make Our Approach Unique



EL closure quantities

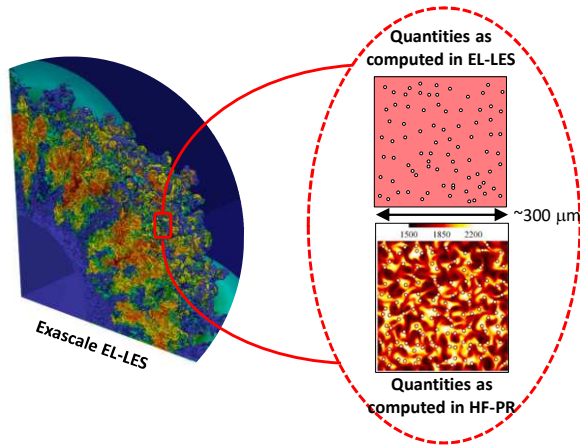
$$\bar{\mathcal{F}}(\bar{\mathbf{u}}(\mathbf{x}, t), \mathbf{v}(j, t), F(j, t), \lambda) = 0$$

- 1) Filtered EL-LES are rigorous. So we seek
Best-possible $F(j, t)$ = Best-possible EL-LES
 - 2) Lagrangian-based closure is ideal for $F(j, t)$
 - 3) $F(j, t)$ has both deterministic (pseudo-turbulence) and stochastic (classic turbulence) contributions
 - Deterministic invertible $\rightarrow$ Epistemic uncertainty
 - Stochastic is non-invertible $\rightarrow$ Aleatoric uncertainty
- Closure must be modelled as a PDF: $P(F)$
- 4) All the closures must be modelled together

CM³CRVE: Representative volume element = one ($\sim 300 \mu\text{m}$)³ grid cell

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Closure Operator Is Very Complex



EL closure quantities modeled as functions of

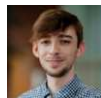
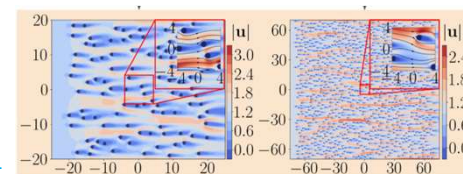
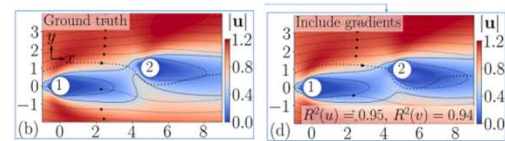
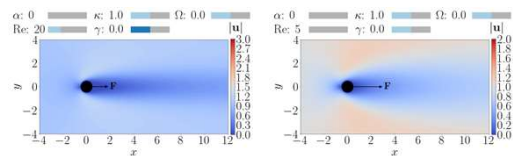
$$\bar{\mathcal{F}}(\bar{\mathbf{u}}(\mathbf{x}, t), \mathbf{v}(j, t), P(F)(j, t), \lambda) = 0$$

- Mesoscale information within RVE (λ):
 - $\bar{\mathbf{u}}, \bar{T}, \bar{p}, \bar{c}_{O_2}, \dots$, particle volume fraction
 - Spatio-temporal gradients (i.e. $\nabla \bar{\mathbf{u}}$ and $\partial \bar{\mathbf{u}} / \partial t$)
- Micro-information of N nearest neighbors (I):
 - Relative position, velocity, rotation
 - Temperature, burn rate, etc.,

There exists a closure operator that we want to model/learn } $\mathcal{G}(P(F), \lambda, I) = 0$

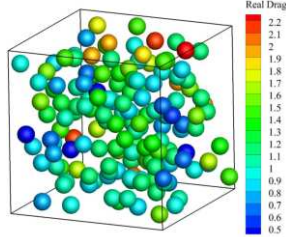
Key Innovation: Modular Operator Superposition (MOS)

- Use ML to rapidly predict the fields around a particle under “all” possible ambient conditions (ML Modules – Aka, Unary Solution)
- Superpose the solution for a local cloud of particles and iterate to convergence (Superposition)
- Embed the superposed-solution into a filtered EL simulation (Embedding)

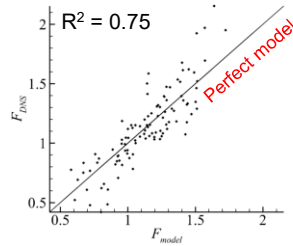


Davin - See T11

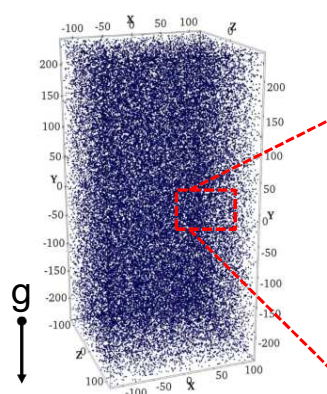
Elementary Modules + Superposition + EL Embedding



Akiki et al 2017

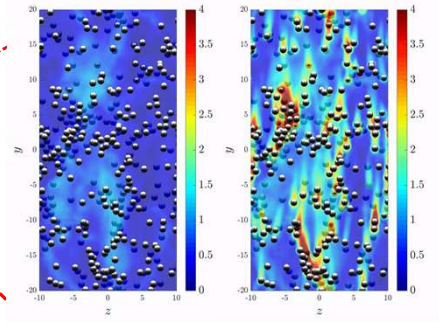
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$$\rho_p/\rho_f = 1.5; Re_p \sim 238$$

Moore & SB 2019
SB et al., 2020

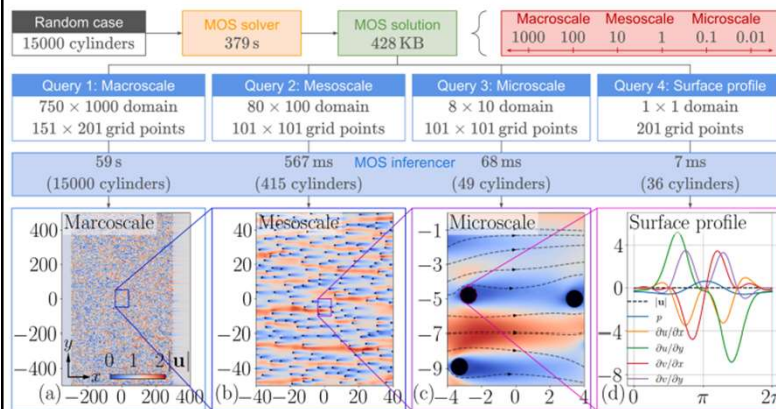
❖ Geometry & Particles

- Domain: $L_x = 240, L_y = 480, L_z = 240$

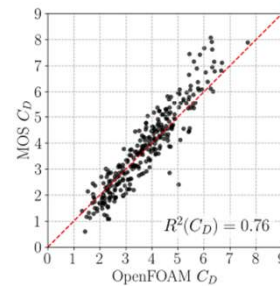
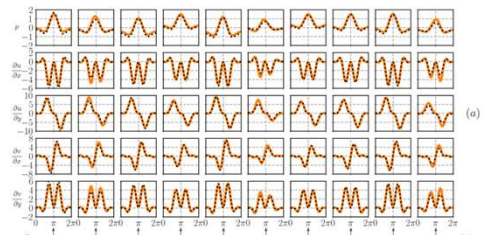
EL - uses
Force
predictionWith flow
prediction
Effective grid = 226 B

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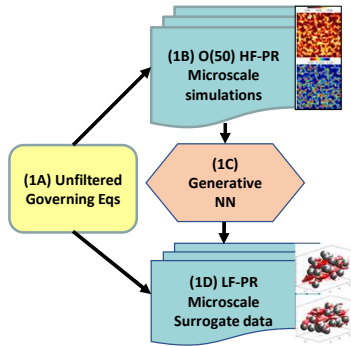
Elementary Modules + Superposition + EL Embedding



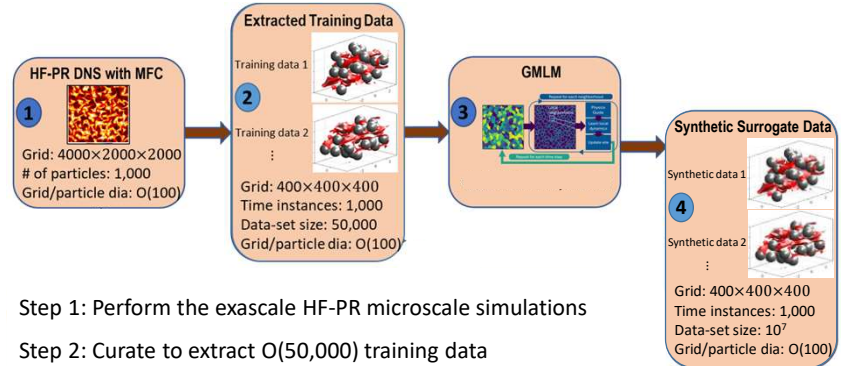
MOS, Liu, K., et al. (2026). JCP, 544, 114435.

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Machine Learning (ML) for Best-Possible EL-LES



Generative ML Model (GMLM) – Training/testing using HF-PR data



Step 1: Perform the exascale HF-PR microscale simulations

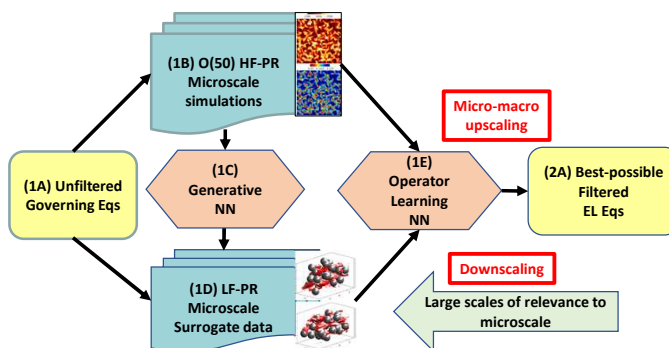
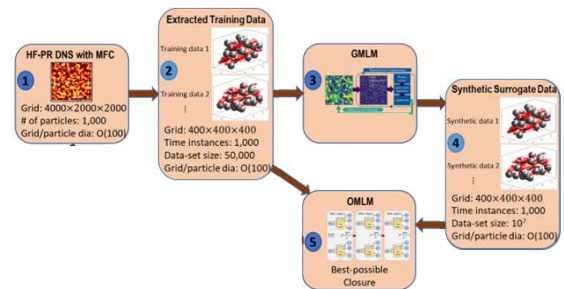
Step 2: Curate to extract O(50,000) training data

Step 3: Train the GLML

Step 4: Deploy the trained GMLM to generate O(10⁷) LF surrogate dataCM³C

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Machine Learning (ML) for Best-Possible EL-LES

EL closure: $\mathcal{G}(P(F), \lambda, I)$ are operators that we are learningSpace of (λ, I) is huge. Includes, gas velocity, temperature, species, and their space-time gradients, all neighboring particlesCM³COperator Learning ML Model (OMLM)
Training/testing using HF-PR & LF-PR data

Step 1: Perform the exascale HF-PR simulations

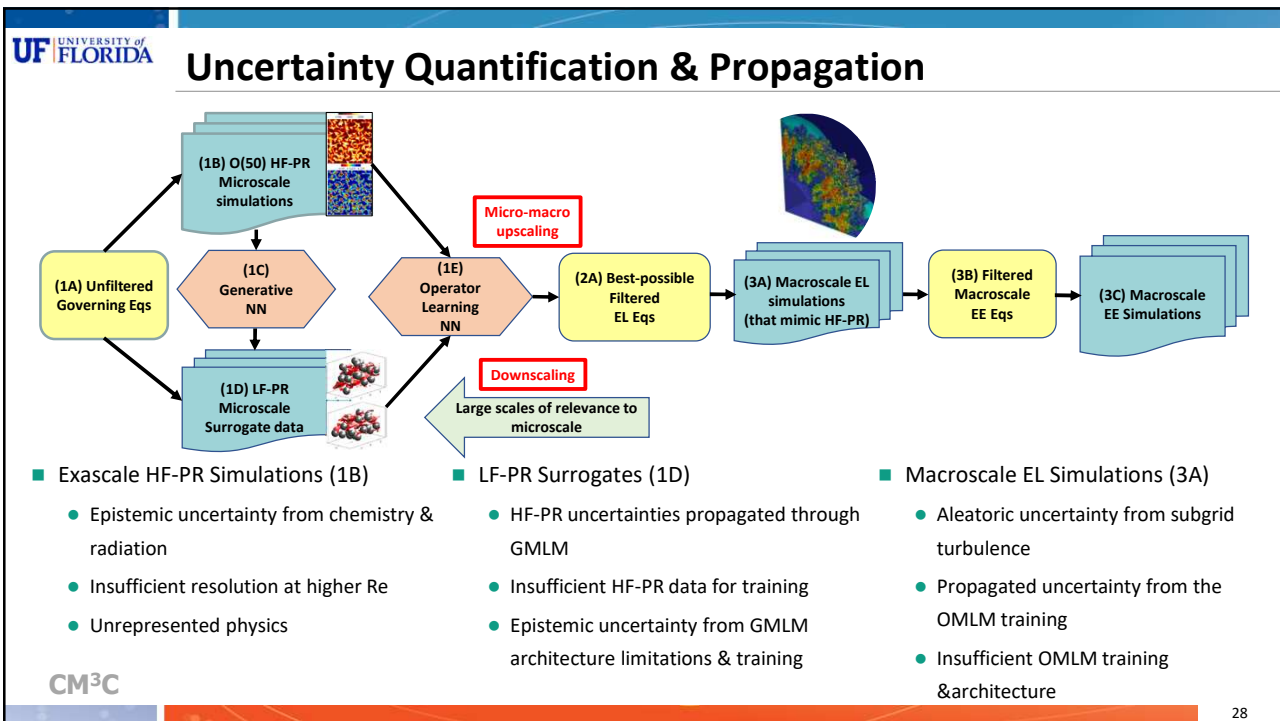
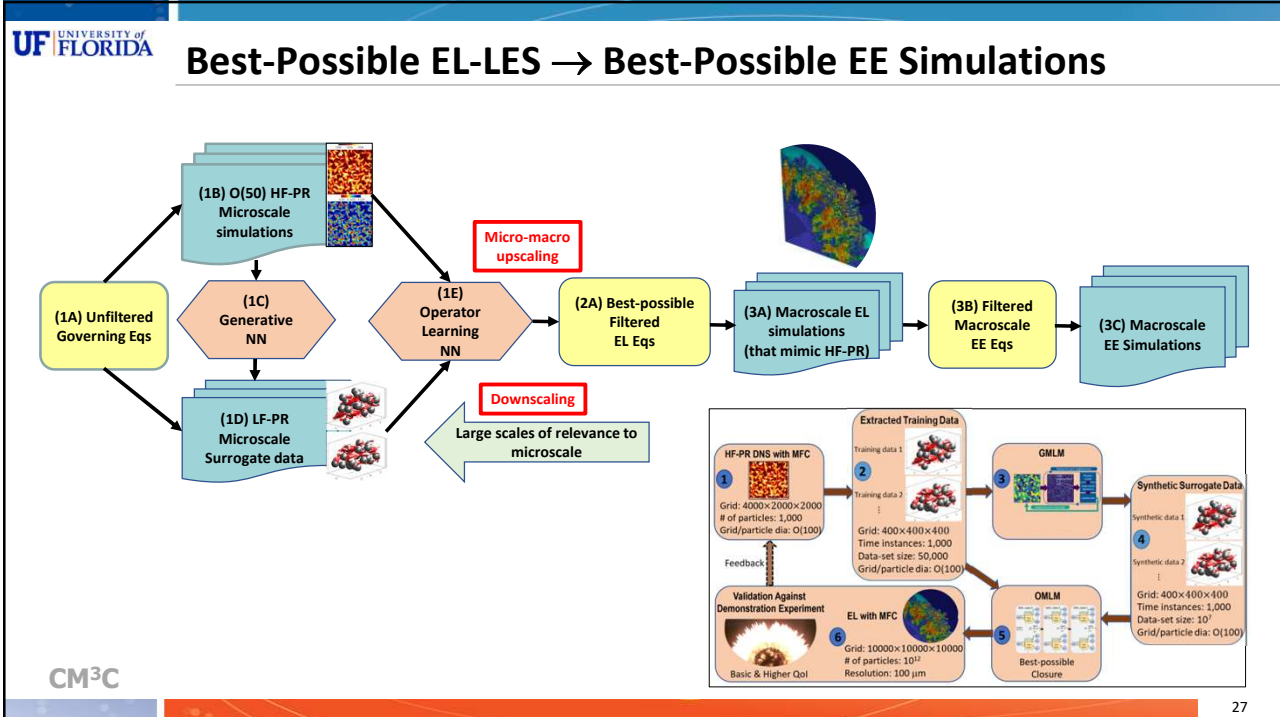
Step 2: Curate to extract training data

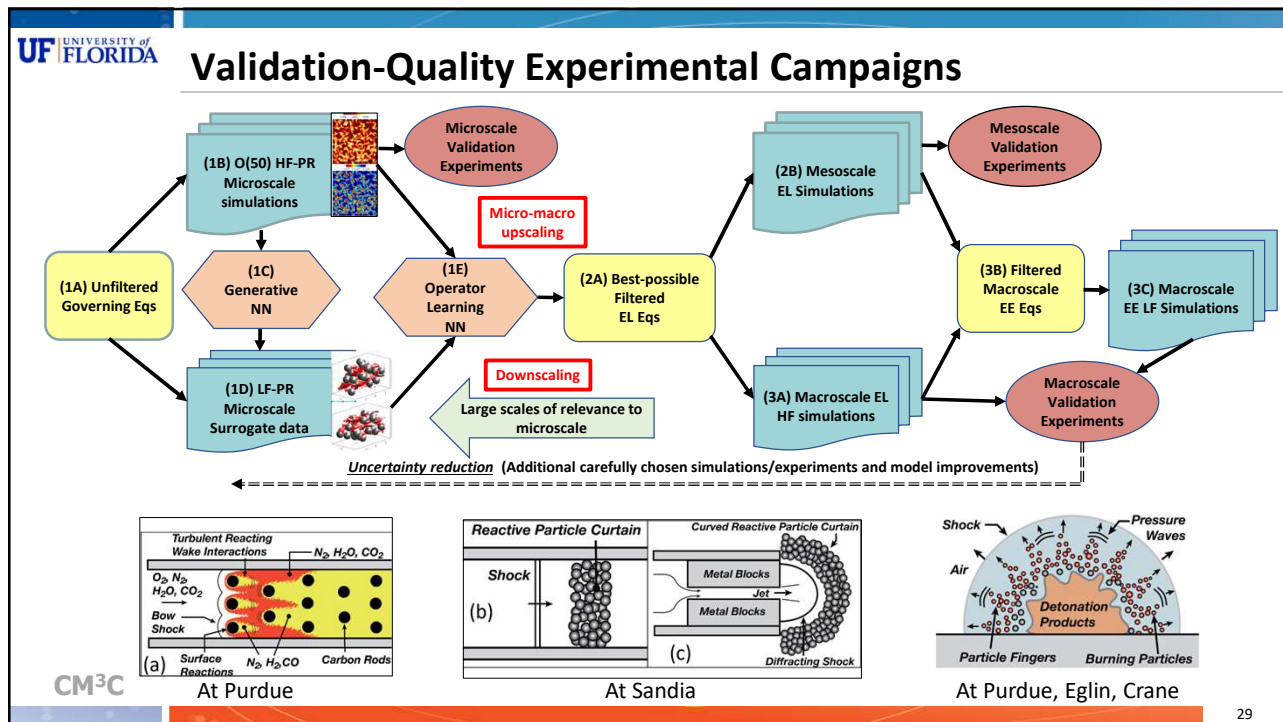
Step 3: Train the GLML

Step 4: Deploy GMLM to generate LF surrogate data

Step 5: Use HF-PR and LF-surrogate to develop OMLM

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Accomplishments To Date

- Putting the contract in place – was not easy
- Recruiting the best talent
 - We have an outstanding group of grad students and postdocs
- MFC training at UF & performing particle-resolved (PR) and EL simulations
 - MFC was/is an awesome code, but was not ready for our problem
 - Performing high-speed PR simulations over $O(10^3)$ particles
 - Capability demonstration + State-of-the-art ML models

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Daniel - See T10

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Accomplishments To Date

- Putting the contract in place – was not easy
- Recruiting the best talent
 - We have an outstanding group of grad students and postdocs



- Performing shock-curtain EL simulations with $O(10^5)$ particles
 - Implemented interpolation, projection, particle-particle collisions
 - Will be soon ready to receive ML-based closure models



Jose - See T3

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Accomplishments To Date

- Putting the contract in place – was not easy
- Recruiting the best talent
 - We have an outstanding group of grad students and postdocs
- MFC training at UF & performing particle-resolved (PR) and EL simulations
 - MFC was/is an awesome code, but was not ready for our problem
 - Performing high-speed PR simulations over $O(10^3)$ particles
 - Capability demonstration + State-of-the-art ML models
 - Performing shock-curtain EL simulations with $O(10^6)$ particles
 - Implemented interpolation, projection, particle-particle collisions
 - Will be soon ready to receive ML-based closure models
- Initiated the carbon rod experiments and preparatory macro experiments
 - Macro simulations helping design of macro experiments



Alex - See T12



Dan - See T6

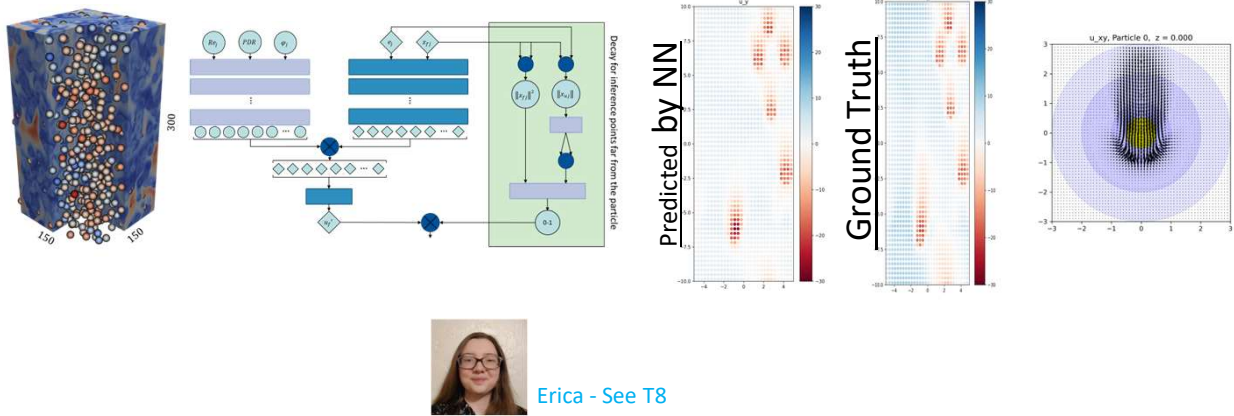
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Accomplishments To Date

- First round of generative ML models are ready (data driven & PINNs)
 - Developed using existing data similar to those being developed

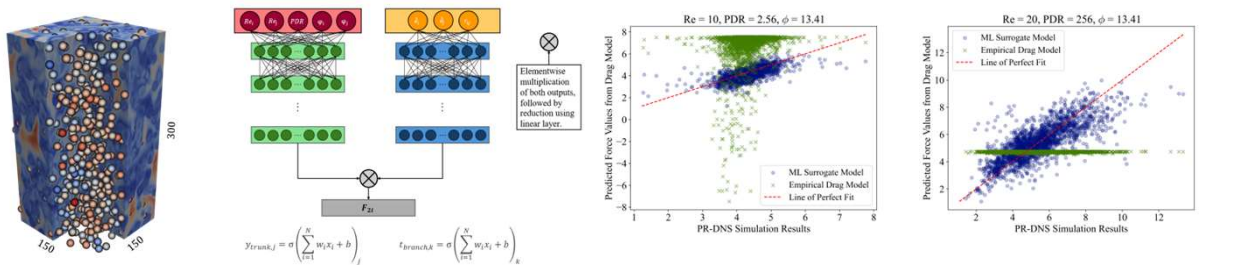


33



Accomplishments To Date

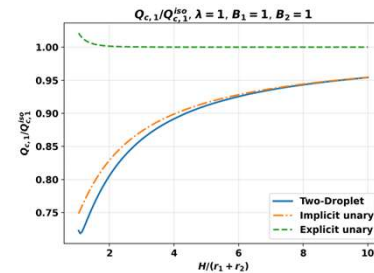
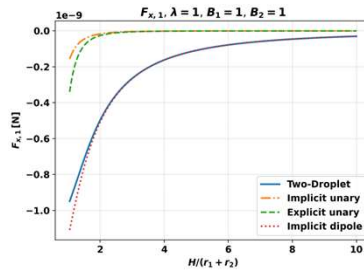
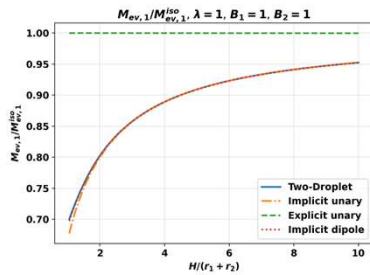
- First round of generative ML models are ready (data driven & PINNs)
 - Developed using existing data similar to those being developed
- First round of operator learning ML models are ready
 - Developed using the same existing data
 - Uses the same neural network configuration as generative ML



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Accomplishments To Date

- Unitary “Generative Solution” implicit superposition in group combustion
 - Very exciting results for Stefan dominated droplet burning
 - Directly applies to all ML modeling efforts
 - Extending the theory to more complex group combustion with PR simulations

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Davin - See T11

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Accomplishments To Date

- Unitary solution & implicit superposition in group combustion
 - Very promising results for Stefan dominated flow
 - Extending to more complex cross flow
 - Will follow-up with PR simulations
- Uncertainty quantification pipeline for ML being formulated and implemented
 - First implementation will be for data-driven generative ML
 - UQ pipeline will be extended for ML models
- Uncertainty quantification an integral part of carbon rod experiments/simulations



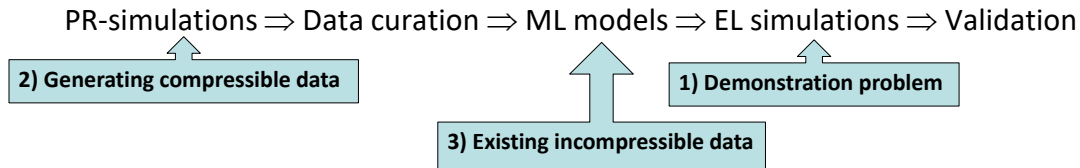
Nam Ho - See T14

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Work/Dataflow and Science Problems

■ Staggered workflow and dataflow



■ Sequence of problems

- ML models with existing incompressible PR-DNS data
- Generating new high-speed and shock particle interaction PR-DNS data
- Next step will be group combustion PR-DNS data

■ Each team's responsibilities are broken into internal workflow

- PR-DNS, ML and UQ Pipelines/tools will be established, so that they can be reused

Biggest Challenges Ahead

- Reduced order chemistry has too much uncertainty
 - Experimental measurements are hard or chemistry is too complex
- Radiation between particles (or rods) is non negligible
- Initial compaction phase introduces complex unaccounted physics
- Load-balanced exascale EL implementation in MFC
- Not enough PR data for Generative ML model
- Implicit superposition model fails for finite Re combustion
- Generative + Operator Learning models introduce too much uncertainty
- Availability of computing resources



Agenda to Follow – What to Watch For

- T2 – MFC for PR-DNS (B.C. Kim)
- T3 – MFC for EL Simulations (Jose)
- T4 – MFC at Exascale (Spencer)

- T5 - Reactive Flow Simulations (Ryan)
- T6 – Experimental plan (Dan)
- T7 – Rigorous EL Closure (Thierry)
- T8 – Generative ML (Erica)
- T9 – Operator Learning ML (Cameron)

- T10 – IB in MFC (Daniel)
- T11 – Group combustion modeling (Davin)
- T12 – Carbon rod experiments (Alex)
- T13 – UQ in ML and PINNs (Isley and Juan)
- T14 – UQ for the center (Nam Ho)

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Questions?



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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T1: Management Plan

Technical Manager: Thomas L. Jackson

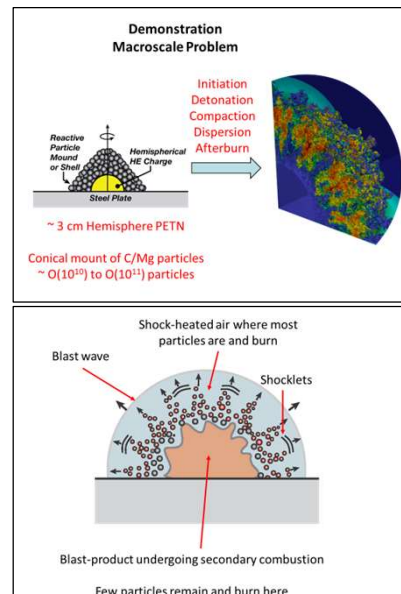
Department of Mechanical and Aerospace Engineering



UF UNIVERSITY of FLORIDA

Outline

- Kickoff Meeting
- Organization
- Partnership management plan

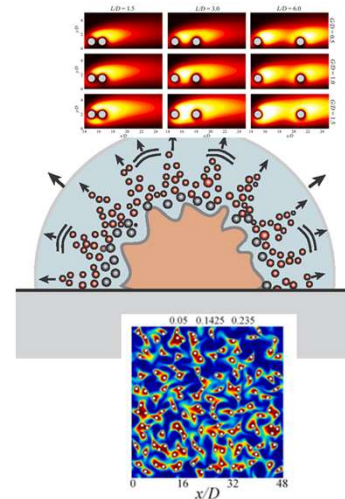


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Internal Kick-off Meeting

- December 23, 2025
- 2-hr Zoom
- 28 in attendance
- Overview presentation
 1. History of DOE/PSAAP program
 2. Big-picture view
 3. Key science questions
 4. Introduction of Teams/Tasks
 5. Purpose of weekly student-led meetings
 6. Outline of short-term tasks/goals (*Bala's talk*)
- Round-robin of introductions by all present



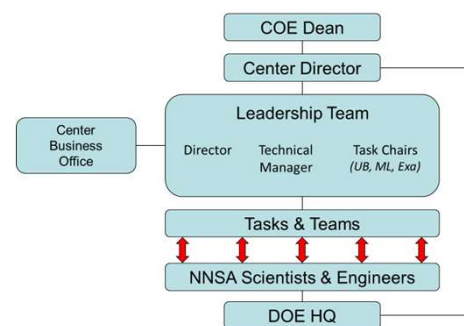
Concept of Group Combustion

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Organizational Chart & Leadership Team

- Primary responsibilities
 - Set priorities consistent with vision and goals
 - Monitor timely progress towards complete operational capability
 - Monitor progress towards integrated simulations
 - Make critical decisions on resource allocation
 - Review and promote effective implementation of NNSA laboratory interaction plan
 - Oversee student internships at NNSA
 - Implementation of educational policies
 - Oversee recruitment, training and placement of students and postdocs



Organizational Chart for CM³C

CM³C

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Technical Manager and Staff

■ Primary responsibilities of Technical Manager

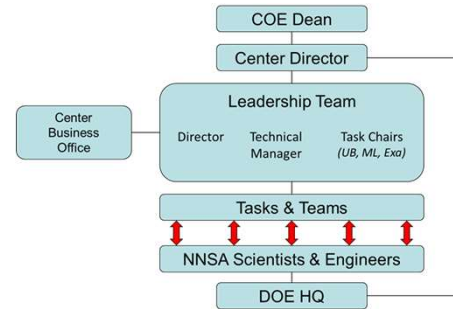
- Day-to-day activities of center
- Budget
- Resource planning/allocation
- Organize all reviews, reports, etc

■ Technical staff – Postdocs (~3)

- Engaged in center-wide activities focused on proposed cross-cutting tasks
- PD1 - physics modules, MFC, exascale simulations
- PD2 – ML
- PD3 - micro/mesoscale simulations, ML, VVUQ

PD1 – Jose Aguilar
PD2 – Erica Lindbeck
PD3 – BC Kim

PD4 – Colton Willhardt
PD5 – Jacob Posey



Postdocs are center resources

Role of Participants

Name	Role	Tasks	Expertise
Balachandar	Director, Phys-S, CD-N, ML	T1, T2, T3	Comp. Multiphase turbulence, ML
Bryngelson	CD-N, CD-CS	T1, T5	Comput. physics, num. methods, HPC
Zare	ML	T2, T5	Machine learning, pattern recognition
Houim	Phys-S, CD-N	T1, T3, T5	Multiphase reactive flow, num. methods
Harley	ML, CD-CS	T2, T5	Large data science, machine learning
Guildenbecher	Phys-E	T1, T3	Multiphase flows, imaging diagnostics
Jackson	Tech. Manager	T1, T3	Combustion, multiphase flow
Kim	VVUQ	T3, T4	Design optimization, sensitivity, UQ
Fairbanks	CS-WF, CD-CS	T4, T5	Computational mathematics, informatics
Postdoc #1	CD-N, Phys-S, CD-CS	T1, T5	Comp. science, physics background
Postdoc #2	ML, CD-CS	T2, T4	ML, physics or CS background
Postdoc #3	CD-CS, Phys-S	T1, T3	HPC expertise in physics simulations

Integration of Tasks and Teams

Tasks

- T1: Exascale simulations and experiments of demonstration problem (TM)
- T2: Integration of ML for scale-bridging (PD2)
- T3: UQ-driven micro and mesoscale investigations (PD3)
- T4: Multi-fidelity data/workflow (PD2)
- T5: Load balancing for exascale performance (PD1)

	Phys-S&E Director, TM Guldenbecher Houim Students	VVUQ/WF Kim, Fairbanks Students	ML Zare, Harley Director Students	CD-N Houim Bryngelson Director Students	CS-Exa Bryngelson Harley Students
Task 1 Lead: TM	X	X		X	X
Task 2 Lead: PD2	X	X	X		X
Task 3 Lead: PD3	X	X	X	X	
Task 4 Lead: PD2	X		X		X
Task 5 Lead: PD1				X	X

Teams

1. Phy-S&E Physics – Simulations and Experiments
2. ML Machine learning
3. CD-N Co-Design – Numerics
4. CS-Exa Computational Science – Exascale
5. VVUQ/WF Verification, Validation and Uncertainty
Quantification/Workflow

The Center will be organized by physics-based tasks and cross-cutting teams, rather than by faculty and their research groups

Tasks and Teams

1. Weekly student/staff-led team interactions
 - a. See the “big picture”
 - b. Ownership of their problem
 - c. Become invested in the success of the center!
 - d. Students are quick to help others
 - e. Sense of belonging
 - f. New students are quickly integrated
2. Team meetings include students, staff, and faculty
3. All staff and large number of graduate students are co-located (4th floor of NEB)
4. Team meetings held in Student Village conference room, 4th floor NEB

Tasks	T1	T2	T3	T4	T5
Teams					
Phys-S&E	X	X	X	X	
ML		X	X	X	
CD-N	X		X		X
CD-CS	X	X		X	X
VVUQ	X	X	X		

Weekly Student-led Team Interactions

Tuesday Meetings - Tasks and Teams

1. Weekly combined in-person (UF) and zoom (Purdue, GT)

a. Noon – 1 pm: **MFC/Physics**

- 3 GT + 6 UF + 3 Purdue

b. 1 pm – 2 pm: **Experiments/Group Combustion**

- 8 Purdue + 6 UF

c. 2 pm – 3 pm: **ML/VVUQ**

- 10 UF

Tasks	T1	T2	T3	T4	T5
Teams					
Phys-S&E	X	X	X	X	
ML		X	X	X	
CD-N	X		X		X
CD-CS	X	X		X	X
VVUQ	X	X	X		

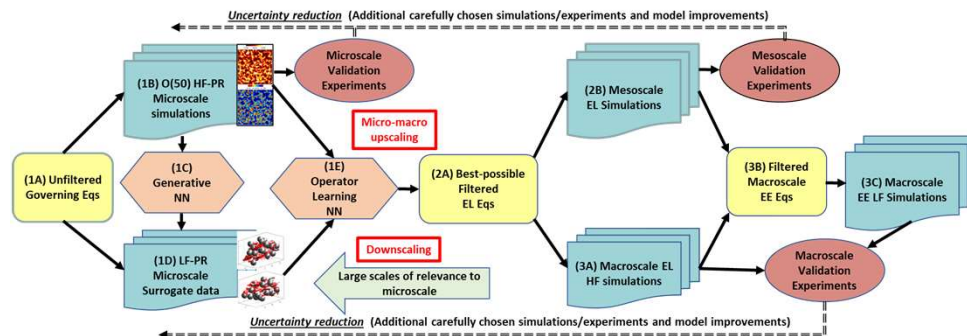
2. However, after TST there will be some re-organization

a. Algorithms for MFC/group combustion

b. Workflow

Weekly Student-led Team Interactions

Integration of Tasks and Teams



HF-PR

- BC Kim
- Mike VandenBoom
- Fahad Nabid
- Davin Anderson

ML

- Isaly Tappen
- Zhihui Tian
- Khanh Tran
- Sam Gallic
- Erica Lindbeck
- Noah Summerlin

EL/EE

- Jose Aguiar
- Thierry Daoud
- Joshua Gillis
- Jacob Posey
- Gregory O'Brien
- Chrissie Spangler

MFC

- Ben Wilfong
- Dan Vickers
- Thierry Daoud
- Joshua Gillis
- Avery Couture
- Matthew VandenBoom

Exp

- Alex Millauer-Hsu
- Colton Willhardt
- Cohen Nunes
- Amogh Natu
- Rees Verleur

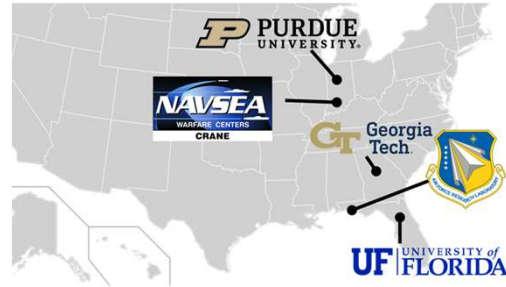
UQ/Workflow

- Matthew Cuffaro
- Joana Bou Barcelo
- Luke Morris
- George Rauta
- Richard Samuelson



Partnership Management Plan

- Center
 - Primarily housed at UF
 - Partnerships – Purdue and GT
- Computational activities
 - UF, GT, and Purdue
 - All students on Slack and MFC workspaces
 - Student-led meetings
- Experimental activities
 - Purdue
 - Close interaction with physics and UQ teams

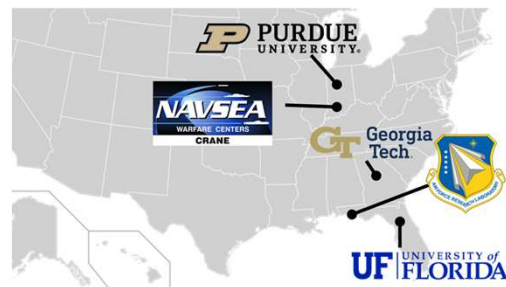
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CM³C Slack

- Allows for everyone to be part of the conversation
- 39 users
- Active channels
 1. [#general](#)
 2. [#gpu-ppiclf](#)
 3. [#lc-users](#) (Livermore Users)
 4. [#machine-learning](#)
 5. [#macro-ee-sims](#)
 6. [#macro-el-sims](#)
 7. [#macro-exp](#)
 8. [#micro-pr-sims](#)
- Frequent crossover with MFC slack



Allows everyone to be part of the conversation

cm3c.slack.com
mflowcode.slack.com

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Tuolumne Users

- ~12 users
 1. One pending approval
- Hipergator
 1. Long turn around times
 2. Better suited for ML and flow visualization
- Tuolumne
 1. Short turn around times
 2. Routine 32 gpu runs
 3. All UF students primarily use Tuolumne
 - *In hindsight this was a blessing!*



Questions?

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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T2: MFC for Particle Resolved DNS

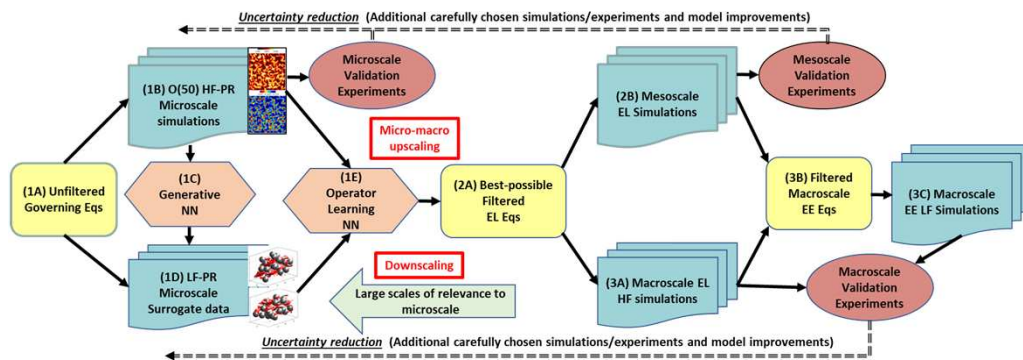
BC Kim, Mike VandenBoom, Fahad Nabid

Department of Mechanical and Aerospace Engineering



UF UNIVERSITY of FLORIDA

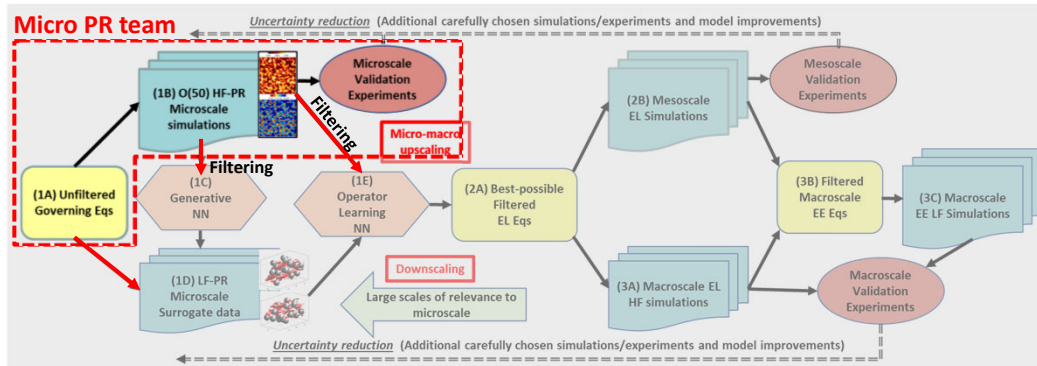
Overall Roadmap & Big Picture



- Macro scale : Euler-Lagrange simulations + ML-based closure models, Euler-Euler Simulations with $\mathcal{O}(10^{11})$ of Particles
- Meso scale : Euler-Lagrange simulations + ML-based closure models
- **Micro scale : High-Fidelity Particle-Resolved simulation with $\mathcal{O}(10^5)$ particles**

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Micro PR team : Our Role & Scope

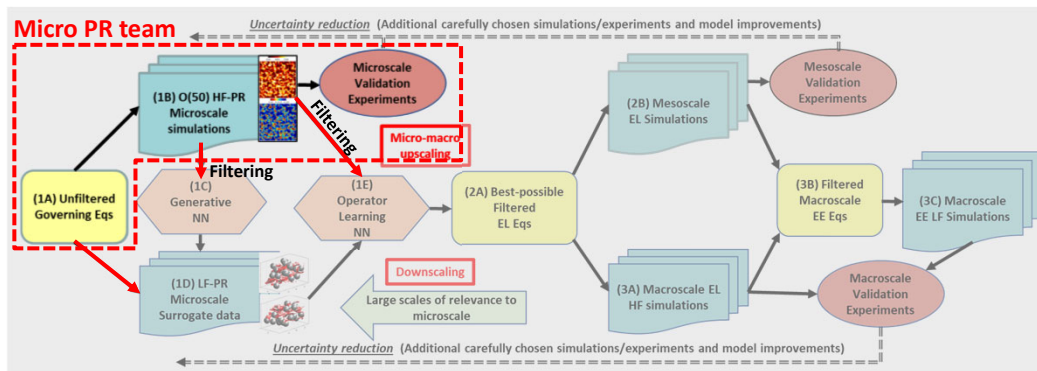


- Macro scale : Euler-Lagrange simulations + ML-based closure models, Euler-Euler Simulations with $\mathcal{O}(10^{11})$ of Particles
- Meso scale : Euler-Lagrange simulations + ML-based closure models
- **Micro scale : High-Fidelity Particle-Resolved simulation with $\mathcal{O}(10^5)$ particles**

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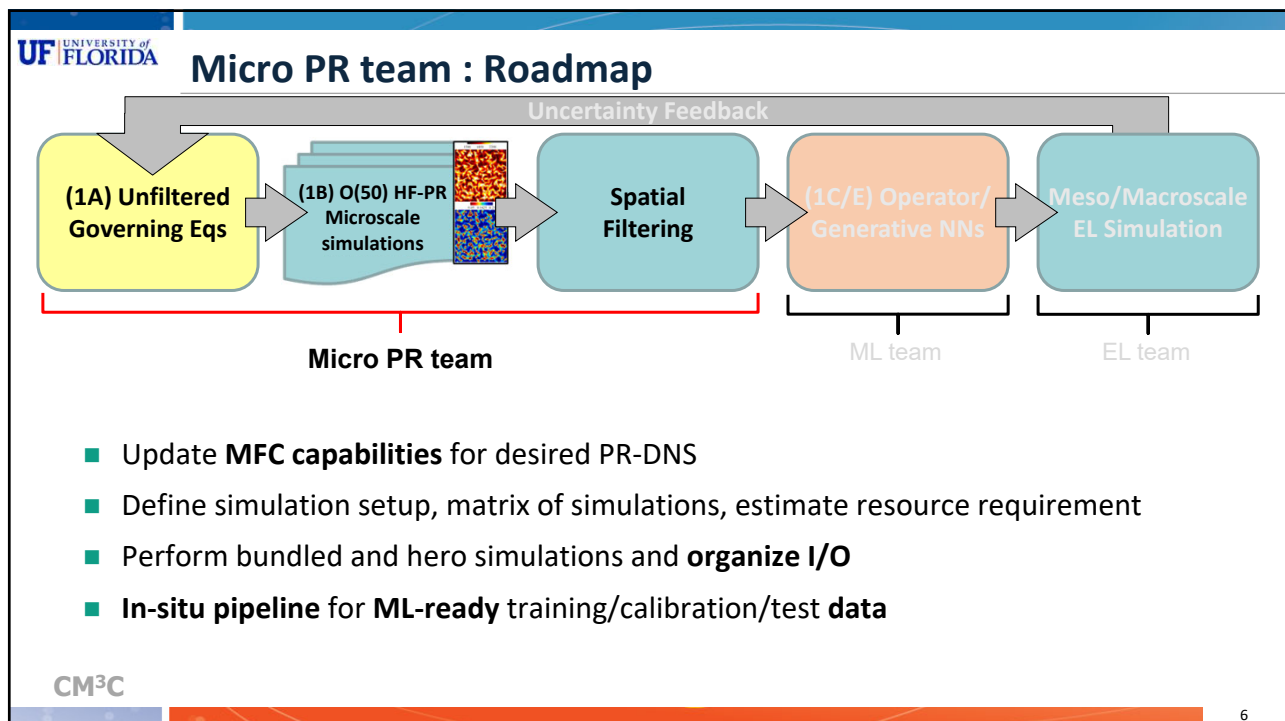
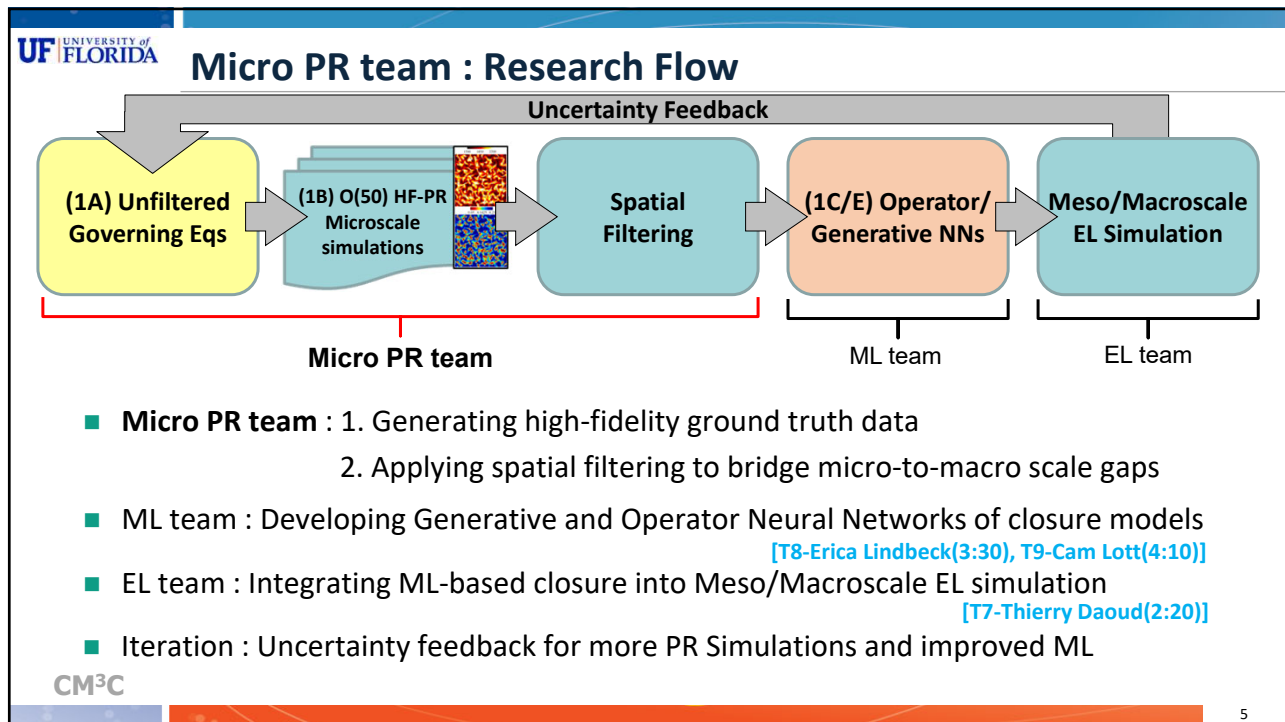
Micro PR team : Current Status



- High-Fidelity Particle-Resolved Simulation with $\mathcal{O}(100+)$ Particles
- In-situ Data Filtering for ML
- Providing Training Data for Surrogate Models

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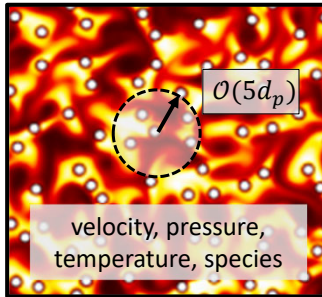


What ML Needs

Generative ML
(data-driven & PINNs)

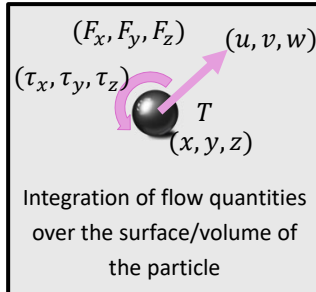
High-resolution flow field centered around particles in regions $\mathcal{O}(5d_p)$

at different time instances

CM³Operator-Learning ML
for Phase-coupling

Complete list of particle-related quantities for each particle

at frequent times

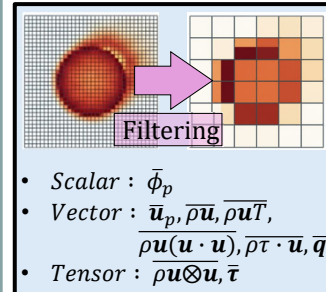


[T10-Daniel Vickers(Tomorrow 9:00AM)]

Operator-Learning ML
for subgrid closure

Complete list of filtered fields (two-orders lower-resolution)

at $\mathcal{O}(10^2)$ time instances



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Novel Implementation of Filter Operation

- Several **filtered quantities** are needed for subgrid closure

Non-reactive flow :

- Scalar: $\bar{\phi}_p, \bar{\rho}, \bar{\rho(u \cdot u)}$
- Vector: $\bar{u}_p, \bar{\rho u}, \bar{\rho u T}, \bar{\rho u(u \cdot u)}, \bar{\rho \tau \cdot u}, \bar{q}$
- Tensor: $\bar{\rho u \otimes u}, \bar{\tau}$

Filter definition

$$\bar{\rho}(x) = \iiint \rho(x') G(x - x') dx'$$

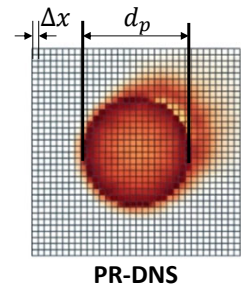
- $G(x - x')$ is a Gaussian filter of width $[\frac{d_p}{2}, \frac{3d_p}{4}, d_p, \frac{3d_p}{2}, 2d_p, 5d_p, \dots]$.

- Challenge:** even the smallest filter $d_p/2 \gg \Delta x$

- Filter size is wider than number of ghost cells in MFC

- Novel Solution :**

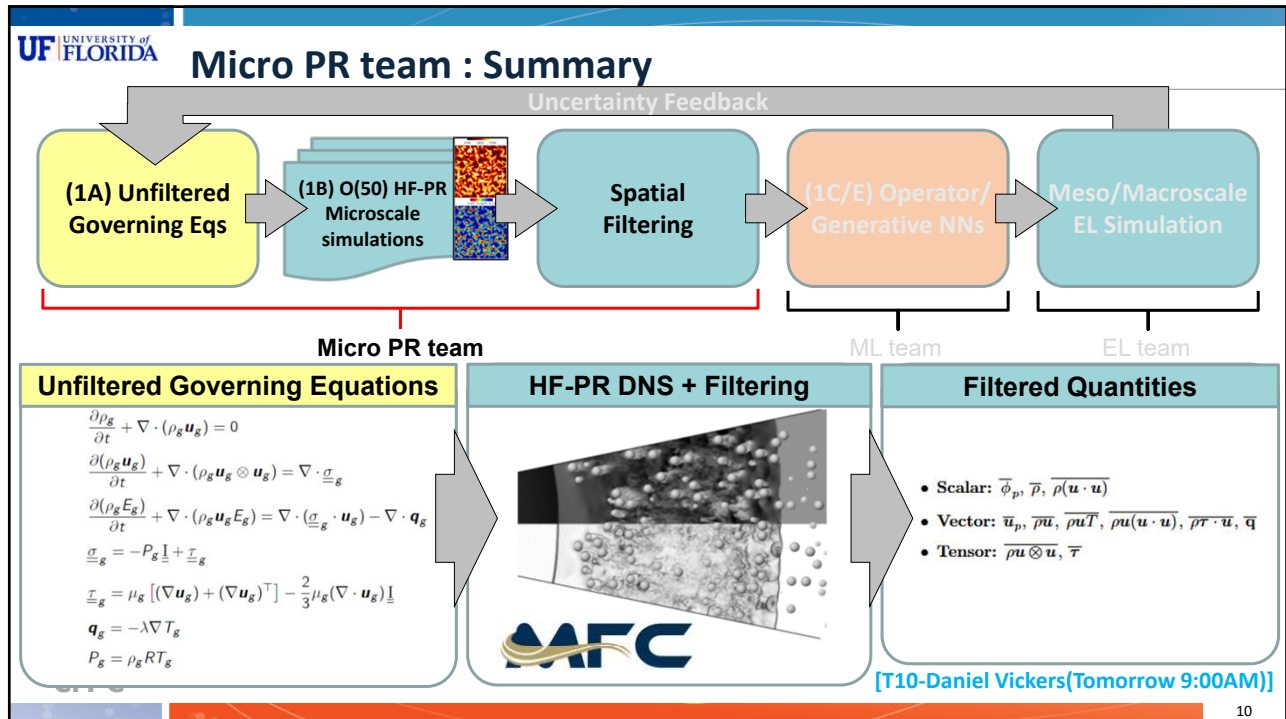
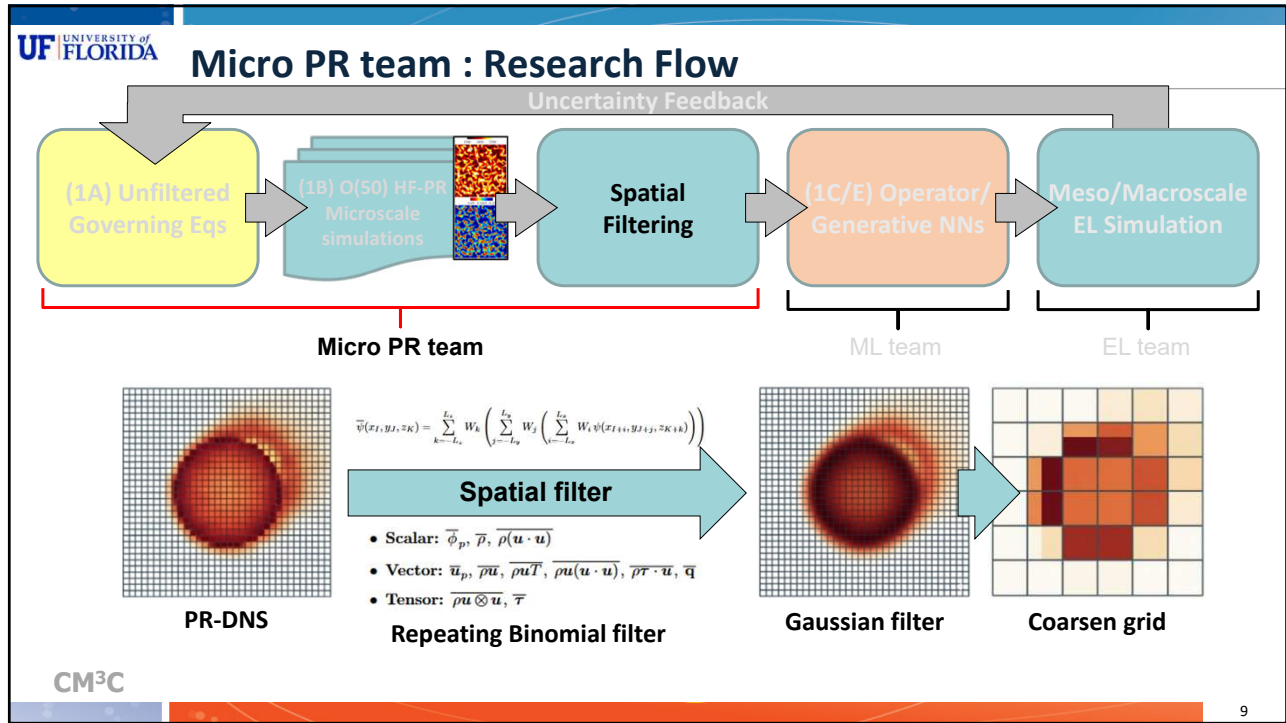
- Gaussian filter of width $d_p/2 = \prod_x \prod_y \prod_z (3/5 - \text{point filters})$
- Requires application of 3/5-point filters multiple times (will be done within MFC)

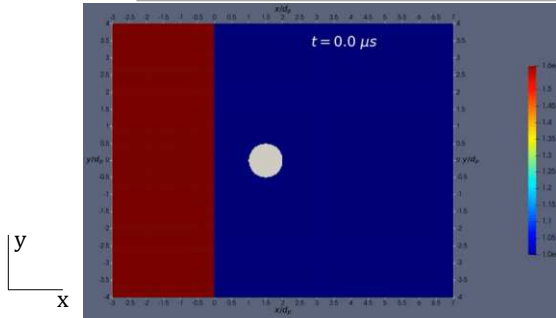


PR-DNS

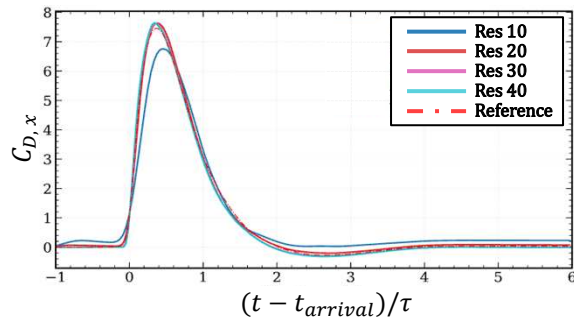
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Results : Single Stationary Particle (Verification) – $M_s=1.22$ Fig. Pressure Contour for $M_s=1.22$ case, (slice at $z=0$). (30 pts/D)Case Setup : Shock-Particle Interaction (**Inviscid**), $M_s=1.22$

Simulation domain	$10D \times 8D \times 8D$			
Particle diameter, D [mm]	0.5			
Grid resolution (Pts/D)	10	20	30	40
Number of Cells	0.64M	5.12M	17.28M	40.96M
GPUs	4	16	16	32
Wall time [m:s]	00:19	00:45	01:58	02:59

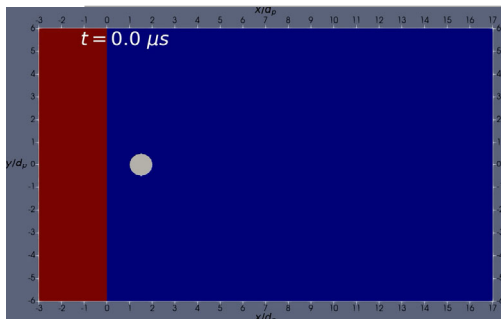


- Inviscid stationary particle is verified
- The $C_{D,x}$ graph follows the trend of Ref. (From Res 20)
- The code remains **highly efficient**, maintaining a **manageable wall time**

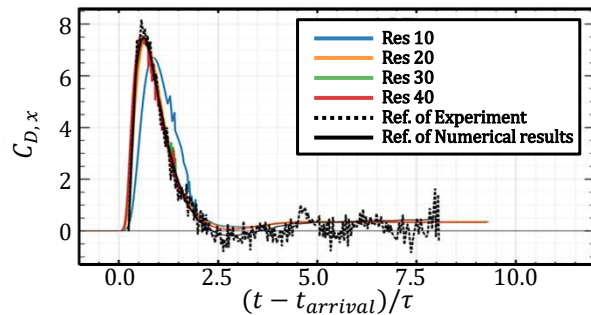
Mehta, Y. et al. (2016). *J. of Applied Physics*, 119(10), 104901.

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Results : Single Stationary Particle (Verification & Validation)

Fig. Pressure Contour for $M_s=1.22$ case, (slice at $z=0$). [Res 20]Case Setup : Shock-Particle Interaction (**Viscous**), $M_s=1.22$

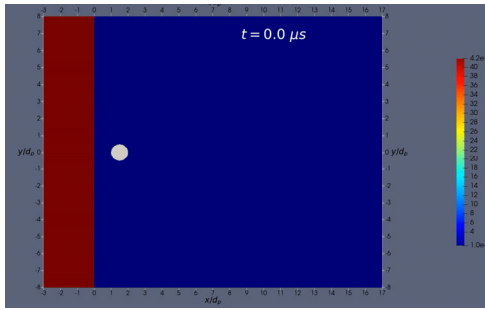
Simulation domain	$20D \times 12D \times 12D$			
Particle diameter, D [mm]	0.5			
Grid resolution (Pts/D)	10	20	30	40
Number of Cells	2.88M	23.04M	77.76M	184.32M
GPUs	4	8	16	16
Wall time [m:s]	0:50	3:80	7:36	20:26



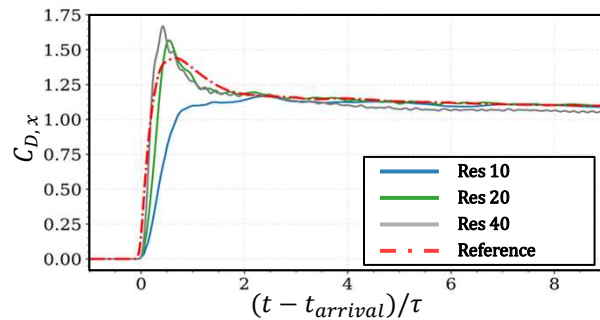
- Viscous stationary particle is validated (From Res 20)
- The $C_{D,x}$ graph follows the trend of Ref.

Sun, M. et al. (2005). *Shock waves*, 14(1), 3-9.

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Results : Single Stationary Particle (Verification) – $M_s=6.0$ Fig. Pressure Contour for $M_s=6.0$ case, (slice at $z=0$). [Res 20]Case Setup : Shock-Particle Interaction (*Inviscid*), $M_s=6.0$

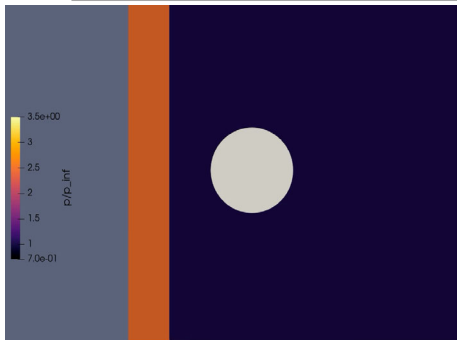
Simulation domain	$20D \times 16D \times 16D$		
Particle diameter, D [mm]	0.5		
Grid resolution (Pts/ D)	10	20	40
Number of Cells	5.12M	40.96M	327.68M
GPUs	32	64	64
Wall time [m:s]	0:39	1:45	14:12

Fig. Drag plot validation for $M_s=6.0$

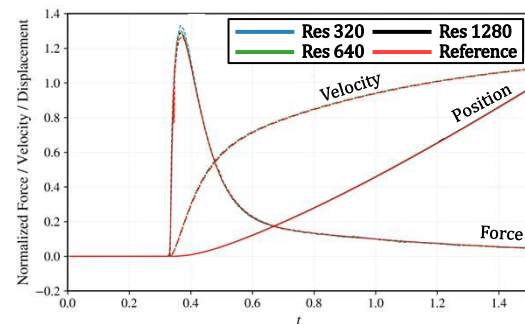
- Inviscid stationary particle is on the verification with grid resolution
- The $C_{D,x}$ graph follows the trend of Ref.
- Higher resolution (Res > 40) is needed

Mehta, Y. et al. (2016). *J. of Applied Physics*, 119(10), 104901.

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Results : Single Moving Cylinder (Verification) – $M_s=1.5$ Case Setup : Shock-Cylinder Interaction (*Viscous*), $M_s=1.5$

Simulation domain	$4D \times 4D$		
Particle diameter, D	1.0		
Reynolds number, Re	2.5×10^6		
Grid resolution (Pts/ D)	320	640	1280
Number of Cells	1.64M	6.55M	26.21M
GPU	16	24	24
Wall time [m:s]	2:09	9:11	49:10



- Verified numerical accuracy through a rigorous grid convergence study
- Performance: Stable computation of complex interaction at $Re = 2.5 \times 10^6$

Henshaw, W. D., & Schwendeman, D. W. (2006). *J. of Comp. Physics*, 216(2), 744–779.

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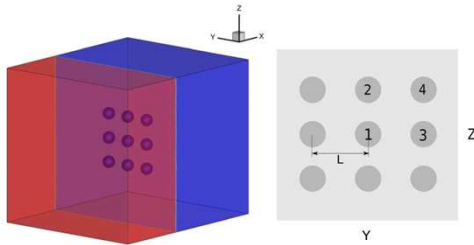
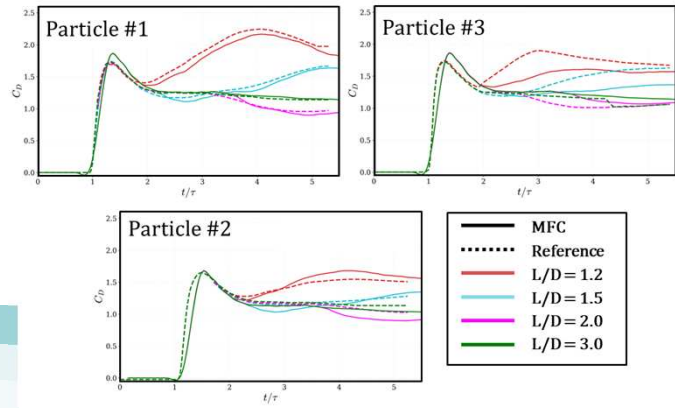
Results : Structured 9-Particle Array (Verification) – $M_s=3.0$ 

Fig. Simulation domain and particle locations

Case Setup : Shock-Particle Interaction (*inviscid*), $M_s=3.0$

Simulation domain	$10D \times 8D \times 8D$			
Particle diameter, D [mm]	0.5			
Grid resolution (Pts/D)	30			
Number of Cells	17.28 M			
Inter-particle spacing, L/D	1.2	1.5	2.0	3.0
GPUs	24			
Wall time [m:s]	2:50			



- **Particle #1 & #2:** High accuracy and consistency for all spacings
- **Particle #3:** Spacing-dependent results; aligns with reference as L/D increases

Mehta, Y. et al. (2016). *J. of Applied Physics*, 119(10), 104901.

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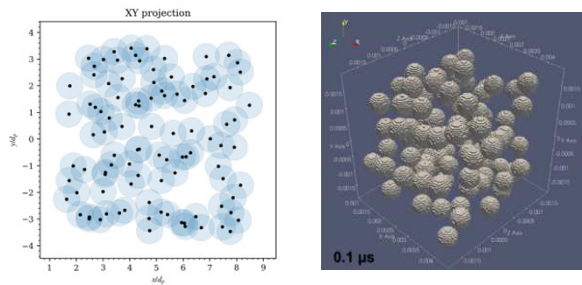
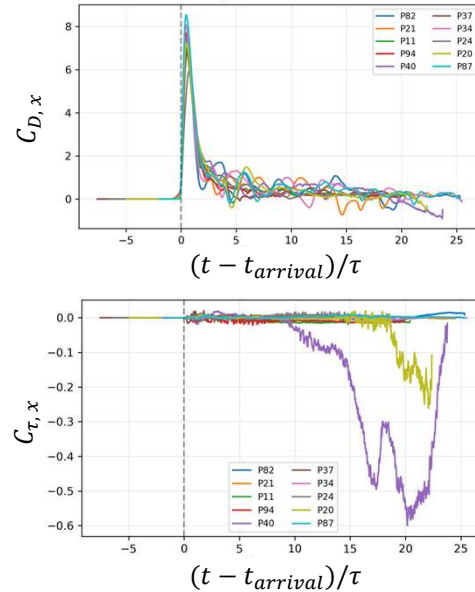
Results : Random moving 100 particles array – $M_s=1.22$ 

Fig. Simulation domain and particle locations

Case Setup : Shock-Particle Interaction (*Viscous*), $M_s=1.22$

Simulation domain	$30D \times 20D \times 20D$	
Particle diameter, D [mm]	0.5	
Grid resolution (Pts/D)	15	30
Number of Cells	40M	320M
GPUs	8	48
Wall time [h:m:s]	1:21:16	1:48:22



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Results : Summary

No. of Particles	Particle motion	Array	Viscosity	Mach number, M_s	Resolution (Pts/D)	Geometric parameters
1	X	-	X	1.22	10, 20, 30, 40	
		-	O	1.22	10, 20, 30, 40	
		-	X	6.0	10, 20, 40	
	O	-	O	1.22	20	
1 (Cylinder)	O	-	O	1.5	320, 640, 1280	
9	X	Structured	X	3.0	30	$L/D = 1.2, 1.5, 2.0, 3.0$
100	X	Random	O	1.22	10, 20, 30	
	O	Random	O	1.22	15, 30	

- **Systematic validation is underway**, covering both inviscid and viscous flows with **various particle layouts** (including structured and random arrays)
- We are now **ready to scale up to $O(1,000)$ particles**, providing high-fidelity "ground truth" data for the ML team's closure models

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Micro-Team: Y1+ Plan

- Towards **exascale simulation** of particle-resolved simulations of **$O(10^4)$ particles** subjected to **explosive conditions** on Tuolumne
 - Capabilities to be implemented: JWL equations of state, detonation products of initial condition, random packing of resolved particles, resolution study
- **Bundled simulations** of high-speed and shock flow over a random array of particles for varying Re , Ma , density ratio, and volume fraction
 - Supply generative and operator learning ML efforts with compressible flow data
- **Build capability** for particle-resolved simulations of **burning** and **reactive flow**
 - Surface and gas-phase reduced order chemistry, stiff chemistry solver, time-dependent particle size/shape, expand the list of filtered quantities for closure model

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Do you have any questions?



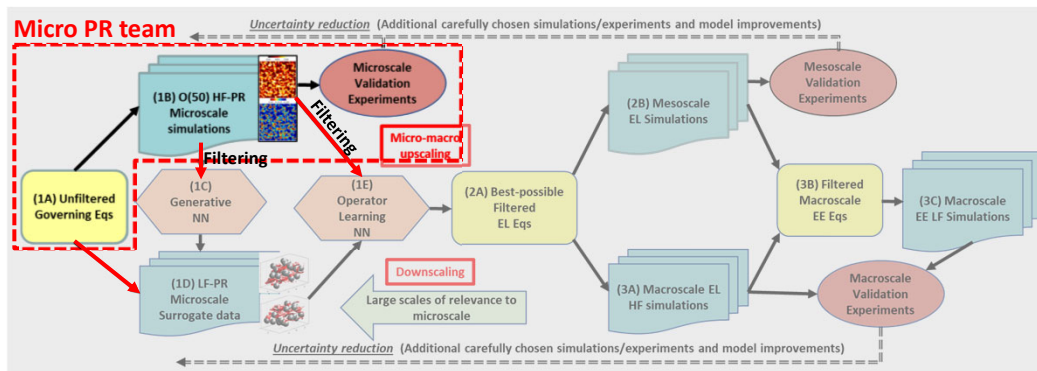
Georgia Tech

UF UNIVERSITY of FLORIDA

P PURDUE UNIVERSITY

UF UNIVERSITY of FLORIDA

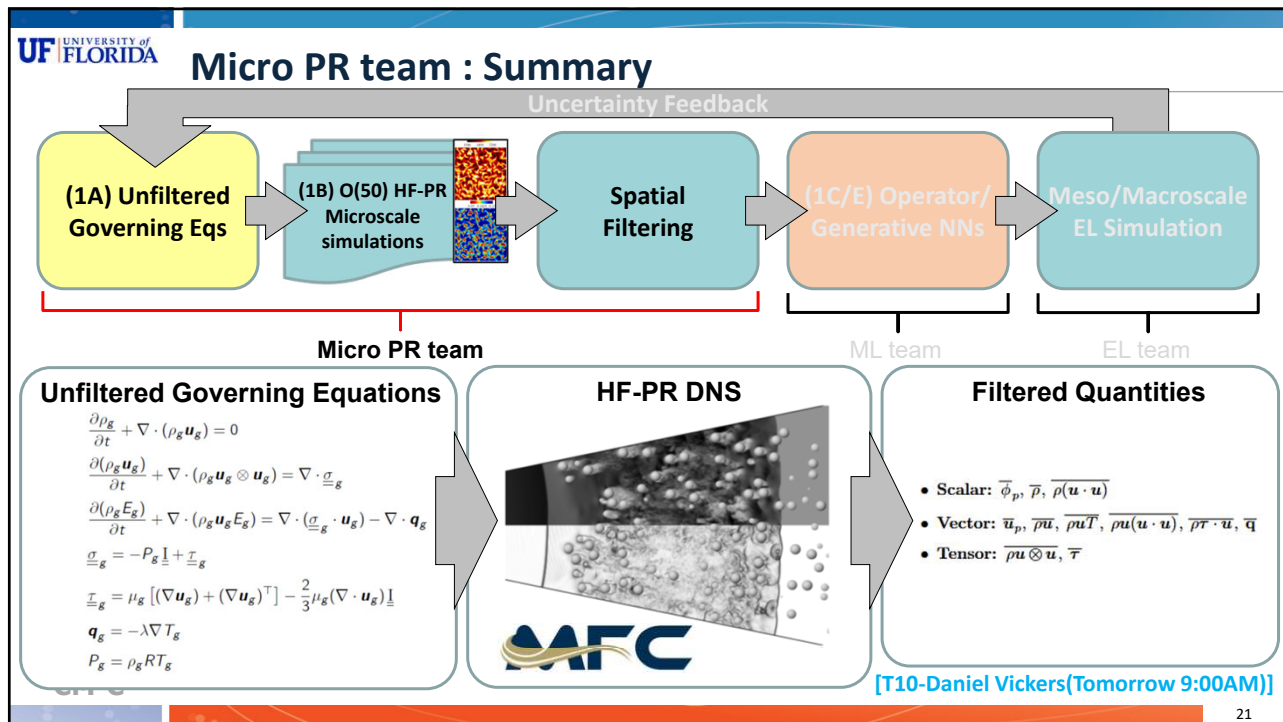
Micro PR team : Current Status



- Macro scale : Euler-Lagrange simulations + ML-based closure models, Euler-Euler Simulations with $\mathcal{O}(10^{11})$ of Particles
- Meso scale : Euler-Lagrange simulations + ML-based closure models
- **Micro scale : High-Fidelity Particle-Resolved simulation with $\mathcal{O}(10^5)$ particles**

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What ML Needs

- **Generative ML (data-driven and PINNs)**
 - High-resolution flow field (velocity, pressure, temperature, species) centered around particles in regions $\mathcal{O}(5d_p)$ at different time instances
- **Operator-Learning ML for phase-coupling**
 - Complete list of particle-related quantities for each particle at frequent times: position, velocity, acceleration, rotation, temperature, force, torque, heat transfer, burn-rate
- **Operator-Learning ML for subgrid closure**
 - Complete list of filtered (two-orders lower-resolution) fields at $\mathcal{O}(10^2)$ time instances: subgrid stress, subgrid thermal flux, subgrid KE flux, subgrid viscous stress flux, etc
- **Other in-situ processing for multiphase physics**
 - Full high-resolution flow field at infrequent times

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What ML Needs

- Generative ML (data-driven and PINNs)
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- Other in-situ processing for multiphase physics
 - Full high-resolution flow field at infrequent times

Force/torque, etc., require integration of flow quantities over the surface/volume of the particle. Performed in-situ. [T10-Daniel Vickers \(Tomorrow 9:00AM\)](#)

What ML Needs

- Generative ML (data-driven and PINNs)
 - High-resolution flow field (velocity, pressure, temperature, species) centered around particles in regions $\mathcal{O}(5d_p)$ at different time instances
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- Other in-situ processing for multiphase physics
 - Full high-resolution flow field at infrequent times

Requires filtering of several flow quantities with varying filter values. Careful implementation of online and offline tasks

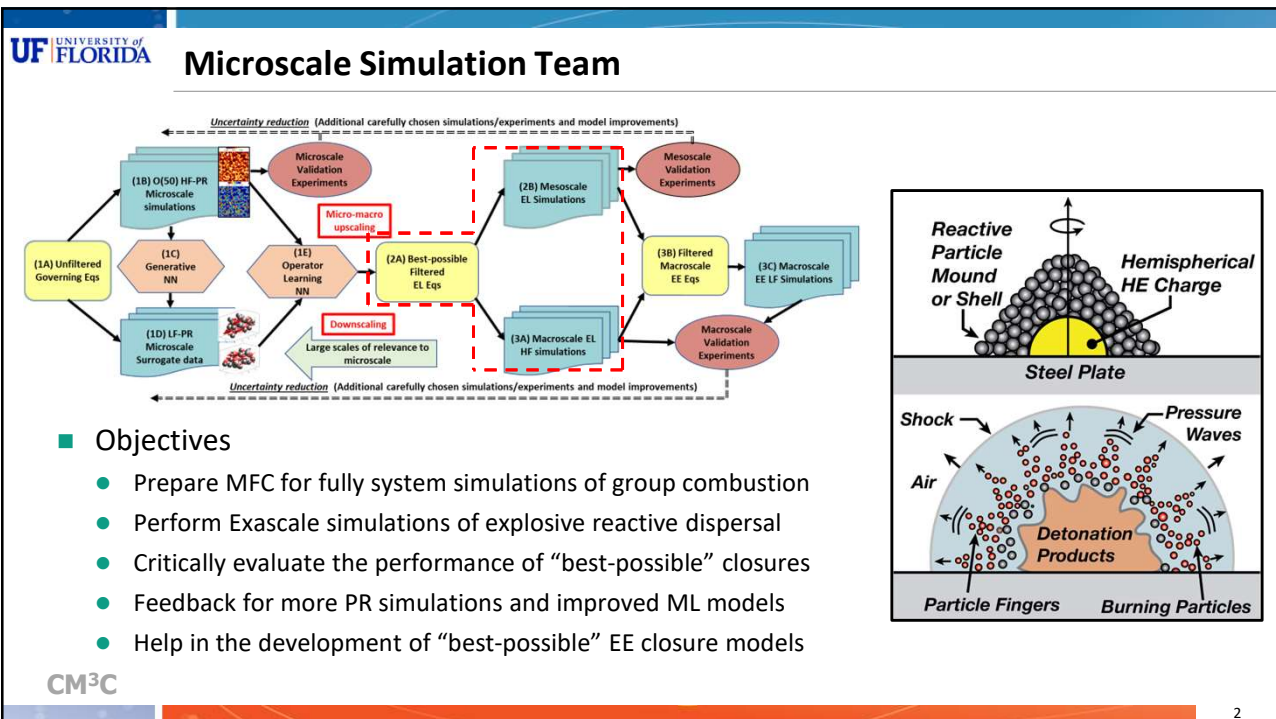
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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T3: MFC for Euler-Lagrange Simulations

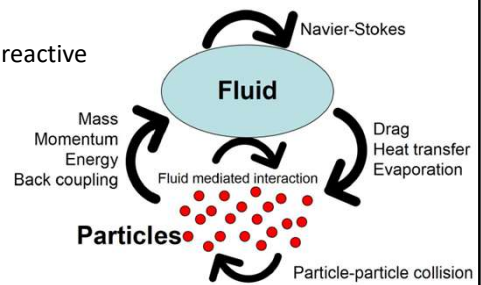
Jose Aguilar

Department of Mechanical and Aerospace Engineering, UF



Motivation for Euler–Lagrange in MFC

- We will be performing predictive simulations of particle-laden compressible flows with:
 - Sub-grid Euler-Lagrange (EL) particles
 - Resolved immersed boundary (IB) particles, much larger than grid
- Relevant to:
 - Shock–particle interactions, multiphase detonations, dispersed reactive flows
- Need:
 - State-of-the-art phase-coupling closure models
 - State-of-the-art subgrid closure models
 - Euler-Lagrange coupling (interpolation & projection)
 - Particle-particle direct & gas-mediated interactions
- Goal:
 - Enable high-fidelity, exascale E-L simulations within MFC



MFC Readiness for EL simulations

- MFC previously had limited Euler-Lagrange capability for Lagrangian particle tracking.

Requirement	Previously	Currently
Lagrangian Particle Architecture: Initialization, per-particle arrays, MPI rank ownership, data output	Fully Implemented	Fully Implemented
Fluid-to-Particle Force Application: Interpolation, force models, heat transfer	Not Yet Implemented	Currently Developing
Particle-to-Fluid Field/Force Spreading: Projecting fields and hydrodynamic forces back to Eulerian field	Not Yet Implemented	Fully Implemented
Lagrangian Particle Motion: Moving particles, MPI particle transfer, boundary conditions	Not Yet Implemented	Fully Implemented
Collision Forces: Inter-particle collisions, wall collisions	Not Yet Implemented	Currently Developing
IB-EL Collision Forces: Resolved particles must be able to collide with sub-grid particles	Not Yet Implemented	Not Yet Implemented
Scaling: MFC with $O(\sim 10^{11})$ moving lagrangian particles	Not Yet Implemented	Currently Developing
Particle Reactions: Particle mass exchange, heat release, chemistry	Not Yet Implemented	Not Yet Implemented

Euler–Lagrange Governing Framework

Eulerian Fluid - 5 Equation Multi-Component Flow Model

- Continuity (one per component):

$$\frac{\partial (\alpha_i \rho_i)}{\partial t} + \nabla \cdot (\alpha_i \rho_i \mathbf{u}) = \frac{\alpha_i \rho_i}{\phi_f} \left[\frac{\partial \phi_p}{\partial t} + \mathbf{u} \cdot \nabla \phi_p \right]$$

Sub-grid closure terms

Additional terms with average fluid velocity field

- Momentum:

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I} - \boldsymbol{\tau}^v) = \frac{\rho \mathbf{u}}{\phi_f} \left[\frac{\partial \phi_p}{\partial t} + \mathbf{u} \cdot \nabla \phi_p \right] + \frac{1}{\phi_f} (p \nabla \phi_f - \boldsymbol{\tau}^v \cdot \nabla \phi_f) - \frac{1}{\phi_f} (\phi_f \mathbf{R}_{sg}) - \frac{1}{\phi_f} \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i) \mathbf{F}'_{gp,i}$$

- Energy:

$$\begin{aligned} \frac{\partial (\rho E)}{\partial t} + \nabla \cdot [(\rho E + p) \mathbf{u} - \boldsymbol{\tau}^v \cdot \mathbf{u}] &= \frac{\rho E}{\phi_f} \left[\frac{\partial \phi_p}{\partial t} + \mathbf{u} \cdot \nabla \phi_p \right] - \frac{p}{\phi_f} (\mathbf{u} \cdot \nabla \phi_f + \nabla \cdot (\phi_p \mathbf{u}_p)) \\ &+ \frac{\boldsymbol{\tau}^v}{\phi_f} : (\mathbf{u} \nabla \phi_f + \nabla (\phi_p \mathbf{u}_p)) - \nabla \cdot \mathbf{q} - \frac{1}{\phi_f} \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i) (W'_{gp,i} + Q'_{gp,i}) \end{aligned}$$

- Particle volume fraction (not explicitly solved):

$$\frac{\partial \phi_p}{\partial t} + \nabla \cdot (\phi_p \mathbf{u}_p) = 0$$

N_p = number of particles
 ϕ_f = fluid volume fraction
 ϕ_p = particle volume fraction

$\mathbf{F}'_{gp,i}$ = force on particle i

$\sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i)$ = projection function

Euler–Lagrange Governing Framework

Lagrangian Particles

- Particle position (Implemented):

$$\frac{d\mathbf{x}_{p,i}}{dt} - \mathbf{u}_{p,i} = 0$$

Particle – particle interaction terms

- Particle velocity (Implemented):

$$m_{p,i} \frac{d\mathbf{u}_{p,i}}{dt} = \mathbf{F}_{gp,i} + \mathbf{F}_{pp,i}$$

$\mathbf{F}_{gp,i}$ = Hydrodynamic force on particle i

- Particle Temperature (In development):

$$m_{p,i} C_{p,i} \frac{dT_{p,i}}{dt} = Q_{gp,i}$$

$\mathbf{F}_{pp,i}$ = Particle-particle collision force on particle i

$Q_{gp,i}$ = Fluid-particle heat transfer to particle i

- Particle Angular Velocity (In development):

$$I_{p,i} \frac{d\boldsymbol{\omega}_{p,i}}{dt} = \boldsymbol{\tau}_{gp,i} + \boldsymbol{\tau}_{pp,i}$$

$\boldsymbol{\tau}_{gp,i}$ = Hydrodynamic torque on particle i

$\boldsymbol{\tau}_{pp,i}$ = Collisional torque due to particle-particle interactions on particle i

Fluid-Particle Coupling

- The current implementation has the hydrodynamic force on particle i as :

$$\mathbf{F}_{gp,i} = \mathbf{F}_{qs,i} + \mathbf{F}_{pg,i} + \mathbf{F}_{am,i}$$

- Quasi-steady drag force: $\mathbf{F}_{qs,i} = 3\pi\mu_{g,i}d_{p,i}(\mathbf{u}_{g,i} - \mathbf{u}_{p,i})\frac{Re_{p,i}}{24}C_{D,i}(Re_{p,i}, M_{p,i}, \phi_{p,i})$
- Pressure Gradient Force: $\mathbf{F}_{pg,i} = -V_{p,i}(\nabla p_g)_i$
- Added Mass Force: $\mathbf{F}_{am,i} = V_{p,i} C_M(M_{p,i}, \phi_{p,i}) \underbrace{\left[\frac{D(\rho_{g,i}\mathbf{u}_{g,i})}{Dt} - \frac{d(\rho_{g,i}\mathbf{u}_{p,i})}{dt} \right]}_{\mathbf{a}_{r,i}}$
- Best available models are used for drag coefficient¹ and added mass coefficient^{2,3}

¹Osnes *et al.* (2023) Comprehensive quasi-steady force correlations for compressible flow through random particle suspension

²Beguin *et al.* (2016) Void fraction influence on added mass in a bubbly flow

³Parmar *et al.* (2008) On the unsteady inviscid force on cylinders and spheres in subcritical compressible flow

Fluid Interpolation to Particle

- Fluid properties ($\mathbf{u}_g, p_g, \rho_g$, etc.) need to be interpolated to the particle's location to obtain local fluid information for force computation**
 - Eulerian field values are for the cell center location
- Interpolated using high-order Barycentric Lagrange interpolation¹ :

$$A(\mathbf{X}) = \frac{\sum_{k=1}^N \sum_{j=1}^N \sum_{i=1}^N \frac{w_i w_j w_k}{\bar{X}_{ijk}} a_{ijk}}{\sum_{k=1}^N \sum_{j=1}^N \sum_{i=1}^N \frac{w_i w_j w_k}{\bar{X}_{ijk}}}$$

$$\bar{X}_{ijk} = (\mathbf{X} - x_i)(\mathbf{X} - y_j)(\mathbf{X} - z_k)$$

$\mathbf{X}$ = particle position

x_i, y_j, z_k = cell center coordinate

a_{ijk} = Eulerian value at cell center

- Weights at each cell in each stencil are pre-computed for each direction as:

$$w_j = \frac{1}{\prod_{k \neq j} (x_j - x_k)}; j, k = 1, \dots, N$$

- Method supports:
 - Variable grid spacing in any direction
 - High-order accuracy:
 - 2nd-order accuracy = 3-point stencil around the particle's cell; N=3
 - 4th-order accuracy = 5-point stencil; N=5

¹Zwick *et al.* (2020) A scalable Euler-Lagrange approach for multiphase flow simulation on spectral elements.

Particle to Fluid Projection

- Particles must be projected back to Eulerian mesh for coupling terms on the RHS of the fluid conservation equations. This includes:

- Particle volume fraction: $\phi_p(\mathbf{x}, t) \approx \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i) V_{p,i}$
- Momentum source terms: $S_{mom,x}(\mathbf{x}, t) \approx \sum_{i=1}^{N_p} -G(\mathbf{x} - \mathbf{X}_i) F_{gp,i,x}$

- A normalized Gaussian kernel function is used to project particle contributions to fluid cells:

$$G(\mathbf{x} - \mathbf{X}_i) = \exp \left(-\alpha \left[\left(\frac{x - X_i}{\Delta_x} \right)^2 + \left(\frac{y - Y_i}{\Delta_y} \right)^2 + \left(\frac{z - Z_i}{\Delta_z} \right)^2 \right] \right)$$

- For each particle i , particle's effect is spread across its host cell and M neighboring cells in each direction
- N is the total number of cells over which the particle's influence is felt in each direction (x,y,z) $M = \frac{N-1}{2}$
 - Here, N can be chosen as $N = 3, 5, 7 \dots$

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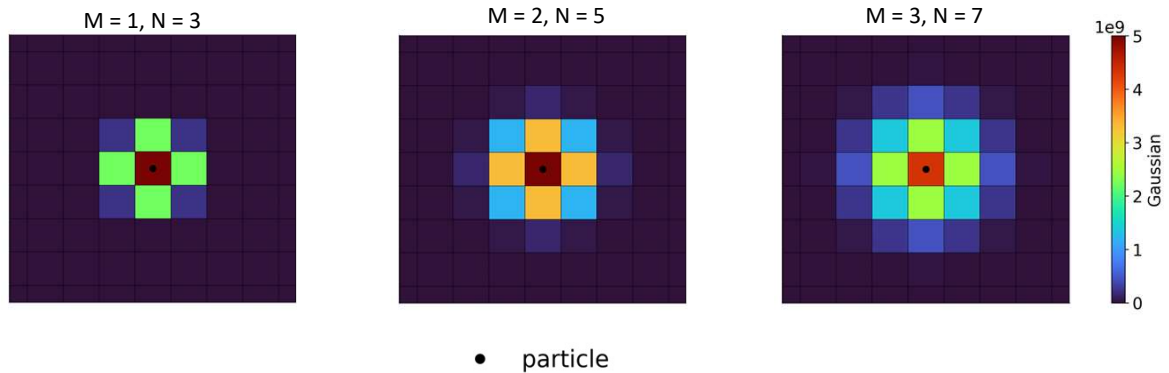
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Particle to Fluid Projection

- Gaussian kernel projection has compact support around particle's cell

- M determines the number of surrounding cells the particle affects
- Larger values for M/N smear out the projected field more, yielding a smoother result
- This comes at a cost, as every particle must be projected to N^3 cells in 3D

$$M = \frac{N-1}{2}$$

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Collision Model

Particle-Particle Collisions: Pairwise interactions with soft-sphere force model

$$\mathbf{F}_{pp,i} = \sum_{\substack{j=1 \\ j \neq i}}^{N_c} (\mathbf{F}_{pp,ij}^n + \mathbf{F}_{pp,ij}^t)$$

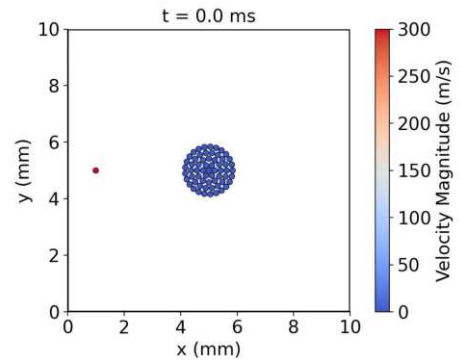
Implemented

In development

$$\mathbf{F}_{pp,ij}^n = -\kappa_n \delta_{ij} \mathbf{n}_{ij} - \eta_n \mathbf{u}_{n,ij}$$

 κ_n = Spring stiffness coefficient δ_{ij} = Overlap distance $\mathbf{n}_{ij}$ = Unit vector from particle i to j η_n = normal force damping coefficient $\mathbf{u}_{n,ij}$ = relative velocity along normal vector

- MPI-aware handling w/ ghost particles (see next slide)



Particle projectile launched at 300 m/s towards cluster of particles

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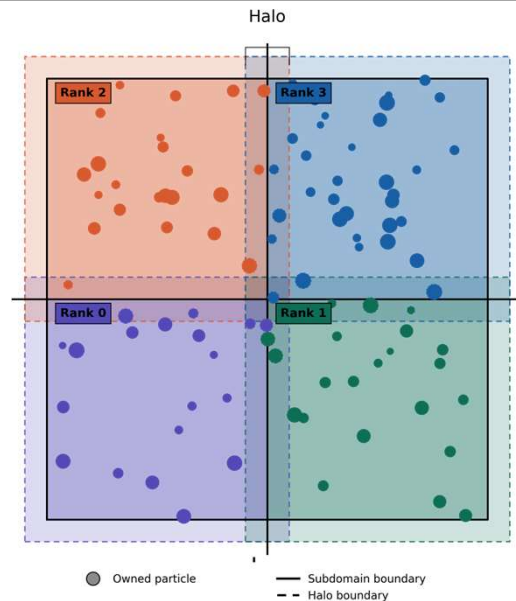
Parallelization

Current MPI Strategy

- MFC decomposes domain among MPI ranks prior to particle reading
- Each rank maintains ownership of particles that are within its domain bounds
- Particles that cross sub-domain boundaries are communicated to the corresponding rank
- Particles within 1 cell into the Halo region of neighboring ranks are communicated as ghost particles only for collision detection

Goal:

- Maintain scalability for large particle counts ($O(\sim 10^6)$, $O(\sim 10^9)$, $O(\sim 10^{11})$)

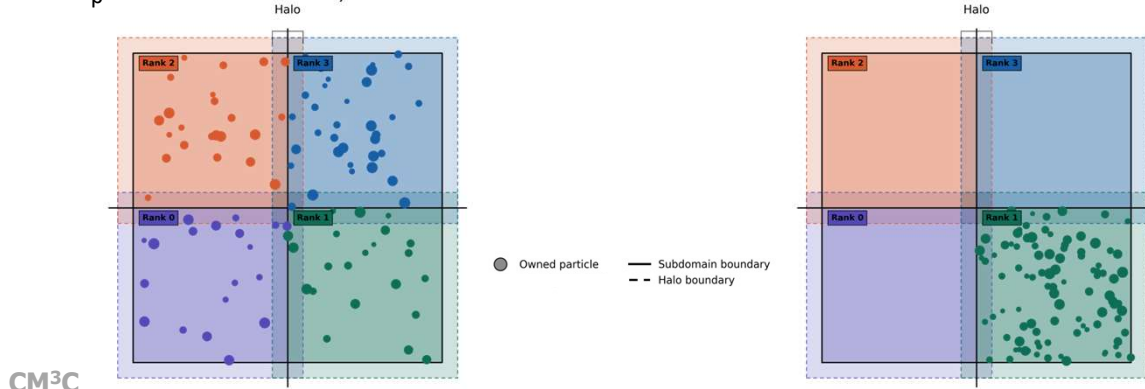
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Parallelization

■ MPI Strategy Concerns

- Current strategy maintains efficient load balancing when the particle distribution is approximately uniform ($N_{p,rank} \sim \frac{N_{p,total}}{N_{ranks,total}}$;
 N_p = Number of Particles)
- With particles sequestering within a single mpi sub-domain, $N_{p,rank 1} = N_{p,total}$. Particles are not efficiently load balanced → One rank can dominate run time



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Parallelization

■ Potential solutions to address load-balancing for E-L particles

➤ Ppiclf Strategy

- Ppiclf**: a parallel particle-in-cell library written in Fortran
- Developed and maintained at UF

Option 1: Adapt Ppiclf's strategy for particle tracking

- Particles are binned among ranks separately from the fluid domain.
- Fluid domain is mapped to each bin

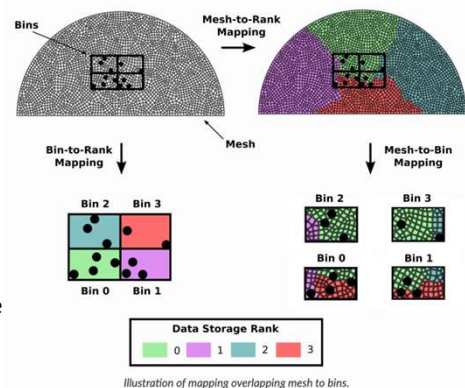
Option 2: Couple MFC with the GPU version of Ppiclf

- GPU-Ppiclf currently being developed
- Current MFC developments would be combined with Ppiclf particle tracking

➤ Will need to evaluate feasibility of options as it relates to:

- Implementation complexity
- Timeline
- Performance

■ More on Ppiclf in T7



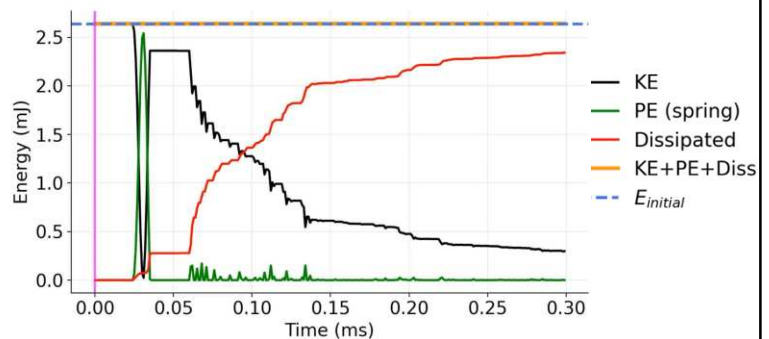
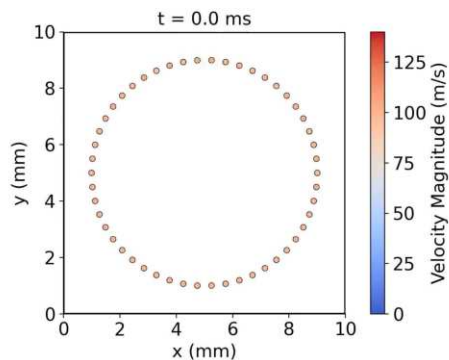
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MFC EL Solver Verification

- Unit testing (regression suite):
 1. Barycentric Lagrange interpolation
 2. Projection operation
 3. MPI particle transfer
 4. Inter-particle collision force and torque
 5. Quasi-steady, added-mass, pressure-gradient, viscous-unsteady, lift force models
 6. Quasi-steady, undisturbed flow, diffusive unsteady torques
 7. Quasi-steady, undisturbed flow, diffusive unsteady heat transfer
 8. Reynolds stress, other sub-grid closure models
- Code verification (regression suite):
 1. Shock-curtain interaction
 2. Spherical explosive dispersal

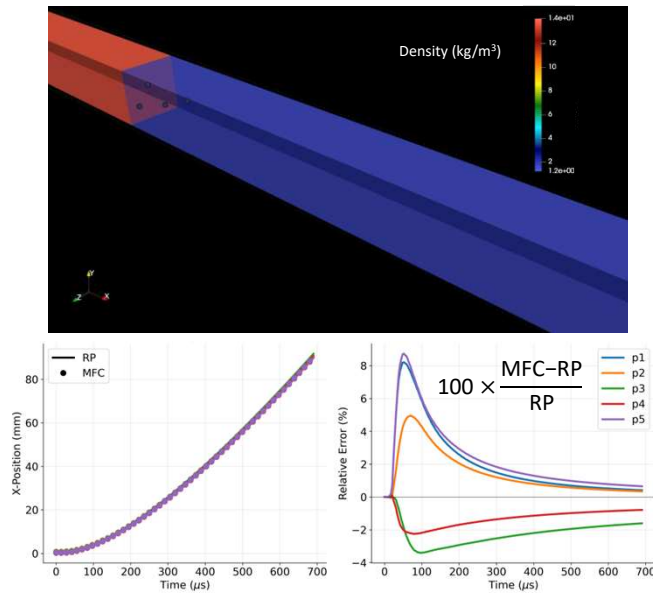
Unit Test – Inter-Particle Collisions

- Initial Code Verification for:
 1. Inter-particle collisions (Normal Component)
- 50 particles configured in a ring launched at 100 m/s towards the ring center
- **Verify:** Collision implementation, energy conservation



Solver Verification – Diaphragm Problem

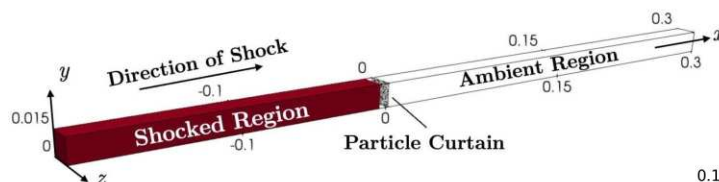
- Initial Code Verification for:
 1. Quasi-steady drag force
 2. Particle integration
 3. MPI particle transfer
 4. Barycentric Lagrange Interpolation
- 5 particles subjected to diaphragm rupture ($P_{\text{high}} \sim 1.2 \text{ MPa}$, $\rho_{\text{high}} \sim 14.1 \text{ kg/m}^3$)
- Comparing MFC with RocFlu w/ Ppiclf (RP) for initial verification

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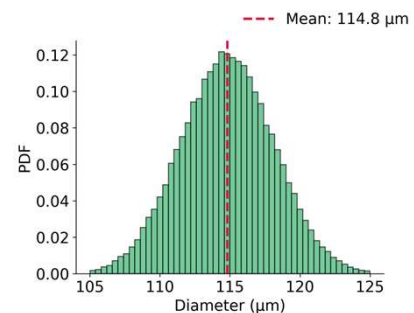
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Solver Validation – Experimental Comparison

- Scaled down setup for comparison to shock-particle curtain experiments of Wagner et al. (2012)



- Discretized with 2000x30x30 cells (1,800,000 grid points)
- 2 mm thick curtain; $\sim 10^5$ particles
- Particle diameters ranging from 105 μm – 125 μm
- Simulation Parameters:
 1. $M_s = 1.66$; $T_{\text{amb}} = 296.4 \text{ K}$; $p_{\text{amb}} = 82.7 \text{ E3 Pa}$
- Volume fraction varying between 26% ($\gamma=0.15$) to 15% ($\gamma=0$)
- Physics considered:
 1. Quasi-steady drag; Pressure gradient; Added mass, Fluid-particle coupling; Inter-particle collisions (normal component)

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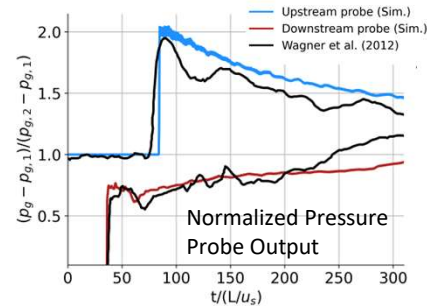
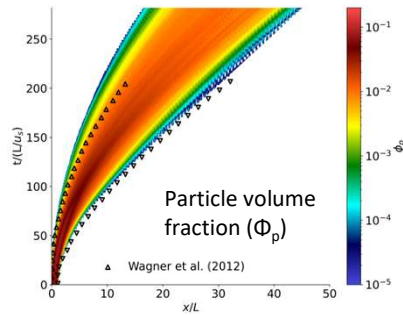
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Solver Validation – Experimental Comparison

- Results show qualitative agreement with expected physics for shock-particle curtain



- Simulation run on LLNL's Tuolumne with 8 AMD MI300A GPUs to $t_{\text{final}} = 1$ ms
- Total compute time: ~ 650 sec



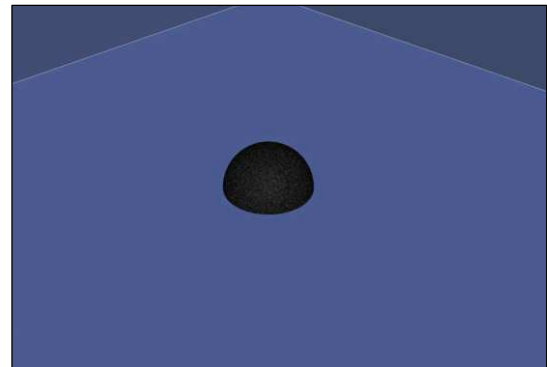
Time (t) : normalized by curtain width ($L=0.002$ m) and shock speed ($U_s = 572.91$ m/s).
Position (x) : normalized by initial curtain width (L)

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Performance and Capability – Hemi-Spherical Blast

- 20 mm diameter hemisphere of post shock air for $M_s = 3$
- 2 mm thick layer of 50 micrometer diameter particles packed at a 20% volume fraction
 - $N_{\text{particles}} = 3,298,361$
- Simulation run on LLNL's Tuolumne with 24 AMD MI300A GPUs to $t_{\text{final}} = 1$ ms
- Total compute time: ~ 4 hours
- Incorporates:
 - Quasi-steady drag, pressure force, added mass force
 - Four-way coupling (particle-fluid and particle-particle interactions)



Hemispherical shell of particles dispersed in post-shock flow

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Summary of Accomplishments

- EL capability integrated into MFC with initial implementation of state-of-the-art force models for particle dynamics
- Implemented:
 1. Interpolation, projection, two-way coupling, MPI particle communication
 2. Efficient collision detection with nearest neighbors
- Performed initial verification of software and particle/coupling physics
- Solver has complete particle workload offloading to GPU-accelerated kernels
- Demonstrated on shock-particle curtain problem and spherical blast

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Future Work

- Implement full range of models for accurate particle dynamics:
 1. Fluctuation forces for quasi-steady drag, added mass, Viscous-unsteady drag, lift force
 2. Quasi-steady/Diffusive unsteady heat transfer
 3. Tangential component of collision forces and hydrodynamic torque forces
 4. Reynolds stress and other closure models
 5. ML-based phase-coupling and subgrid closure models (to be developed at the center)
- Account for real gas and reactive physics
 - JWL EOS is implemented and other complex EOS
 - Gas-phase chemistry and droplet/particle burn-rate
 - Detonation products as initial condition
- Rigorous code verification
- Scalability for $O(10^{11})$ particles
- Detailed experimental validation with exascale simulation setup

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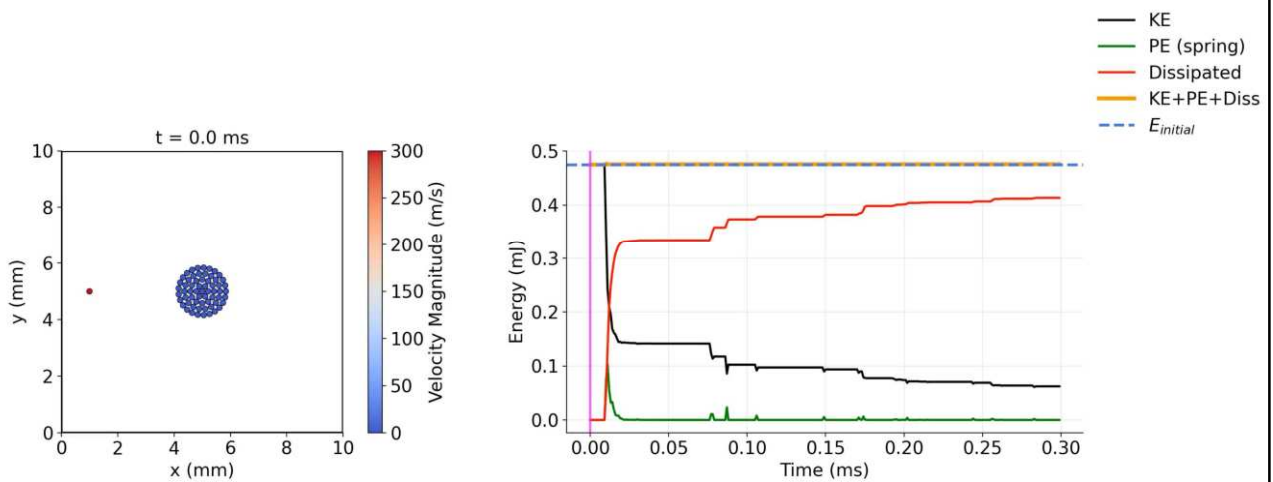
22

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Questions?



Backup – Energy Conservation for Particle Projectile Case



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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T4: Extreme-scale Many-physics CFD

Spencer H. Bryngelson

Georgia Tech

*School of Computational Science and Engineering
Guggenheim School of Aerospace Engineering
Woodruff School of Mechanical Engineering*



Today

Software Strategy

Performance Strategy

Center Integrations

Tooling vignettes

Future challenges

The tool for the job: MFC

3

What is it anyway?
[pre-PSAAP IV]

4

Brief Overview

Scalable, fast, portable
Many-physics

Used by
2 PSAAP IV centers
1 MURI
Many research groups

Part of mission-critical test suites
HPE/Cray, Oak Ridge[OLCF],
AMD, SPEChpc, NVIDIA



I'm MIT Open Source!

An exascale compressible multiphase and multiphysics flow code.

GitHub Download v5.1.5 Quick Start
Documentation Papers



Hold DOE ALCC & INCITE allocations exceeding
10M Frontier GPU hrs/yr

Bryngelson et al. (2021), Radhakrishnan et al. (2024), Wilfong et al. (2026) 5

Pushing the computing boundary

Scalable and fast depends on..
hardware
software
compiler bugs?!

MFC *already* scales to 100% El Capitan
[AMD MI300A, conducted pre-classification]



Fastest Supercomputer in the World, 2 ExaFLOPs

Example Pull Request

GPU Offload Strategy: Going Fast Everywhere

Directives: OpenACC & OpenMP support [abstracted away]

On Tuo/ElCap

Compilers: Support Cray CCE and AMD Flang

USM mode: We use it! No host copies [OMP only]

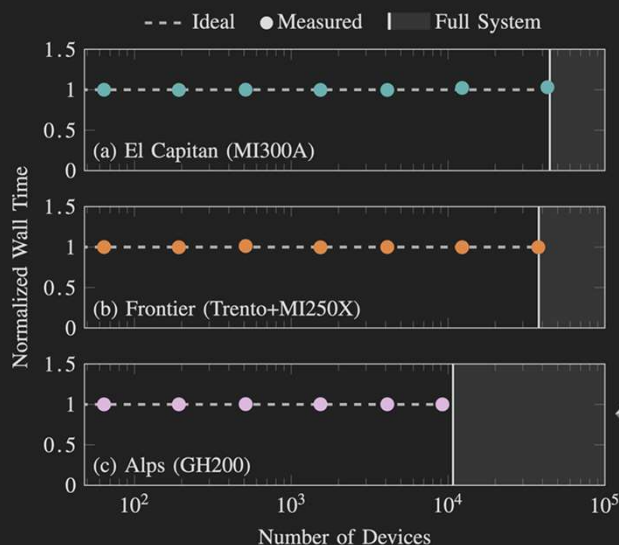
Can use OpenACC, it's fast too!

On Frontier

Maintain state-of-the-art performance (we use UVM)

7

Performance: Weak scaling



Lawrence Livermore
National Laboratory

OAK RIDGE
National Laboratory

CSCS
CSCS Alps

MFC
mflowcode.github.io

Ideal scaling

~1.5 EFLOPS

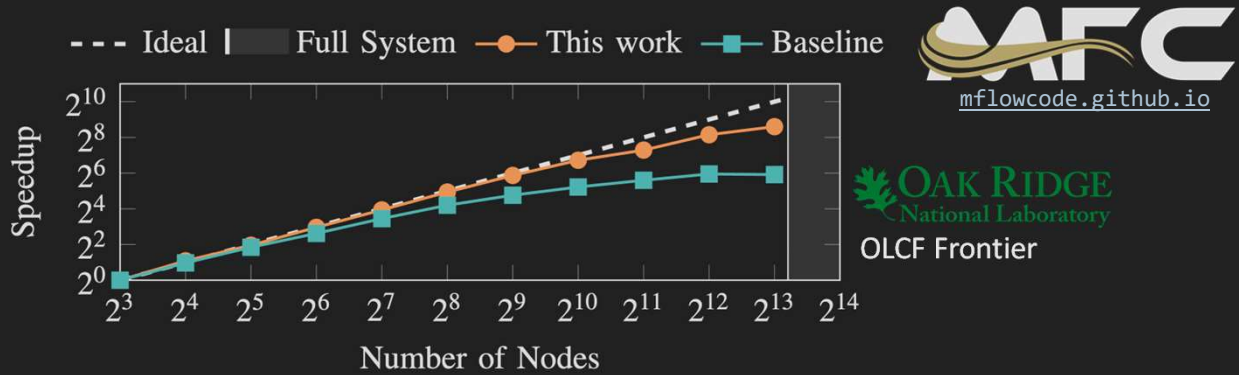
200T points [Frontier]
(1 quadrillion DoF)

~160T points [El Capitan]

~25× largest ever

Wilfong et al. (2025), Radhakrishnan et al. (2026) 8

Performance: Strong scaling



MFC
mflowcode.github.io

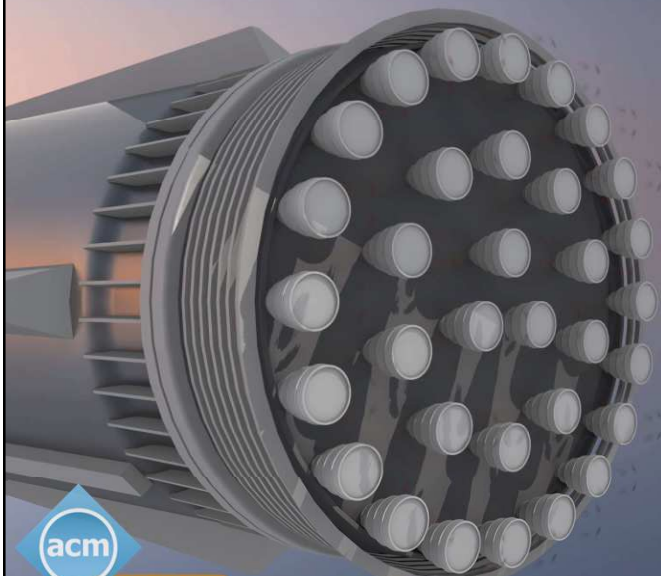
OAK RIDGE
National Laboratory
OLCF Frontier

SoA: Greater than 95% perf. penalty ✗
Us: Less than 50% perf. penalty ✓

So...
160B grid point
sim. in <1 min!

Wilfong et al. (2025), Radhakrishnan et al. (2026) 9

Wilfong et al., SC25



acm

2025 ACM
Gordon Bell
Prize Finalist

Sim of SpaceX Super Heavy
2-species (gas/gas) Mach 10 exhaust

Largest flow sim
 200T pts., 1
 quadrillion DoF
Fastest
compressible sim.
 5x faster

MFC

Gordon Bell Prize Winners Push Open Science Boundaries With NVIDIA-Powered Supercomputers

Five finalists for the esteemed high-performance computing award have achieved breakthroughs in climate modeling, fluid simulation and more with the Alps, JUPITER and Perlmutter supercomputers — with two winners taking home the prize.

November 20, 2025 by [Dean Hays](#)



H. Le Berre
SpaceX



R. Budiardja
Oak Ridge



B. Dorschner
NVIDIA



Brian Cornille
AMD



N. Tselepidis
NVIDIA



S. Abbott
HPE/Cray

Industry + Lab Partners



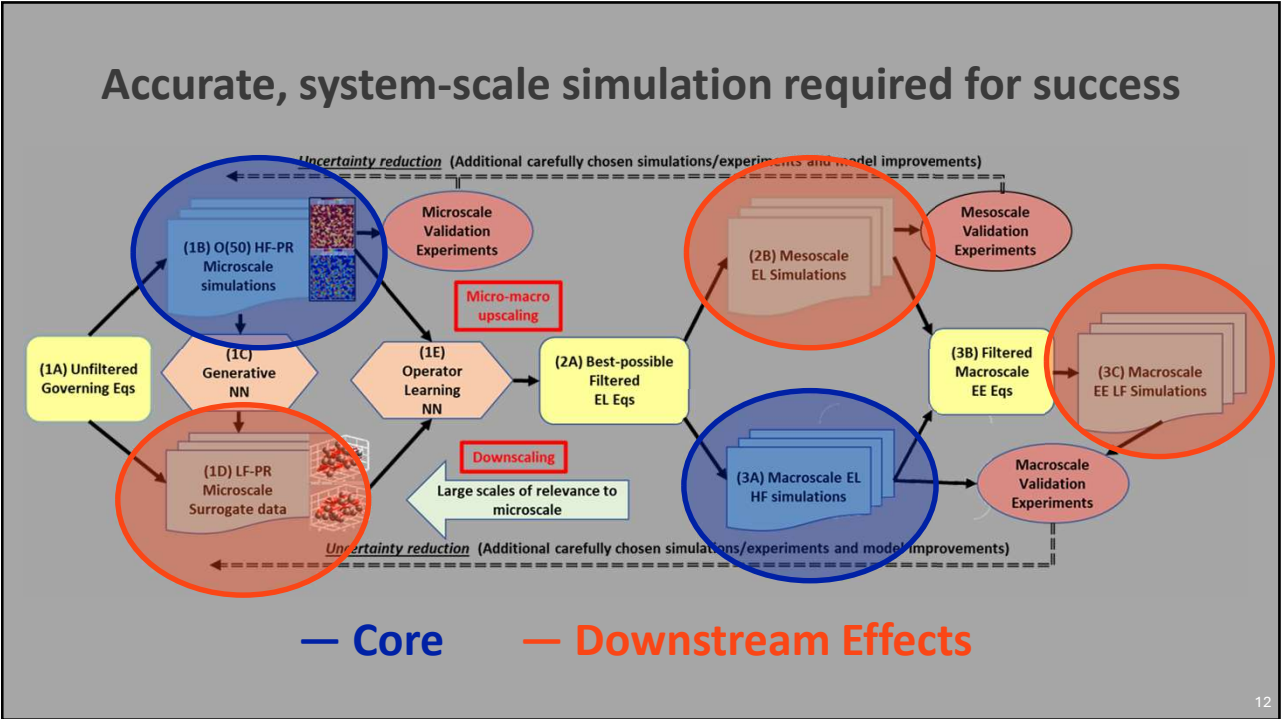
Gordon Bell finalist team pushes scale of rocket simulation on El Capitan

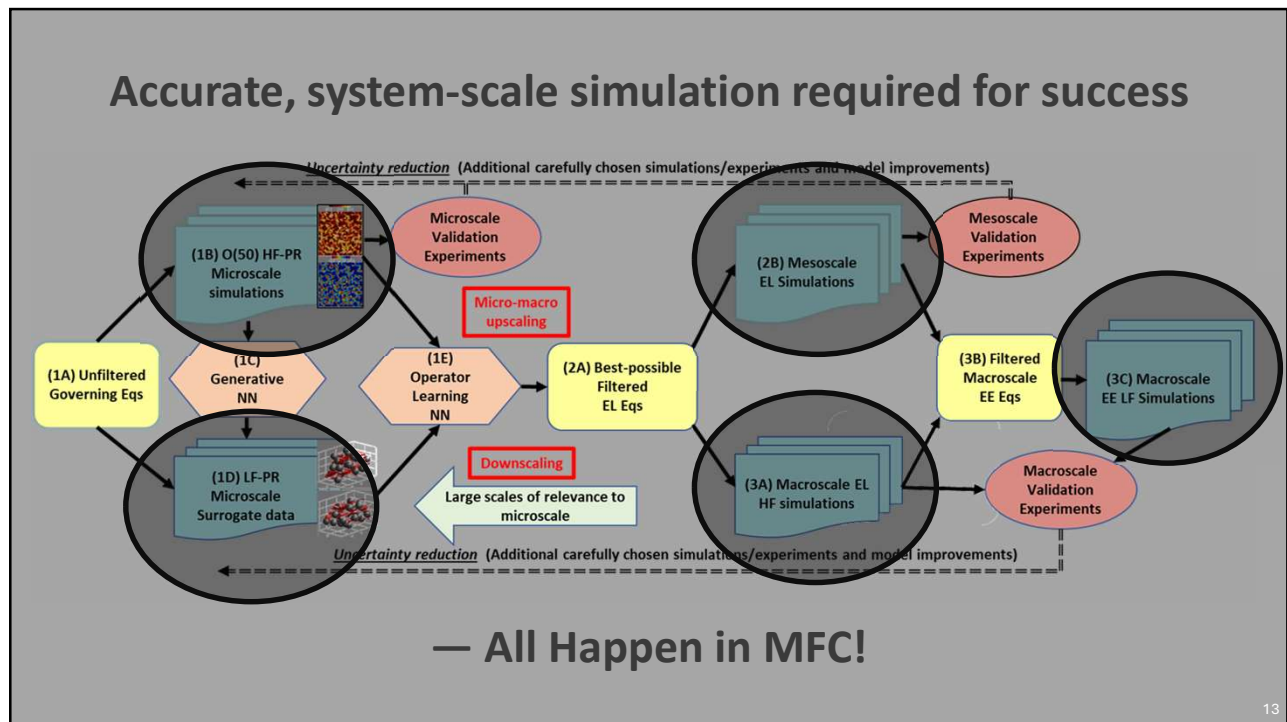


Shock Treatment for CFD Simulations

By Coury Turezyn | July 23, 2025 | Featured, Technology | 9 min read

NVIDIA, AMD, HPE/Cray, LLNL, ORNL Partnerships Key Enablers





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How we're getting there

Jose Aguilar [UF] + Ben Wilfong [GT]: Euler–Lagrange (forces, collisions, ...)

Joshua Gillis [UF] + Ben Wilfong [GT]: EL verification

BC Kim [UF]: Filtering, overseeing particle-resolved team + transitions

Fahad Nabid [UF] + Dan Vickers [GT]: Particle-resolved data generation for ML

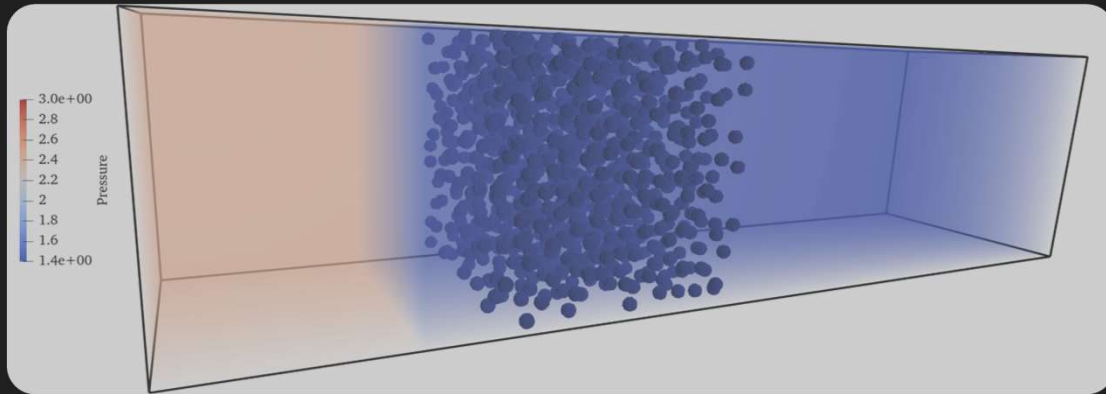
Mike VandenBoom [UF] + Dan Vickers [GT]: Verification for flow over particles

Davin Anderson [UF] + Dimitrios Adam [GT]: Group combustion models

State as of Mar. 2026
Post TST – Regrouping teams

14

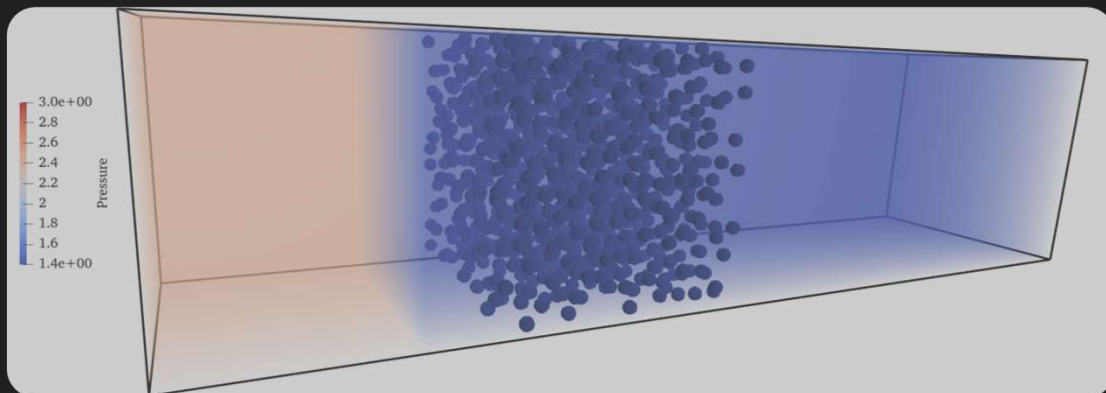
Performance Vignettes: Particle-laden resolved



Rather coarse grid... still, 1K "resolved" particles; *single* A100 GPU node

15

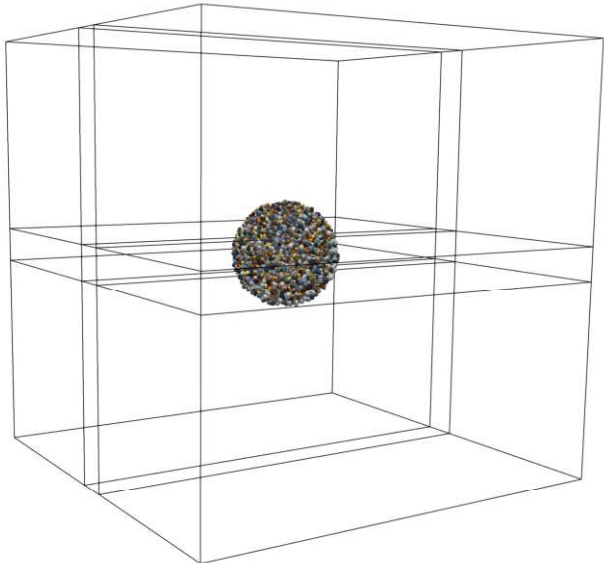
Performance Vignettes: Particle-laden resolved



With our method, **>10M resolved particles** on leadership machines
See Dan Vickers [GT] talk later for details!

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Performance Vignettes: Euler–Lagrange Particles



Lines show GPU boundaries. Sim. by B. Wilfong

Euler–Lagrange sub-grid closure

Handle bubbles/particles/etc.

Works at scale across GPU domains

Work in progress [discussed later!] :

Collision models

Clustering

Reactions

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Neural Nets! How we handle...

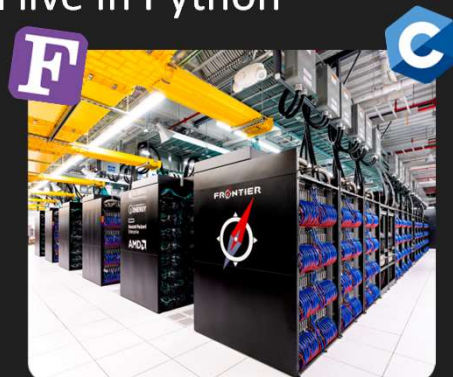
MFC/CFD codes live in “low”-level codebases

Neural networks useful, but often live in Python



?

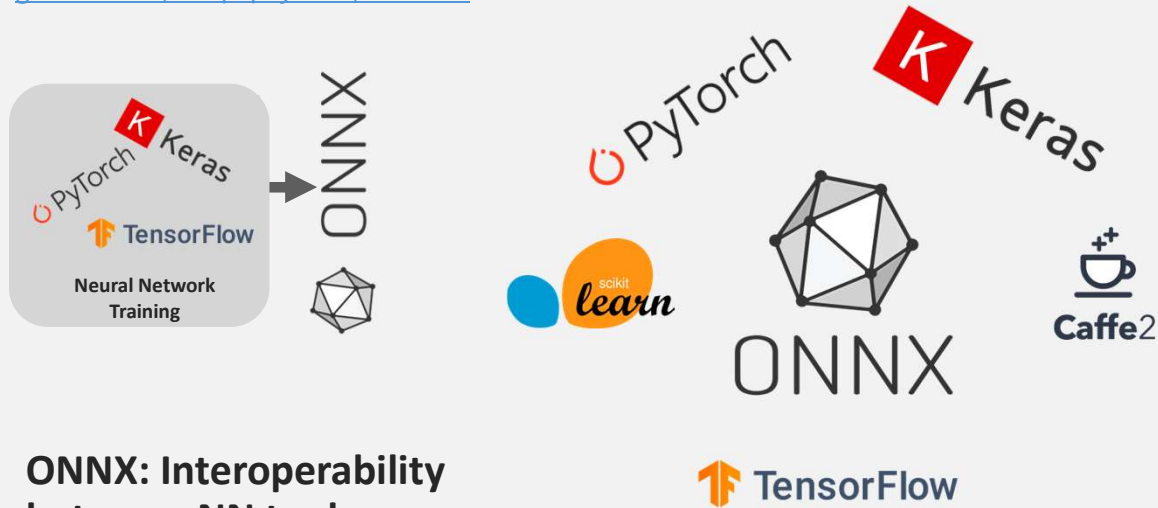
A tool to bridge
this gap



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roseNNa: Pipeline

github.com/comp-physics/rosenna



19

roseNNa: Pipeline

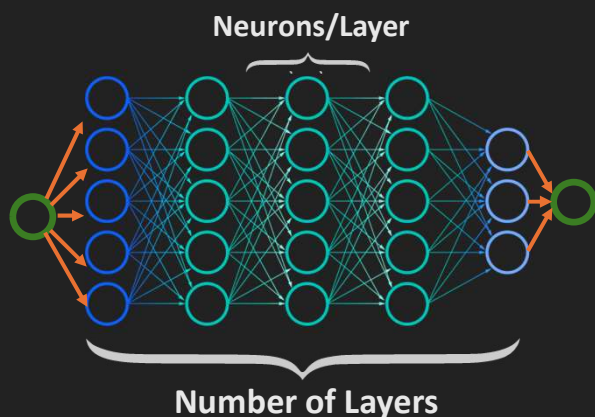
github.com/comp-physics/rosenna



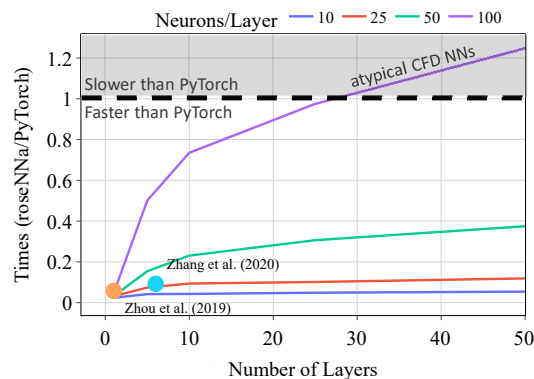
20

roseNNa: MLPs

github.com/comp-physics/rosenna



Bati & Bryngelson *CPC* (2024)



How is it faster than PyTorch?

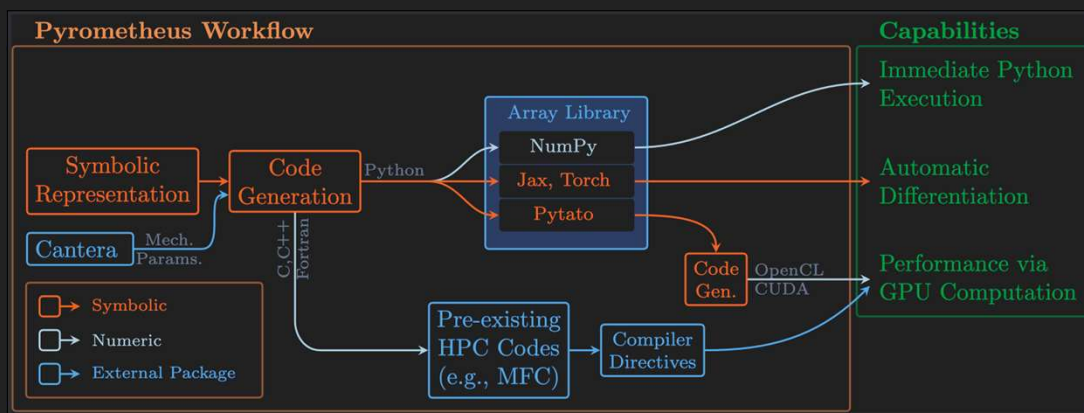
NNs for CFD are shallow
Optimized, hand-written code
quicker for small BLAS3 ops.

21

Pyrometheus: Integrating chemistry

github.com/pyrometheus

Use similar thinking for *thermochemistry*

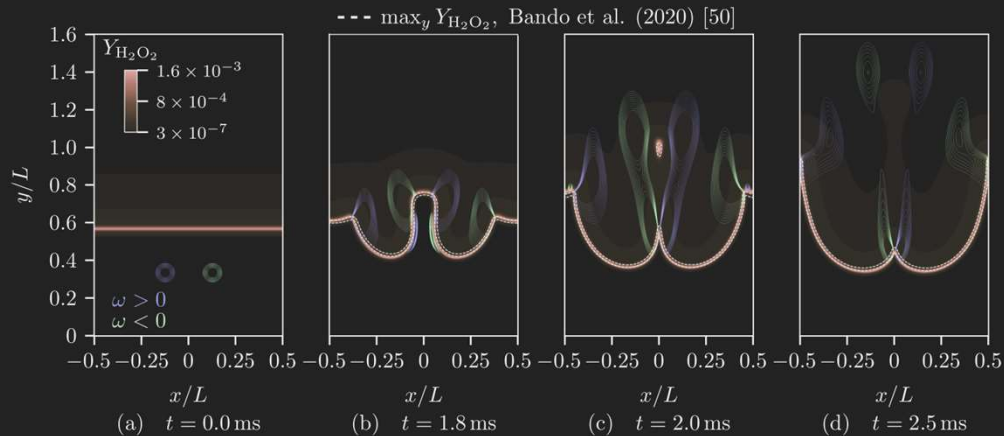


Integrate complex reactions. Easy, fast, scalable, portable, sensitivities, ML, ...

22

Pyrometheus: Integrating chemistry

github.com/pyrometheus

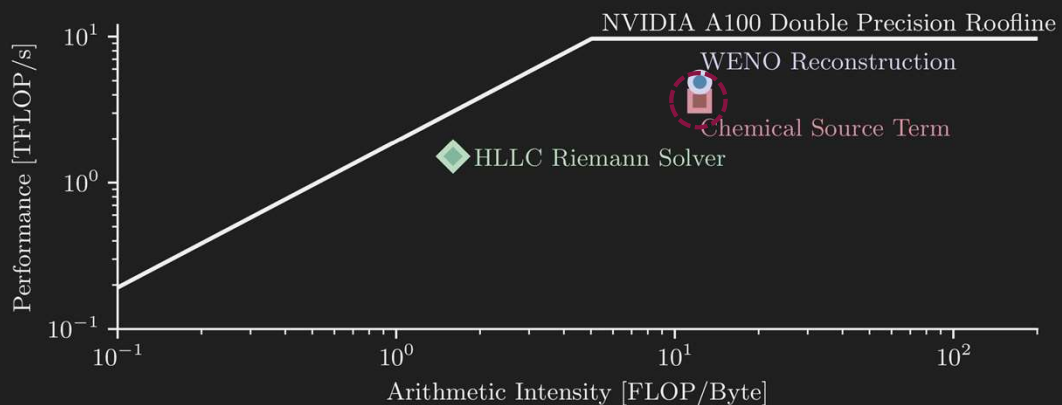


MFC vs. Ref.: Hydrogen–air flame–vortex interaction

23

Pyrometheus: Integrating chemistry

github.com/pyrometheus



Roofline. Compute Bound! Auto-generate fast GPU kernels.

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Looking Forward

Physics + Numerics

More models and methods needed!

Challenges readily manageable

Data analysis

Rabbit filesystem burst output

HPC

No compromises on speed or scaling

Particle trace-based load balancing

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*Do you have any
questions?*



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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T5: Reactive Flow Simulations for Group Combustion

Co-Principal Investigator: Ryan W. Houim

School of Mechanical Engineering, Purdue University

School of Aeronautics and Astronautics, Purdue University



CM³C Purdue Reactive Flow Physics Simulation Team



Chrissie Spanger, 2nd year PhD student

- Expected graduation ~2030
- Worked on EoS for explosive dispersal simulations
- Leveraged student working primarily on AFRL projects



Gregory O'Brien, 2nd year PhD student

- Expected graduation ~2030
- Performing explosive dispersal simulations
- Will transition to C-rod simulations when model is ready.
- Supported by PSAAP



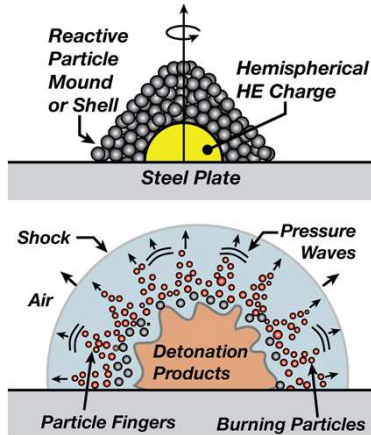
Jacob Posey, Postdoc

- Supporting simulations and model development
- Prior AFRL Intern through the Scholars Program
- Partially supported by PSAAP (~15%)

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CM³C Reactive Flow Simulation Plans

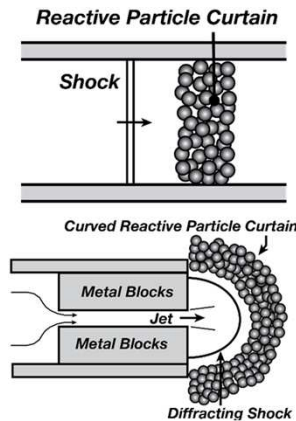
Demonstration Scale



E-E simulations performed to assist experimental team.

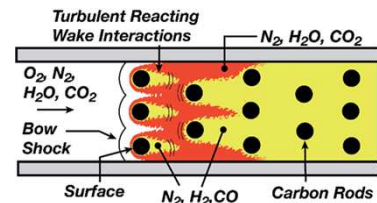
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Mesoscale



Discussion with SNL personnel on proposal development.

Microscale

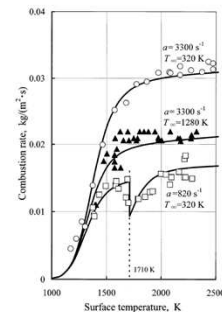
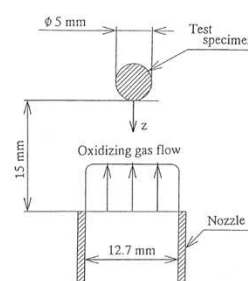
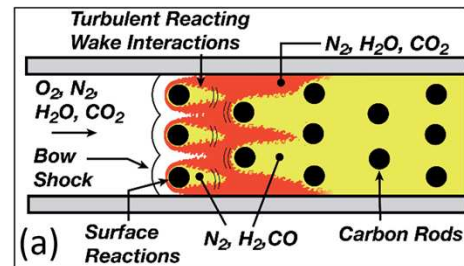


Focus of current effort.

3

Carbon Rod Model: Goals

- Develop and reduce the interface-resolved carbon burning model
 - Same model for particles and rods: just different geometry
 - Help with chemical mechanism reduction.
 - Compare oxidation in wet and dry air.
- Validate burning against experimental measurements
 - Makino *et al.*, *Combust Flame*, 2003
 - Guildenbecher group at Purdue.



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Microscale Carbon Rod/Particle: Governing Equations

- Level Set/IBM used to separate gas and solid (carbon) regions
- Solid Region: Isothermal or Moving Heat Equation

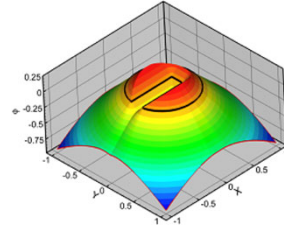
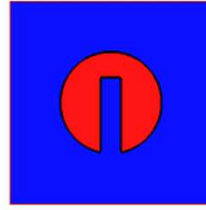
$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \frac{1}{\rho C} \nabla \cdot \lambda \nabla T + \frac{\dot{q}_{\text{gen}}}{\rho C}$$

- Gas Region: Reactive N-S, with Mixture-averaged molecular transport

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot [\rho Y_i (\mathbf{v} + \mathbf{V}_i)] = \dot{\omega}_i,$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) + \nabla p = \nabla \cdot \sigma,$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot [\mathbf{v}(\rho E + p)] = -\nabla \cdot \mathbf{q} + \nabla \cdot (\sigma \cdot \mathbf{v}),$$

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Diffusion Velocity

$$\mathbf{V}_i = \hat{\mathbf{V}}_i - \sum_{i=1}^{N_s} Y_i \hat{\mathbf{V}}_i.$$

$$\hat{\mathbf{V}}_i = -\frac{D_{i,\text{mix}}}{X_i} \nabla X_i,$$

$$D_{i,\text{mix}} = \frac{1 - Y_i}{\sum_{j=1}^{N_s} X_j / D_{ij}},$$

$$\mathbf{q} = \rho \sum_{i=1}^{N_s} Y_i \mathbf{V}_i h_i - \lambda \nabla T,$$

$$\sigma = \mu \left[\nabla \mathbf{v} + \nabla \mathbf{v}^T - \frac{2}{3} \nabla \cdot \mathbf{v} \right]$$

$$p = \rho R u T \sum_{i=1}^{N_s} \frac{Y_i}{MW_i},$$

$$E = \sum_{i=1}^{N_s} Y_i \left(h_{f,i}^0 + \int_{T_{\text{ref}}}^T C_{p,i}(\theta) d\theta \right) - \frac{p}{\rho} + \frac{1}{2} \mathbf{v} \cdot \mathbf{v},$$

5

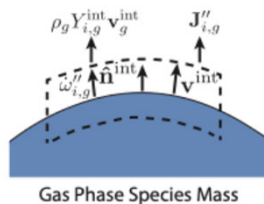
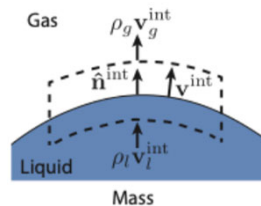


Surface Boundary Conditions

- Mass Balance:** Gives gas-phase Stefan ("blowing") velocity due to phase change.
- Species Mass Balance:** The mass production rate due to surface reaction is balanced with species mass advection and diffusion
- Energy Balance:**
 - 1st step: Assume isothermal to compare to experiments
 - 2nd step: fully couple energy equations

Surface Energy Balance

$$\lambda_g \frac{\partial T_g}{\partial n} - \lambda_c \frac{\partial T_c}{\partial n} = \dot{m}_c'' (H_c - H_g) + \sigma_{nn} u_g$$

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Definitions

$$u = \mathbf{v} \cdot \hat{\mathbf{n}}$$

$$\mathbf{v}_{\text{int}} \cdot \hat{\mathbf{n}} = u_c - r_b$$

$$\dot{m}_c'' = \rho_c r_b$$

Surface Mass Balance

$$u_g = u_c + \dot{m}_c'' \left(\frac{1}{\rho_g} - \frac{1}{\rho_c} \right)$$

Surface Species Mass Balance

$$\dot{\omega}_i'' = \rho_g Y_i (r_b + V_i)$$

$$\sum \dot{\omega}_i'' = \dot{m}_c''$$

Temperature Continuity and no-slip at Surface

$$T_g = T_c = T_{\text{int}} \quad v_g = v_l = v_{\text{int}}$$

6

Carbon Surface Chemistry

- Originally from Bradley *et al.* ProCI, 1985. $C(s) + \frac{1}{2}O_2 \rightarrow CO,$ (15)
- Accounts for wet and dry oxidation of carbon $C(s) + CO_2 \rightarrow 2CO,$ (16)
- Involves 5 reactions and 8 gas species $C(s) + H_2O \rightarrow CO + H_2,$ (17)
- $C(s) + OH \rightarrow CO + H,$ (18)
- $C(s) + O \rightarrow CO,$ (19)

Table 1
Kinetic parameters for heterogeneous reactions Eqs. (15)–(19) taken from Bradley *et al.* [49].

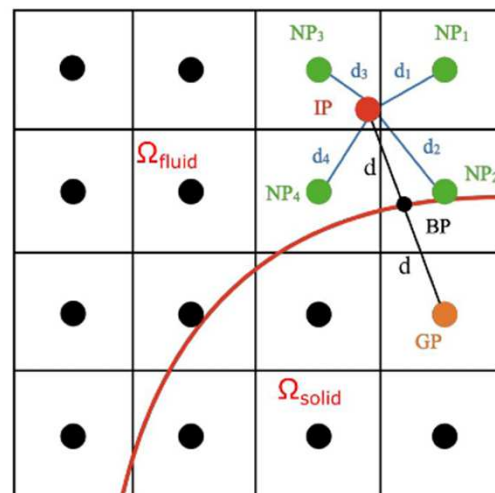
r	B_r	E_r [kJ/mol]	A_r	R_r [kg/(m ² s)]
15a	0	125.6	2400 [kg/(m ² s atm)]	$R_{15} = \frac{K_{15a}P_{O_2}\xi}{1+K_{15a}P_{O_2}} + K_{15c}P_{O_2}(1-\xi)$ $\xi = 1 / (1 + \frac{K_{15d}}{K_{15c}P_{O_2}})$
15b	0	-17.17	21.3 [1/atm]	
15c	0	63.64	0.535 [kg/(m ² s atm)]	
15d	0	406.1	18.1×10^6 [kg/(m ² s)]	
16	0	285	9×10^3 [kg/(m ² s atm ^{0.5})]	$R_{16} = K_{16}P_{CO_2}^{0.5}$
17	0	288	4.8×10^5 [kg/(m ² s atm ^{0.5})]	$R_{17} = K_{17}P_{H_2O}^{0.5}$
18	-0.5	0	361 [kg K ^{0.5} /(m ² s atm)]	$R_{18} = K_{18}P_{OH}$
19	-0.5	0	665.5 [kg K ^{0.5} /(m ² s atm)]	$R_{19} = K_{19}P_O$

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Surface Boundary Condition Implementation

- Ghost-fluid method (GFM):
 - Interpolate data on gas-phase grid to image point (IP)
 - *Inert BCs*: Fill ghost point (GP) by direct copying or reflecting IP values.
- Need to implement reactive GFM
 - Need to apply surface species mass balance.
 - Image point is distance d away from surface $\Rightarrow$ cannot use surface balance conditions directly!
 - Need to build a surface wall model (analogous to wall model in turbulence)

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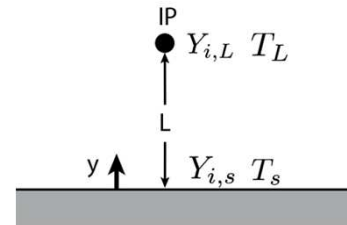
8



Reactive Wall Model: Assumptions and Governing Equation

Assumptions:

- Quasi-steady
- Fickian Diffusion
- Constant ρD
- No gas-phase reactions between surface and IP



Species mass equation and BCs:

Species mass equation between surface and IP

$$\dot{m}_c'' \frac{dY_i}{dy} = \frac{d}{dy} \left(\rho D \frac{dY_i}{dy} \right)$$

Continuity equation between surface and IP

$$\dot{m}_c'' = \text{const}$$

Surface species mass balance

$$\dot{\omega}_i'' = \dot{m}_c'' Y_{i,s} - \rho D \left. \frac{dY_i}{dy} \right|_s$$

Known mass fraction at IP determined from interpolation

$$Y_i(L) = Y_{i,L}$$

Surface reaction model

$$\dot{\omega}_i'' = f(Y_s, T_s, p_s, \dots)$$

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Reactive Wall Model: Solution

Integrating twice and applying BCs gives system gives

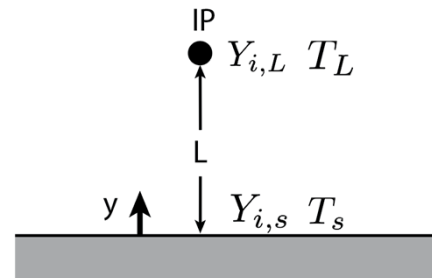
$$Y_{i,s} = \frac{\dot{\omega}_i''}{\dot{m}_c''} + \left(Y_{i,L} - \frac{\dot{\omega}_i''}{\dot{m}_c''} \right) e^{-\frac{\dot{m}_c'' L}{\rho D}}$$

with the Surface reaction model

$$\dot{\omega}_i'' = f(Y_s, T_s, p_s, \dots) \quad \dot{m}_c'' = \sum \dot{\omega}_i''$$

Difficult to solve nonlinear algebraic equations for surface mass production rate and mass fractions

- Needs Brent's method if only 1 surface reaction
- Multiple surface reactions almost always diverge to non-physical solutions.



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Wall Model Results with 1 Reaction: $C + 1/2O_2 \rightarrow CO_2$

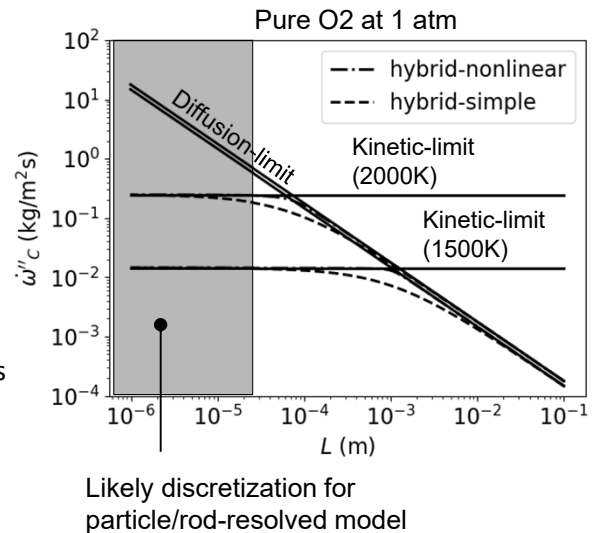
Compared several models:

1. Full model (hybrid-nonlinear)
2. Diffusion-limited ($Y_{O_2,s} = 0 \Rightarrow \dot{m}''_{diff} = \frac{2MW_c \rho D}{MW_{O_2} L} Y_{O_2,L}$)
3. Kinetic-limited: $\dot{m}''_{kin} = \dot{\omega}''_i(Y_{i,L}, T_s, p_s)$
4. Simplified hybrid: $\dot{m}''_{hyb} = \frac{\dot{m}''_{diff} \dot{m}''_{kin}}{\dot{m}''_{diff} + \dot{m}''_{kin}}$

Can likely assume kinetic limited surface reactions based on resolution requirements

- 2 mm-diameter rod with 100 – 200 grid points across the diameter gives: $\Delta x \sim 10$ to $20 \mu m$

Simplified model overexaggerates the width of the transition region

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Wall Model Results: Full Surface Reaction Mechanism

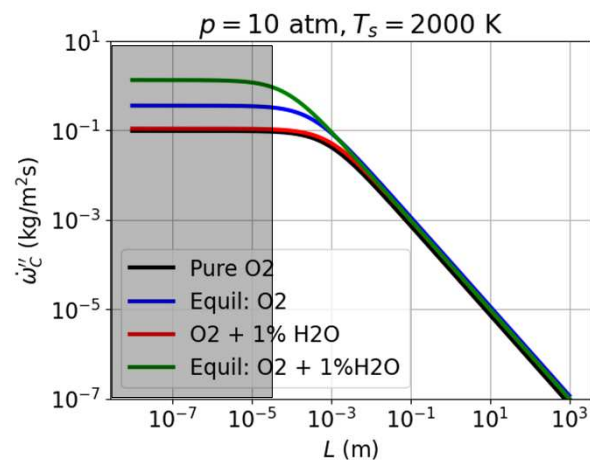
Wall model with full reaction mechanism diverges with more than 1 surface reaction.

- $\Rightarrow$ Lacks robustness and is not a good candidate for CFD
- Results with multiple oxidizers use the simplified hybrid model

Results shown for pure O₂, 99%O₂ and 1% H₂O (by mole) and both mixtures at chemical equilibrium (TP)

- Still likely kinetic-limited at grid resolution

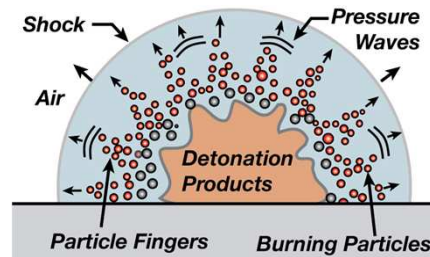
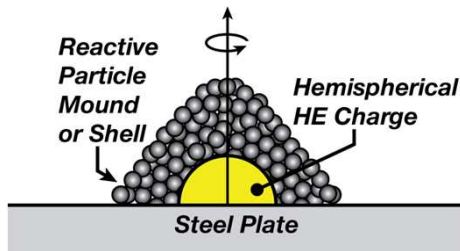
Dissociation of H₂O into OH and O has a significant effect on surface burning rate at higher temperatures

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Macroscale Simulations for Experimental Design



- Simulate explosive dispersal, ignition, and combustion of Mg particles by PETN with the HyBurn code to aid design of experiments at Purdue
- Eulerian model used for the multiphase mixture and programmed burn for the explosive
- Simplified approach for thermodynamics and reactions based on our previous work with Al particles

Houim & Oran, *J. Fluid Mech.* (2016)

Houim, *Shock Waves*, (2021)

Posey *et al.*, *Phys. Fluids* (2021)

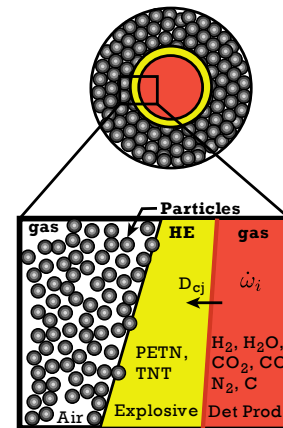
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Three Phase Eulerian-Eulerian Multiphase Flow Model

- The models and algorithms developed in-house¹⁻³
- The three-phases are: gas, explosive, and particles (Eulerian)
- A diffuse interface method (DIM) is used for reactive gas and high-explosive (HE)
 - Novel detonation burn model converts HE to products at the CJ composition^{1,3}
- An Eulerian multiphase model is used for the particles²
 - Accounts for phase change, drag, heat transfer, collisions, and compaction
 - Particles burn via an empirical burning model
- Turbulence modeled using ILES⁴



¹Houim, *Shock Waves* (2021)

²Houim, Oran, *J. Fluid Mech.* (2016)

³Egeln, *et al.*, *Phys. Fluids*, (2023)

⁴Thornber, *et al.*, *J. Comp. Phys.*, (2008)

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Kuhl & Khasainov JWL EOS and Lumped Species Model

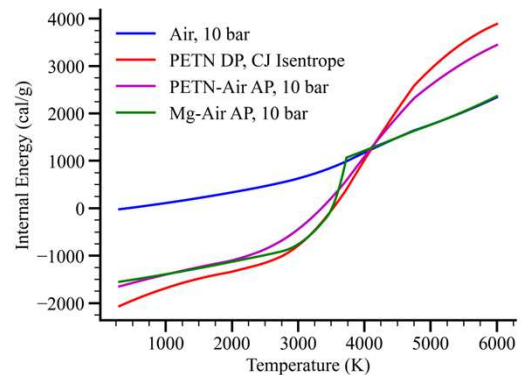
- The Kuhl variant of the JWL equation of state is used (parameters from PETN at 1.6 g/cc)

$$p_g = A \left(1 - \frac{\omega \rho_g}{R_1 \rho_o} \right) \exp \left(-\frac{R_1 \rho_o}{\rho_g} \right) + B \left(1 - \frac{\omega \rho_g}{R_2 \rho_o} \right) \exp \left(-\frac{R_2 \rho_o}{\rho_g} \right) + \rho_g R_{mix} T_g$$

- Lumped species approach used for major groups

- Air
- Detonation products (DP)
- DP-air afterburning products (AP)
- Mg-Air combustion products (Mg AP)

- Internal energies for each species are piecewise polynomial fits to equilibrium thermodynamic data

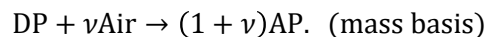
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Gas-Phase and Particle Combustion Model

- The gas-phase after burning reaction is assumed to be mixing limited

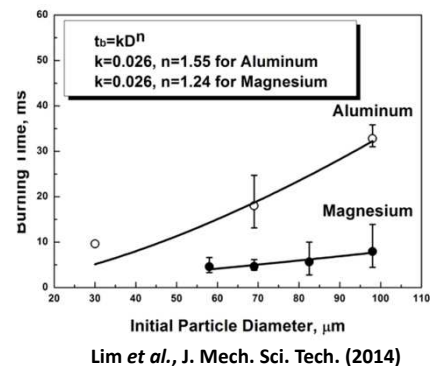


- Few correlations exist for Mg burning in air.

- $t_b = K d_p^{1.24}$ Lim *et al.*, J. Mech. Sci. Tech. (2014)
- Phase change source term: $\dot{M}_p = \frac{3\alpha_p \rho_p}{t_b} \frac{Nu}{2}$

- Ignition delay time for Mg dust clouds

- $t_{ign} = A \left(\frac{d_p}{17} \right)^n p_g^m \exp \left(\frac{E_a}{RT_g} \right)$ Medvedev *et al.*, Combust. Explos. Shock Waves (1982)
- Scalar transport equation for an ignition parameter, f
- Particles burn when $f \geq 1$

Lim *et al.*, J. Mech. Sci. Tech. (2014)

Ignition parameter transport

$$\frac{\partial \alpha_p \rho_p f}{\partial t} + \nabla \cdot (\alpha_p \rho_p f \mathbf{u}_p) = \frac{\alpha_p \rho_p}{t_{ign}}$$

Kuhl *et al.*, Combust. Explos. Shock Waves (2010)CM³C

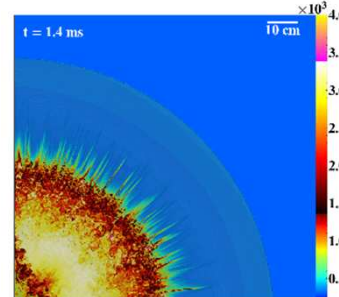
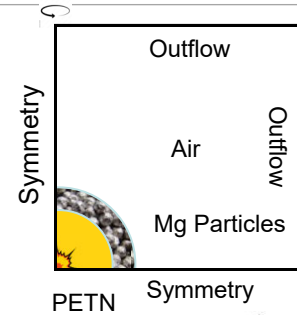
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Explosive Particle Dispersal Initial Conditions

- Axisymmetric to allow for faster computation times
 - 1 m by 1 m domain
 - Non-reflecting Outflow BCs used on outer boundaries
- PETN hemisphere diameter: 12 mm
- Mg particle shell thicknesses: 1, 5, 10 mm
- Initial particle shell volume fraction: 50%
- Mg particle diameters: 10, 45 μm (monodisperse)

- Adaptive Mesh Refinement with AMReX
 - 8192² grid at finest level (8 levels)
 - 122 μm grid at finest level.

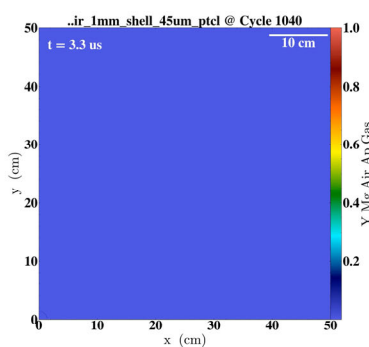
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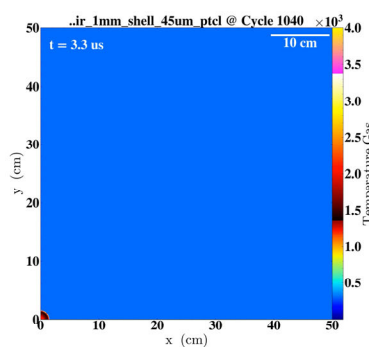


Results: 1 mm shell with 45 μm -diameter particles

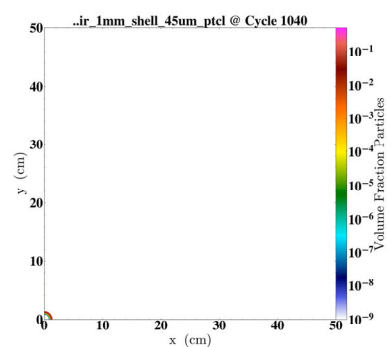
**Mg-Air Combustion
Product Mass Fraction**



Gas-Phase Temperature



Mg Particle Volume Fraction



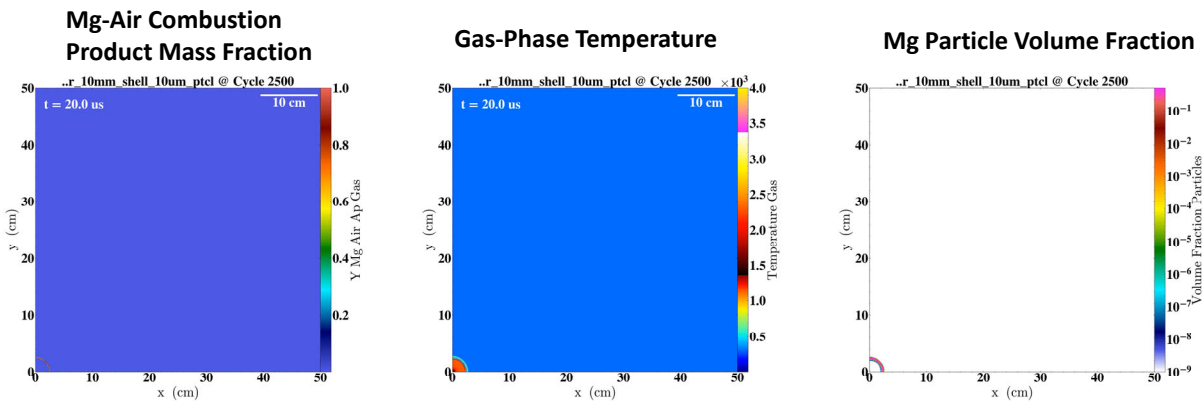
- Some of the Mg particles ignites rapidly in the post-shock air
- Many of the particles are ejected into the air without igniting
- Likely need to consider anerobic Mg burning in this case

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Results: 10 mm shell with 10 μm -diameter particles



- Majority of the Mg burning is after $\sim 150 \mu\text{s}$
- Need to perform analysis on this case... don't see this burning mechanism with Al particles
- Likely due to larger Stokes numbers and lower ignition delay of the burning Mg particles

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Work for the next 6 months

- Microscale (Carbon Rods)
 - Implement reactive BC (kinetic-limited) into the IBM model in HyBurn to shake out model
 - Setup isothermal simulations and compare model to existing experimental results
 - Test reduced chemical kinetic models
 - Assist with implementation into MFC
- Macroscale
 - Implement Carbon particle burning model
 - Perform simulations with C-particles
 - Analyze results and assist experimental team as needed
- Mesoscale
 - Continue collaborating with SNL and send them preliminary results as needed for their proposal development

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Questions?



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SPARE

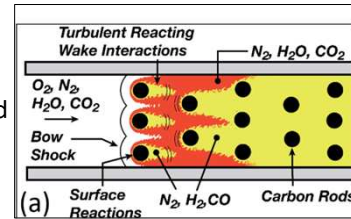




Microscale Simulations – Goals and Approach

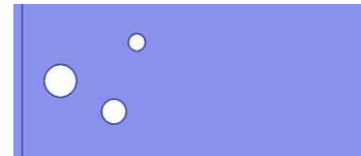
Science goals:

1. Understand the role of burning wake interactions on particle combustion in compressible flows
2. Understand the unsteady effects such as ignition and flame spread across rows of particles
3. Develop sub-grid models of group combustion that can be used in E-L and E-E models



Modeling Approach: Resolve details of the flow around each burning particle/droplet or rod

- Fixed carbon rod configuration has a 2D analogy and allow ease of designing experimental diagnostics and 3D simulations details
- Arrays of burning Mg droplets to capture effects of volatile combustion and breakup



Resolved simulation of a shock wave interacting a vaporizing nitromethane droplet

- Over 10^{11} grid points to resolve flow around many droplets and capture wake interactions $\Rightarrow$ **Exascale is a must!**

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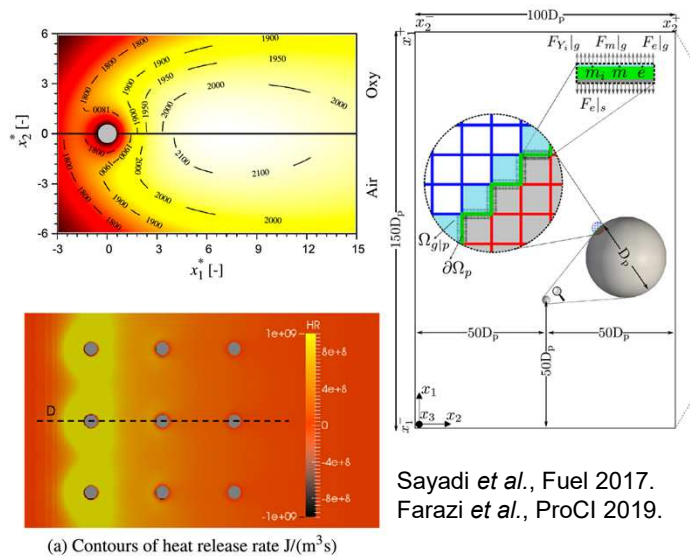


Carbon Particle Simulations: Prior Work

Heinz Pitsch's group performed a series of simulations for interface-resolved carbon particle simulations.

- Used immersed boundary methods
- Used detailed gas and surface chemistry
- Coupled solid heat equation to reacting gas
- Restricted to Low Mach laminar flow

They looked at effect of single particles as well as arrays of particles



Sayadi *et al.*, Fuel 2017.
Farazi *et al.*, ProCI 2019.

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Three Phase Eulerian-Eulerian Multiphase Flow Model

Explosive Phase – Mass and Volume Fraction

$$\frac{\partial \alpha_c \alpha_h \rho_h}{\partial t} + \nabla \cdot (\alpha_c \alpha_h \rho_h \mathbf{u}_c) = -\dot{M}_{h \rightarrow g} \quad \frac{\partial \alpha_h}{\partial t} + \nabla \cdot (\alpha_h \mathbf{u}_c) = \alpha_h \nabla \cdot \mathbf{u}_c - \frac{\dot{M}_{h \rightarrow g}}{\rho_h}$$

Gas Phase – Species Mass

$$\frac{\partial \alpha_c \alpha_g \rho_g Y_i}{\partial t} + \nabla \cdot [\alpha_c \alpha_g \rho_g Y_i (\mathbf{u}_c + \mathbf{V}_{i,g})] = \alpha_c \alpha_g \dot{\omega}_i + \dot{M}_{i,d \rightarrow g} + \dot{M}_{i,h \rightarrow g}$$

Continuous Phase – Mixture Momentum and Energy

$$\frac{\partial \alpha_c \rho_c \mathbf{u}_c}{\partial t} + \nabla \cdot (\alpha_c \rho_c \mathbf{u}_c \mathbf{u}_c) = -\alpha_c \nabla p_c + \alpha_c \nabla \cdot \boldsymbol{\sigma}_c + \mathbf{S}_{\text{mom}}$$

$$\frac{\partial \alpha_c \rho_c E_c}{\partial t} + \nabla \cdot [\alpha_c \mathbf{u}_c (\rho_c E_c + p_c)] = -p_c \nabla \cdot (\alpha_s \mathbf{u}_s) - \alpha_c \nabla \cdot \mathbf{q}_c + \alpha_c \nabla \cdot (\boldsymbol{\sigma}_c \cdot \mathbf{u}_c) + \mathbf{S}_{\text{en}}$$

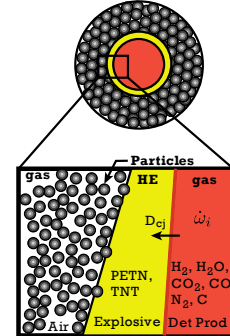
Granular Phase - Aluminum Particles

$$\frac{\partial \alpha_s \rho_s}{\partial t} + \nabla \cdot (\alpha_s \rho_s \mathbf{u}_s) = -\dot{M}_{\Delta 1}$$

$$\frac{\partial \alpha_s \rho_s \mathbf{u}_s}{\partial t} + \nabla \cdot (\alpha_s \rho_s \mathbf{u}_s \mathbf{u}_s) + \nabla p_s + \nabla p_{\text{fric}} = -\alpha_s \nabla p_g - \mathbf{S}_{\text{mom}}$$

$$\frac{\partial \alpha_s \rho_s E_s}{\partial t} + \nabla \cdot (\alpha_s \rho_s E_s \mathbf{u}_s) = -p_s \nabla \cdot \mathbf{u}_s + \mathbf{S}_{\text{gran en}}$$

$$\frac{\partial \alpha_s \rho_s e_s}{\partial t} + \nabla \cdot (\alpha_s \rho_s e_s \mathbf{u}_s) = \mathbf{S}_{\text{sens en}}$$



Volume Fractions

$$\alpha_g + \alpha_h = 1$$

$$\alpha_c + \alpha_s = 1$$

Continuous Mixture

$$\rho_c = \alpha_g \rho_g + \alpha_h \rho_h$$

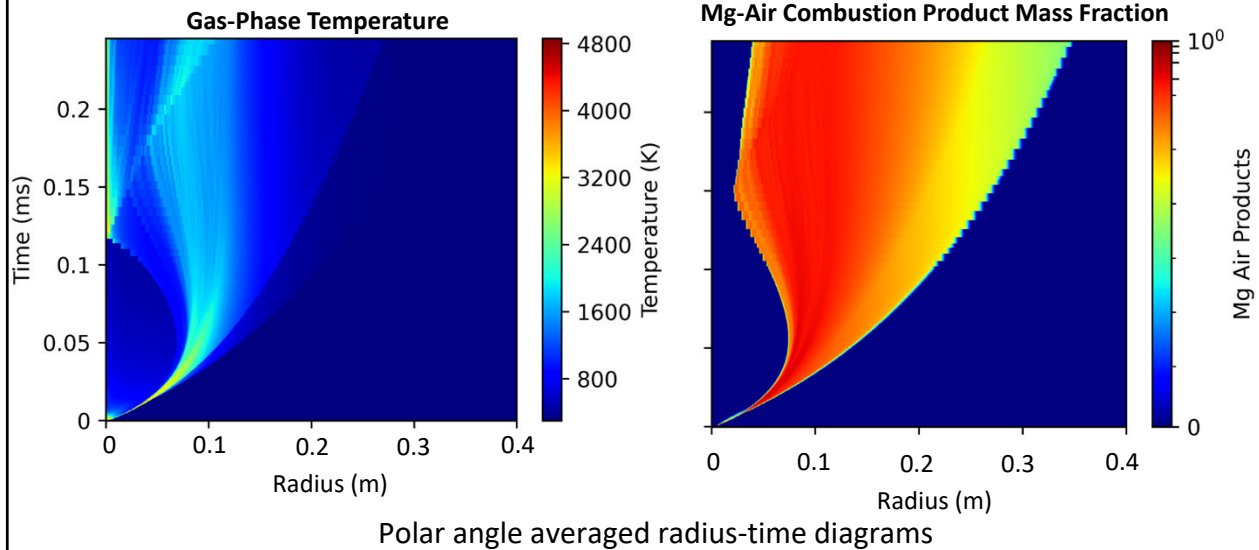
$$\rho_c E_c = \alpha_g \rho_g E_g + \alpha_h \rho_h E_h$$

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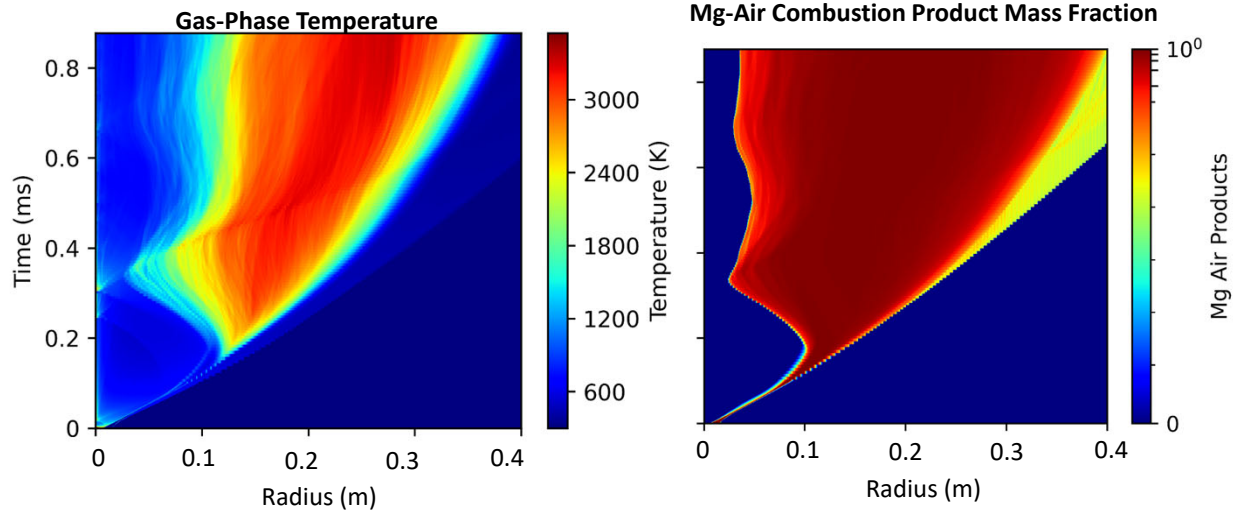
Results: 1mm shell with 45 μm-diameter particles

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Results: 10mm shell with 10 μ m particles



Polar angle averaged radius-time diagrams

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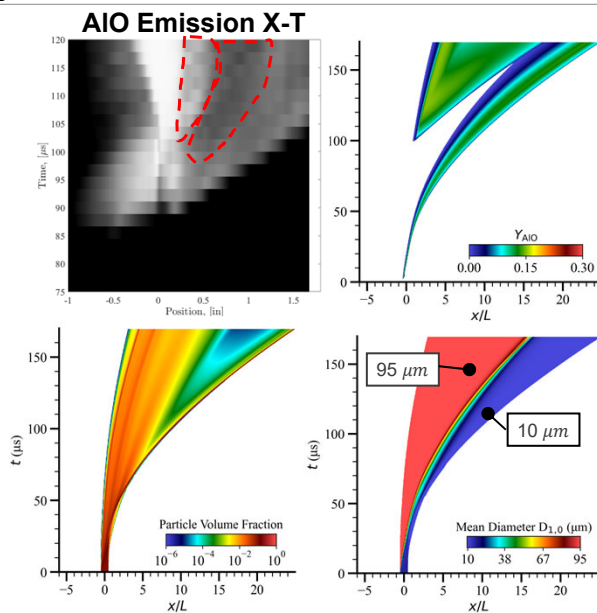


Mesoscale – Preliminary Reactive Al Particle Curtain Results

- Working with SNL to develop proposal ideas to fund these experiments.
 - An LDRD attempted but didn't pan out
 - SNL working on other DoD (e.g., JMP) funding vehicles
- Setup:
 - Mach 11.2 shock in air at 3240 Pa
 - Bidisperse Mix: 25% 10 μ m and 75% 95 μ m
 - 10% Al particle volume fraction
- AIO imaging and spectrometer indicates Al combustion.
- Our new polydisperse E-E model* qualitatively captures these experimental observations.

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*Fox, et al., *Int. J. Multiphase Flow*, 2024



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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T6: Overview of Experiments and Progress to Date

Co-Principal Investigator: D. Guildenbecher

School of Mechanical Engineering, Purdue University



Prof. Daniel R. Guildenbecher



2011-2015:
Senior Member of the
Technical Staff



2016-2021:
Principal Member of
the Technical Staff



2022-2023:
Distinguished Member
of the Technical Staff



2023-present:
Associate Professor
School of Mechanical Engineering

**Multiphase Flows
in National Security Missions**

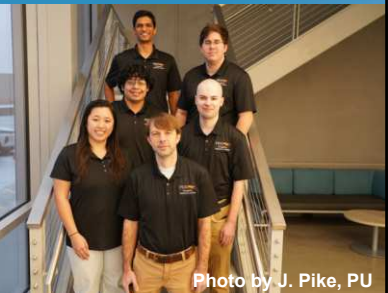
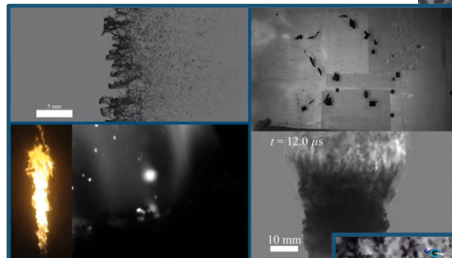
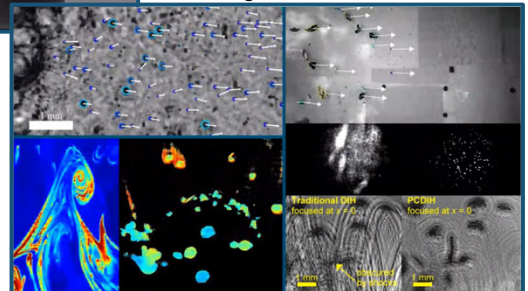


Photo by J. Pike, PU

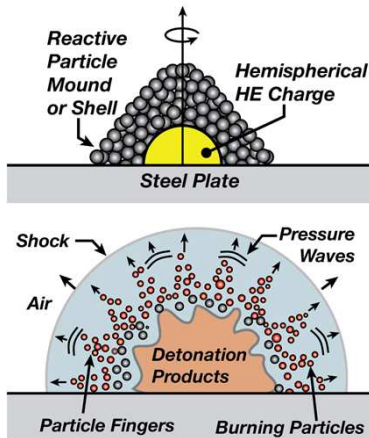
**Advanced Experimental
Diagnostics**



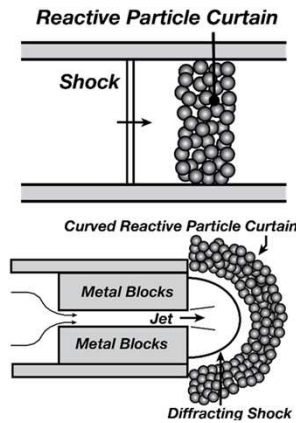
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CM³C Experimental Plans

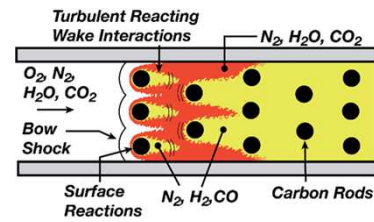
Demonstration Scale



Mesoscale



Microscale

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CM³C Experimental Schedule

Year 1-2 Tasks:

- Establish the demonstration problem → Explosive tests to be performed at Purdue using our existing facilities and diagnostics
- Lay the foundation for HF-PR validation → Design/build carbon rod experiments and validate single rod results against published literature

Task	Year 1	Year 2	Year 3	Year 4	Year 5
Explosive demonstration using standard imaging and pressure gauges		▲	▲		
Single rod experiments validated against published literature		▲			
Design/build upgraded multi-rod configurations					
Multi-rod group combustion measurements				▲	
Sub-scale testing of blast-hardened LAS/emission/extinction gauges					
Design/build multi-rod configurations with spacing guided by UQ team					
Multi-rod, multi-parameter experimental measurements guided by UQ team					▲
Demonstration experiments with LAS and other advanced diagnostics					▲
Final Publication/documentation					▲

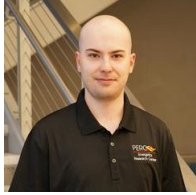
▲ Indicate major milestones, see T1 presentation by S. Balachandrar

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CM³C Experimental Team



Cohen Nunes, 3rd year PhD student

- Expected graduation Spring or Summer 2027
- Leading optical diagnostics for demonstration scale experiments
- Supported by PSAAP



Bekah Travis, 4th year DPhD student

- Expected graduation Spring or Summer 2027
- Leading explosive physics for demonstration scale experiments
- NDSEG funded and SNL graduate student intern



Alex Millauer-Hsu, 1st year DPhD student

- Expected graduation ~2030
- Leading microscale c-rod experiments
- Supported by PSAAP



Colton Willhardt, Postdoc

- Supporting experimental operations
- Prior SNL graduate student intern
- Partially supported by PSAAP (~15%)

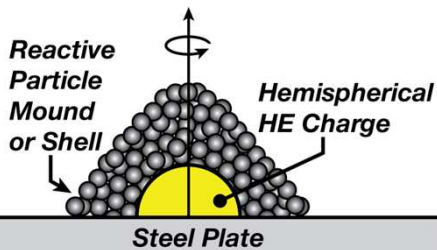
Assisted by several others who are supported on related projects: Nick Gurdian-Aponte (UGrad research for credit), Rees Verleur (3rd year DPhD, supported by NNSA/SSAP), Amogh Natu (2nd year DPhD, supported by AFRL), Amy Chang (2nd year PhD, supported by NASA), Daniel Struk (DPhD starting Fall 2026, supported by fellowship and ONR)

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CM³C Demonstration Problem



Charge: 3 cm diameter hemisphere of PETN ($C_5H_8N_4O_{12}$)

Particles: Free-standing mounds of carbon and magnesium particles

- Carbon reacts at the surface. Magnesium reacts in a vapor flame

Diagnostics: **QoI1:** high-speed imaging for blast wave locations; **QoI2:** pencil gages for pressure; **QoI3:** visible and X-ray extinction for particle concentration

Advantages:

1. Well-defined initial conditions, free of uncertainty due to case confinement
2. PETN equations of state (EoS) are publicly available, and PETN produces little soot
3. We have extensive modeling and experimental experience with this base charge



Ongoing Metalized Charge Experiments At Purdue (Courtesy S. Son, R. Travis)

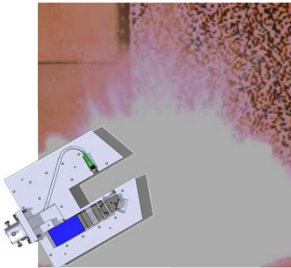
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CM³C Demonstration Problem

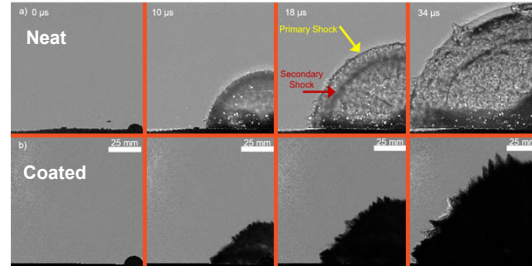
Our diagnostic advancements will enable measurements of **QoI4**: temperature variations due to combustion and heat release, **QoI5**: fingering instabilities, and **QoI6**: PDFs of pressure and particulate volume fraction

Blast-Hardened Laser Absorption Spectroscopy



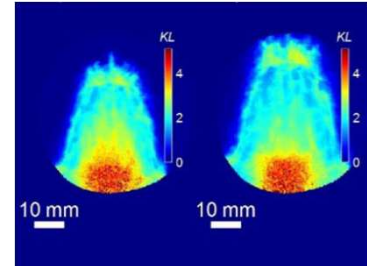
Travis *et al.* (2025) *Work In Progress*

Quantitative Schlieren



Nunes and Guildenbecher *et al.* (2025) *AIAA SciTech*

Extinction Imaging



Saltzman and Guildenbecher *et al.* (2023) *Propellants Explos. Pyrotech.*

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CM³C Demonstration Problem

Planned DoD partnerships create extensive opportunities for larger scale demonstration experiments



Naval Surface Warfare Center,
Crane Division, Crane, Indiana

- Indoor and outdoor explosive test facilities focused on expeditionary warfare missions
- Extensive collaborations with Purdue



Georgia Tech



Air Force Research Laboratory,
Munitions Directorate, Eglin AFB

- National leaders in munitions development and predictions
- Large explosive test facilities
- Extensive collaborations with UF

We are not dependent on collaborations, but will seek to maximize them for broad impact



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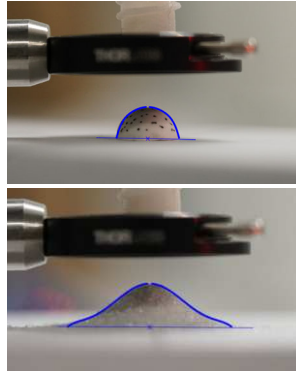
Demonstration Problem: Progress To Date

We have verified the free-standing mound configuration in static tests

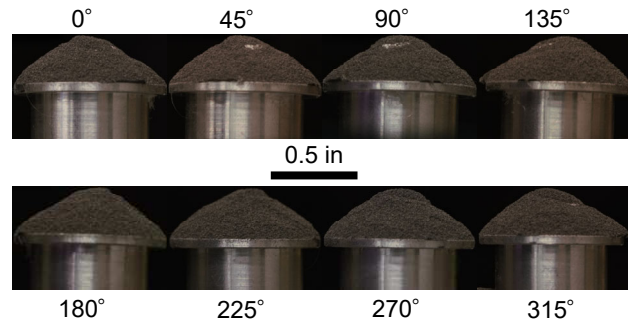
Raw Image



Contour Recognition



Work performed by Navi Singh
US Army Westpoint Cadet
and Summer 2025 research intern



Takeaway: Possible but likely problematic for larger, outdoor shots

Concerns include (i) uneven coating, (ii) shot-to-shot variability, (iii) time on range needed to coat the charges, and (iv) disruption by breezes before initiation

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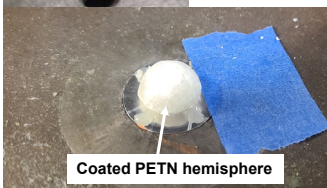
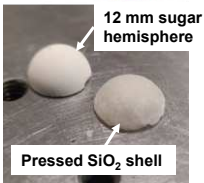


Demonstration Problem: Progress To Date

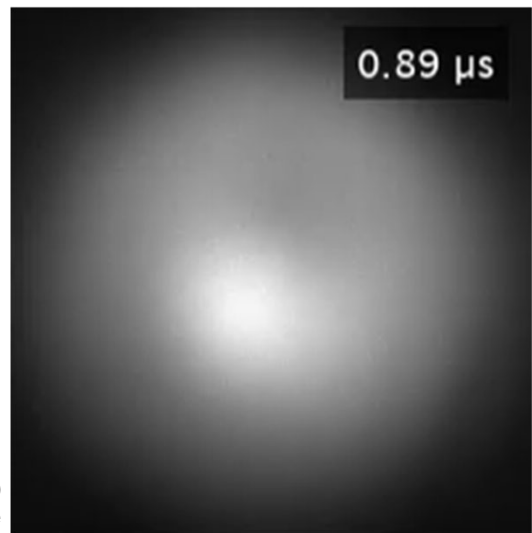


Nunes *et al.* (2025) AIAA SciTech demonstrated use of Kinetic Sand™ as an unconfined hemispherical shell
We have now advanced to custom shells using pre-characterized particles

- 100 nm flake, 800 nm spherical, and 10 μm spherical SiO_2 particles
- Polydimethylsiloxane (PDMS) binder (silicon-based organic polymer)
- Size distributions quantified in advance using a Malvern Mastersizer



Diffuse Back-Illumination
Extinction Imaging (DBI-EI)
of an SiO_2 coated hemisphere



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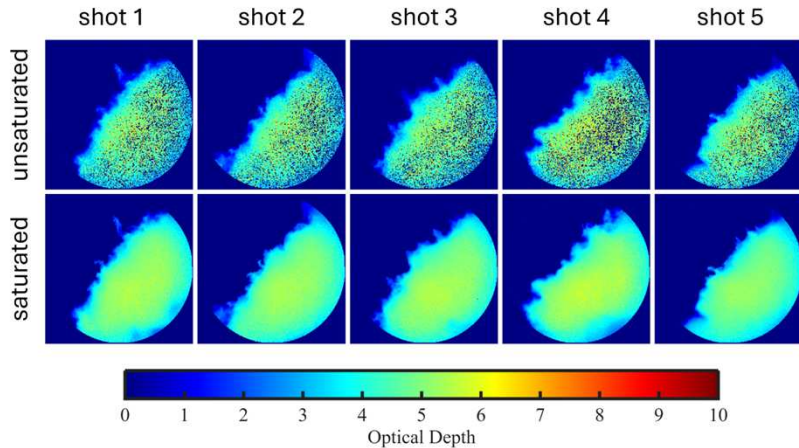
This work was supported by an 2023-2025 SNL Academic Alliance LDRD led by Adam Hammond-Clements

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Demonstration Problem: Progress To Date

DBI-EI Quantifies the Light Extinction, $OD = -\ln(I/I_0)$



OD is a function of:

- Particle concentration, particle size, index-of-refraction, optical configuration, and *multi-scattering*
- Out new theory in Nunes *et al.* (2026) *Optics Express* should allow us to account for multi-scattering and achieve quantitative assessment of particle concentrations
- Future possibility to integrate this as a post-processor of simulations akin to our prior work on fireball temperatures.

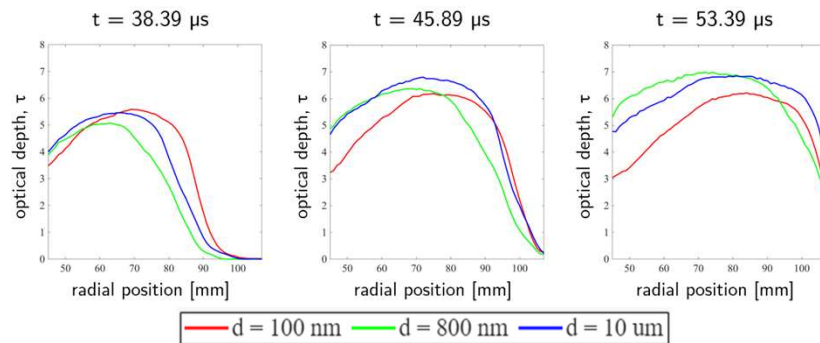
CM³C This work was supported by an 2023-2025 SNL Academic Alliance LDRD led by Adam Hammond-Clements

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Demonstration Problem: Progress To Date

Even for Non-Reacting Particles, We see some Surprising Transport Phenomena



Azimuthally averaged LOS extinction
at several times since initiation

Transport of intermediate particle size (**green**) does not clearly interpolate between the large (**blue**) and small (**red**) particles

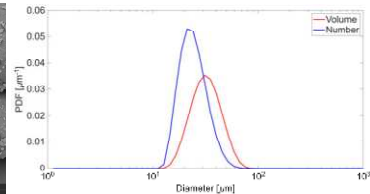
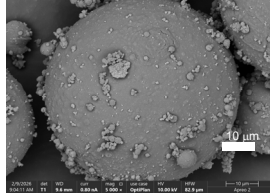
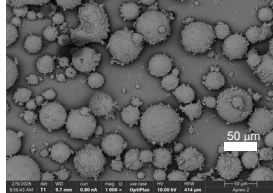
- Caveat: Measurement uncertainties may be a contributing factor
- Ongoing efforts are improving data processing and interpretation of results

CM³C This work was supported by an 2023-2025 SNL Academic Alliance LDRD led by Adam Hammond-Clements

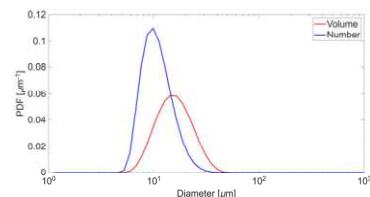
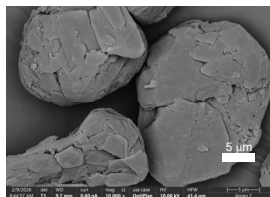
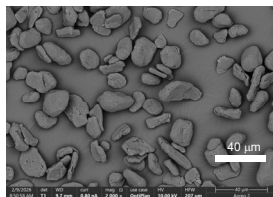
12

Demonstration Problem: Progress To Date

First Reactive Particle Test Campaign is Planned at Purdue this Month



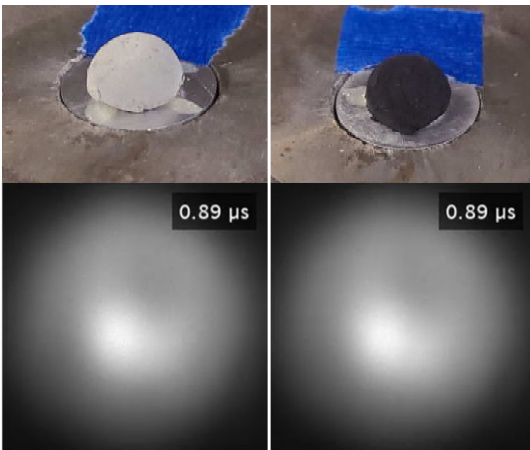
Nominally 45 μm Magnesium Particles Purchased from US Nanomaterials



Nominally 25 μm "Spherical" Carbon Particles Purchased from US Nanomaterials

Demonstration Problem: Progress To Date

First Reactive Particle Test Campaign is Planned at Purdue this Month



Prior proof-of-concept demonstrations of Al (left) and carbon black (right)+PDMS coated PETN hemispheres



Initial demonstration of a 10 mm thick Mg shell showing (left) the outer diameter and (right) the indentation for the 12 mm diameter PETN hemisphere

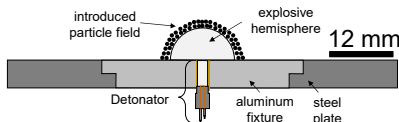


Demonstration Problem: Progress To Date

First Reactive Particle Test Campaign is Planned at Purdue this Month

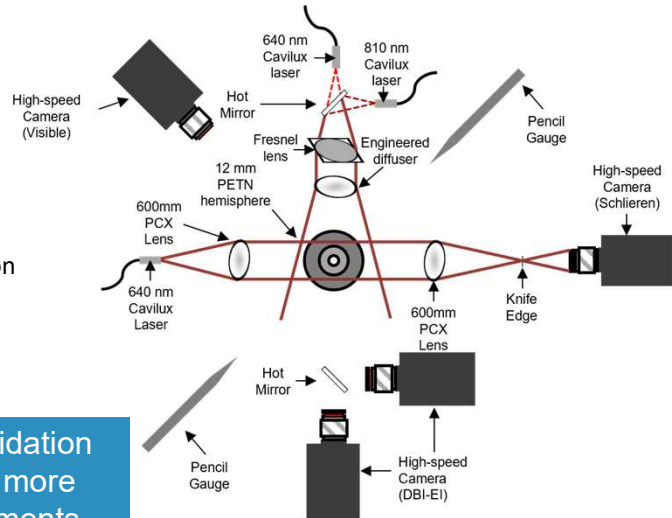
12 mm diameter PETN charges, coated with Mg/PDMS and C/PDMS shells

- Size distributions, shell thickness, and volume fractions quantified in advance
- **QoI1:** high-speed schlieren imaging for blast wave locations
- **QoI2:** pencil gages for pressure
- **QoI3:** visible extinction for particle concentration



Success will provide initial validation data and inform planning for more complete larger-scale experiments

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Demonstration Problem: Progress To Date

Intent is to do Larger-Scale Experiments at a Partner Organization, hopefully Early FY27

We are currently actively discussing with:

- **NSWC Crane Indiana:** POC Eric Scheid. They have both indoor and outdoor test arenas and experienced explosive operators. They are currently reviewing the feasibility of PETN pressing and financial ability to support their team.
- **AFRL Eglin Florida:** POC Don Littrell. They supported the UF PSAAP-II center and likely the nations best blast characterization arena. We are currently collaborating on a white paper for AFRL support of PSAAP-IV shots
- **SNL Albuquerque:** POC Ryann Marinis. SNL has been conducting similar PETN hemisphere experiments stemming from prior collaborations with this team. Supporting project may be unable to perform our desired experimental scales until ~FY27 or 28.
- **LLNL:** POC has been identified, we are trying to engage in conversation.



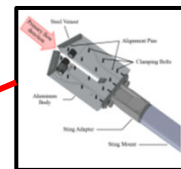
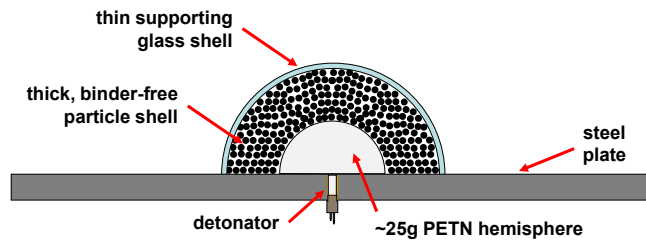
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Demonstration Problem: Progress To Date

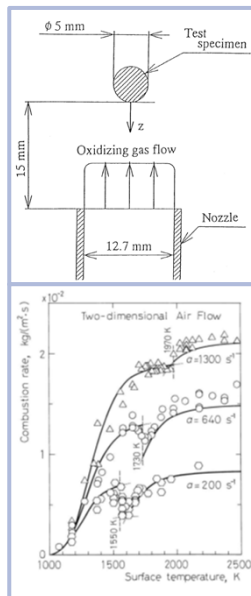
At the larger-scale, we hope to reduce uncertainty in the particle shell

- No PDMS binder with poorly known chemistry
- Larger charge reduces relative influence of initiation spot size
- Thicker shell such that most particles are not crushed by the shock waves
- Desired diagnostics: (i) high-speed imaging, (ii) pressure gauges, (iii) x-ray or visible extinction
- We hope to engage Chris Murzyn of SNL with his blast-hardened laser absorption gauges, likely leveraging current JMP funding

CM³CMurzyn et al. (2024) *J. Appl. Phys*

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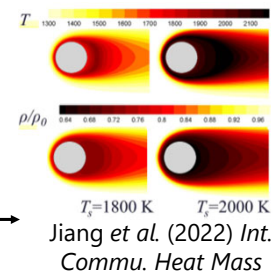
CM³C Microscale Problem



The combustion of carbon rods has been extensively studied by Makino *et al.* 1990-2011

- 5 mm Ø rods, Joule heated to ignition
- Surface temperatures measured by radiography
- Regression rates measured by videography

- Results have been used to validate HF-PR simulations of isolated particles



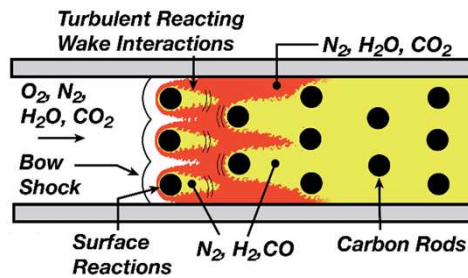
Again, proven experiments with a foundation of existing data

We plan to acquire new experimental data for several V&V objectives:

1. Validate HF-PR simulations of *group combustion of particle arrays*
2. Quantify uncertainty in chemical kinetics and validate HF-PR modeling assumptions (such as negligible radiation effects)
3. Support uncertainty reduction via computational informed experiments (relay validation)



CM³C Microscale Problem



Conditions: Flow velocities up to Mach 1.5, 300 K static temperature (500 K stagnation temperature)

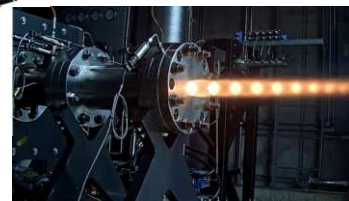
- Equivalent to leading particles in a multiphase blast propagating into the surrounding air

Diagnostics: (i) High-speed videography of regression rates, (ii) IR pyrometry of surface temperature, (iii) Laser absorption spectroscopy (LAS) of gas products, (iv) Laser induced fluorescence (LIF) of intermediates

Enabled by the world leading high enthalpy infrastructure at the Maurice J. Zucrow Laboratories



8 lbs/sec, 1500 F heated air system



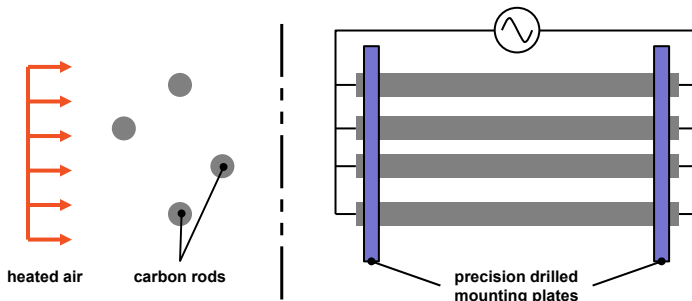
ramjet testing at Zucrow

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CM³C Microscale Problem



Experimental Concept:

- Carbon rods mounted in precision drilled plates for flexible 2D configurations
- Electrically heated to ignition
- Side-view imaging for straight forward measures of diameter changes

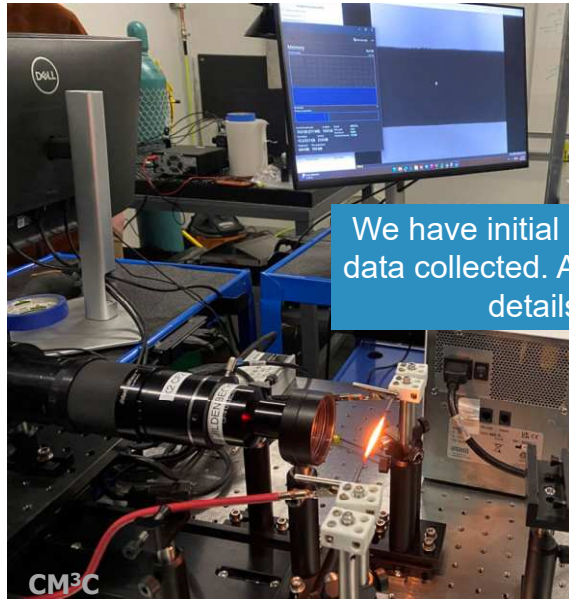
Experimental Plans:

1. First experiments will validate single rod measurements against published data
2. Quantify clusters of several closely-spaced rods that induce wake-wake and group combustion dynamics
3. Investigate geometries and quantities from our large collection of HF-PR simulations which most significantly impact the uncertainty budget (relay validation)

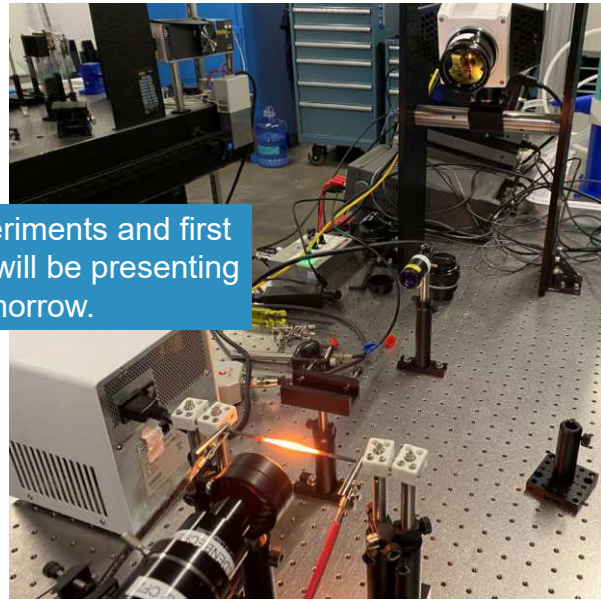
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Microscale Problem: Progress To Date

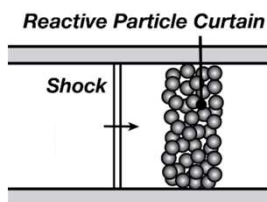


We have initial experiments and first data collected. Alex will be presenting details tomorrow.



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CM³C Mesoscale Problem

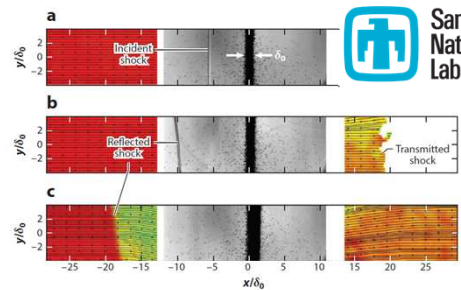


Mesoscale experiments will validate simulation of :

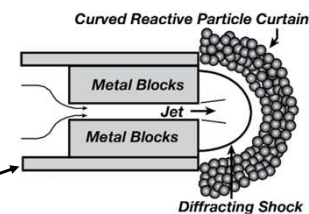
- Dense particle combustion
- Turbulence and transport
- Clustering/voids

Proposed experiments will build on current efforts to include:

- Reactive curtains of carbon and magnesium particles
- Post-shock temperatures >2500 K in the SNL High-Temperature Shocktube (HST) will ignite particles in a strong convective flow
- Imaging, LAS, CARS, PIV, and other diagnostics that leverage a long history of PU and SNL partnerships
- A novel curved curtain would further induce fingering instabilities



Capecelatro and Wagner (2024)
Annu. Rev. Fluid Mech.



Collaborations will further our long track record of joint experiments and student exchange

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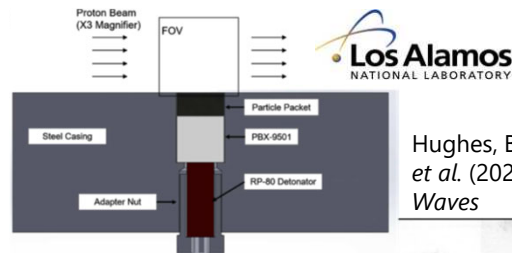


CM³C Mesoscale Problem

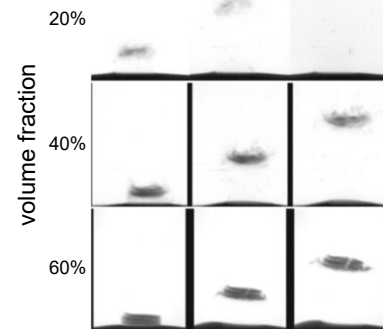
This team also has a long-history of experimental collaboration with LANL, LLNL, and other NNSA sites

- Example: P-Rad measures of explosive particle dispersal
 - Proposed experiments will consider reactive C and Mg particles
- Broad partnership opportunities for (i) reactive ejecta, (ii) carbon ablatives, (iii) explosive device testing, (iv) and other NNSA missions

We are not dependent on collaborations, but will seek to maximize them for broad impact



Hughes, Balachandar, *et al.* (2021), *Shock Waves*



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Mesoscale Problem: Progress To Date

We have made the least progress here.

- Hoped for LDRD ideas and JMP proposals at SNL have not yet gained traction
 - We will continue to work with SNL partners (e.g., Kyle Daniel, Justin Wagner, Daniel Richardson) for upcoming proposal opportunities
- Perhaps PSAAP student internships could be an impetus for collaboration. Cohen Nunes is most likely to do an internship in the short term, but detailed conversations on that have yet to begin.

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Summary

We are making reasonable progress along the planned research timeline.

■ **Current pain point/risks:**

1. We continue to debate the best way to construct the demonstration charges
2. Mesoscale problem is not currently supported at our lab partners
3. Funding is healthy for now, but experimental always carry risks of unforeseen costs

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Questions?



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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T7: Rigorous EL Closure & Exascale Implementation

T. Daoud, A. Couture, M. VandenBoom

Department of Mechanical and Aerospace Engineering

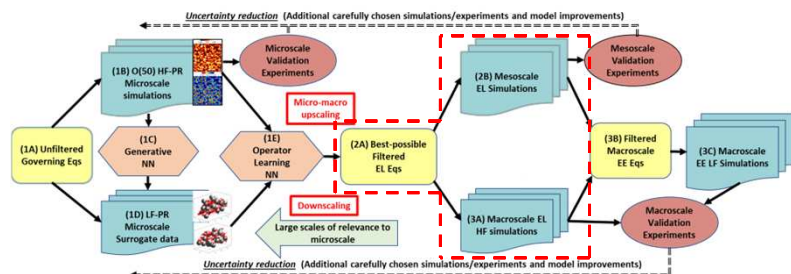
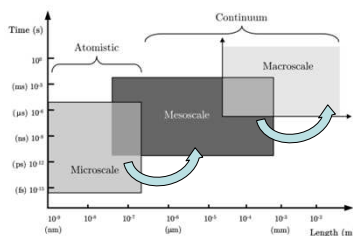
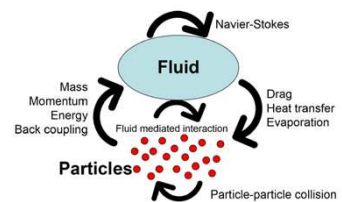


UF UNIVERSITY of FLORIDA

Big Picture

■ Coupled system

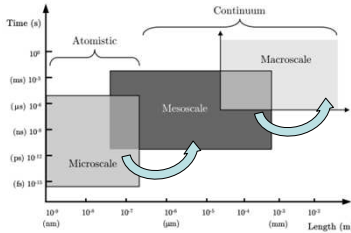
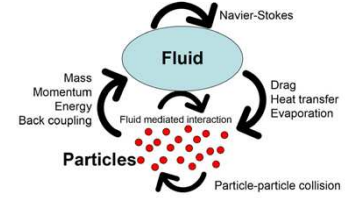
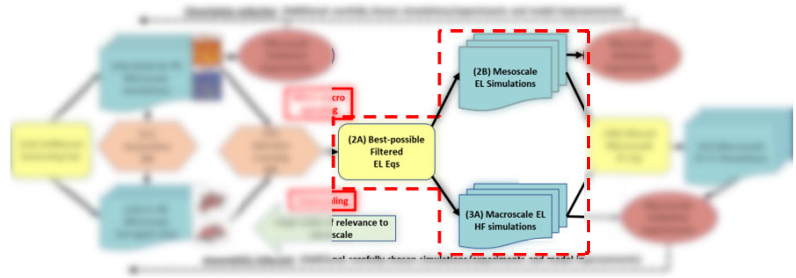
- PR-DNS simulations feed data into Machine Learning (ML)
- ML feeds in closure models to the Euler-Lagrange simulations
- EL simulations enable studying large scale problems
- Validation & UQ of EL results trigger more PR-DNS and better ML models

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Big Picture

■ Coupled system

- PR-DNS simulations feed data into Machine Learning (ML)
- ML feeds in closure models to the Euler-Lagrange simulations
- EL simulations enable studying large scale problems
- Validation & UQ of EL results trigger more PR-DNS and better ML models

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Rigorous Filtered EL Equations

■ Exact coarse-grained gas-phase equations from filter operation

$$\begin{aligned}
 \frac{\partial(\alpha_i \rho_i)}{\partial t} + \nabla \cdot (\alpha_i \rho_i \mathbf{u}) &= \frac{\alpha_i \rho_i}{\phi_f} \left[\frac{\partial \phi_p}{\partial t} + \mathbf{u} \cdot \nabla \phi_p \right] \\
 \frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I} - \boldsymbol{\tau}^v) &= \frac{\rho \mathbf{u}}{\phi_f} \left[\frac{\partial \phi_p}{\partial t} + \mathbf{u} \cdot \nabla \phi_p \right] - \frac{1}{\phi_f} (\phi_f \mathbf{R}_{sg}) - \frac{1}{\phi_f} \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i) \mathbf{F}'_{gp,i} \\
 \frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + p) \mathbf{u} - \boldsymbol{\tau}^v \cdot \mathbf{u}] &= \frac{\rho E}{\phi_f} \left[\frac{\partial \phi_p}{\partial t} + \mathbf{u} \cdot \nabla \phi_p \right] - \frac{p}{\phi_f} (\mathbf{u} \cdot \nabla \phi_f + \nabla \cdot (\phi_p \mathbf{u}_p)) \\
 &\quad + \frac{\boldsymbol{\tau}^v}{\phi_f} : (\mathbf{u} \nabla \phi_f + \nabla (\phi_p \mathbf{u}_p)) - \nabla \cdot \mathbf{q} - \frac{1}{\phi_f} \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i) (W'_{gp,i} + Q'_{gp,i})
 \end{aligned}$$

Sub-grid Phase-coupling From rigorous derivation Phase-coupling

$$\sum_i \alpha_i = 1; \quad \rho = \sum_i \alpha_i \rho_i$$

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Governing Equations – Lagrangian Particles

- Particles are treated as Lagrangian masses, with time evolution governed by

$$\frac{d\mathbf{x}_{p,i}}{dt} - \mathbf{u}_{p,i} = 0,$$

Phase-coupling

$$m_{p,i} \frac{d\mathbf{u}_{p,i}}{dt} = \boxed{\mathbf{F}_{gp,i}} + \boxed{\mathbf{F}_{pp,i}},$$

Inter-particle

$$m_{p,i} C_{p,i} \frac{dT_{p,i}}{dt} = \boxed{Q_{gp,i}},$$

$$I_{p,i} \frac{d\boldsymbol{\omega}_{p,i}}{dt} = \boxed{\boldsymbol{\tau}_{gp,i}} + \boxed{\boldsymbol{\tau}_{pp,i}}.$$

Phase-coupling force terms:

$$\mathbf{F}_{gp,i} = \mathbf{F}_{pg,i} + \mathbf{F}_{qs,i} + \mathbf{F}_{qs,i}'' + \mathbf{F}_{am,i} + \mathbf{F}_{am,i}'' + \mathbf{F}_{vu,i} + \mathbf{F}_{L,i}$$

Phase-coupling heat-transfer terms:

$$Q_{gp,i} = Q_{uu,i}^{gp} + Q_{qs,i}^{gp} + Q_{du,i}^{gp}$$

Phase-coupling torque terms:

$$\boldsymbol{\tau}_{pp,i} = \sum_j \boldsymbol{\tau}_{pp,c,i,j} + \boldsymbol{\tau}_{pp,r,i,j}$$

$$\boldsymbol{\tau}_{gp,i} = \boldsymbol{\tau}_{gp,un,i} + \boldsymbol{\tau}_{gp,h,i}$$

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Subgrid Closure Terms

- The subgrid terms can be calculated using

Pseudo-turbulent to mean flow kinetic energy

$$\hat{k}(\phi_p, \text{Re}_m, M_p) = \hat{k}(\phi_p, \text{Re}_m) \left[1 + \hat{k}^{M_p}(\phi_p, M_p) \right]$$

Anisotropy parallel component

$$b_{\parallel,\parallel}(\phi_p, \text{Re}_m, M_p) = b_{\parallel,\parallel}(\phi_p, \text{Re}_m) \left[1 + b_{\parallel,\parallel}^{M_p}(\phi_p, M_p) \right]$$

Mean flow Reynold Stress Tensor

$$\tilde{R}_{ij} = 2k \left(b_{ij} + \frac{\delta_{ij}}{3} \right) \quad \langle \tilde{\mathbf{R}} \rangle_p = \begin{bmatrix} R_{\parallel,\parallel} & 0 & 0 \\ 0 & R_{\perp,\perp} & 0 \\ 0 & 0 & R_{\perp,\perp} \end{bmatrix}$$

Lagrangian model for Reynolds Stress

$$\frac{\tilde{R}_{\parallel,\parallel}^p}{\langle \tilde{R}_{\parallel,\parallel} \rangle_p} = 1 + 0.064 \frac{C_D'}{\langle C_D \rangle_p} + \xi_{\parallel}(\phi_p, M_p)$$

$$\frac{\tilde{R}_{\perp,\perp}^p}{\langle \tilde{R}_{\perp,\perp} \rangle_p} = 1 + A_3 \frac{C_D'}{\langle C_D \rangle_p} + \xi_{\perp}(\phi_p, M_p)$$

Pseudo-turbulent heat flux

$$\mathbf{Q}_{sg} = \langle \mathbf{u}_g' \mathbf{T}' \rangle = \boldsymbol{\alpha}_{PT} \cdot \nabla T_g \quad \alpha_{\parallel,\parallel} = \frac{2\tilde{\text{Re}}_p(\tilde{\text{Re}}_p + 1.4) \text{Pr}^2 \exp(-0.002089\tilde{\text{Re}}_p) [\phi_g(-5.11\phi_p + 10.1\phi_p^2 - 10.85\phi_p^3) + 1 - \exp(10.96\phi_p)]}{3\pi\tilde{\text{Nu}}_p \left(1.17\phi_p - 0.2021\phi_p^{1/2} + 0.08568\phi_p^{1/4} \right) \phi_g^2 \left[1 - 1.6\phi_p\phi_g - 3\phi_p\phi_g^4 \exp\left(-\tilde{\text{Re}}_p^{0.4}\phi_p\right) \right]}$$

$$\boldsymbol{\alpha}_{PT} = \mathbf{Q} \boldsymbol{\alpha}_{PT}^{\dagger} \mathbf{Q}^T$$

$$\boldsymbol{\alpha}_{PT}^{\dagger} = \begin{bmatrix} \alpha_{\parallel,\parallel} & 0 & 0 \\ 0 & \alpha_{\perp,\perp} & 0 \\ 0 & 0 & \alpha_{\perp,\perp} \end{bmatrix}$$

$$\alpha_{\perp,\perp} = \left(\frac{R_{\perp,\perp}}{R_{\parallel,\parallel}} \right) \alpha_{\parallel,\parallel} = \left(\frac{3b_{\perp,\perp} + 1}{3b_{\parallel,\parallel} + 1} \right) \alpha_{\parallel,\parallel}$$

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EL Closure Modeling – Key Considerations

- EL governing equations are exact
 - “Best-possible closure models” $\Rightarrow$ “best-possible EL simulation”
 - Aleatoric + epistemic uncertainties $\Rightarrow$ Only statistical replication of PR-DNS
- Our closure models are Lagrangian $\Rightarrow$ I.e., per particle
- Standard approach has many short comings
 - Only mean effect $\Rightarrow$ every particle in a finite-volume cell gets the same closure
 - Often viscous-unsteady, torque, subgrid flux, etc., models are ignored
 - Models developed under entirely different conditions are extrapolated & used
- Our target closure models are deterministic and neighbor-informed
 - Currently only quasi-steady and added-mass are neighbor informed
- Deterministic models are physics-based and therefore are correlated

Counter Questions

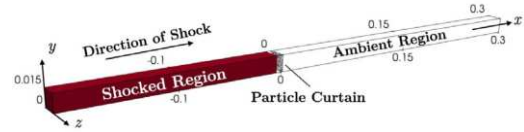
- Do we really need “best-possible closure models”?
- Aren’t mean models good enough?
- If particle-to-fluid density is large, do we need to care about pressure-gradient and added-mass forces?
- Is neighbor-induced force fluctuation important in explosive dispersal?
- Are heat transfer models important for particle motion?
- How large is particle rotation in shock-driven flows?
- What is the main source of particle rotation?
- Are subgrid closures important?

Simple Test Problem: Shock-Curtain Interaction

- There is no quantitative evidence in the literature about the importance of each closure model
- Simulated a test matrix of 13 cases in which models are progressively turned on

$$\mathbf{F}_{gp,i} = \mathbf{F}_{pg,i} + \mathbf{F}_{qs,i} + \mathbf{F}_{qs,i}'' + \mathbf{F}_{am,i} + \mathbf{F}_{am,i}'' + \mathbf{F}_{vu,i} + \mathbf{F}_{L,i}$$

$$Q_{gp,i} = Q_{uu,i}^{gp} + Q_{qs,i}^{gp} + Q_{du,i}^{gp}$$



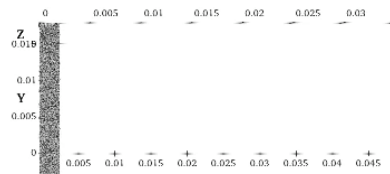
Case	QS-Avg	QS-Fluct	PG	AM	VU	Collision	Torque Model	Lift	Heat Transfer	Work	Comment
0	Osnes	No	No	No	No	Normal	No	No	No	Yes	Baseline case
1	Osnes	Yes	No	No	No	Normal	No	No	No	Yes	Effect of QS Fluct
2	Osnes	Yes	Yes	No	No	Normal	No	No	No	Yes	Effect of PG Force
3	Osnes	Yes	Yes	Avg	No	Normal	No	No	No	Yes	Effect of AM Force
4	Osnes	Yes	Yes	Fluct	No	Normal	No	No	No	Yes	Effect AM fluctuation
5	Osnes	Yes	Yes	Fluct	Yes	Normal	No	No	No	Yes	Effect of viscous unsteady
6	Osnes	Yes	Yes	Fluct	Yes	Tangential	Collision	No	No	Yes	Effect of tangential collision on rotation
7	Osnes	Yes	Yes	Fluct	Yes	Tangential	Collision + QS + UT	No	No	Yes	Effect of QS & undisturbed torques
8	Osnes	Yes	Yes	Fluct	Yes	Tangential	Collision + QS + UT	Yes	No	Yes	Effect of lift force
9	Osnes	Yes	Yes	Fluct	Yes	Tangential	Collision + QS + UT	Yes	QS	Yes	Effect of quasi-steady heat transfer
10	Osnes	Yes	Yes	Fluct	Yes	Tangential	Collision + QS + UT	Yes	QS + DU	Yes	Effect of diffusive-unsteady heat transfer
11	Osnes	Yes	Yes	Fluct	Yes	Tangential	Collision + QS + UT	Yes	QS + DU	No	Effect of not projecting work exchange
12	Osnes	Yes	Yes	Fluct	Yes	Tangential	Collision + QS + UT	Yes	QS + DU	Partially	Effect of not projecting dissipative work

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Best Available Closure Models

- Comprehensive paper in which more than 15 models have been investigated and quantified their contribution to the various dynamics
- One of the earliest works to include neighbor-induced quasi-steady drag and added-mass



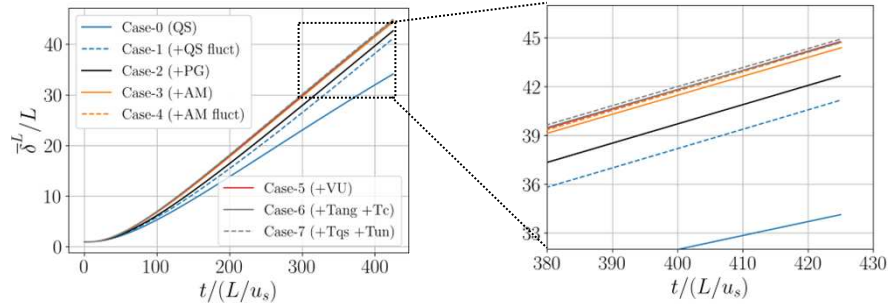
Model	Source	Input
Quasi-steady average	Loth <i>et al.</i> [39] Osnes <i>et al.</i> [51] Parmar <i>et al.</i> [58] Gidaspow [24]	$\text{Re}_{p,i}, M_{p,i}, \phi_{p,i}, C_{D,i}$
Quasi-steady fluctuation	Osnes <i>et al.</i> [51]	$\text{Re}_{p,i}, M_{p,i}, \phi_{p,i}, \sigma_{D,i}, \sigma_{T,i}, \tau_{F,i}$
Added mass average	Béguin <i>et al.</i> [16] Parmar <i>et al.</i> [57]	$V_{p,i}, C_M, \rho_{g,i}, u_{g,i}, u_{p,i}$
Added mass fluctuation	Briney <i>et al.</i> [14]	$V_{p,i}, C_{M,std}, \eta_i, \mathbf{a}_{r,i}, \mathbf{a}_{r,j}, \mathbf{B}_{ij}^A, \mathbf{B}_{ij}^F$
Viscous-unsteady	Mei and Adrian [46]	$\text{Re}_{p,i}, \phi_{p,i}, K_{vu}$
Lift	Saffman [5] and Magnus [38]	$\text{Re}_{p,i}, C_{L,i}^*, \omega_{p,i}, \Omega_{g,i}$
Collision	Balachandar [5]	$x_{p,i}, x_{p,j}, u_{p,i}, u_{p,j}, \omega_{p,i}, \omega_{p,j}$
Quasi-steady torque	Finn <i>et al.</i> [23]	$\theta_{r,i}, I_{g,i}, I_{p,i}, \omega_{p,i}, \omega_{p,j}$
Quasi-steady heat	Gunn <i>et al.</i> [25] Sun <i>et al.</i> [68]	$k_{g,i}, \text{Nu}, T_{g,i}, T_{p,i}$
Diffusive-unsteady heat	Balachandar and Ha [7]	$k_{g,i}, K_{du}, T_{g,i}, T_{p,i}$
Work coupling	Balachandar [5]	$u_{g,i}, u_{p,i}, \mathbf{F}_{gp,i}', \mathbf{F}_{qs,i}'$

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Closure Models

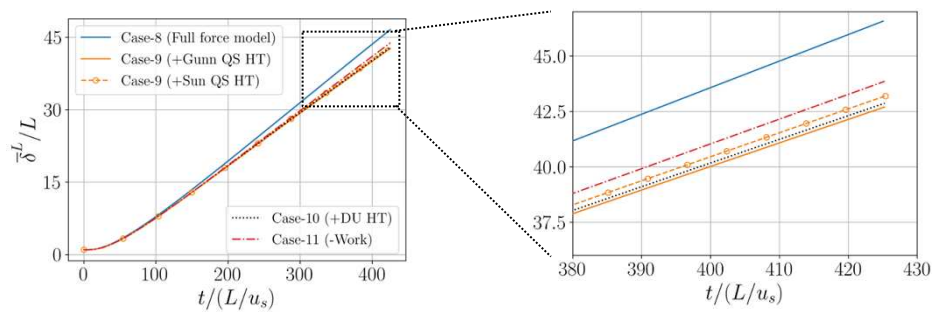
- More closure models $\Rightarrow$ more particle spreading

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Closure Models

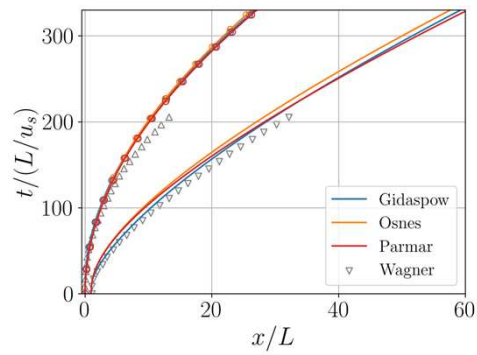
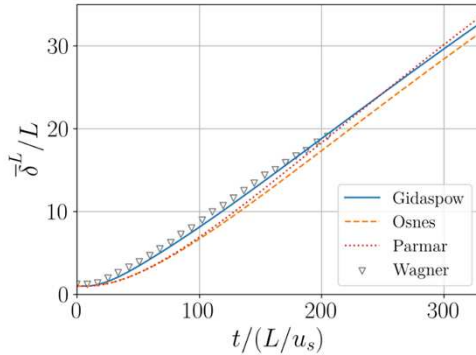
- More closure models $\Rightarrow$ more particle spreading
- Inclusion of heat transfer $\Rightarrow$ less curtain spreading
- Excluding work coupling $\Rightarrow$ more curtain spreading

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Closure Models

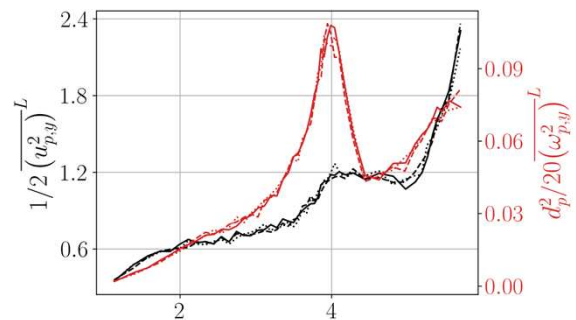
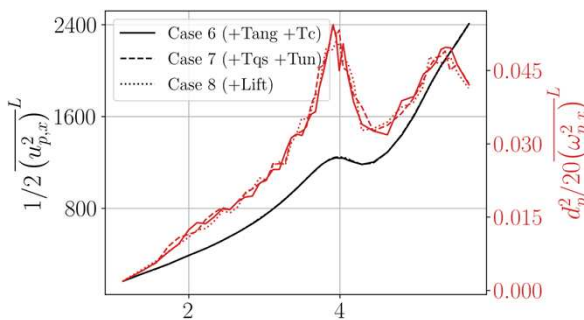
- More closure models $\Rightarrow$ more curtain spreading
- Inclusion of heat transfer $\Rightarrow$ less curtain spreading
- Excluding work coupling $\Rightarrow$ more curtain spreading
- Minimal sensitivity to QS model choice

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Closure Models

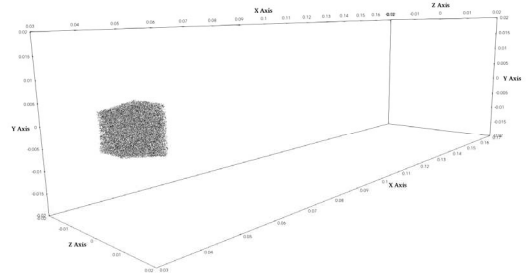
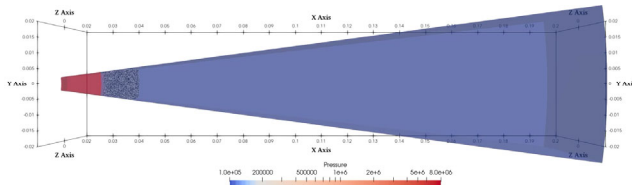
- Particle rotation motion is mainly induced by tangential collisions
- The rotational particle motion has small effects on the dynamic features
- Rotational motion is generally lower than 30% of translational motion

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Capabilities - Conical Setup

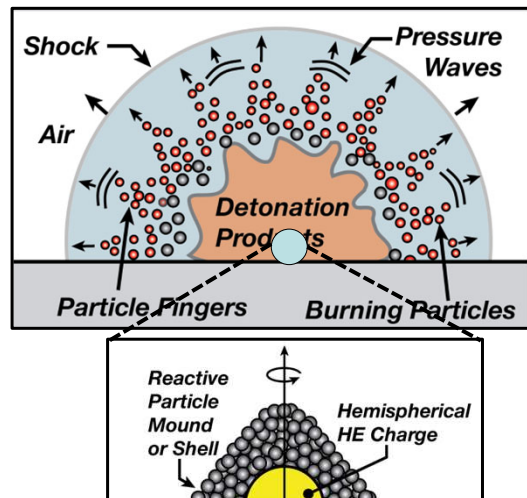
- Ability to simulate cone slice out of the hemi-sphere problem for computational efficiency while preserving geometry features
- Current animations are for high-pressure air, future work to include explosives

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EL Simulation at Exascale is Challenging

- All the particles are initially in a small region
- Even at a later time particle and fluid domains do not overlap
- Particles are in constant motion
- Load-balanced partitioning of Lagrangian data is challenging
 - Localized approach – in the same rank as local fluid
 - Joint-workload approach – will not work
 - Bin-based approach – independent partitioning

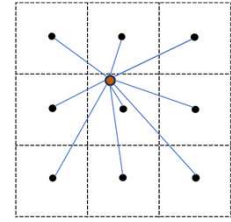
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Eulerian and Lagrangian Data-structure Coupling

- Interpolation: 27 neighboring cells (3D) data are interpolated

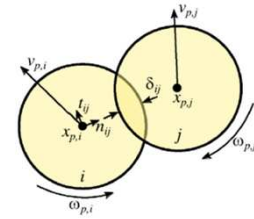
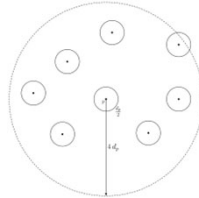
$$\psi_i^L = \frac{1}{\sum_{j=1}^{N_c} w_{i,j}} \sum_{j=1}^{N_c} \psi_j w_{i,j} \quad \text{where} \quad w_{i,j} = \frac{1}{\Delta d_{i,j}^2 + \epsilon}$$



- Projection: Gaussian function is used

$$w_{i,j} = V_j (1 - \phi_{p,j}) \exp \left(-\alpha_c \left[\left(\frac{x - X_i}{\Delta x} \right)^2 + \left(\frac{y - Y_i}{\Delta y} \right)^2 + \left(\frac{z - Z_i}{\Delta z} \right)^2 \right] \right) \quad \psi_j = \frac{1}{\left(\sum_{j=1}^{N_c} w_{i,j} \right)} \sum_{i=1}^{n_p} \psi_i^L w_{i,j}$$

- Collisional: Search neighboring region within $4 \times d_{p,max}$



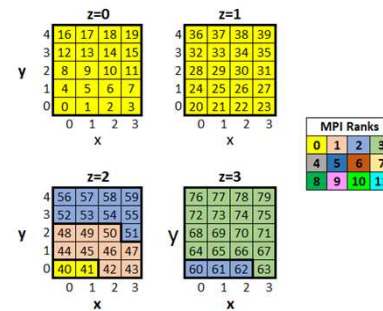
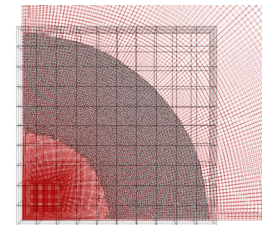
Bin-Based Strategy

- A Cartesian Bin structure is laid on the particle domain and used to partition particles evenly as over the ranks
- Bins identified by tuple (ib, jb, kb)
- Partition by space-filling curve (classical math problem)
 - Morton Z-order, Hilbert curve, recursive bisection, etc
 - Simple approach: longest, second longest, ...

$$s = ib + N_{x,b} jb + N_{x,b} N_{y,b} kb$$

- Cartesian 3D bins serve as easy intermediary

- For each particle & fluid cell easy find its bin



z continues 4:25 ...

Bin-Based Strategy

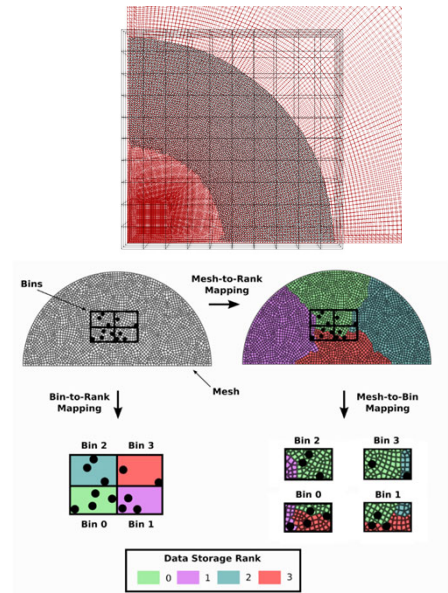
■ Each MPI rank owns

- A fraction of fluid cells + ghost cells
- A fraction of particles + ghost particles
- Overlap fluid cells (for interpolation/projection)

Data transfer

■ Maps as go between (particles, cells, bins, ranks)

- Particle-to-Bin (size: $\mathcal{O}(N_p/N_r)$ ints, frequency: every time, cost: $\mathcal{O}(N_p/N_r)$)
- Overlap Cell-to-Rank (size: $\mathcal{O}(N_{oc})$ ints, frequency: every bin update, cost: $\mathcal{O}(N_{oc})$)
- Bin-to-Rank (Only global data structure – size: $\mathcal{O}(N_b)$ ints)
- Particle-to-Cell

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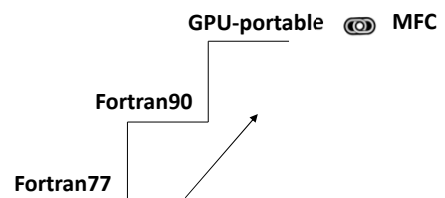
ppicIF Plan

■ Current efforts are update ppicIF to Fortran90

1. Allow wider range of build systems to compile code (nvhpc on HiPerGator, Cray compilers on Tuolumne).
2. Dynamically allocated arrays instead of static.
3. Usage of modules instead of large common blocks.

■ This step was crucial prior the transition to GPU

1. Switch to non-blocking data transfers.
2. Better compiler optimization.
3. Re-writing of some code algorithms to benefit from parallelization.

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CM³C

Questions?



CM³C

CM³C: Center for Multiscale Modeling of Multiphase Combustion

T8: Generative ML Models

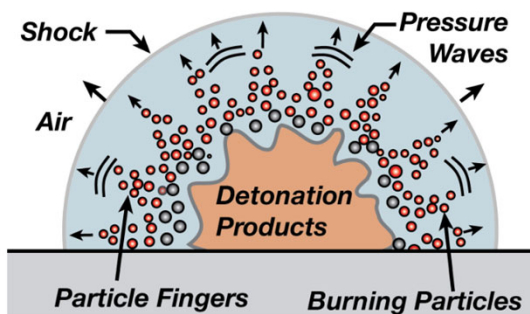
Erica Lindbeck

University of Florida

April 1, 2026



Project Overview

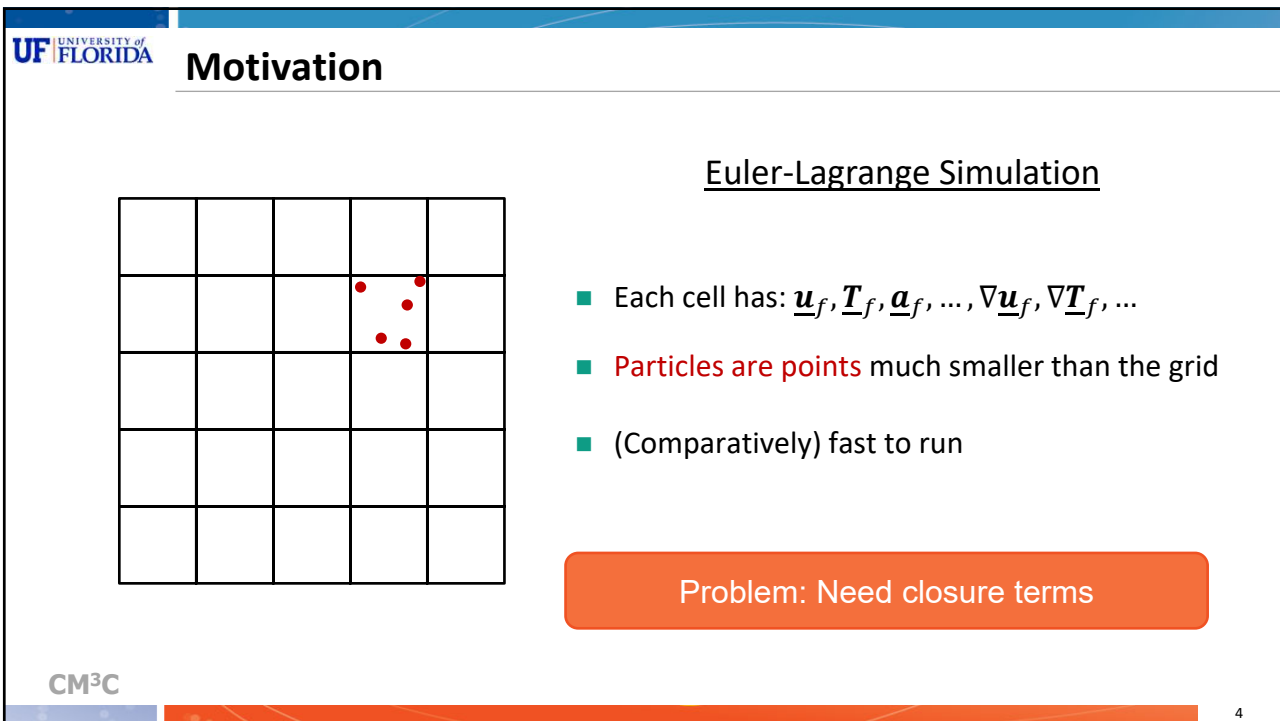
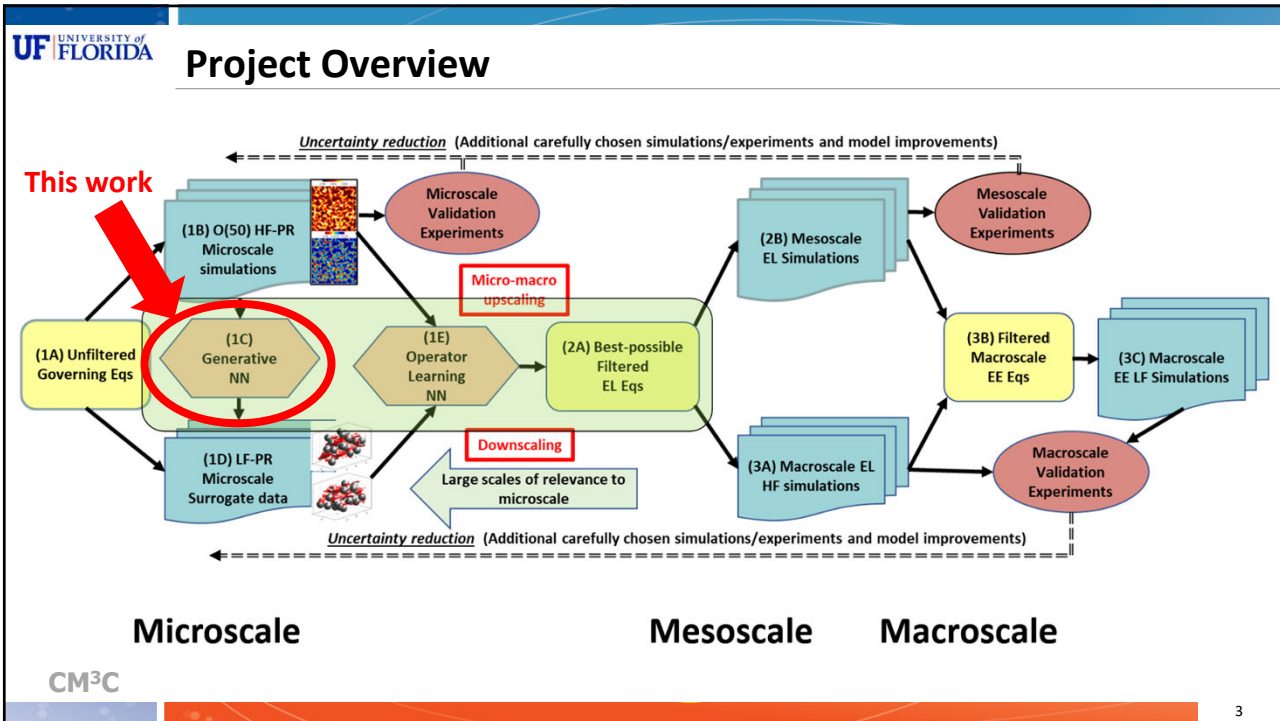


Large Combustion Events

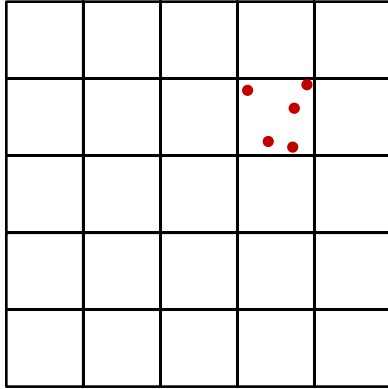
- Dangerous
- Cannot generate experimentally
- Simulations must be very large-scale

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2



Motivation



Euler-Lagrange Simulation

Fluid motion described by:

$$\underline{a}_f = \frac{\partial \underline{u}_f}{\partial t} = \dots = - \sum_{j=1}^{N_p} \underline{F}_j + \nabla \cdot (\underline{R})$$

Reynolds stress

Computed from fluid behavior
(later: Operator Learning)

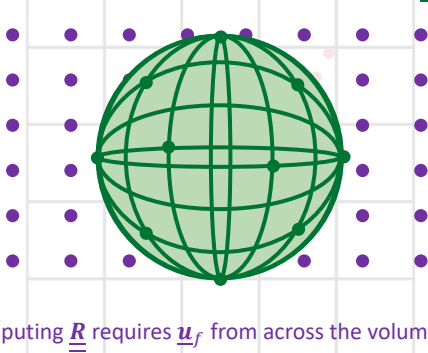
Particle motion described by:

$$m_j \frac{\partial \underline{u}_{p_j}}{\partial t} = \underline{F}_j + \text{gravity}$$

Problem: Need closure terms

Motivation

Lebedev points around a particle can be used to approximate an integral to compute $\underline{F}_j$ from $\underline{p}_f$



Computing $\underline{R}$ requires $\underline{u}_f$ from across the volume

Euler-Lagrange Simulation

Fluid motion described by:

$$\underline{a}_f = \frac{\partial \underline{u}_f}{\partial t} = \dots = - \sum_{j=1}^{N_p} \underline{F}_j + \nabla \cdot (\underline{R})$$

Reynolds stress

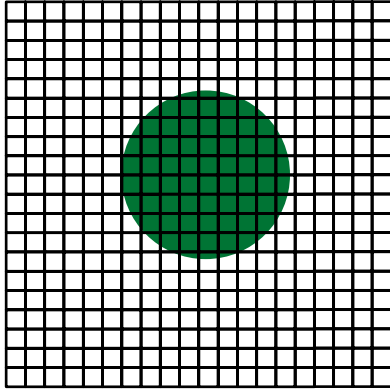
Computed from fluid behavior
(later: Operator Learning)

Particle motion described by:

$$m_j \frac{\partial \underline{u}_{p_j}}{\partial t} = \underline{F}_j + \text{gravity}$$

Problem: Need closure terms

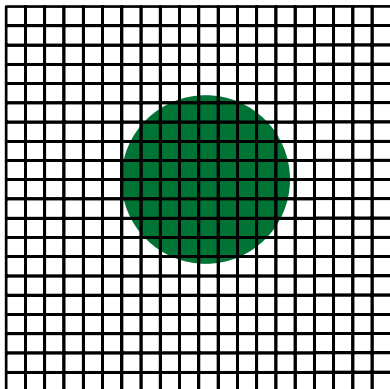
Motivation



Particle Resolved Simulation

- Each cell has: $\underline{u}_f, \underline{T}_f, \underline{a}_f, \dots, \nabla \underline{u}_f, \nabla \underline{T}_f, \dots$
- Particles are spheres much bigger than the grid
- Can find closure terms

Motivation

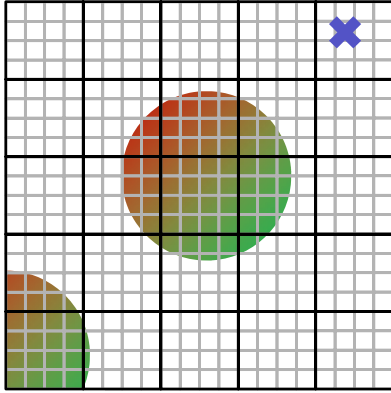


Particle Resolved Simulation

- Each cell has: $\underline{u}_f, \underline{T}_f, \underline{a}_f, \dots, \nabla \underline{u}_f, \nabla \underline{T}_f, \dots$
- Particles are spheres much bigger than the grid
- Can find closure terms

Problem: Slow

Motivation



Machine Learning Goals

- Use inputs available in Euler-Lagrange simulation
- Predict $\underline{u}_f$ and $\underline{p}_f$ anywhere in the simulation
- Use to generate closure training data

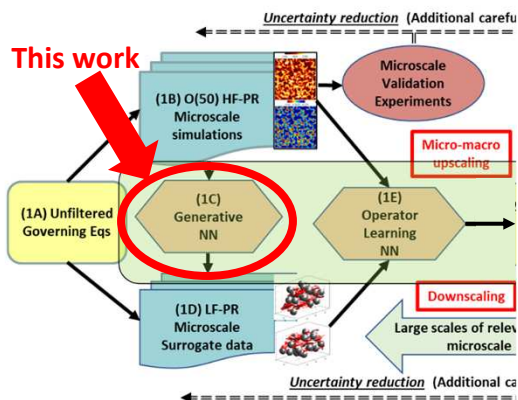


Result: Fast large-scale simulations!

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Motivation



Machine Learning Goals

- Use inputs available in Euler-Lagrange simulation
- Predict $\underline{u}_f$ and $\underline{p}_f$ anywhere in the simulation
- Use to generate closure training data



Result: Fast large-scale simulations!

Microscale

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Problem Setup

“Possible” Inputs:

$\underline{u}_f$ = average velocity of fluid

$\underline{a}_f$ = average acceleration of fluid

$\frac{\rho_p}{\rho_f}$ = particle density ratio

For all particles:

φ_j = particle volume fraction around particle j

$\underline{x}_j$ = position of particle j

$\underline{u}_j$ = velocity of particle j

$\underline{a}_j$ = acceleration of particle j

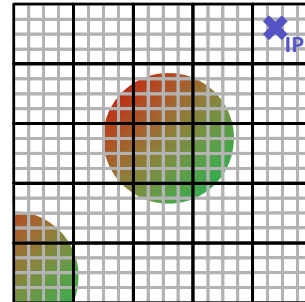
$\underline{x}_{ip}$ = location of inference point

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Desired Outputs:

$\underline{u}_f(\underline{x}_{ip})$ = fluid velocity at inference point

$p(\underline{x}_{ip})$ = pressure at inference point



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Problem Setup

Picking one “Reference Particle” RP:

$\underline{u}_f$ = average velocity of fluid **relative to RP**

$\frac{\rho_p}{\rho_f}$ = particle density ratio

$\varphi_j = \varphi_{RP}$ = particle volume fraction

For **neighboring** particles:

$\underline{x}_j$ = position of particle j **relative to RP**

$\underline{u}_j$ = velocity of particle j **relative to RP**

$\underline{a}_j$ = acceleration of particle j

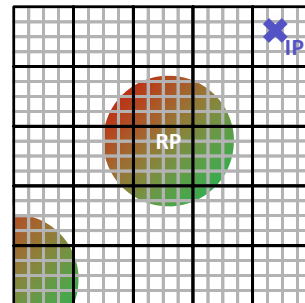
$\underline{x}_{ip}$ = location of inference point **relative to RP**

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Desired Outputs:

$\underline{u}_f(\underline{x}_{ip})$ = fluid velocity at inference point

$p(\underline{x}_{ip})$ = pressure at inference point



Neighbor j

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Problem Setup

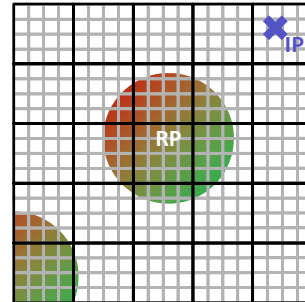
Break down $\underline{u}_f$

$$\underline{u}_f(\underline{x}_{ip}) = \langle \underline{u} \rangle + \sum_{j=1}^{N_p} \underline{u}_1(\text{particle}_j, \underline{x}_{ip}) + \dots$$

Average velocity

Change in velocity due to wake from particle j (Unary model)

Desired Outputs:

 $\underline{u}_f(\underline{x}_{ip})$ = fluid velocity at inference point $p(\underline{x}_{ip})$ = pressure at inference pointCM³C

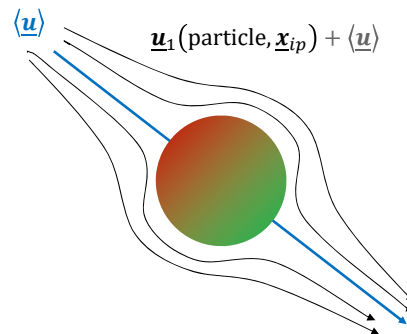
13

Problem Setup

Break down $\underline{u}_f$

$$\underline{u}_f(\underline{x}_{ip}) = \langle \underline{u} \rangle + \sum_{j=1}^{N_p} \underline{u}_1(\text{particle}_j, \underline{x}_{ip}) + \dots$$

Average velocity

Change in velocity due to wake from particle j (Unary model)What does $\underline{u}_1$ look like?CM³C

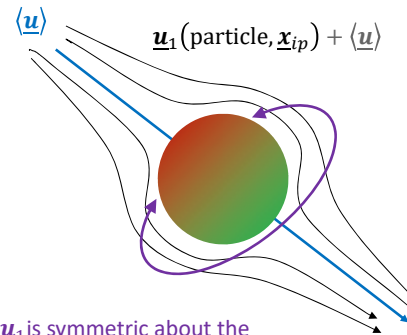
14

Problem Setup

Break down $\underline{u}_f$

$$\underline{u}_f(\underline{x}_{ip}) = \langle \underline{u} \rangle + \sum_{j=1}^{N_p} \underline{u}_1(\text{particle}_j, \underline{x}_{ip}) + \dots$$

Average velocity

Change in velocity due to wake from particle j (Unary model)What does $\underline{u}_1$ look like?

$\underline{u}_1$ is symmetric about the axis through the particle in the direction of $\langle \underline{u} \rangle$

CM³C

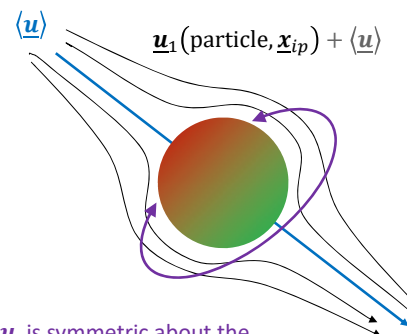
15

Problem Setup

Break down $\underline{u}_f$

$$\underline{u}_f(\underline{x}_{ip}) = \langle \underline{u} \rangle + \sum_{j=1}^{N_p} \underline{u}_1(\text{particle}_j, \underline{x}_{ip}) + \dots$$

Average velocity

Change in velocity due to wake from particle j (Unary model)What does $\underline{u}_1$ look like?

$\underline{u}_1$ is symmetric about the axis through the particle in the direction of $\langle \underline{u} \rangle$

Option 1: Use e3nn for symmetry

Option 2: Change of basis to cylindrical r' and z' CM³C

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Problem Setup

Break down $\underline{u}_f$

$$\underline{u}_f(\underline{x}_{ip}) = \langle \underline{u} \rangle + \sum_{j=1}^{N_p} \underline{u}_1(\text{particle}_j, \underline{x}_{ip}) + \dots$$

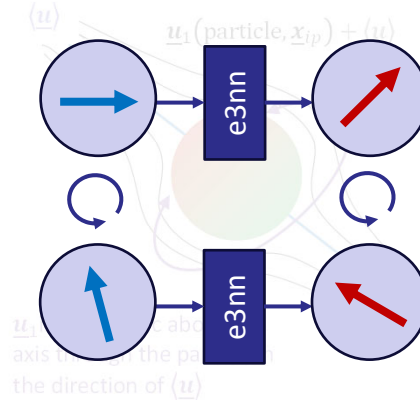
Average velocity

Change in velocity due to wake from particle j (Unary model)

Option 1: Use e3nn for symmetry

Option 2: Change of basis to cylindrical r' and z'

e3nn: Equivariant to rotations in $\mathbb{R}^3$



Problem Setup

For each "Reference Particle" RP:

$\underline{u}_f$ = average velocity of fluid **relative to RP**

$\frac{\rho_p}{\rho_f}$ = particle density ratio

$\varphi_{RP} = \varphi$ = particle volume fraction (assumed average)

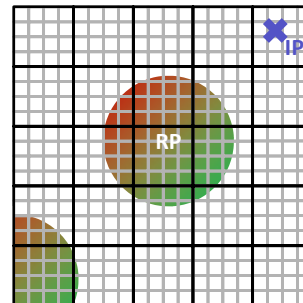
$\underline{x}_{ip}$ = location of inference point **relative to RP**

Desired Outputs:

$\underline{u}_1(RP, \underline{x}_{ip})$ = **change in fluid velocity at inference point based on RP**

$p(\underline{x}_{ip})$ = pressure at inference point

Neighbor j



Problem Setup

For each “Reference Particle” RP:

$\underline{e}_{RP}$ = unit vector in direction of **RP relative velocity**

Re_{RP} = Reynolds number of **RP** (proportional to $|\underline{u}_f|$)

$\frac{\rho_p}{\rho_f}$ = particle density ratio

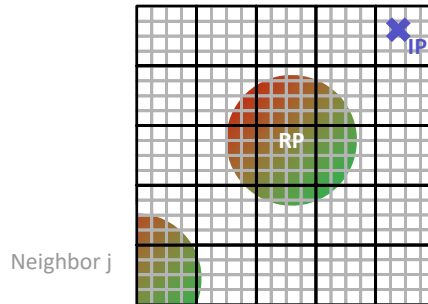
$\varphi_{RP} = \varphi$ = particle volume fraction (assumed average)

$\underline{x}_{ip}$ = location of inference point **relative to RP**

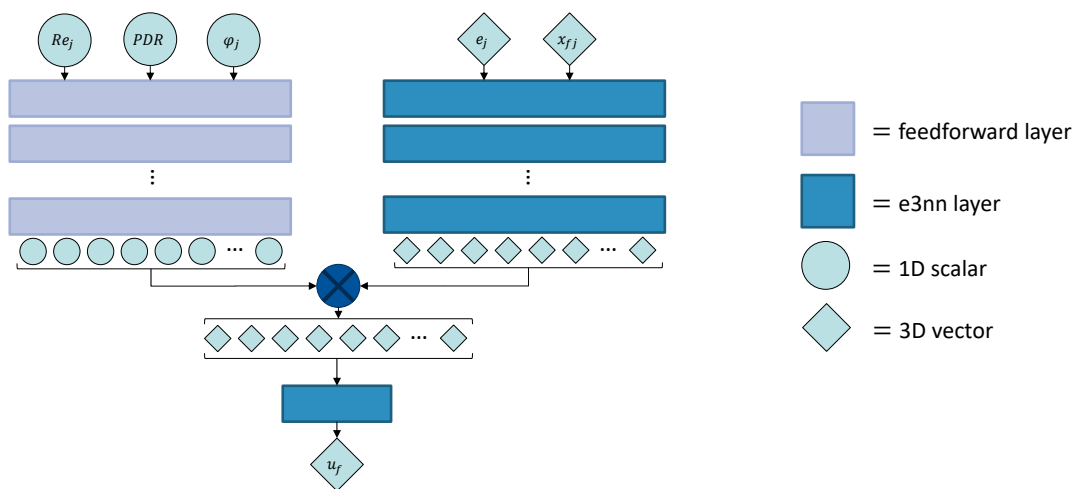
Desired Outputs:

$\underline{u}_1(RP, \underline{x}_{ip})$ = **change in fluid velocity at inference point based on RP**

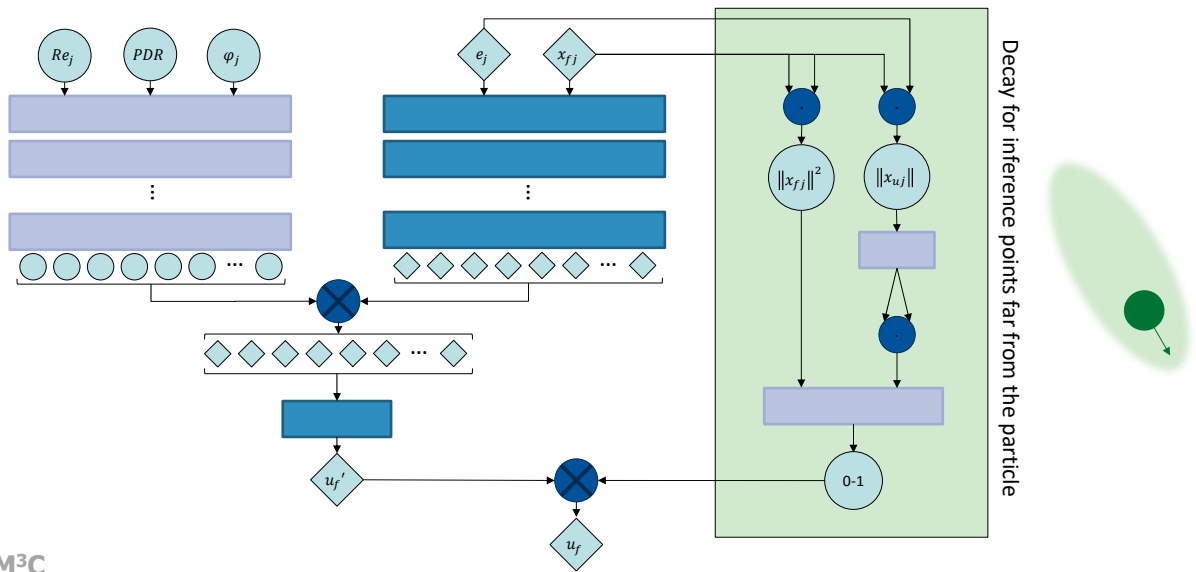
$p(\underline{x}_{ip})$ = pressure at inference point



Machine Learning Architecture



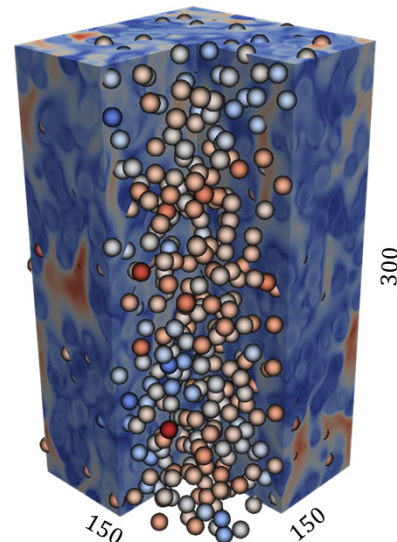
Machine Learning Architecture



Simulated Data

Particle-Resolved Simulation (General Info)

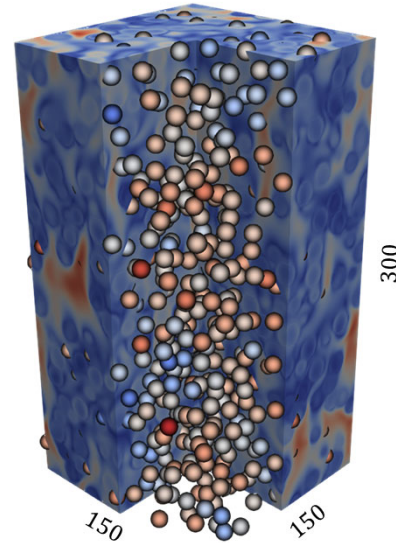
- Particles falling in otherwise stationary fluid
- Triply periodic boundaries
- Rigid bodies, no rotation
- Data are non-dimensionalized
 - Length scale: d (particle diameter)
 - Velocity scale: d / ν (diameter / kinematic viscosity)
- Mesh dimensions: $150 \times 300 \times 150$
 - $10 \times 20 \times 10$ particle diameters
- Variety of Re , PDR , and average ϕ



Simulated Data

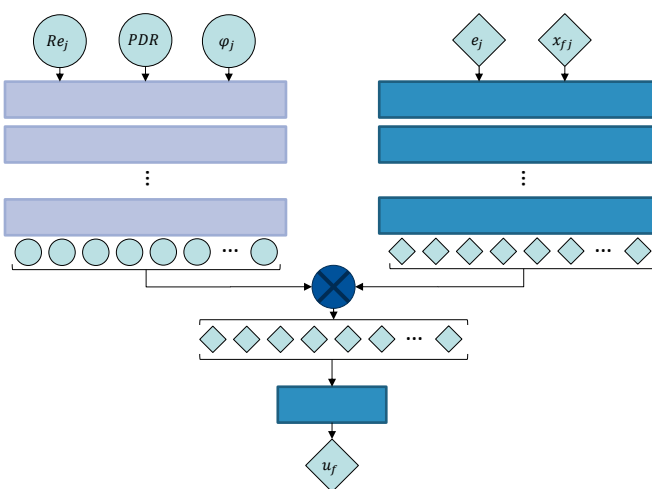
ML-Used Data

- One simulation (one Re , PDR, and average φ)
- 64 particles
- Training data
 - t_0
 - Subsampled by factor of 5 per-axis
 - Samples only from within $r = d, 2d$, or $3d$ of any particle centers
- Validation data
 - t_1 , otherwise same conditions as training
- Test data
 - t_2
 - Subsampled by factor of 5 for test sets within radius of 5 particles, by 15 for entire simulation test set
 - Samples from within $r = d, 2d$, or $3d$ of particle centers, or entire simulation

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Hyperparameter Tuning



Hyperparameters Searched

- Training data selection radius
 - $d, 2d, 3d$
- Number of layers
 - 3, 5, 7, 9, 11
- Layer size
 - 25, 50, 100, 150, 200, 250
- Decay term
 - Included or excluded

Hyperparameters Fixed

- Learning rate and LR decay
- Batch size (100)
- # epochs (30)
- Activation functions

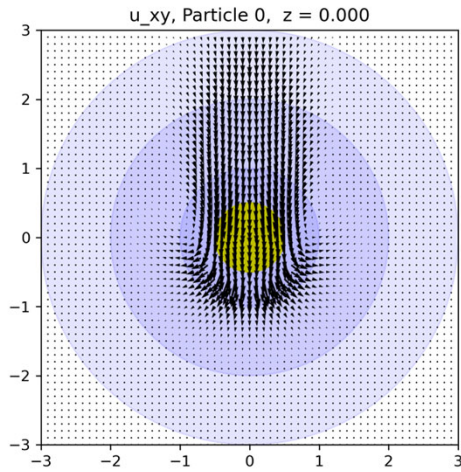
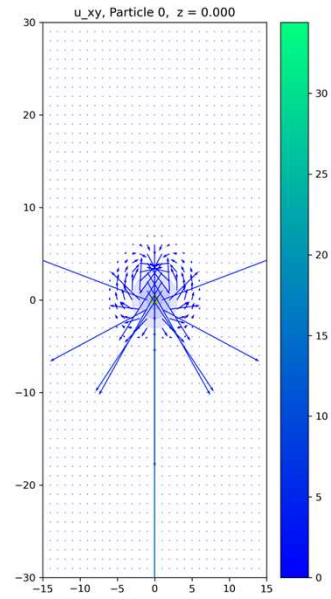
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Results

1-Particle Wake

with distance-based decay

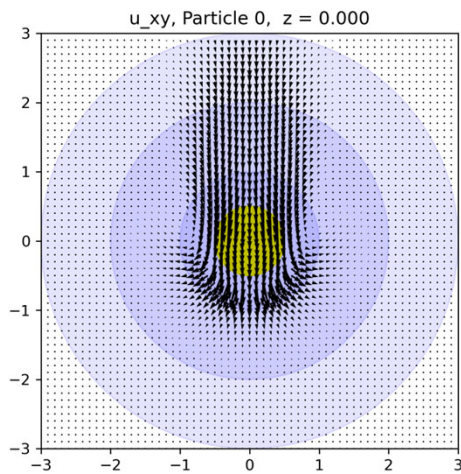
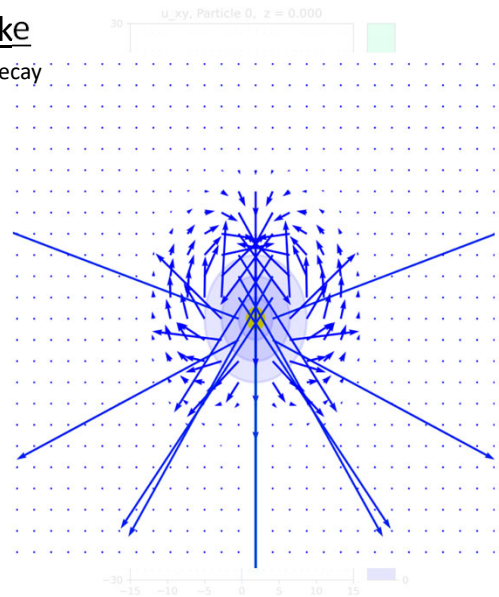
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Results

1-Particle Wake

with distance-based decay

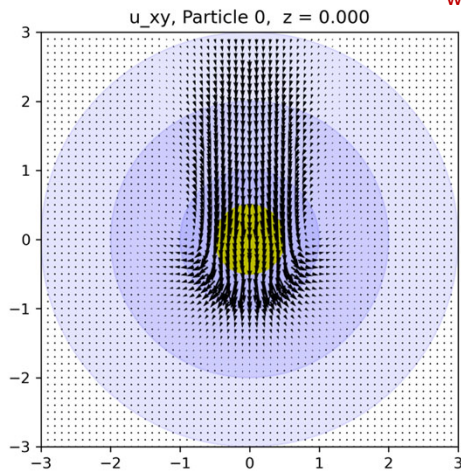
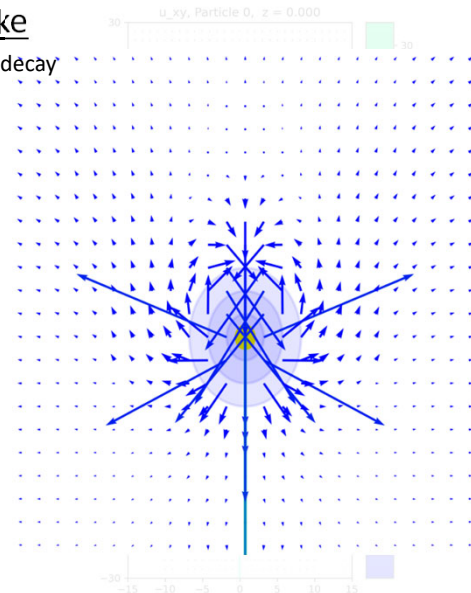
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Results

1-Particle Wake

without distance-based decay

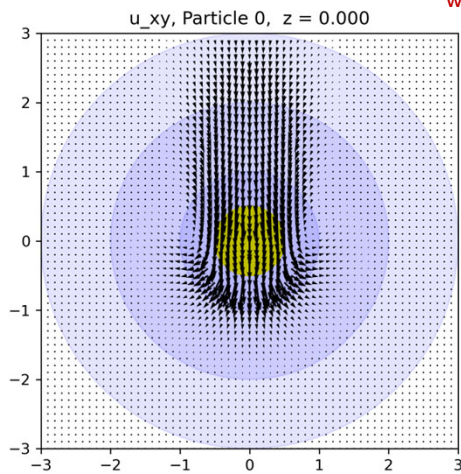
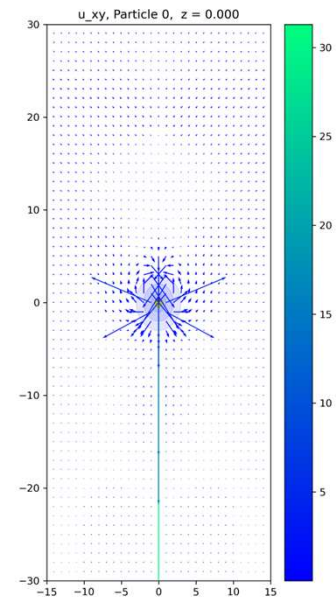
CM³C

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Results

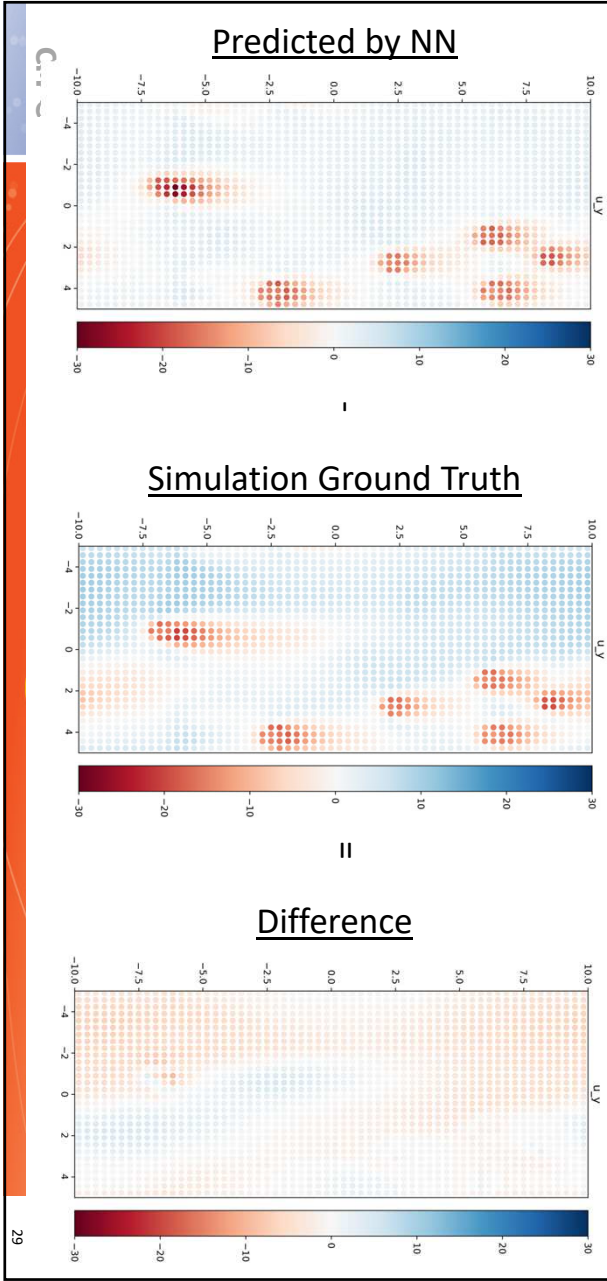
1-Particle Wake

without distance-based decay

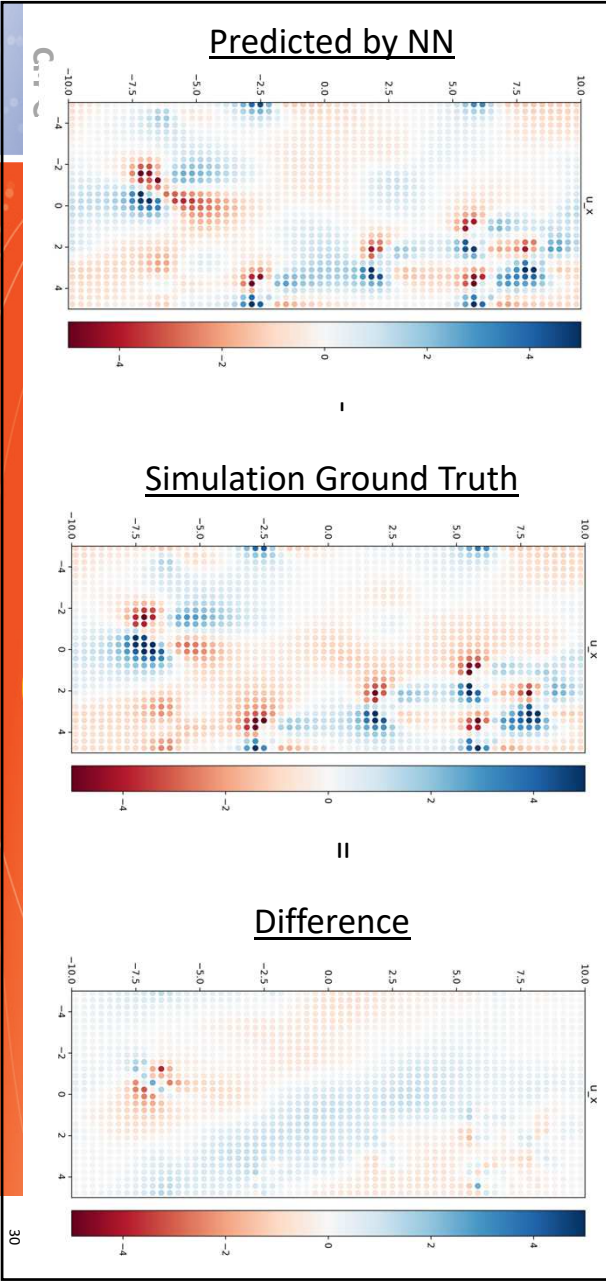
CM³C

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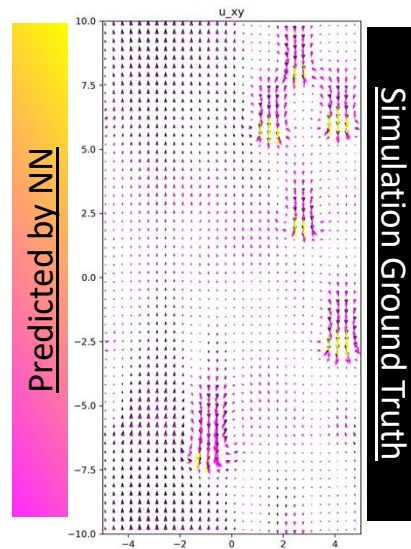
Results



Results



Results

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Results

Max train radius	Max test radius	Test R ²	Test MSE loss	Baseline (mean guess) loss
d	d	0.892	2.206	34.486
d	2d	0.783	2.383	11.260
d	3d	0.743	2.409	9.378
d	all	0.674	3.945	12.109
2	d	0.905	1.947	34.486
2d	2d	0.839	1.765	11.260
2d	3d	0.816	1.726	9.378
2d	all	0.751	3.003	12.109
3d	d	0.909	1.875	34.486
3d	2d	0.837	1.788	11.260
3d	3d	0.814	1.740	9.378
3d	all	0.753	2.987	12.109

Trends

- Training out to $r = 3d$ gives a best-case for each test set
- Training on bigger radii tended to improve nearer to particles
- (Not indicated) Only some best cases used distance decay
- R^2 limited to at most around 0.9

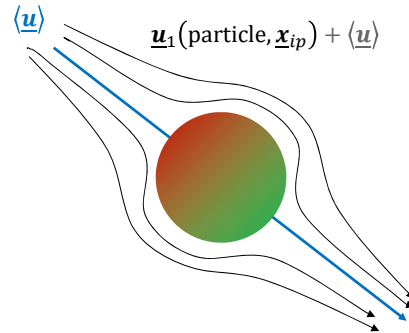
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Discussion

Next Steps

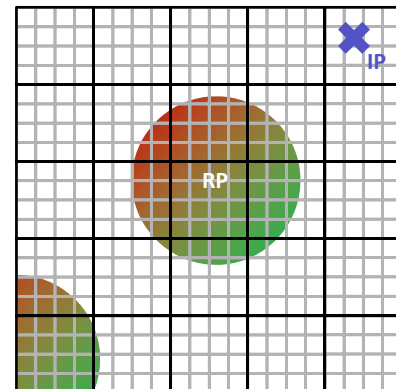
- Train on multiple Re, PDR, and φ
 - Simulations exist but need to be processed
- Include pressure prediction and particle rotation
- Compare predictions of data-driven GML (T8: Erica), PINN-based GML (T13: Juan), and OLML (T9: Cam) for the same problem
- Perform uncertainty quantification (T13: Isaly)



Discussion

Plan for Y1+

- Prepare the data-driven GML pipeline
 - Work with PR-DNS data for compressible flow being developed
- Extend the framework for calculating phase-coupling and subgrid closure quantities for OLML
- Automated uncertainty quantification for data-driven and PINN-based GML



Neighbor j

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Questions?



CM³C

CM³C: Center for Multiscale Modeling of Multiphase Combustion

T9: ML Operator Learning Overview

Presented By: Cam Lott

Department of Mechanical and Aerospace Engineering

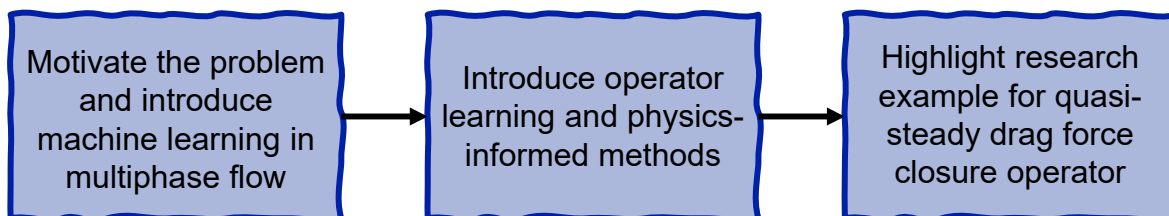


Introduction and Outline

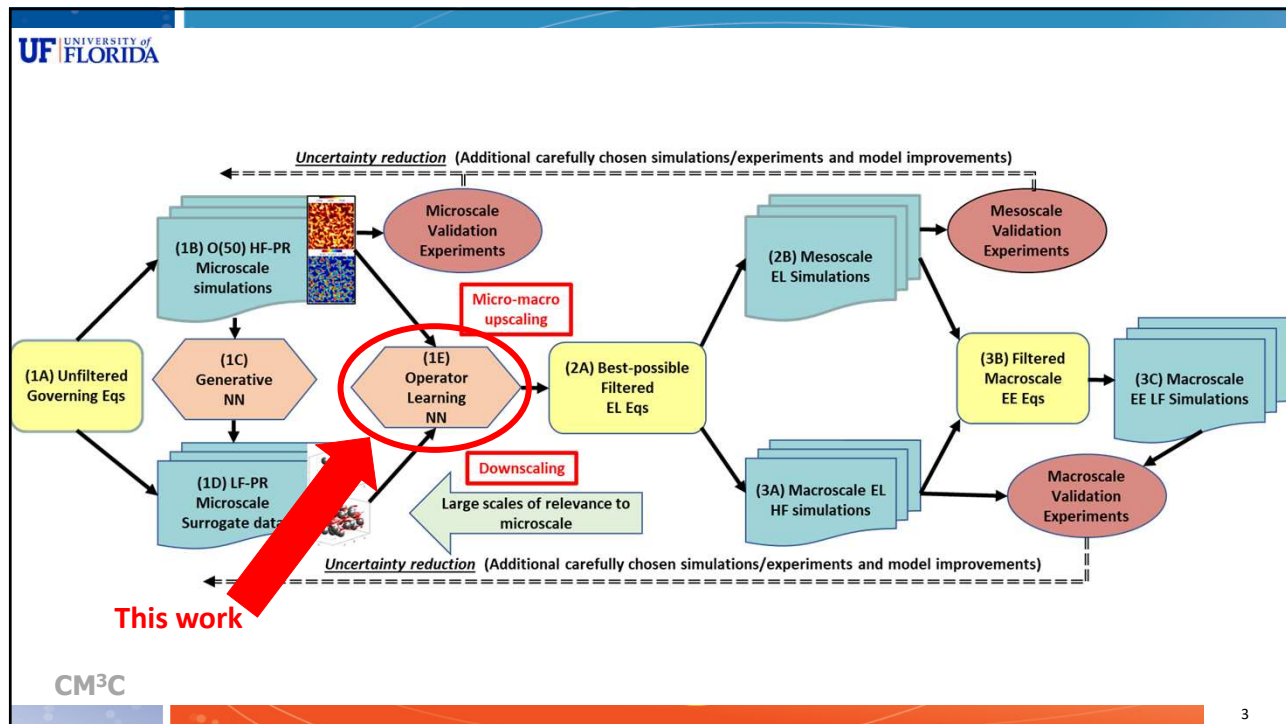
About Me:

- Second year Ph.D. under Dr. Bala
- Use machine learning to develop deterministic closures
- Worked with/at Los Alamos last summer

Presentation Outline



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Multiphase Flow Equation Closures

- Running fully-resolved simulations is possible, but only at the microscale
- Going to the meso and macroscales introduce new terms that require closure

Continuous Phase Equations:

$$\nabla \cdot \mathbf{u} = 0, \quad \rho \phi_f \frac{D\mathbf{u}_f}{Dt} - \nabla \cdot \boldsymbol{\sigma}_f = -\frac{1}{V} \sum_i \mathbf{F}_{fp} + \nabla \cdot \mathbf{R}_{sg}$$

Particle Equation of Motion (Single Particle):

$$m_p \frac{d\mathbf{u}_p}{dt} = \mathbf{F}_{fp} = \mathbf{F}_{qs} + \mathbf{F}_{sg} + \mathbf{F}_{am} + \mathbf{F}_{vu} + \mathbf{F}_{col}$$

- E.g., the quasi-steady force, $\mathbf{F}_{qs}$, has been derived for slowly-varying or uniform flow assuming no neighbors and single, isolated particles. This misses some important things:
 - Particle-fluid-particle interactions
 - Finite volume fraction effects (flow is no longer “undisturbed”)

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Quasi-Steady Drag Force Example Closure

- These closures fall into three broad categories: (1) Deterministic (2) Stochastic and (3) Mean
- The most well-known models just give an ensemble averaged force
- We do not have crucial particle-fluid-particle fluctuations, and we cannot solve for individual particle forces

Example of a popular empirical drag model (Tenneti, et. al 2011):

- $F_{isol}(Re_m)$ represents isolated particle contributions
 - $F_\phi(\phi)$ captures volume fraction corrections
 - $F_{\phi,Re_m}(\phi, Re_m)$ captures the coupled concentration and inertia effects

$$F(\phi, Re_m) = \frac{F_{isol}(Re_m)}{(1 - \phi)^3} + F_\phi(\phi) + F_{\phi,Re_m}(\phi, Re_m),$$

$$F_\phi(\phi) = \frac{5.81\phi}{(1 - \phi)^3} + \frac{0.48\phi^{\frac{1}{3}}}{(1 - \phi)^4},$$

$$F_{\phi,Re_m}(\phi, Re_m) = \phi^3 Re_m \left(0.95 + \frac{0.61\phi^3}{(1 - \phi)^2} \right)$$

Summary of Closure Models

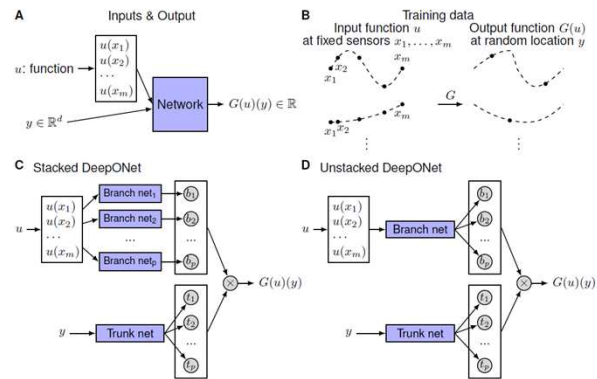
- Full list of closure quantities to be modeled
 1. Interphase coupling: force, torque, heat transfer, burn-rate
 2. Subgrid: Reynolds stress, subgrid thermal flux, subgrid viscous flux, subgrid diffusive flux
 3. These quantities are correlated – must be modeled together
- Full list of input parameters
 1. Mesoscale gas properties: cell-averaged velocity, acceleration, pressure gradient, vorticity, temperature
 2. Particle properties velocity, acceleration, angular velocity, angular acceleration, temperature
 3. Neighbor properties (microscale): position, velocity, acceleration, angular velocity, angular acceleration, temperature
- Model philosophy
 1. Lagrangian-based (per particle)
 2. Deterministic – every neighbor's effect will be accounted for deterministically (not in a mean or statistical sense)
 3. Each closure model is treated as an "Operator, which will be learnt"
 4. Must be data-driven – Governing equations of closure quantities are unknown
 5. Every available physics and symmetries must however be used

Operator Learning Framework

- An operator is some kind of function that maps inputs to some kind of output

$$G: \{Re, \phi, PDR, \hat{e}_i, r_i, \dots\} \rightarrow \{\text{subgrid}(\mathbf{Re}_{sg}), \text{interface}(\mathbf{F}_{qs})\}$$

- Operators in classical physics fit a specific functional form
 - NNs do not assume any functional form, the network learns this
 - We do assume a structured class of functions via the NN architecture

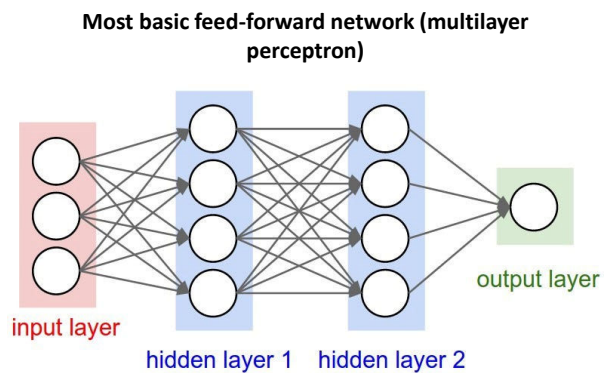


Various DeepONet Architectures
(Lu and Karniadakis 2020)

Traditional Machine Learning Approaches

- ML approaches in fluid dynamics research can use a few different methods including, but not limited to,

- Supervised Learning-
 - Regression problems
 - Support vector machines
 - Random Forests
- Unsupervised Learning-
 - POD/PCA
 - Autoencoders
 - Self-organizing maps
- Semi-Supervised Learning-
 - Generative Adversarial Networks (GANs)
 - Reinforcement Learning



$$L[y, \phi(x, y, w)] = \|y - \phi(x, y, w)\|^2$$

Common loss function used in regression problems

Physics-Based Innovations & Improvements

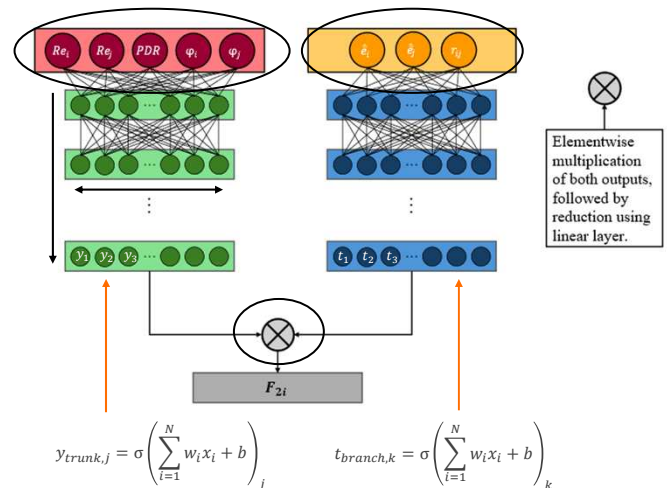
- Our approach is not just an MLP, innovation comes in three parts:
 - Hierarchical expansion (N-Body interactions represented pair-wise or higher-order)

$$F_i[M_2, M_3, 3] = \underbrace{F_{1i}(\text{Re}_i, \phi_i)}_{F_i[M_2, 2]} + \sum_{j=1}^{M_2} F_{2i}(\text{Re}_i, \phi_i, r_j) + \sum_{j=1}^{M_3-1} \sum_{k=j+1}^{M_3} F_{3i}(\text{Re}_i, \phi_i, r_j, r_k)$$

- Modal expansion, left and right network- we are trying to find our modal coefficients on the left network and the modes on the right
- Very interpretable and allows domain mappings to be developed
- Invariance and equivariance captured through symmetry-preserving network *e3nn*
 - I.e., if inputs are rotated, outputs should be rotated

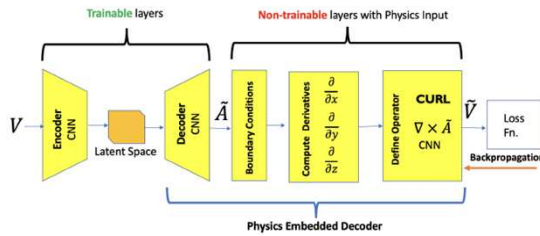
Neural Network Architecture

- PyTorch model inspired by *DeepONet*
- Trunk network takes 5 scalar inputs:
 1. Reynolds number, volume fraction, and particle density ratio
- Branch network takes 3 vector inputs:
 1. Unit vector of velocity, and distance
- Branch and trunk network run 6 layers deep with 250 neurons per layer
- Elementwise multiplication for scalars and vectors followed by a final linear reduction layer to obtain the force vector
- Equivariance captured through E(3) group rotation and reflection symmetric network, *e3nn*



Physics-Informed Machine Learning

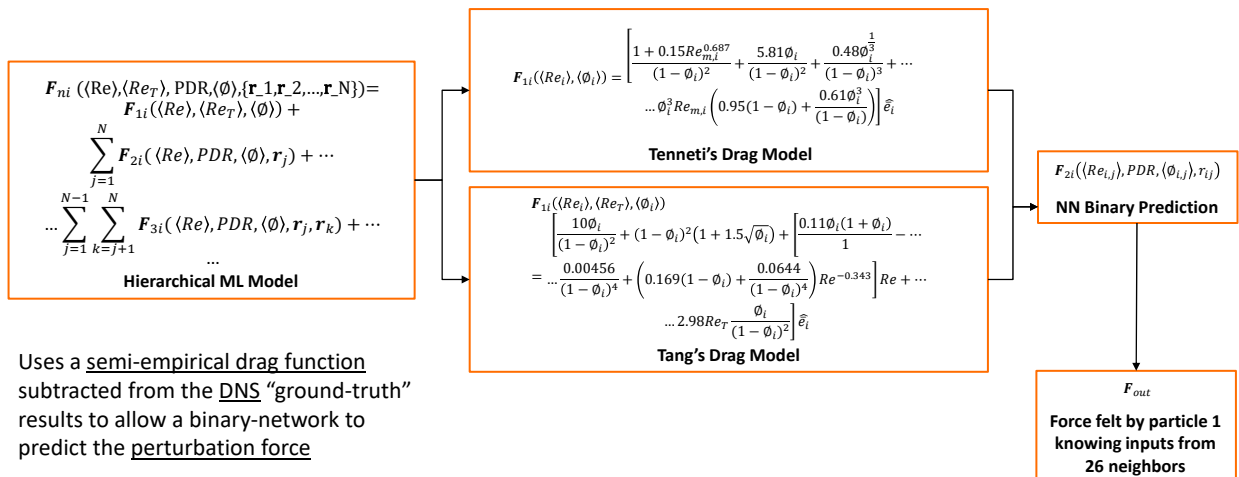
- PINNs are very popular for nonlinear fluids NN regression problems
- Embedding governing physics directly into training is very efficient
 - i.e., enforce conservation laws without requiring labeled data
- Hard constraints** like forcing cons. of mass (Mohan et. al 2023) stabilize the loss, but may have trouble with scaling [restrict the solution space]
- Soft constraints** (loss penalties) scale better but can become unstable



One example of a hard constraint PINN that satisfies divergence of velocity (Mohan et. al 2023)

- Network encodes fine-scale velocity to coarse-grained potential
- Operator fit to take curl of potential which satisfies divergence-free velocity field

Our Physics-Informed Loss and Inputs



Current Application of Operator Learning

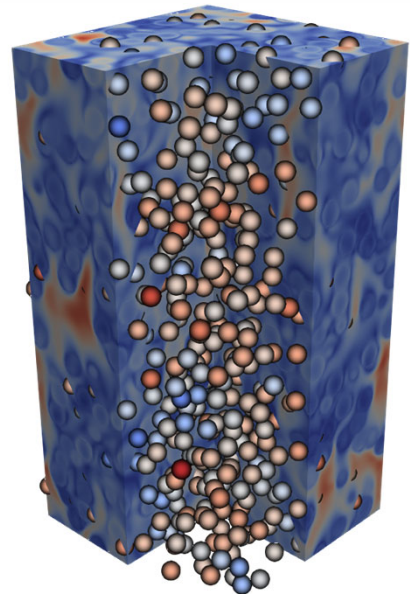
Problem Setup

- We have a triply periodic domain-
 1. Particles going out from any one side come back in from the complementary side
 2. Size: $-1 \text{ cm} \times 1 \text{ cm}$, $-2 \text{ cm} \times 2 \text{ cm}$, $-1 \text{ cm} \times 1 \text{ cm}$

x

y

z
- Particles are spherical with a fixed diameter of 0.2 cm
- Each particle influences its surrounding neighbors via fluid-mediated interactions, i.e. wakes, turbulent vortices, etc.
- Predicting the force that each particle experiences is difficult due to this process
 1. Current best methods are statistically based in averages
 2. We use ML to learn this individual particle force



PR-DNS High Fidelity Data

- Particle-resolved direct numerical simulations (PR-DNS) were performed for a range of parameters
- Each of the 48 simulations had a different number of timesteps
- Particle data was output at each step for each particle
 - Number of particles for each volume fraction are 64, 216, 512, and 1000 corresponding to 1.68%, 5.65%, 13.21%, and 26.22%
 - Transient simulation
- Particle properties output (left to right)-
 - Timestep
 - X, Y, Z location
 - V_x, V_y, V_z
 - F_x, F_y, F_z - Total force = Hydrodynamic Force + Body Force + Collision Force
 - H_x, H_y, H_z – Hydrodynamic force only

$$Re = \frac{\rho UL}{\mu}$$

$$\emptyset = \frac{V_p N_p}{V_t}$$

$$PDR = \frac{\rho_s}{\rho_f}$$

Reynolds Number, <i>Re</i>	Volume Fraction, ϕ	Particle Density Ratio, <i>PDR</i>
{10, 20, 60}	{1.68%, 5.65%, 13.21%, 26.22%}	{2.56, 12, 55, 256}

[illegible]

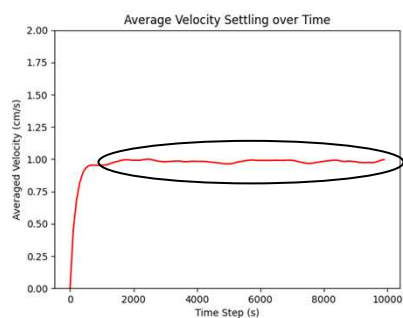
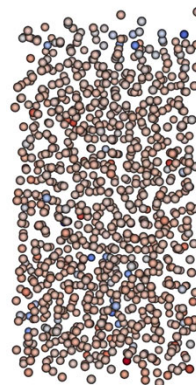
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Steady State Verification

- Needed to ensure steady state was reached for each simulation
 - Compared ensemble averaged velocity per time step to determine this

$\langle V - \langle V \rangle \rangle \approx 0$ over a sufficiently long time-period

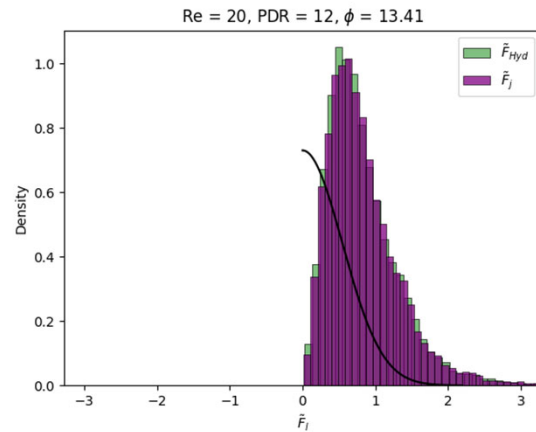
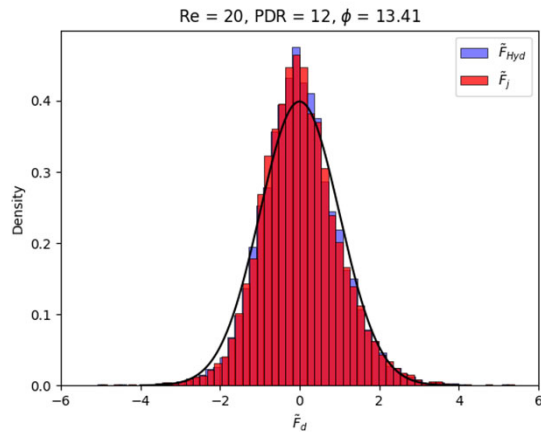
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- Video of freely falling particles in the triply periodic box.
- As the time step moves forward, the simulation converges to a stable solution
- The particles affect each other throughout the entire simulation

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Data Statistics and Validation

- Wanted to make sure data was not corrupt, so many probes were done:
- Looked at PDFs from data to determine mean and variance between forces

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Data Statistics and Validation

Structured as:

- ↑ Reynolds Number,
- ↑ Particle Density Ratio (PDR),
- ↑ Volume Fraction

- Congruency between datasets without overlap in parameters was seen
- Average drag force seen increases with increasing volume fraction and Reynolds number

Case	Target Reynolds Number	Achieved Reynolds Number	PDR	Volume Fraction	Hydro Drag Force Mean
1	10	18.612754	2.56	1.68	2.711152813
2	10	17.681774	12	1.68	2.616950185
3	10	20.567156	55	1.68	2.643498455
4	10	17.870576	256	1.68	2.588853104
5	10	9.502736	2.56	5.65	2.860345116
6	10	9.862162	12	5.65	2.907498819
7	10	10.130728	55	5.65	2.836137798
8	10	9.799004	256	5.65	2.889745263
9	10	10.45395035	2.56	13.41	3.804359679
10	10	10.51535376	12	13.41	4.178493826
11	10	11.1712576	55	13.41	4.199423118
12	10	10.634556	256	13.41	4.336530581
13	10	12.86864468	2.56	26.22	6.45414894
14	10	12.55138875	12	26.22	6.631649929
15	10	11.93026723	55	26.22	7.959897666
16	10	8.6166378	256	26.22	10.66980134
17	20	18.79464	2.56	1.68	2.793206727
18	20	18.70954	12	1.68	2.789782964
19	20	18.942546	55	1.68	2.792919196
20	20	18.332308	256	1.68	2.751641492
21	20	18.5125997	2.56	5.65	3.480900124
22	20	18.925922	12	5.65	3.609820275
23	20	19.250252	55	5.65	3.511928279
24	20	19.612152	256	5.65	3.604354764
25	20	19.10206017	2.56	13.41	4.883962236
26	20	22.16309374	12	13.41	5.050126163
27	20	20.00482	55	13.41	5.05403413
28	20	18.91084	256	13.41	5.398033121
29	20	20.44894927	2.56	26.22	8.908653548
30	20	18.41667588	12	26.22	10.28187438
31	20	18.6769704	55	26.22	11.26979211
32	20	17.10295	256	26.22	12.26255362

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Data Statistics and Validation

Important statistics highlighted:

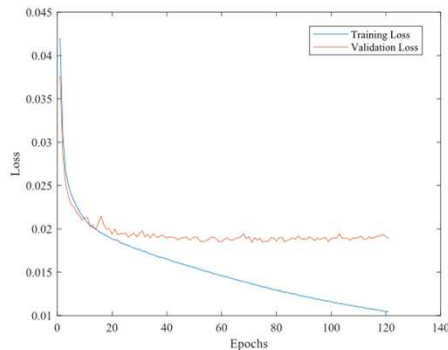
Standard deviation in velocity and acceleration decreases with increasing particle density ratios

Makes sense as $PDR \rightarrow \infty$ represents more static particles

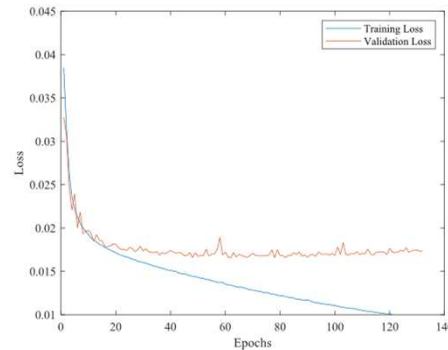
Avg Vx	Avg Vy	Avg Vz	Std Vx	Std Vy	Std Vz	Avg ax	Avg ay	Avg az	Std ax	Std ay	Std az
0.00449537	18.612754	-0.07026341	1.077215712	3.295369226	1.022400975	-0.00028361	-0.0009815	4.71E-05	0.316994092	0.969009837	0.306167981
-0.05236121	17.681774	-0.01398561	0.929112708	3.19540325	0.948194073	3.89E-05	0.000246	0.00030615	0.277266667	0.92141498	0.281304737
0.05830457	20.567156	-0.00201379	0.928542553	4.120955801	0.819068437	1.37E-06	0.0013695	2.98E-05	0.273293407	1.1575625	0.254594406
0.02196979	17.870576	-0.04332882	0.636642639	1.667245461	0.603316891	2.79E-05	0.00062	7.32E-05	0.195489496	0.483402633	0.176555546
0.03248509	9.502736	0.02869161	1.259617698	2.5339649	1.258919491	-0.00015818	0.0002195	-0.00053281	0.39585329	0.729778908	0.389238484
0.12613239	9.862162	0.06745028	1.164701526	2.589778385	1.337335782	2.75E-05	0.000977	0.0003254	0.352703226	0.728290516	0.391037182
-0.11548018	10.130728	-0.20013287	1.200272821	2.233672874	1.146643741	0.00031707	-0.001151	-0.00023727	0.373561134	0.653961998	0.344524016
0.02658218	9.799004	-0.03887612	0.676030962	1.130466084	0.70280954	3.29E-05	8.00E-06	0.00011745	0.210643602	0.334325291	0.222297515
-0.06396371	10.45395035	-0.07129861	2.423172778	4.880352371	2.882275599	0.00013111	-0.00026181	0.00185792	0.763958714	1.398611615	0.836720313
0.01572849	10.51535376	0.04803151	2.202111639	3.83996936	2.029401034	-0.00208719	-0.0020138	-0.00047244	0.6941433	1.128366901	0.656235675
-0.01974017	11.1712576	-0.20509115	1.676387275	3.039000085	1.665167397	-0.00042139	-0.0028801	-0.0005736	0.533721003	0.843381802	0.532643351
0.06587731	10.634556	-0.26379513	0.915314268	1.698799495	0.931428057	2.36E-05	-9.00E-05	-0.00018794	0.296618207	0.435817481	0.292590022
0.01597984	12.86864468	-0.12705276	3.862977857	6.52939874	3.739652005	-0.00379912	-0.00448339	0.00754594	1.19311278	1.751930715	1.186054191
-0.13093353	12.55138875	-0.07028848	3.271715605	6.539195124	3.226932361	-0.00505389	0.00228423	-0.00273292	1.026820046	1.670683232	0.979693104
0.06740187	11.93026723	-0.80149582	2.293130662	4.897088549	2.870818349	-0.00103973	-0.00125872	0.0016791	0.733132962	1.236095676	0.789164189
0.01058916	8.6166378	-1.93700917	1.317094577	2.084332331	1.507833003	-0.00045188	0.000239	-0.00111459	0.425949575	0.490649253	0.429313784
0.06352224	18.79464	0.01012225	1.08983785	3.74253297	1.183720921	-0.00023034	-0.000117	-0.00035024	0.289330277	0.932606303	0.319720344
0.04661736	18.70954	0.00341543	0.753307337	3.017729257	1.01452773	-0.00010487	0.0071465	-8.04E-06	0.204178916	0.737501614	0.264836579
-0.0299126	18.942546	0.03946426	0.855807393	3.221064054	0.856730994	0.00015979	-0.000251	-3.14E-05	0.260966406	0.862022031	0.25637757
-0.00191032	18.332308	-0.04908784	0.596806821	1.821561647	0.699621487	-1.37E-06	0.000259	-2.48E-05	0.184926172	0.533600527	0.216855301
0.04883484	18.5125997	-0.01336377	2.380407422	5.516243906	2.277049862	-0.00057363	-0.00241099	-0.00011007	0.750694742	1.662712757	0.70815436
-0.00332948	18.925922	0.04408662	2.054555165	5.273965537	2.1622216296	-0.00096649	-0.0008035	-9.51E-06	0.660197035	1.591958407	0.682205741
-0.00699851	19.250252	-0.15048008	1.824371159	3.216831052	1.6122216296	-0.00022684	-0.0014425	0.00029829	0.588171586	0.989684334	0.521021169
0.11096786	19.612152	-0.11246425	1.187434304	2.024618801	1.107452302	4.43E-05	-0.000368	0.00020517	0.383495967	0.637416967	0.359960036
-0.12383816	19.1026017	-0.03682971	4.116604751	7.361816844	4.128015717	-4.88E-05	0.0041272	-0.00230424	1.355445645	2.34446501	1.294552612
-0.01337788	22.16309374	0.00940736	3.721876469	7.486996356	4.050430358	0.00207418	0.00563115	0.00111864	1.208902198	2.183480336	1.250297012
0.00461204	20.00482	-0.39924732	2.290841619	4.194961946	2.508371644	0.00116362	0.002233	-0.00195558	0.770161628	1.236255519	0.792678881
0.12420986	18.91084	-0.58396512	1.485424871	2.202547002	1.453613351	-0.00020025	-0.0009045	0.00015455	0.501411078	0.651195838	0.486585579
0.13893084	20.44894927	-0.23257907	5.897320657	9.717779064	6.471102149	-0.01313718	-0.00844755	-0.0084034	3.690168513	5.486840183	3.926198436
-0.03148422	18.41667588	-0.22067931	4.22433039	6.616571501	4.817787284	-0.00172979	0.0031922	-0.0032117	5.286718695	7.429167596	5.501505292
0.07509775	18.6769704	-1.1115301	2.907813619	6.677580059	3.598278353	-0.00037646	-0.00789094	-0.00132156	3.739510092	6.219656182	4.083950275
0.00785036	17.10295	-1.23910179	1.451194689	2.140692313	1.516372988	-0.00015632	-0.00072	0.00135203	1.890413909	2.296340106	1.928444919

Training and Validation Cost Output

- K-Fold cross validation used to ensure model reliability over entire data-space
- Two different models' loss functions plotted with Tenneti and Tang's empirical drag force used as $F_{1i}(\langle Re_i \rangle, \langle \phi_i \rangle)$
- K-Fold 1 out of 5 shown for ease of viewing – Other 4 folds produced similar results

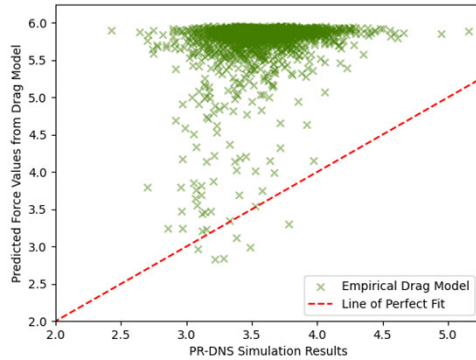


Tang Model's Loss From Fold 1/5



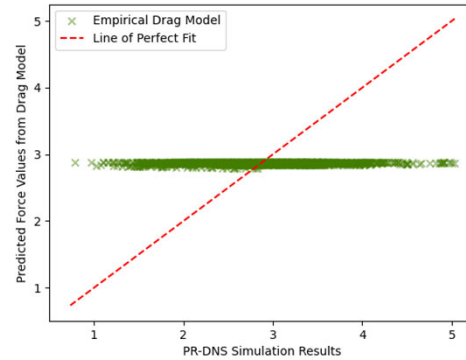
Tenneti Model's Loss From Fold 1/5

Mean Closure – Predictive Capability



Data output from simulation with parameters:

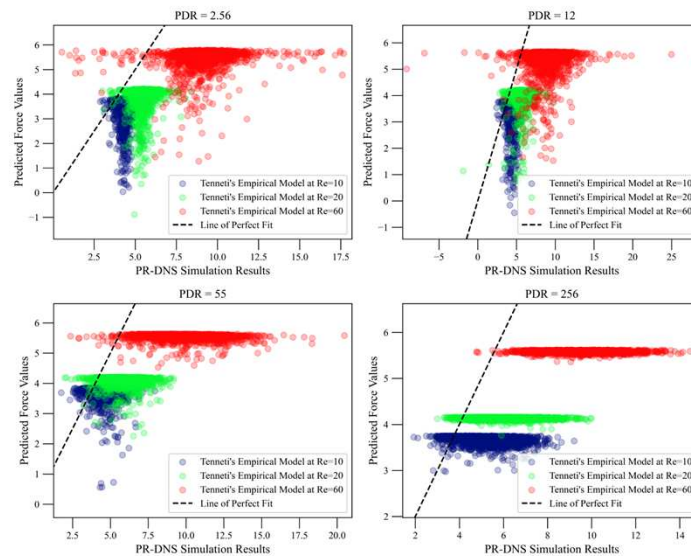
- Reynolds number- 20
- Volume fraction- 5.65%
- Particle density ratio- 2.56



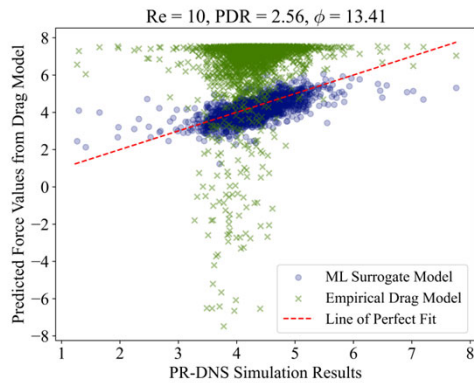
Data output from simulation with parameters:

- Reynolds number- 10
- Volume fraction- 5.65%
- Particle density ratio- 256

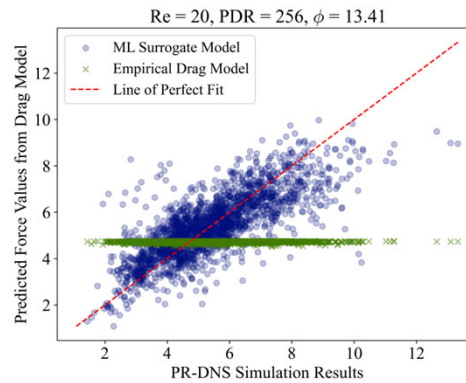
Mean Closure – Predictive Capability



Mean vs Deterministic Closure – Predictive Capability

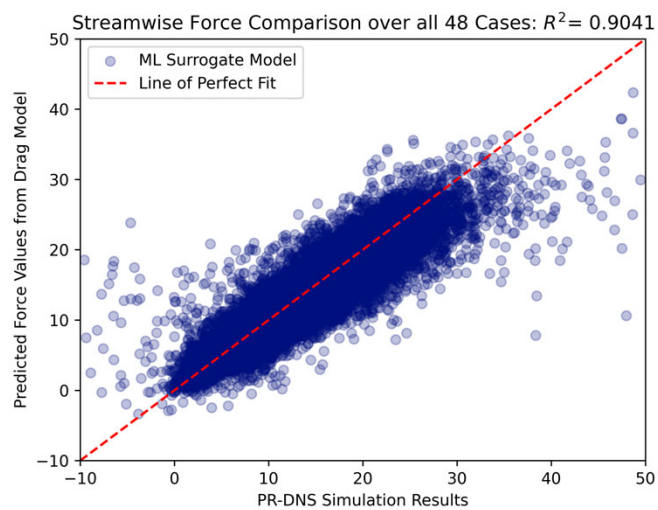


Even though the empirical drag model performs poorly over these parameters, the functional form fit by the NN can generalize well here



Here, the empirical model does much better, yet the NN still acts to pull individual forces per particle out given these parameters

ML Operator-Learned Deterministic Closure



Summary & Future Work

- The above is a major step in multiphase flow closure
 1. But must be extended to include particle rotation, Mach number effect
- Building a general Operator-Learning modeling framework, to be used for:
 1. Torque, heat-transfer, burn-rate, other inter-phase closure modeling
 2. Reynolds stress and other subgrid closure models
- Interpret the models and their correlations
- Explore other (e.g., GNN) network architectures for performance improvement
- Work with UQ team to quantify uncertainty ([T13 for more details](#))
- Make these ML models available for EL simulations & testing

Questions?

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CM³C: Center for Multiscale Modeling of Multiphase Combustion

T:10 Immersed Boundary Development in MFC

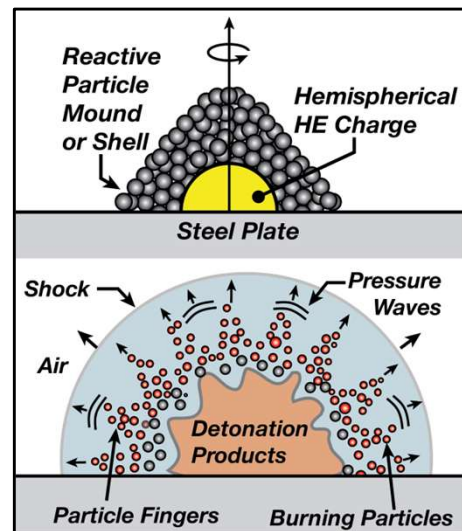
Daniel J. Vickers

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How are we Using Immersed Boundaries?

- We will be simulating particles/debris in the post-shock flow, including:
 - Sub-grid/Euler-Lagrange (EL) particles
 - Resolved immersed boundary (IB) particles, much larger than a grid cell
- Particles will interact with each other and the flow to accurately capture the physics
- Previously, MFC supported stationary IBs, but not motion or IB-EL interactions
- This presentation will cover the capability improvements made to MFC for particle-laden flows



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Required Software Features Overview

- I determined derived requirements for running the simulations of interest

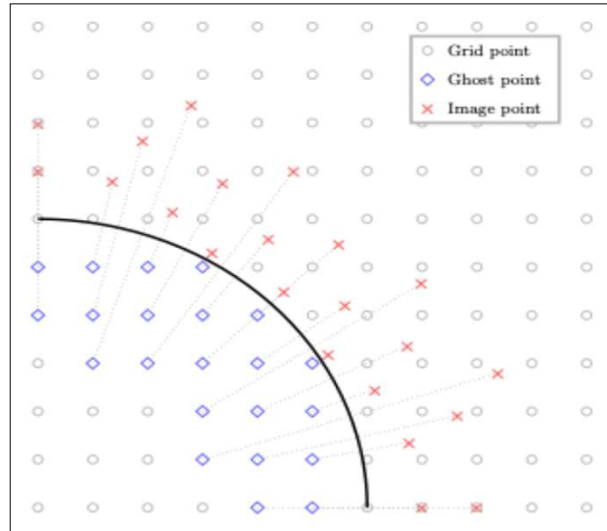
Requirement	Previous Capability	Current Capability
Solid-to-Fluid Force Spreading: MFC must be able to handle force spreading onto the fluid at the IB interface point.	Already Supported in MFC	Already Supported in MFC
Fluid-to-Solid Force Spreading: MFC must be able to compute and apply forces on immersed boundaries from the fluid.	Not Yet Implemented	Fully Implemented
Immersed Boundary Motion: IBs need to be able to move through the fluid	Not Yet Implemented	Fully Implemented
IB-IB Collision Forces: Resolved particles need to be able to collide with each other and the walls of the simulation	Not Yet Implemented	Fully Implemented
IB-EL Collision Forces: Resolved particles must be able to collide with sub-grid particles	Not Yet Implemented	Not Yet Implemented
Scaling: MFC must be able to run up to 1000 moving resolved particles at once.	MFC Could Run Up to 10 Particles	Currently Developing
Reacting Boundaries: MFC needs to support several boundary conditions on IB surfaces, particularly boundaries the can react with interesting chemistry and temperature changes	Not Yet Implemented	Not Yet Implemented

Outline

- ➡ ■ Ghost Point Interface
- Immersed Boundary Motion
- Immersed Boundary Forces
- Collisions
- Scaling

Pre-Existing Ghost Point Interface Method

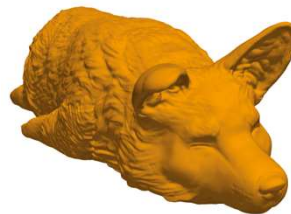
- MFC already supported a ghost-point interface method for stationary IBs
 - Grid points close to the surface of the IB (ghost points) have their values modified to resolve the boundary
 - Momentum and Energy are set to get the correct derivatives near the boundary
 - Normal vectors and levelset (distance to the IB surface) for each ghost point are required

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Ghost-Point Optimization

- We began by optimizing existing code
 - Previously all IB projection and levelset compute was performed on the CPU.
 - I added several optimizations to a GPU implementation of the projection algorithm
- Shown is a stress-test of a ~2 million vertex STL model ray traced on the GPU
 - Mach 0.3 Flow
 - 400x200x200 grid cells on 2 GPUs
- IB was resolved in 80 seconds
 - Previously, a comparable run would have taken ~1000 CPU hours to process



1.94 Million Vertices



Real corgi, for scientific reference purposes (~2e19 Vertices)

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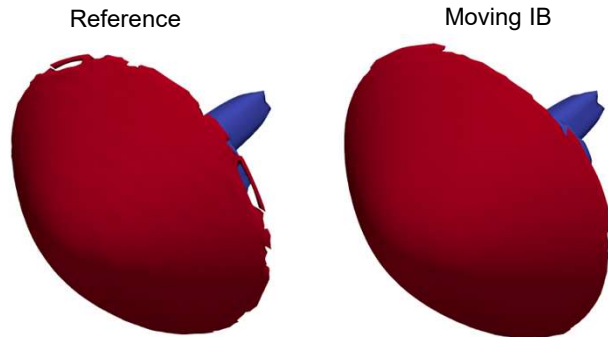
- Ghost Point Interface
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- Moving IBs required two primary changes:
 1. Ghost points needed to be recomputed during each RK substep
 - a. Reprojecting the new IB locations onto the grid
 - b. Recomputing the levelset (ℓ) and levelset normal ($\vec{n}$) values for each IB
 - c. Computing the location of all ghost points and reflected image points
 2. Modifying the fluid values set at the ghost points to take IB motion into account.
- Velocity values are set based on translational & rotational motion
- Pressure of each ghost point (P_{GP}) is determined via an adiabatic approximation across the boundary and the image point pressure (P_{IP}):

$$1. \quad P_{GP} = P_{IP} / \left[1 - \frac{2\ell\gamma}{(\gamma-1)T_{IP}} \frac{d\vec{v}}{dt} \cdot \vec{n} \right]$$

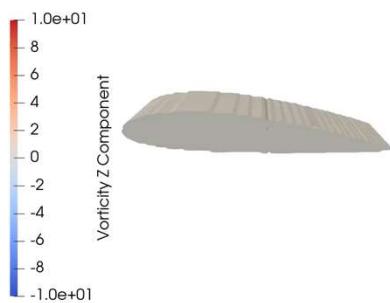
Motion Example: Hypersonic Sphere

- We first compared directly with MFC
- Shown, we compare two different configurations of the same case
 - In the reference, the IB is stationary in a Mach 2.0 flow
 - In the moving case, the IB is moving at Mach 2.0 in a stationary fluid
 - Contours are of constant density (1.0 and 1.5)



Motion Example: Pitching Airfoil

- We also ran over a pitching airfoil
 - Flow is Mach 0.4, and the airfoil pitches as a cosine wave
 - Contours are spanwise vorticity



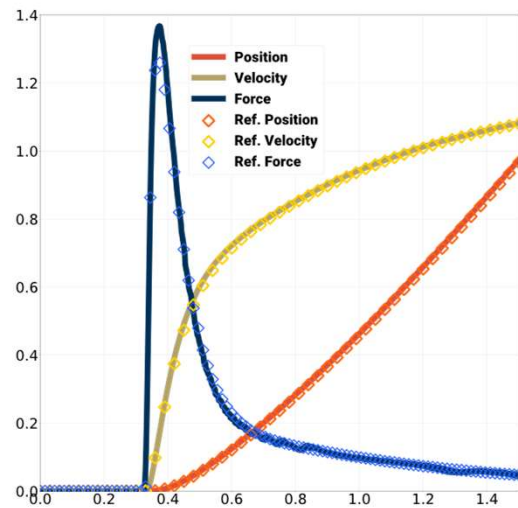
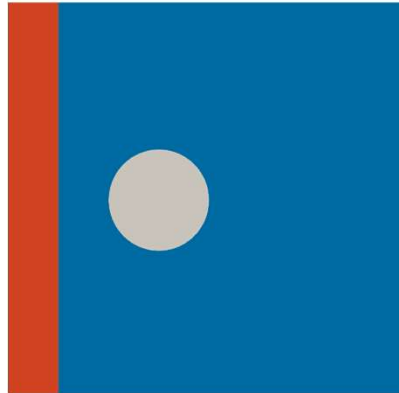
- Ghost Point Interface
- Immersed Boundary Motion
- ➡ ■ Immersed Boundary Forces
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- Pressure and viscous forces/torques are computed via a volume integral
 1. $\vec{F}_f = \iiint_{\partial V_P} \nabla \cdot (-p\vec{I} + \vec{\tau}) dV \approx \sum_k \nabla \cdot (-p\vec{I} + \vec{\tau}) \Delta V_k$
 2. $\vec{T}_v = \iiint_{\partial V_P} \nabla \cdot [r \times (-p\vec{I} + \vec{\tau})] dV \approx \sum_k \nabla \cdot [r \times (-p\vec{I} + \vec{\tau})] \Delta V_k$
- Summing over the volume of the IB and then performing a parallel reduction sum was faster than performing the surface integral variant of these equations

Two-Way Coupling Example: Shock Cylinder

- We compared the 2-way coupled code to a reference case by Henshaw et al
- A 2D cylinder is hit with a Mach 1.5 shock wave
- Cylinder experiences a large pressure force to the right

*Color indicates fluid density

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Outline

- Ghost Point Interface
- Immersed Boundary Motion
- Immersed Boundary Forces
- ➡ ■ Collisions
- Scaling

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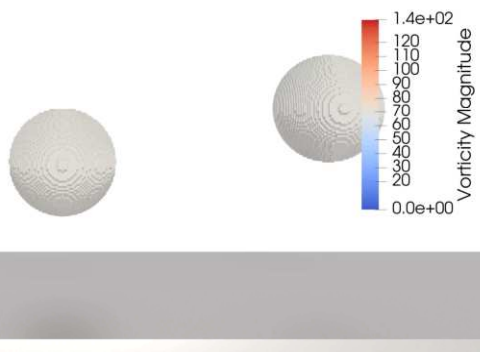
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Collision Forces

- We detect collisions via a filter placed around each ghost point
- Colliding spheres experience forces via the soft-sphere collision model
 1. $F_c = -k\delta\vec{n} - \eta\vec{u}$
 2. δ is the overlapping distance
 3. $\vec{u}$ is the direction tangential to the velocities of the particles
 4. k and η are the spring stiffness and damping term, computed from the effective mass (m_e), collision time (τ_c) and coefficient of restitution (e)
 - a. $k = \frac{m_e(\pi^2 + 1)(e)}{\tau_c}$
 - b. $\eta = -2 \ln(e) m_e$

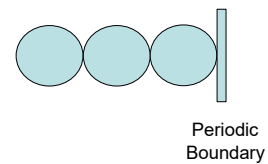
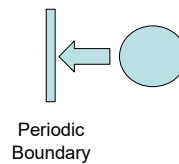
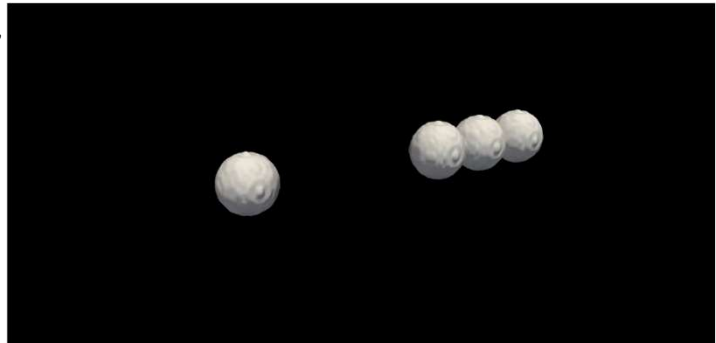
Collision Example: 3D Spheres

- Soft-sphere collisions were implemented for walls and IB-IB collisions
- Show here is a glancing collisions of spheres, which rebound into the wall



Collision Example: Periodic Newton's Cradle

- Collisions work with periodicity, demonstrated here
 - One moving sphere on the left, travels left
 - The moving sphere periodically collides with the right stationary sphere
 - Contours are density



Outline

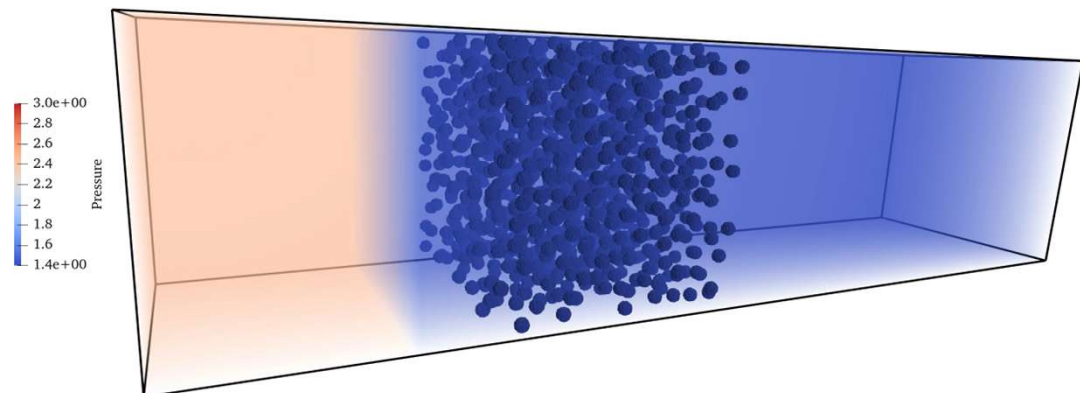
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- ➡ ■ Scaling

Memory Optimization

- For convenient code development, MFC previously allocated $m \times n \times p$ arrays to hold the levelset and normal vectors for each IB
 1. Memory scaled linearly with the number of IBs
 2. One could run up to ~ 100 particles before memory limited simulation size
 3. This is not suitable for scaling to thousands of particles
- To overcome this limitation, I refactored all IB memory allocation, including:
 1. Moving the IB code from a pre-processing step to simulation
 2. Swapping all compute to use the new indexing
 3. Encoding some of our data so that individual indices contained multiple IB parameters
- We also refactored our toolchain, because it was slowing down under processing the case parameters for the number of IBs in larger simulations

Scaling Example: 1000 Spherical Particles

- Demonstrating this optimization, I ran a case with 1000 resolved particles on only 4 ranks
 - Grid resolution of $800 \times 200 \times 200$ and particles have diameters of 10 grid cells
 - Mach 1.5 pressure wave
 - Particles can collide with each other as well as the y and z slip walls





Summary

- In this presentation, I demonstrated:
 - Fluid-to-Solid force spreading via a ghost-point interface method
 - Extensions to the code were made to support particles in motion as well as performance optimizations
 - Solid-to-Fluid force spreading via volume integrals
 - Collision forces supported for spherical particles colliding with walls and each other
- I also showed preliminary results for scaling, including a 1000 particle case run on 4 GPUs
 - More scaling/load-balancing improvements will be made before this is merged into MFC master
- Future development will include integrating IB-IB collisions with subgrid EL particles and adding reacting boundaries

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Questions?



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T11: Group Combustion Binary-Unary E-L Modelling Approach

Davin Anderson

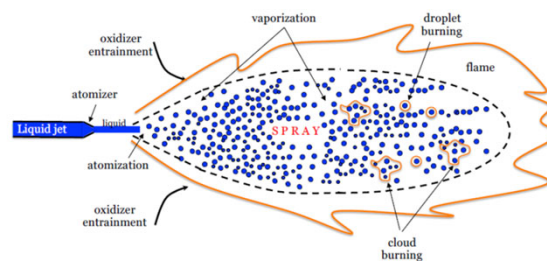
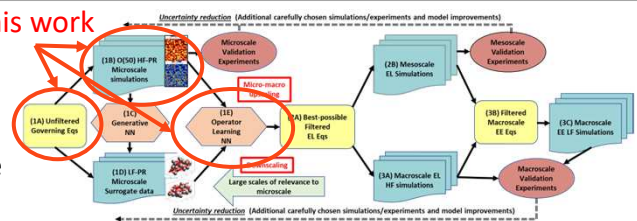
Department of Mechanical and Aerospace Engineering, UF



Introduction

- Droplet combustion is well understood with investigation of evaporation trends that date back to the late 1800s
- Current approaches to group combustion use averaged models to capture the neighboring effects
 - Decoupled from force/Reynolds stress models
- Our goal is to develop “best-possible” closure models that accurately account for neighbor interactions
- We are building a rigorous theoretical framework

This work

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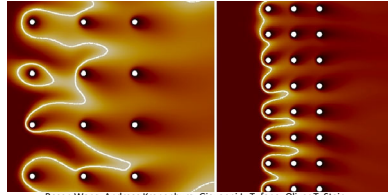
Rigorous Framework for N-Droplet Group Combustion

Fully Coupled problem

Governing Eqn: $\mathcal{N}(\psi = \mathbf{u}, T, p, Y) = 0$

On the droplet: $\psi|_{S_i} = \psi_i$ for $i = 1, 2, \dots, N$

Far field: $\psi|_{\infty}$



Bosen Wang, Andreas Kronenbourg, Giovanni L. Tufano, Oliver T. Stein,
Fully resolved DNS of droplet array combustion

- Droplets interact
- Finite Re cross flow
- Complex chemistry

Unary problem

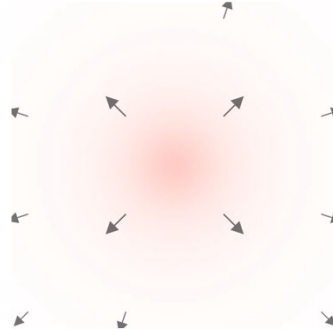
Governing Eqn: $\mathcal{N}(\psi_i^I) = 0$

On the droplet: $\psi_i^I|_{S_i} = \psi_i$

Far field: $\psi_i^I|_{\infty}$

Unary Superposition: $\sum_{i=1}^N \psi_i^I$

Integrated effect of j^{th} droplet on i^{th} : $[\bar{\psi}_j^I]_i$



- One droplet at a time
- Superpose solutions
- Explicit vs Implicit

On the surface of i^{th} droplet:

$$\psi_i + \sum_{\substack{j=1 \\ j \neq i}}^N [\bar{\psi}_j^I]_i$$

Neighbor effect

- Goes back to pairwise interaction (Akiki et al 2017)

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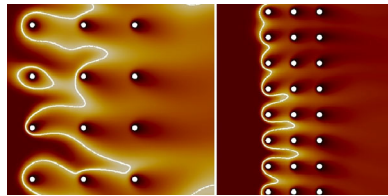
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Bosen Wang, Andreas Kronenbourg, Giovanni L. Tufano, Oliver T. Stein,
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- Droplets interact
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- Complex chemistry

Binary problem

Governing Eqn: $\mathcal{N}(\psi_{i,j}^{II}) = 0$

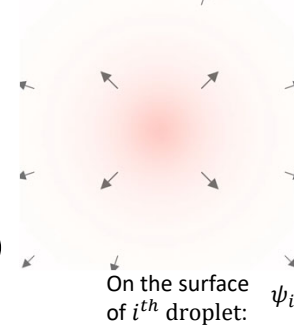
On the droplets: $\psi_{i,j}^{II}|_{S_{i,j}} = \psi_{i,j}$

Far field: $\psi_{i,j}^{II}|_{\infty}$

Binary Superposition:

$$\sum_{i=1}^N \psi_i^I + \sum_{i=1}^N \sum_{j=i+1}^N (\psi_{i,j}^{II} - \psi_i^I - \psi_j^I)$$

Effect of (i, j) droplet pair on k^{th} : $[\bar{\psi}_{i,j}^{II}]_k$



- Droplet pair at a time
- Superpose pairs
- Explicit vs Implicit

On the surface of i^{th} droplet:

$$\psi_i + \sum_{\substack{l=1 \\ l \neq i}}^N \sum_{\substack{m=l+1 \\ m \neq i}}^N ([\bar{\psi}_{l,m}^{II}]_i - [\bar{\psi}_l^I]_i - [\bar{\psi}_m^I]_i)$$

Neighbor effect

- Hierarchical extension of (Akiki et al 2017)

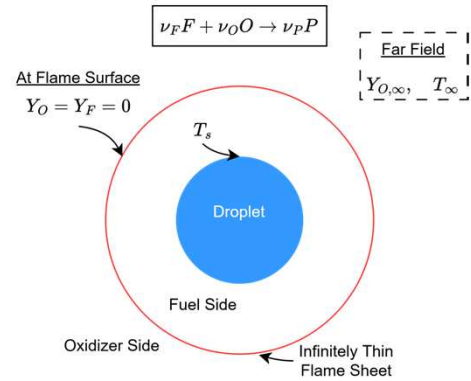
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Classical Unary Problem and Solution

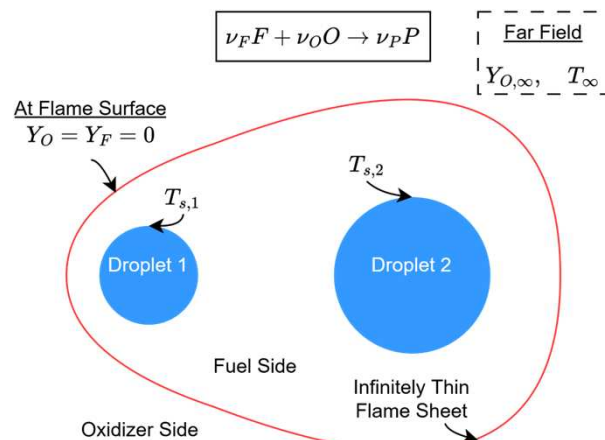
Assumptions:

1. Infinitely thin flame sheet
2. One step global reaction: $\nu_F F + \nu_O O \rightarrow \nu_P P$
3. Gas phase is quasi-steady, potential & axisymmetric
4. Stefan flow dominates
5. Unitary Lewis numbers
6. Constant thermophysical properties
7. Fixed prescribed surface temperatures
8. Neglect gravity, thermal radiation, Soret diffusion, Dufour effects, and Marangoni effects



Unary problem solution is 1D & easy

Binary Problem Setup



Recent work of Li and Zhang, JFM (2024) extended here to include more quantities of interest

Problem Governing Equations & Boundary conditions

$$\rho_g \mathbf{u} = -\nabla \phi, \quad \nabla \cdot (\rho_g \mathbf{u}) = 0 \Rightarrow \nabla^2 \phi = 0$$

$$\nabla \cdot (\beta_k \nabla \phi + \rho_g D \nabla \beta_k) = 0, \quad k \in \{FO, TF, TO\}$$

$$\beta_{FO} = \frac{Y_F}{W_F v_F} - \frac{Y_O}{W_O v_O},$$

$$\beta_{TF} = \frac{c_{p,g} T_g}{Q_{rxn}} + \frac{Y_F}{W_F v_F}, \quad \beta_{TO} = \frac{c_{p,g} T_g}{Q_{rxn}} + \frac{Y_O}{W_O v_O}$$

$$B_{M,i} = \frac{Y_{F,s,i} + \varphi_{stoich}}{Y_{O,\infty} - Y_{F,s,i}}, \quad \varphi_{stoich} = \frac{W_F v_F}{W_O v_O}$$

Boundary Conditions

$\xi = \xi_i: \quad Y_F = Y_{F,s,i}, \quad Y_O = 0, \quad T = T_{s,i},$ $\beta_{FO,s,i} = \frac{Y_{F,s,i}}{W_F v_F}, \quad \beta_{TF,s,i} = \frac{c_{p,g} T_{s,i}}{Q_{rxn}} + \frac{Y_{F,s,i}}{W_F v_F}, \quad \beta_{TO,s,i} = \frac{c_{p,g} T_{s,i}}{Q_{rxn}}$ $\phi_{s,i} = \rho_g D \ln(1 + B_{M,i}), \quad \text{Clausius-Clapeyron relation:}$ $Y_{F,s,i}(T_{s,i}) = \frac{W_F}{W_{\text{mix}}} \exp \left[\frac{q_L}{R_u} \left(\frac{1}{T_B} - \frac{1}{T_{s,i}} \right) \right]$	$\xi \rightarrow \infty$ $Y_F = 0, \quad Y_O = Y_{O,\infty}, \quad T = T_{\infty},$ $\beta_{FO} = -\frac{Y_{O,\infty}}{W_F v_F}, \quad \beta_{TF} = \frac{c_{p,g} T_{\infty}}{Q_{rxn}}, \quad \beta_{TO} = \frac{c_{p,g} T_{\infty}}{Q_{rxn}} + \frac{Y_{O,\infty}}{W_O v_O}$ $\phi \rightarrow 0$
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Series Solution for Potential & Asymptotic Solution for T

1) Exact series solution for mass scalar potential ϕ :

$$\phi(\xi, \theta) = \sqrt{2X(\xi, \theta)} \sum_{n=0}^{\infty} A_n e^{\lambda_n \xi} + B_n e^{-\lambda_n \xi}, \quad \lambda = n + \frac{1}{2},$$

$$A_n = \frac{\phi_{s,1} e^{(2n+1)\xi_1} - \phi_{s,2}}{e^{(2n+1)\xi_1} - e^{(2n+1)\xi_2}}, \quad B_n = \frac{e^{(2n+1)\xi_1} (\phi_{s,2} - \phi_{s,1} e^{(2n+1)\xi_2})}{e^{(2n+1)\xi_1} - e^{(2n+1)\xi_2}}.$$

2) Only asymptotic approximation for β_{TO} and β_{TF} :

$$\beta_k \sim \beta_{k,\infty} - \left[\frac{\beta_{k,\infty} - \beta_{k,s,1}}{L(\phi_{s,1}) - 1} (1 - \mathcal{B}(\xi)) + \frac{\beta_{k,\infty} - \beta_{k,s,2}}{L(\phi_{s,2}) - 1} \mathcal{B}(\xi) \right] (L(\phi) - 1), \quad k = TF, TO.$$

3) Numerical solution using bi-spherical coordinates

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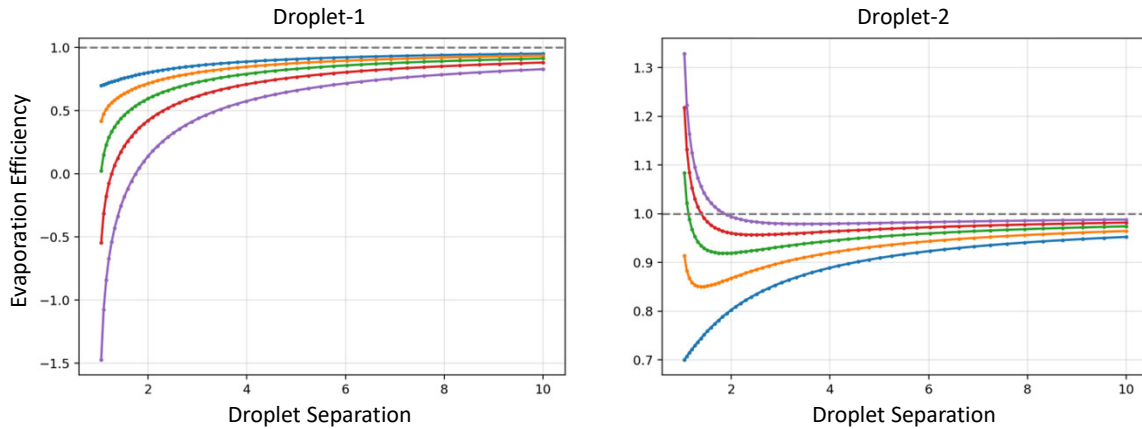
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Two Droplet Results

Radius ratio = $r_2/r_1 = 1$ Spaling Number = $B_1 = f(T_{s,i})$ $B_1 = 1$

B2 values

$B_2 = 1$ $B_2 = 1.495$ $B_2 = 2.403$ $B_2 = 4.314$ $B_2 = 10$

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Drop-1's burn-rate always ↓ – even condensation occurs. Drop-2's evaporation can ↑

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Quantities of Interest

- Total evaporation rate $M_{ev} = \oint \dot{m}_{ev,i} dS$
- Evaporative heat rate $Q_{ev} = \frac{q_l}{w_F} M_{ev}$
- Conductive heat transfer rate $Q_c = \oint \dot{q}_{c,i} dS$
- Net axial force $F_{x,i} = \oint p_{dyn,i}(\theta)(\mathbf{n} \cdot \mathbf{e}_x) dS$

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Implicit Unary Superposition

- Unary superposition – A very simple idea. Just add the well-known solutions of an isolated droplet

e.g. two droplets and two surface spherical harmonic modes

$$\phi(\mathbf{x}) \approx \sum_{i=1}^2 \sum_{m=0}^{M_\phi} c_{i,m}^{(\phi)} G_i^{(m)}(\mathbf{x}), \quad M_\phi = 0 \text{ (monopole)}, \quad M_\phi = 1 \text{ (monopole+dipole)}$$

- Monopole & dipole solutions – They are a hierarchy of more accurate “Generative Solutions” of one-drop problem – Instead of ML here they are analytical

$$\text{Monopole solution: } G_i^{(0)}(\mathbf{x}) = \frac{R_i}{r_i}, \quad \text{Dipole solution: } G_i^{(1)}(\mathbf{x}) = \frac{R_i^2 \nu_i}{r_i^2}$$

Implicit Unary Superposition for ϕ

- Explicit superposition – each superposed solution are based on specified boundary condition; superposition will violate the boundary conditions
e.g. two droplets and two surface spherical harmonic modes

$$\phi \approx \sum_{i=1}^2 \phi_i^I$$

- Implicit superposition – superposed solution must satisfy the boundary conditions, a simple way to allow interaction between the drops

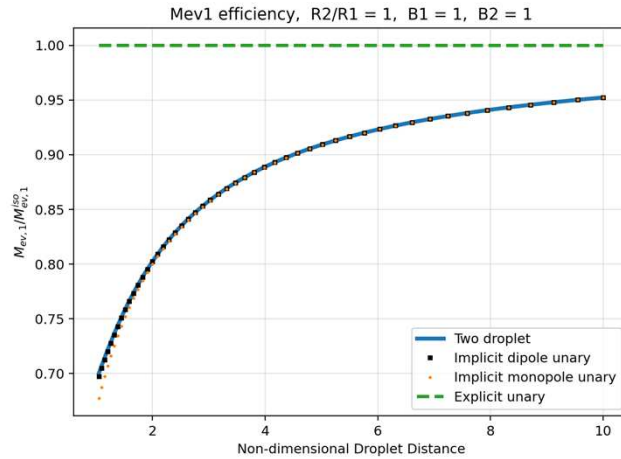
$$\phi^{(M+D)}(\mathbf{x}) \approx \alpha_1 G_1^{(0)}(\mathbf{x}) + \beta_1 G_1^{(1)}(\mathbf{x}) + \alpha_2 G_2^{(0)}(\mathbf{x}) + \beta_2 G_2^{(1)}(\mathbf{x})$$

2-droplet
Monopole + Dipole
superposition

$$A_\phi^{(M+D)} \begin{bmatrix} \alpha_1 \\ \beta_1 \\ \alpha_2 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \phi_{s,1} \\ 0 \\ \phi_{s,2} \\ 0 \end{bmatrix}, \quad A_\phi^{(M+D)} = \begin{bmatrix} 1 & 0 & M_{12}^{00} & M_{12}^{01} \\ 0 & 1 & M_{12}^{10} & M_{12}^{11} \\ M_{21}^{00} & M_{21}^{01} & 1 & 0 \\ M_{21}^{10} & M_{21}^{11} & 0 & 1 \end{bmatrix}$$

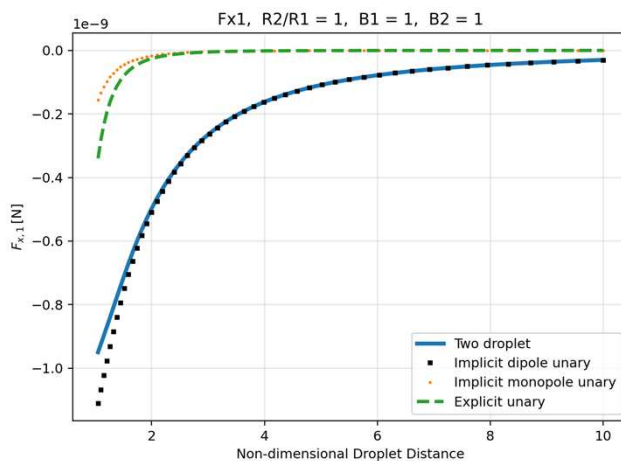
Power of Implicit Unary Superposition - Evaporation

- Explicit unary superposition does not differ from isolated values
- Implicit monopole superposition is good
- Implicit dipole superposition amazingly good



Power of Implicit Unary Superposition - Force

- Explicit unary superposition results in a repulsive net force
- Implicit monopole superposition is no better
- Implicit dipole superposition good, but not perfect
- Pressure force is nonlinear in velocity



Implicit Unary Superposition for β_{TO}

Same idea for superposition of ϕ

- Explicit superposition – each superposed solution are based on specified boundary condition; superposition will violate the boundary conditions

$$\beta_{TO}^{exp} \approx \beta_{TO,\infty} + \sum_{i=1}^2 (\beta_{TO,i}^I - \beta_{TO,\infty})$$

- Implicit superposition – superposed solution must satisfy the boundary conditions, a simple way to allow interaction between the drops

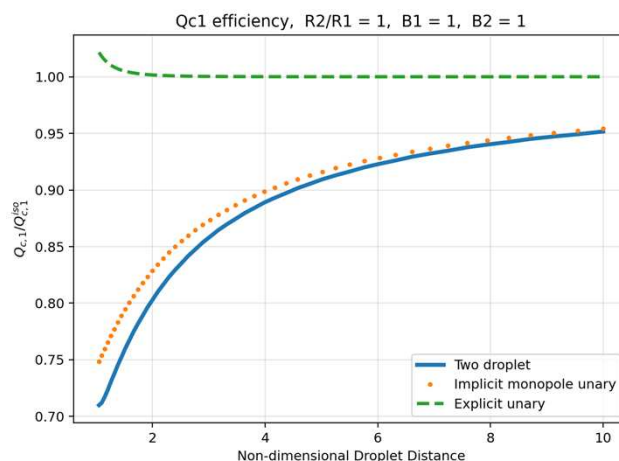
$$\beta_{TO}^{imp} = \beta_{TO,\infty} + \sum_{i=1}^N \gamma_i G_i$$

G_i is found from known solutions

γ_i is the chosen implicitly so $\langle \beta_{TO} \rangle_{S_i} = \beta_{TO,s,i}$

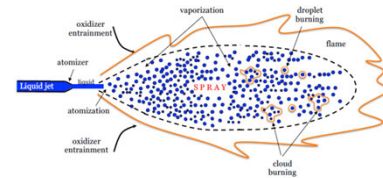
Power of Implicit Unary Superposition – Conduction HT

- Explicit superposition does not work for $\beta_{TO} \rightarrow Q_c$
- Implicit monopole superposition works relatively well
- Monopole + Dipole may improve results even further



Summary, Future Work and Challenges

- Implicit multi-pole superposition is amazingly powerful to solve N-droplet (or N-body) interaction problems
- Analytical simplicity has allowed detailed examination. But the conclusions apply to all the ML modeling efforts
- Extend implicit superposition to clouds of $O(10^4)$ droplets and investigate group combustion in a manner never done before
- MFC Simulations
 - Transfer knowledge from theory to more realistic droplets
 - Finite Re, non-unit Lewis number, complex chemistry



Questions?

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T12: Carbon Rod Experiments

Alexander Millauer-Hsu

School of Mechanical Engineering

Purdue University



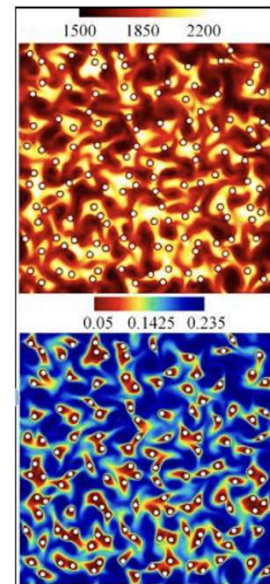
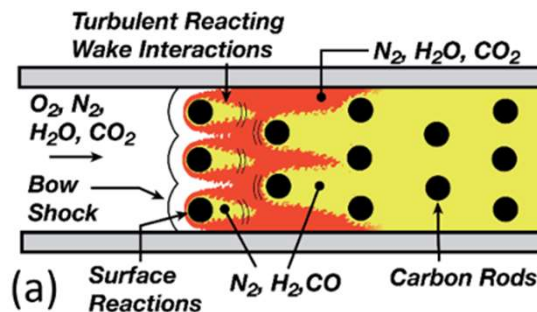
Overview

Model validation:

- Chemistry
- Multi-rod interactions

Rod experiments and 2D simulations form a feedback loop:

- Uncertainty quantification
- Results on each side will inform changes/adjustments on the other

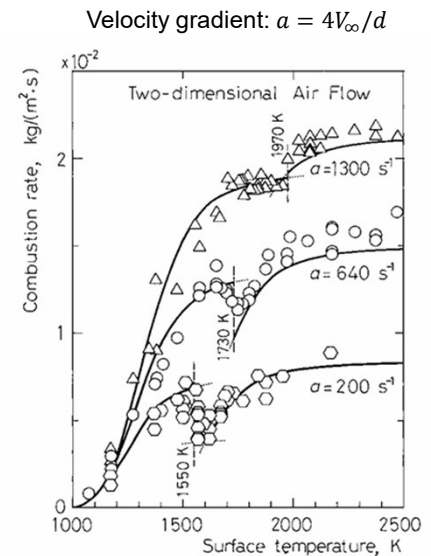
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Single Carbon Rod Combustion: Theory

Theory developed in a series of papers by A. Makino predicts:

- Little to no combustion below $T_s = 1000\text{K}$
- Positive correlation between flow velocity and reaction rate
- An abrupt shift in combustion behavior at a critical T_s
 - “Ignition surface temperature”: CO begins reacting in the gas phase, reducing surface reaction rate
 - Reaction shifts from kinetic to diffusion limited
- Ignition surface temperature shifts right with increasing velocity
 - For high velocity flows, the CO flame never ignites, the reaction is always diffusion limited



Makino, Namikiri, and Araki (1998)

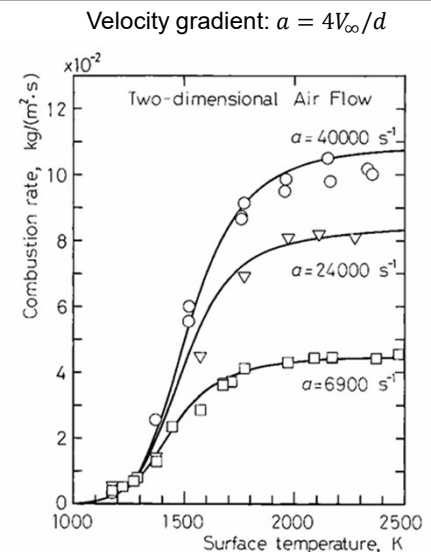
3



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Makino, Namikiri, and Araki (1998)

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Single Carbon Rod Combustion: Existing Methods

Makino performed experiments to validate his models:

- Free jet of oxidizer
 - Pure O_2 , CO_2 , H_2O , as well as dry and humid air
 - Velocities in the range of $0.25 - 50 \text{ ms}^{-1}$
- Joule-heated carbon rod
 - Diameters in the range of $5 - 20 \text{ mm}$
- Combustion rate measured by visually tracking radial regression
 - Corrections made for density change
- Temperature measurements:
 - Surface temperature: radiation thermometer
 - Gas-phase temperature: N_2 CARS

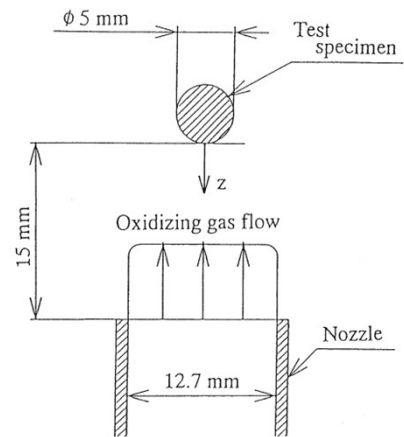


Fig. 2. Combustion field in the forward stagnation region of a graphite rod.

Makino, Namikiri, and Kimura (2003)

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5



Staged Approach

Testing will be divided into stages on two fronts:

- Rod count
 - Begin with single rod experiments
 - Adapt setup for multi-rod arrays
- Air flow
 - Begin with experiments in quiescent conditions
 - Introduce a small air jet to the existing setup to perform subsonic tests with low temperature air
 - Rework setup to leverage the Zucrow air plant for high velocity tests with high-temperature air

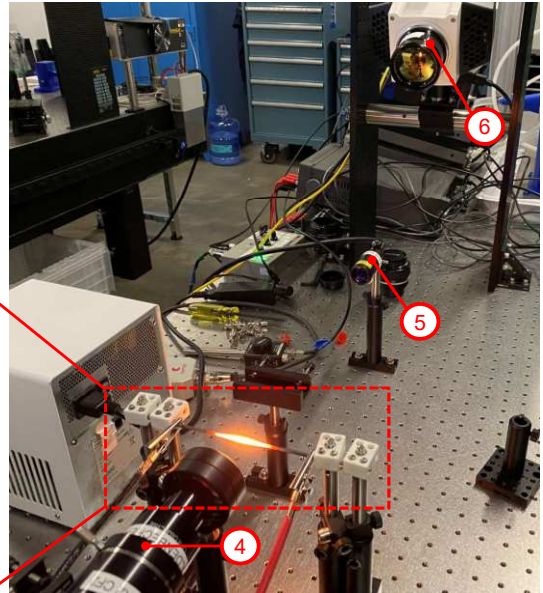
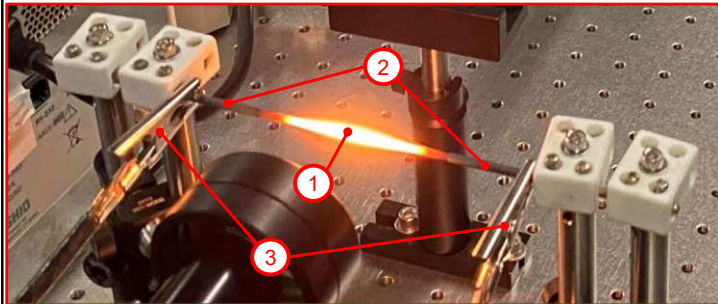
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6



Current Setup

- ① Graphite Rod
- ② Copper Electrode
- ③ Lead to Power Supply
- ④ High Magnification Lens
- ⑤ Cavitlux Pulsed Laser
- ⑥ Telops IR Camera
- ⑦ Imperx High Resolution Camera (shown on next slide)

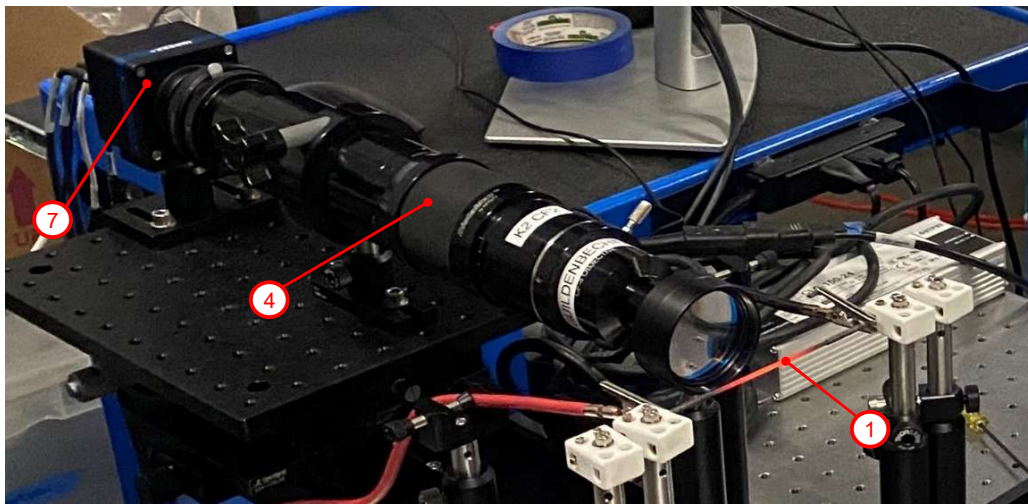


7



Current Setup

- ① Graphite Rod
- ④ High Magnification Lens
- ⑦ Imperx High Resolution Camera



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8

Free Jet vs. Wind Tunnel

- The initial vision was to use a confined configuration. Is this the best option?
- Previous work (at significantly lower velocities) has employed a free jet.

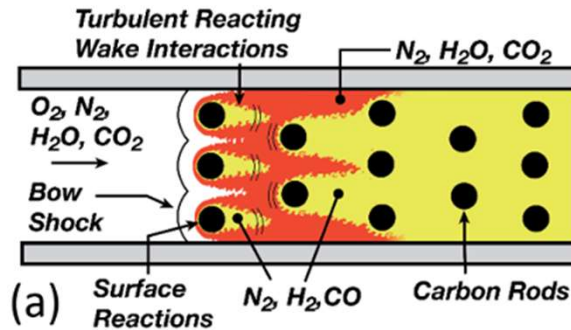
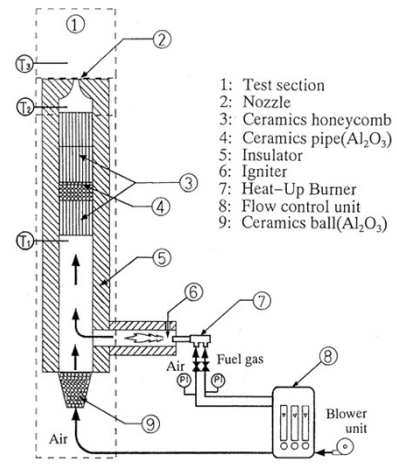
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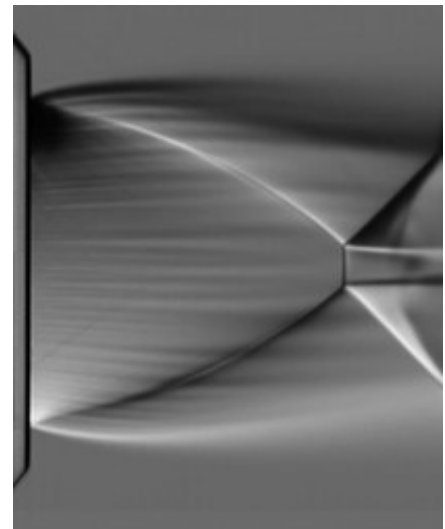
Fig. 1. Schematic drawing of the high-temperature airflow generator.

Makino, Namikiri, and Kimura (2003)

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Free Jet vs. Wind Tunnel

- Optical access is severely limited in a confined (wind tunnel) configuration
- Free jet configuration: a cone of uniform supersonic flow may be maintained at the nozzle exit
 - The nozzle must be sized such that the full rod array fits within this space
 - Leaves ample room for optical diagnostics



A cone of supersonic flow at the exit of a free jet (Edgington Mitchell, 2019)

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Non-Uniform Cooling

Initial trials with airflow indicate that uneven cooling will be a key concern in designing the full setup.

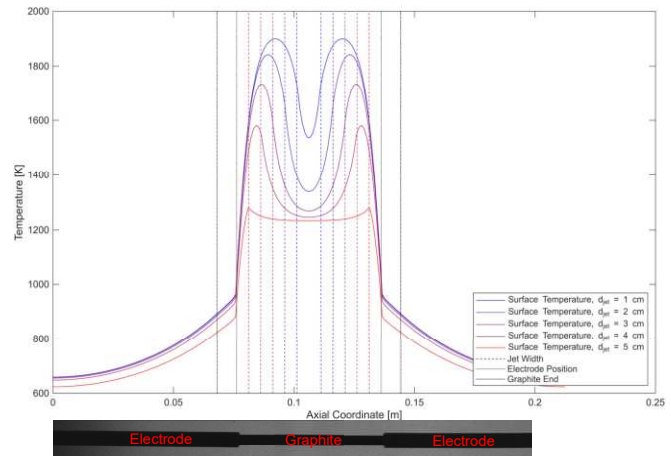
Excessive temperatures outside the zone of interest could:

- 1) Melt the electrodes
- 2) Invalidate gas phase measurements such as LAS, which must be taken along the length of the rod



Emissions imaging from an initial trial with airflow and pencil lead (clay/oil-bound graphite)

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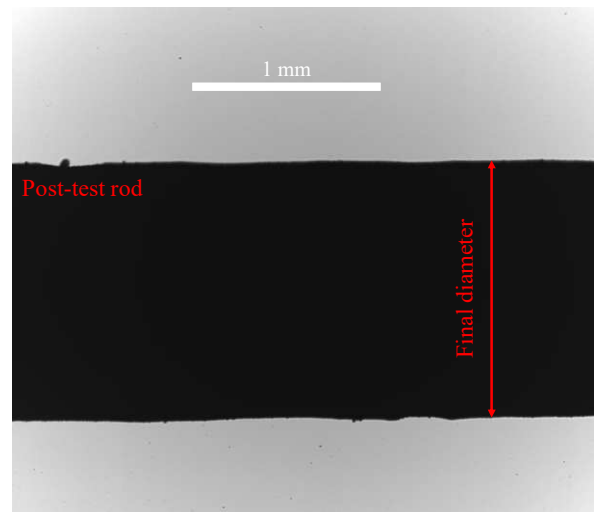
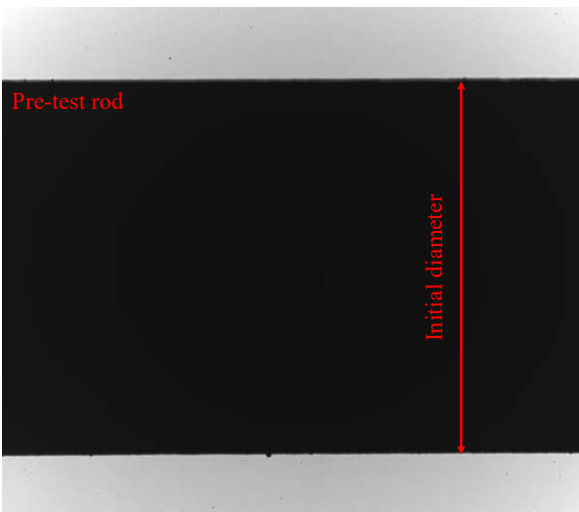


Simulated temperature gradients for various jet diameters, with 100ms^{-1} air-flow and a 1ms^{-1} N_2 co-flow

11

Methods: Combustion Rate Measurements

Similarly to Makino, we estimate combustion rate by tracking radial regression:



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Methods: Combustion Rate Measurements

Two confounding factors for estimating combustion rate from radial regression:

- Thermal expansion
 - The coefficient of thermal expansion is very small for graphite
 - Negligible contribution when the rate of combustion is high
 - No contribution when temperature is steady
- Density change
 - Some combustion will not contribute to a visible change in diameter
 - Internal combustion: diffusion of oxidizer and products through the outer layer
 - Pitting and internal voids

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Methods: Temperature Measurements

2-color pyrometry:

- Point measurement
- Emissivity does not need to be known, as temperature is found using the ratio of spectral intensities. Relies solely on gray-body assumption.

IR camera imaging

- Spatially resolved
- Relies on knowledge of the emissivity
 - For preliminary data we have assumed $\epsilon = 1$



IR imaging for a typical run (16x speed).

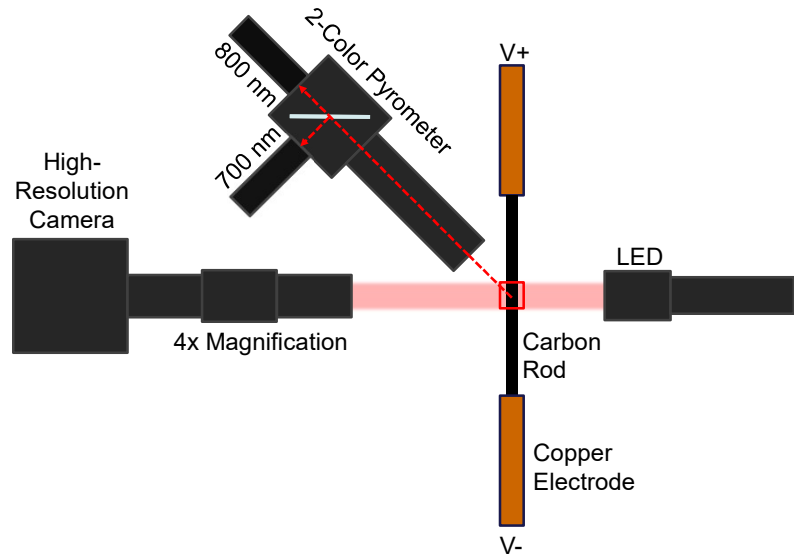
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Pyrometer-Based Setup

Some initial tests were run using 2-color pyrometry:

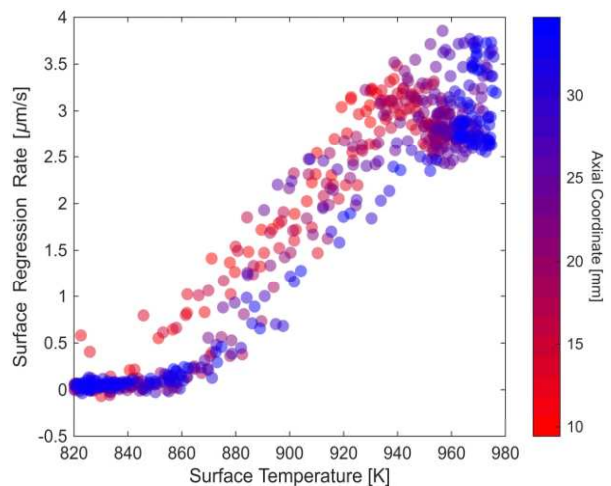
- Aperture and gains are manually adjusted for the expected temperature
- Temperature range was too broad for a single aperture/gain
- Automatic gain adjustment would be required to make this setup practical
- May be revisited in the future

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Preliminary Data: Surface Regression

- Data taken with a wide-angle lens
 - Spatially and temporally resolved diameter and temperature measurements for the full length of the rod
 - Diameter measurements are less precise than they would be with the high-magnification lens
- Measurable combustion begins at $\sim 850\text{K}$
 - Assuming $\epsilon = 1$.
 - Typical literature values: $900 - 1000\text{K}$
- Radial regression, $v_s = -\Delta r / \Delta t$, is sensitive to noise in r .



Radial regression rates for a sub-sample of axial positions along the rod

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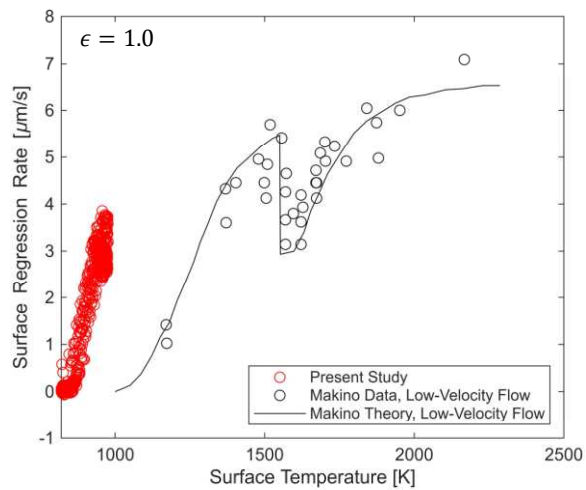
Preliminary Data: Surface Regression

- **Higher temperature experiments will be required for direct comparisons with data from Makino et al.**

- Our chosen grade of graphite seems to ignite at a lower surface temperature

- **Potential sources of deviation:**

- We use a denser, less porous graphite
- Natural convection is uncontrolled and not quantified
- Assumed emissivity value



Measured radial regression rates compared with data from Makino et al.

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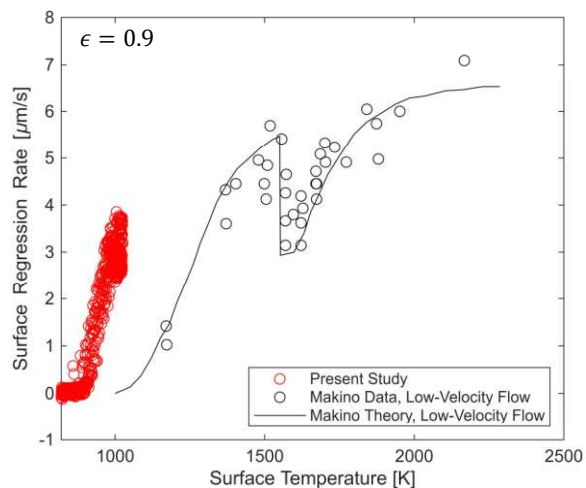
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- **Potential sources of deviation:**

- We use a denser, less porous graphite
- Natural convection is uncontrolled and not quantified
- Assumed emissivity value
 - Adjusting to $\epsilon = 0.9$ does not account for the full discrepancy.



Measured radial regression rates compared with data from Makino et al.

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Planned Developments

- New electrodes are currently being manufactured
 - Improved electrical contact and oxidation resistance
 - Locate the rod firmly in place while accommodating thermal expansion
 - Provide lengthwise optical access for LAS
- A new subsonic heated air system is ready for implementation
- LabVIEW integration for temperature control
- N_2 co-flow system for better temperature regulation

Long term:

- Supersonic air system for use with the heated air plant at Zucrow.
- Multi-rod electrode assembly

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Expected Timeline

April:

- Full single-rod test campaign, including initial experiments with heated sub-sonic air

May-June:

- Development of multi-rod setup and supersonic air system

July-August:

- First experiments with multi-rod arrays

Timing to be determined:

- Implementation of LAS in the single rod setup, and subsequently the multi-rod setup

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Questions?

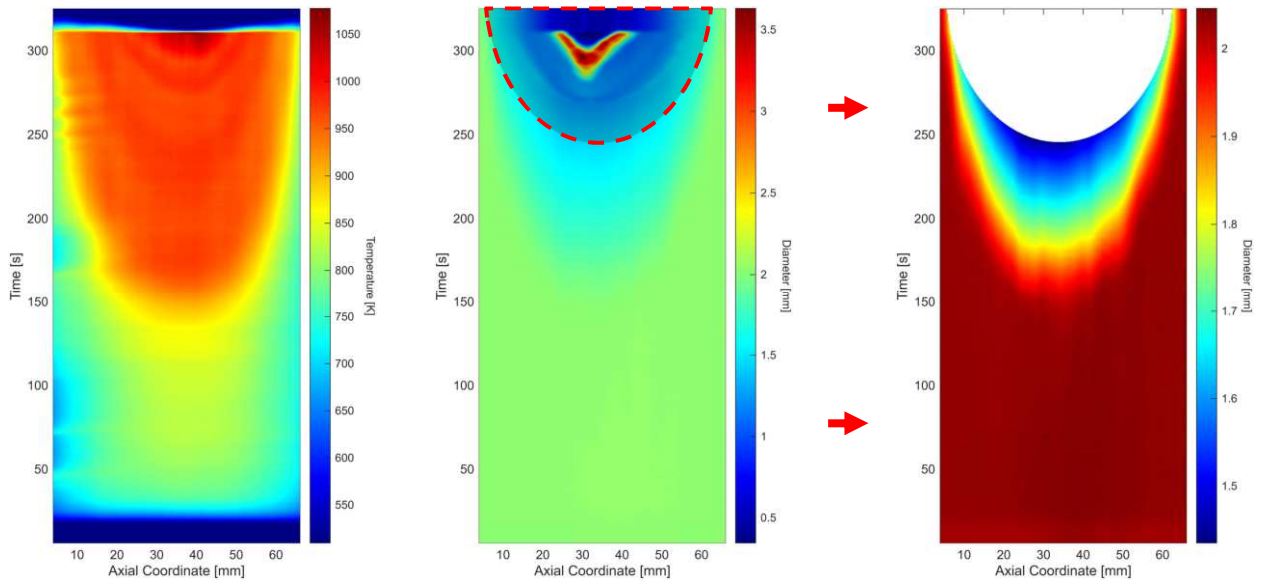


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Additional Slides

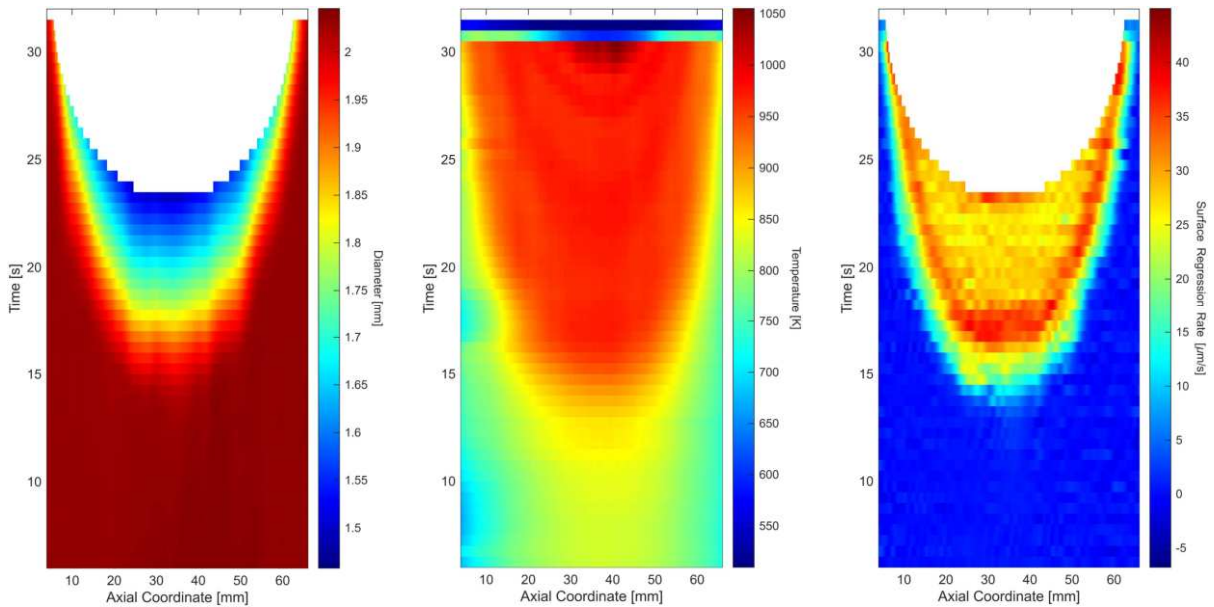


Preliminary Data: Raw Data



23

Preliminary Data: Data Processing



24

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CM³C: Center for Multiscale Modeling of Multiphase Combustion

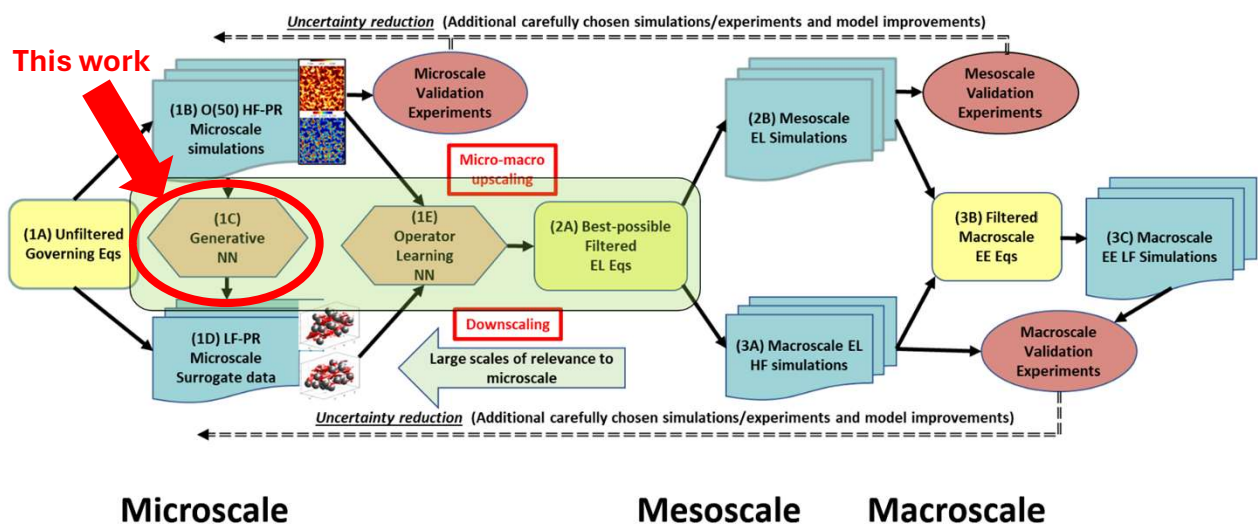
T13: Physics Informed NN

Juan Diaz

Department of Electrical and Computer Engineering



Project Overview

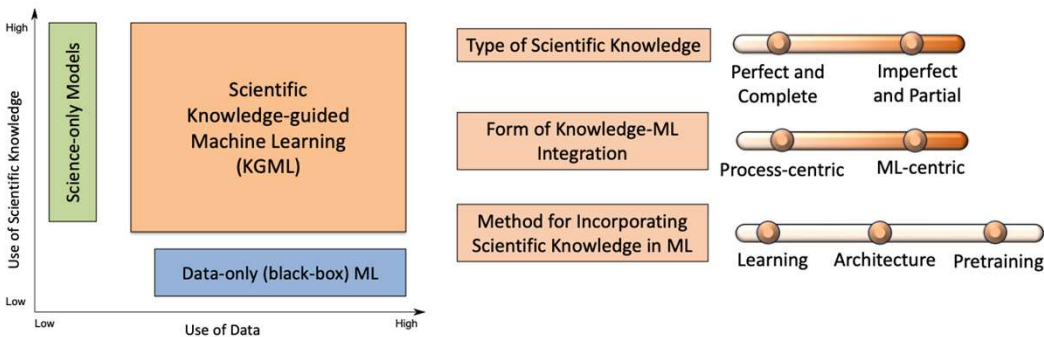


Comparing Networks

- Physics Informed Neural Networks (PINNs) map inputs to solution.
- Neural Operators map functions to functions.

Property \ Model	Traditional NNs	PINNs	Neural Operators
Data Driven?	✓	✗	✓
Discretization Invariance?	✗	✗	✓
Is the output a function?	✗	✓	✓
Can query the output at any point?	✗	✓	✓
Can take the input at any point?	✗	✓	✓
Universal Approximation?	✗	✓	✓

Knowledge Informed Machine Learning (KGML)



Knowledge-guided Machine Learning: Current Trends and Future Prospects
Anuj Karpatne, Xiaowei Jia, and Vipin Kumar
arXiv preprint arXiv:2403.15989, 2024

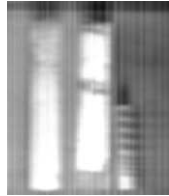
Knowledge Informed Machine Learning



Field collected soil cores

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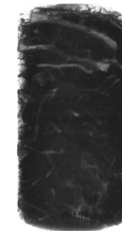
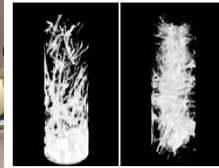
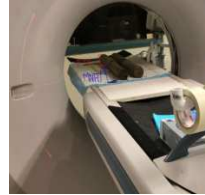
Scanned with Back-scatter X-ray System



Soil cores being washed and root measurements collected for ground-truth comparison



Scanned with X-ray CT in Columbia, MO (and Texas)



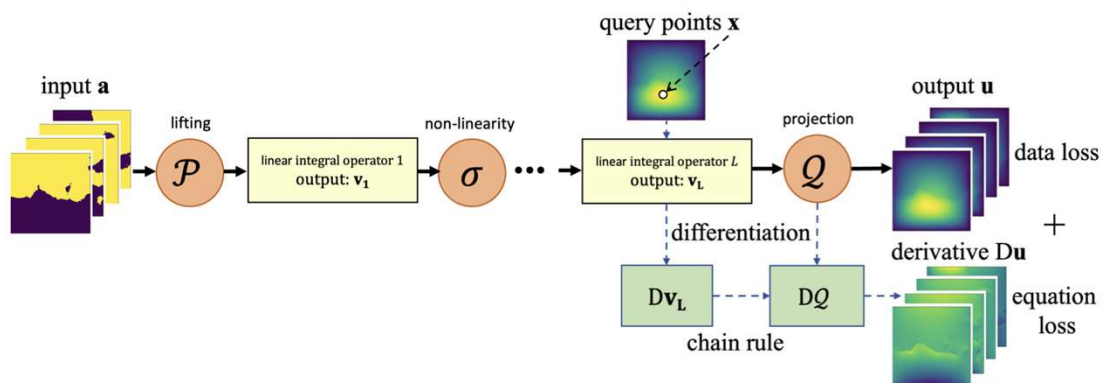
Real Soil Core



Modeled Core

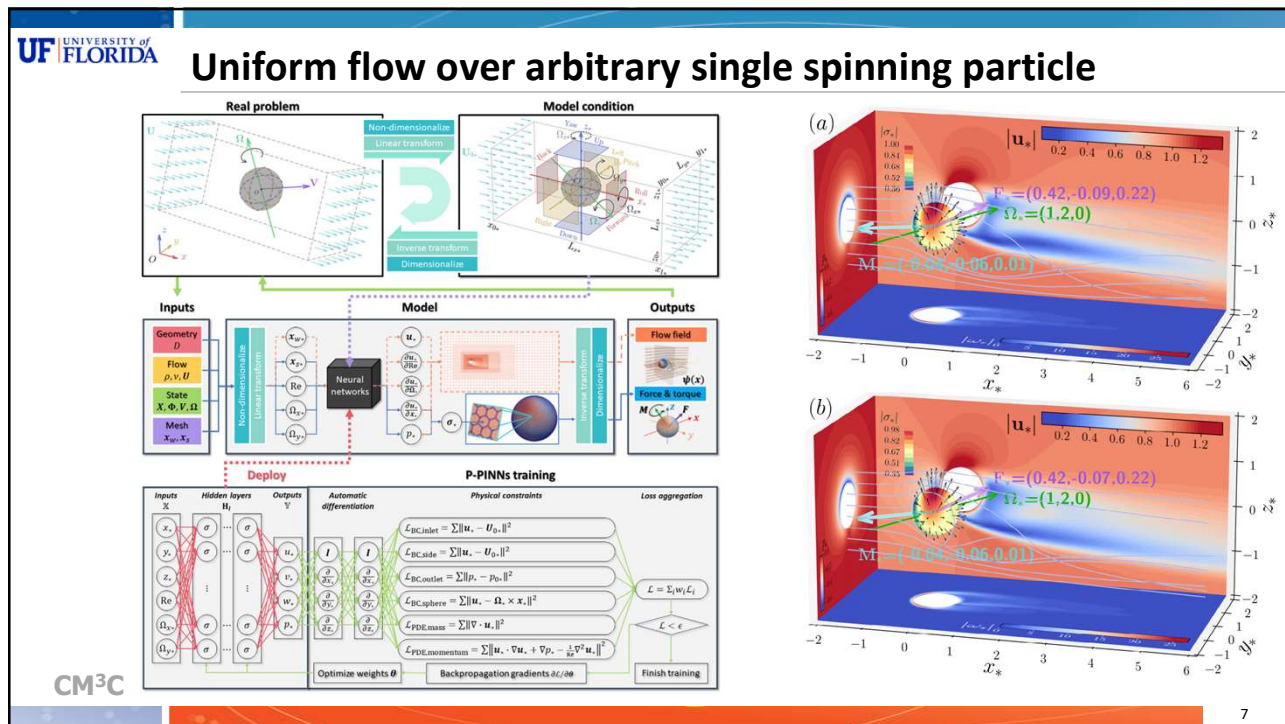
5

Hybrid Approach

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Physics-Informed Neural Operator for Learning Partial Differential Equations
Zongyi Li*, Hongkai Zheng*, et. all

6



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Questions?

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Georgia Tech.

UF UNIVERSITY of FLORIDA

P PURDUE UNIVERSITY.

NASA

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CM³C Center for Multiscale Modeling of Multiphase Combustion

T13: Uncertainty Quantification in Deep Learning

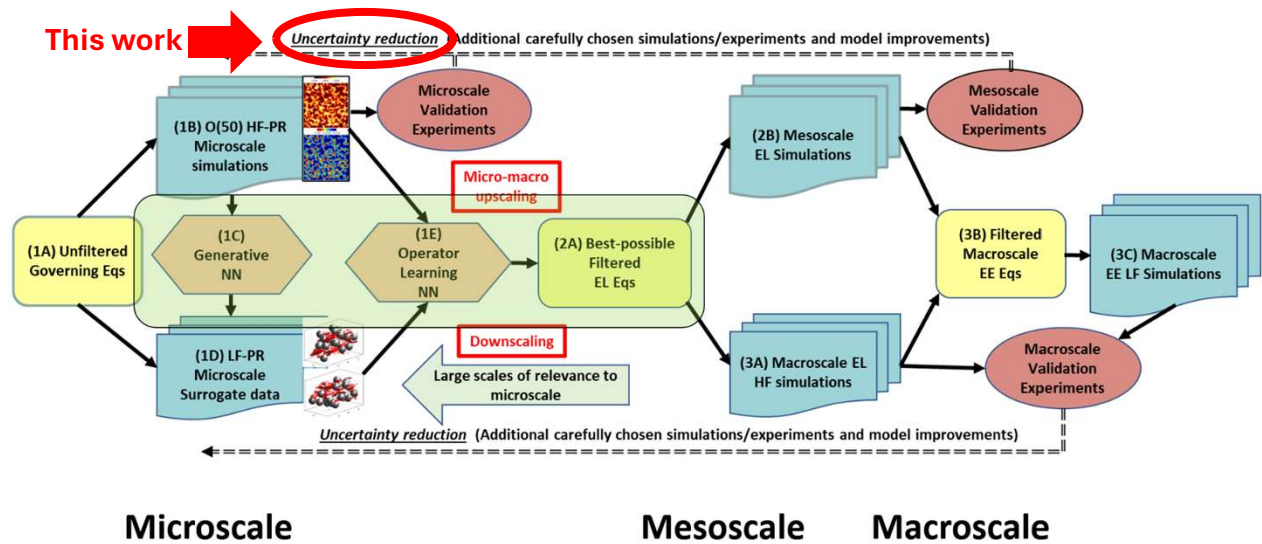
Isaly Tappan

Department of Electrical and Computer Engineering

April 2, 2026



Project Overview



Outline

1. Motivation: Sources of uncertainty in the deep learning pipeline
2. Uncertainty quantification methods for deep learning
 1. Example results highlighting the strengths and weaknesses of different methods
3. Uncertainty quantification results from generative machine learning data
4. Open questions

Uncertainty in Deep Learning (DL)

Deep Learning Pipeline:

1. Data and knowledge acquisition
2. Model specification and hyperparameter selection
3. Model training
4. Inference (deployment)

Goal: Fit the flexible deep learning model to the underlying true function using data and knowledge.

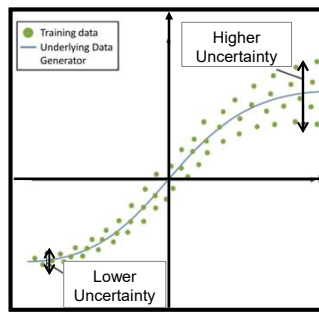
Uncertainty in Deep Learning (DL)

Deep Learning Pipeline:

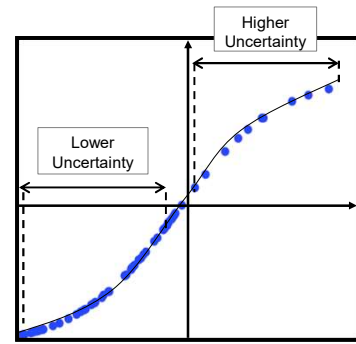
1. Data and knowledge acquisition
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3. Model training
4. Inference (deployment)

Uncertainty Sources:

- Measurement errors (noise, device error)
- Limited information (e.g, distribution, sampling frequency)



Heteroskedasticity



Sampling Frequency

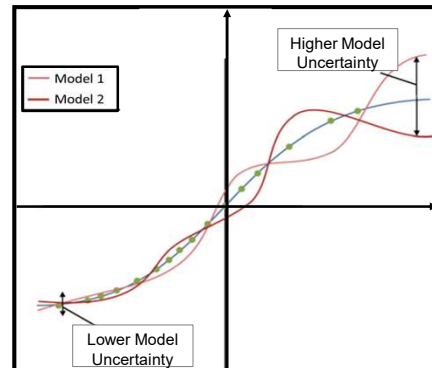
Uncertainty in Deep Learning (DL)

Deep Learning Pipeline:

1. Data and knowledge acquisition
2. Model specification and hyperparameter selection
3. Model training
4. Inference (deployment)

Uncertainty Sources:

- Model type
- Model structure (type of layers, number of layer, nodes, etc.)
- Model complexity (number of parameters)



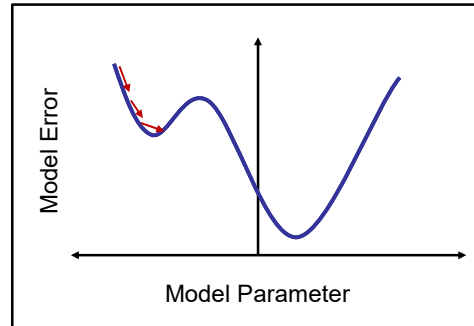
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Uncertainty Sources:

- Number of epochs
- Batch size
- Optimizer
- Learning rate



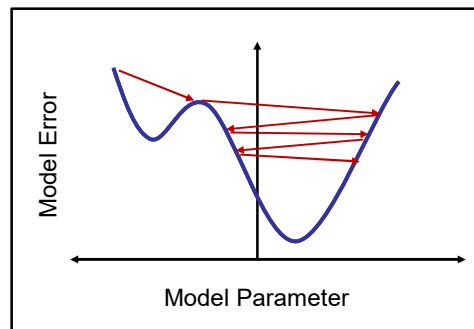
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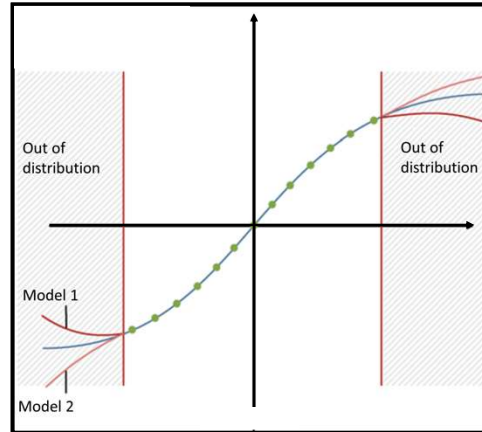
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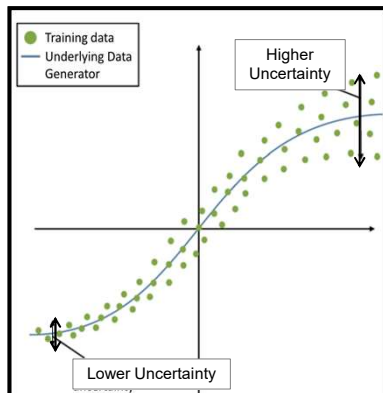
Distribution shift, real-world variability, performance metrics



Figures adapted from: Gawlikowski, J., Tassi, C.R.N., Ali, M. et al. A survey of uncertainty in deep neural networks. *Artif Intell Rev* 56 (Suppl 1), 1513–1589 (2023)

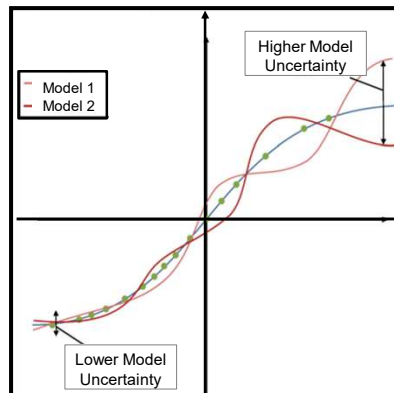
Uncertainty in Deep Learning (DL)

Measurement errors (noise, device error), sampling frequency



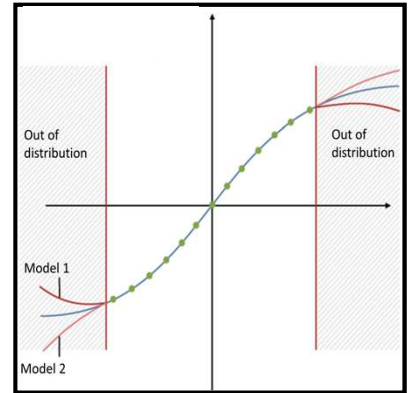
Data Acquisition

Model type, complexity, structure, training



Model Development

Distribution shift and real-world variability after deployment



Model Deployment

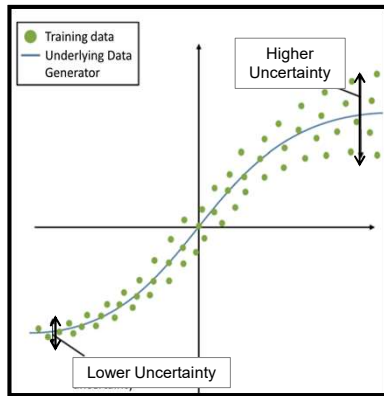
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Uncertainty in Deep Learning (DL)

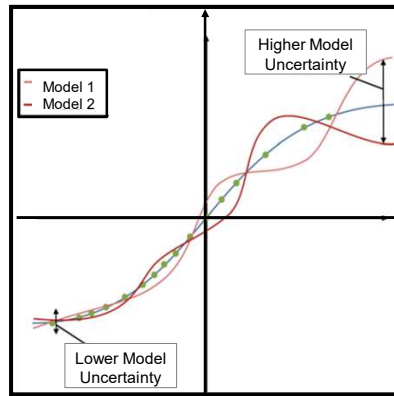
Measurement errors (noise, device error), sampling frequency

Model type, complexity, structure, training

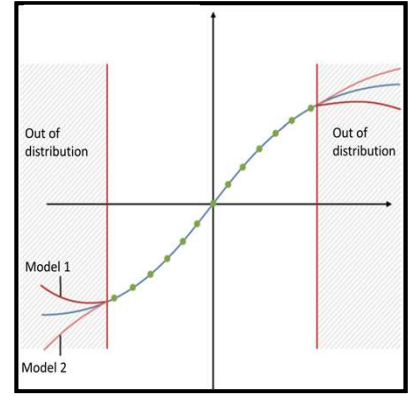
Distribution shift and real-world variability after deployment



Data Acquisition



Model Development



Model Deployment

Aleatoric: irreducible noise in the data

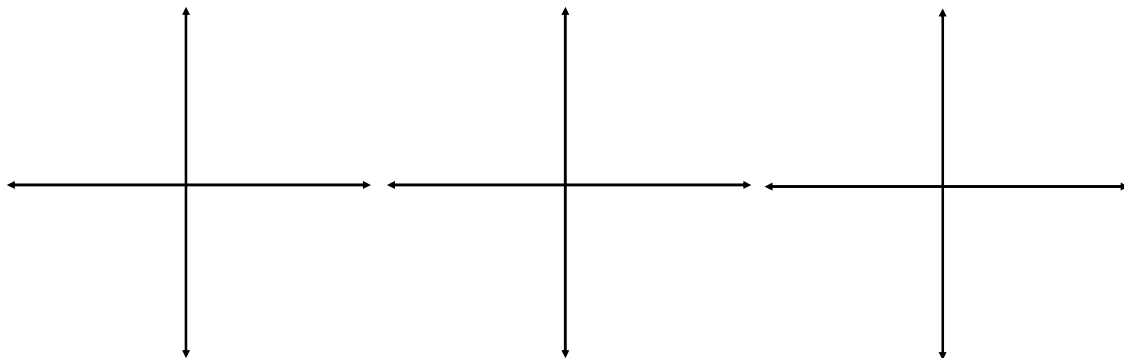
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Epistemic: reducible error in the model (can theoretically be reduced with more information)

Figures adapted from: Gawlikowski, J., Tassi, C.R.N., Ali, M. et al. A survey of uncertainty in deep neural networks. *Artif Intell Rev* 56 (Suppl 1), 1513–1589 (2023)

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Uncertainty Propagation



Figures adapted from: Gawlikowski, J., Tassi, C.R.N., Ali, M. et al. A survey of uncertainty in deep neural networks. *Artif Intell Rev* 56 (Suppl 1), 1513–1589 (2023)

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Uncertainty in Deep Learning (DL)

Deep Learning Pipeline:

1. Data acquisition
2. Model specification and hyperparameter selection
3. Model training
4. Inference (deployment)
 - a) Uncertainty quantification framework

Recap of Uncertainty Sources:

- Measurement errors (noise, device error)
- Limited information (e.g, distribution, sampling frequency)
- Model type, complexity, structure
- Model training
- Distribution shift and real-world variability after deployment

Aleatoric: irreducible noise in the data

Epistemic: reducible error in the model (can theoretically be reduced with more information)

Uncertainty in Deep Learning (DL)

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Uncertainty in Deep Learning (DL)

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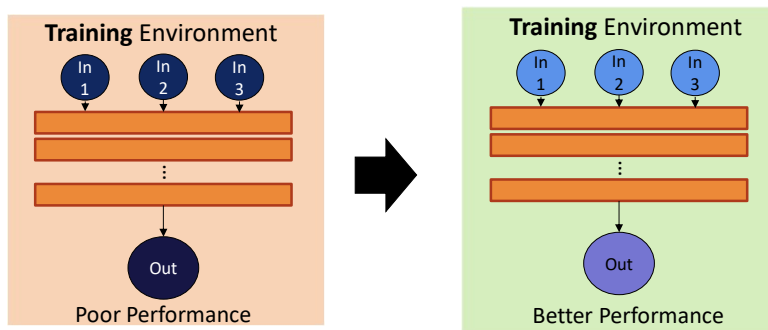
Plan: Transitions etc with pictures

Uncertainty Sources:

- Training data
 - Data distribution
 - Measurement errors (noise, limited information, e.g. resolution)
- Processing the training data
- Model architecture (number of parameters and model complexity)
- Model training
- Inference: real-world variability (distribution shift and measurement error)

Challenges in DL Upon Deployment

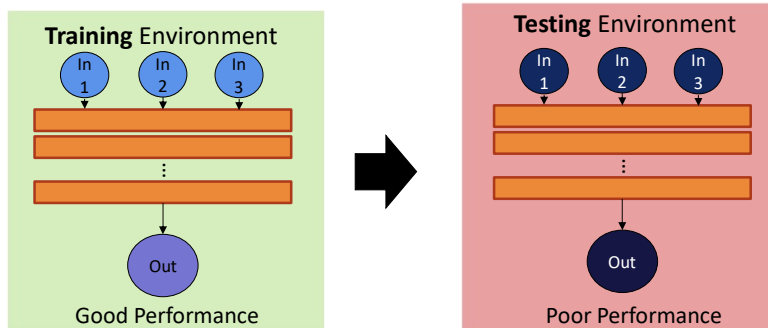
Model uncertainty can theoretically be alleviated via data pre-processing, improving model architecture and training, or gathering more information.



Challenges in DL Upon Deployment

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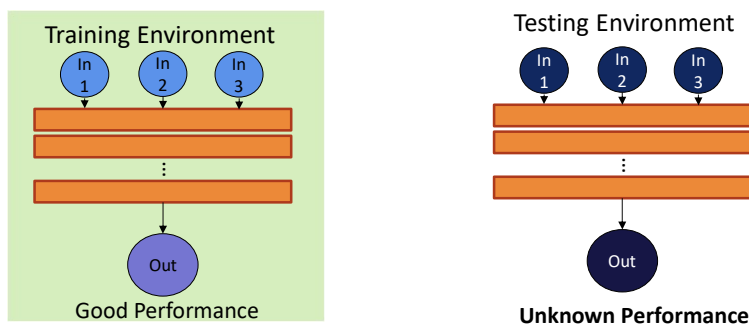
Great performance during validation does not guarantee great performance upon deployment.



Challenges in DL Upon Deployment

Model uncertainty can theoretically be alleviated via data pre-processing, improving model architecture and training, or gathering more information.

Great performance during validation does not guarantee great performance upon deployment.



Standard DL models lack the inherent capability to communicate uncertainty.

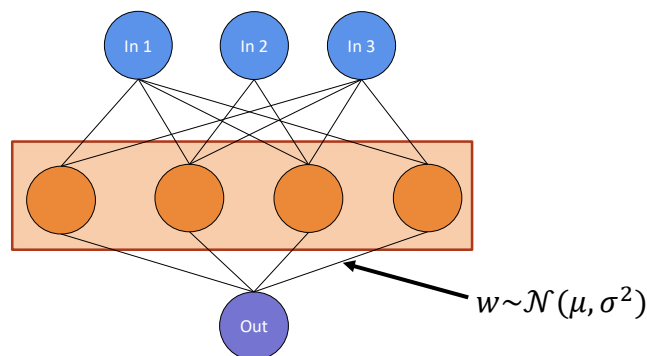
Uncertainty Quantification in DL

Goal: Include an uncertainty estimate with each prediction.

- **Bayesian:** Learn a posterior distribution over the model parameters rather than a maximum likelihood estimate
- **Ensembles:** Train multiple models with different initializations
- **Deterministic Methods:** A predicted uncertainty estimate is added to the list of outputs

Types of Uncertainty Quantification for DL

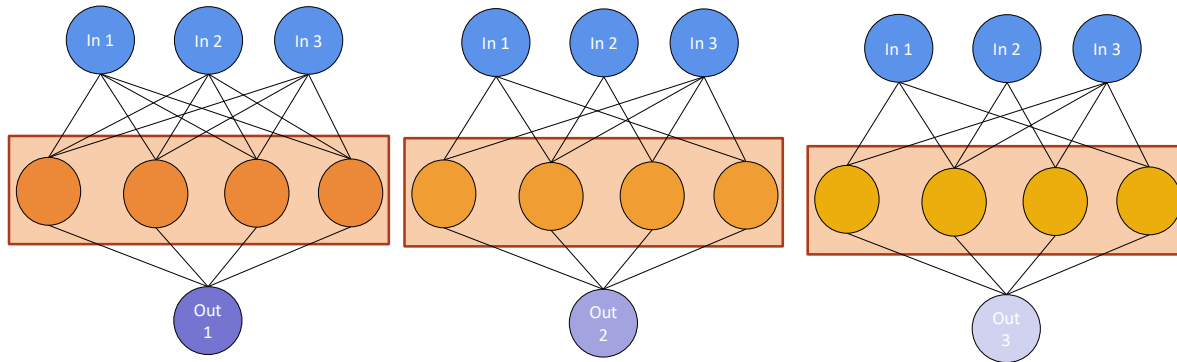
Bayesian: Learn a posterior distribution over the model parameters rather than a maximum likelihood estimate



Uncertainty Estimate: Variance of the outputs across multiple stochastic runs.

Types of Uncertainty Quantification for DL

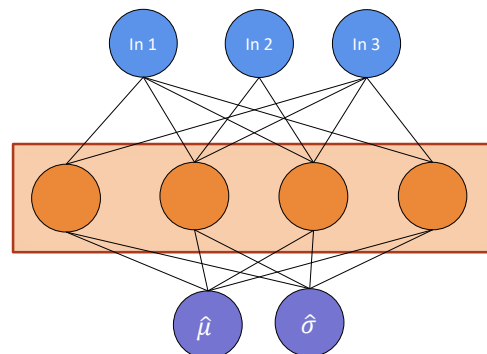
Ensembles: Train multiple models with different initializations



Uncertainty Estimate: Variance of the outputs across the ensemble

Types of Uncertainty Quantification for DL

Deterministic Methods: The model is taught to predict a density rather than a point estimate



Uncertainty Estimate: The predicted standard deviation

Toy Problem: Half Moons

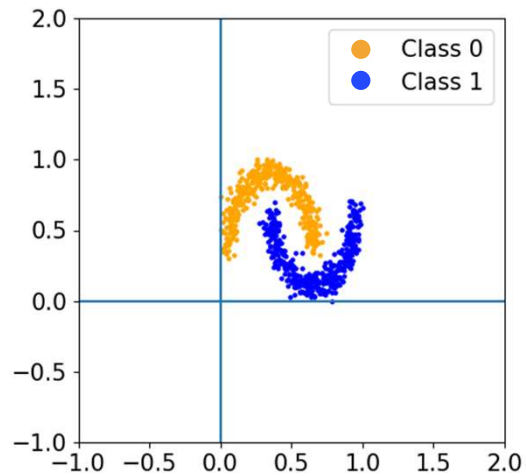
Data (binary classification):

- $\mathbf{x} \in \mathbb{R}^2$
- $y \in \{0,1\}$

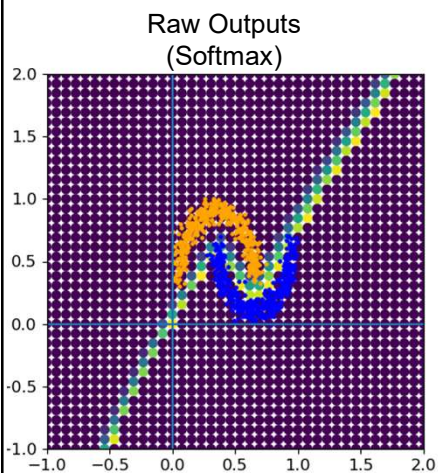
Architecture:

Multi-Layer Perceptron (feed-forward neural network)

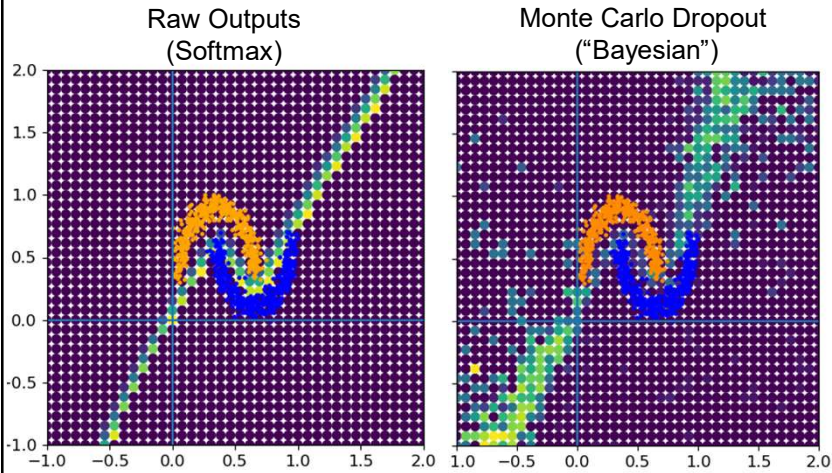
- 3 layers



Toy Problem: UQ Methods Applied to Half Moons

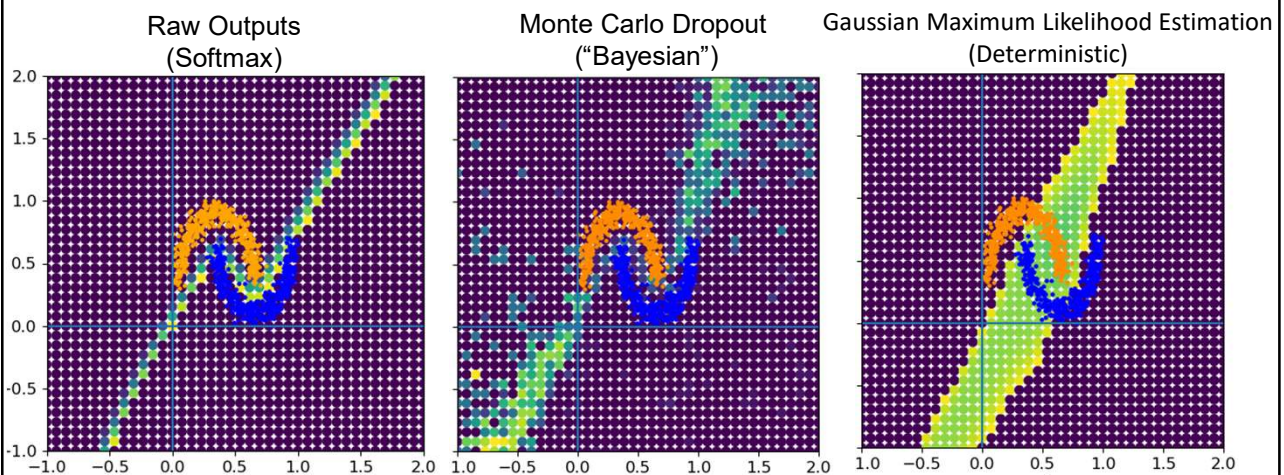


Toy Problem: UQ Methods Applied to Half Moons

CM³C

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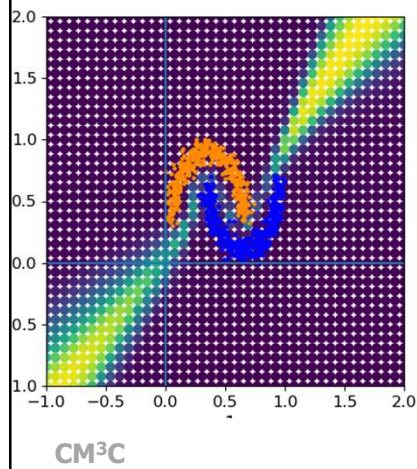
Toy Problem: UQ Methods Applied to Half Moons

CM³C

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Toy Problem: UQ Methods Applied to Half Moons

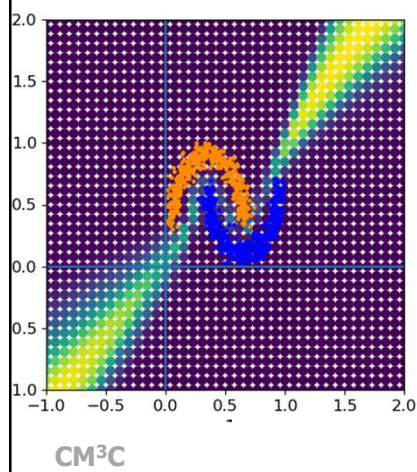
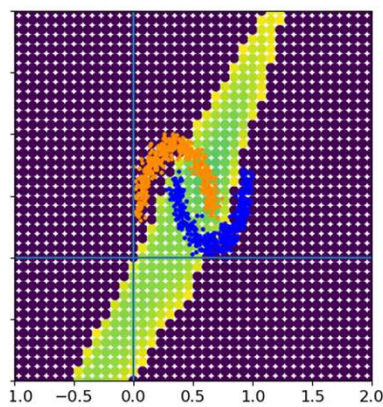
Ensemble



27

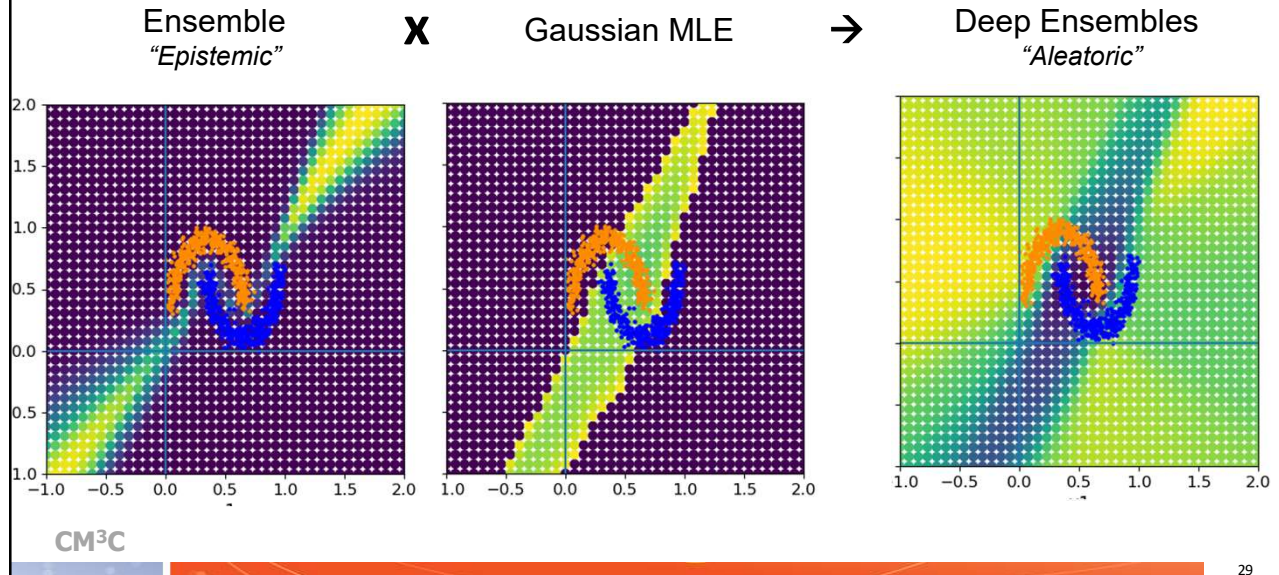
Toy Problem: UQ Methods Applied to Half Moons

Ensemble

Gaussian MLE
(Deterministic)

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Toy Problem: UQ Methods Applied to Half Moons

Application of Ensembling UQ to Generative Modeling: Data Review

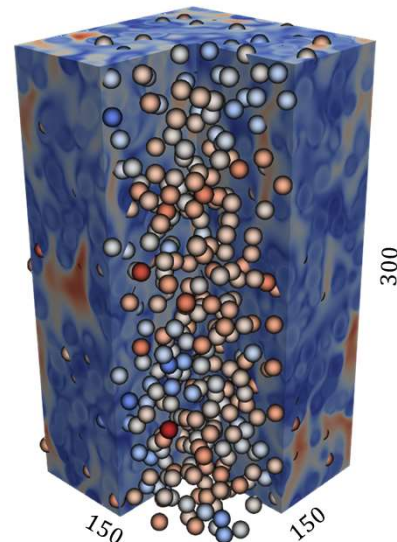
Review from Dr. Lindbeck

Training:

- One simulation (one Re, PDR, and average φ)
- 64 particles
- Data: $\mathbf{t}_0$
 - Samples only from within $r = 1d, 2d, \text{ or } 3d$ of any particle centers

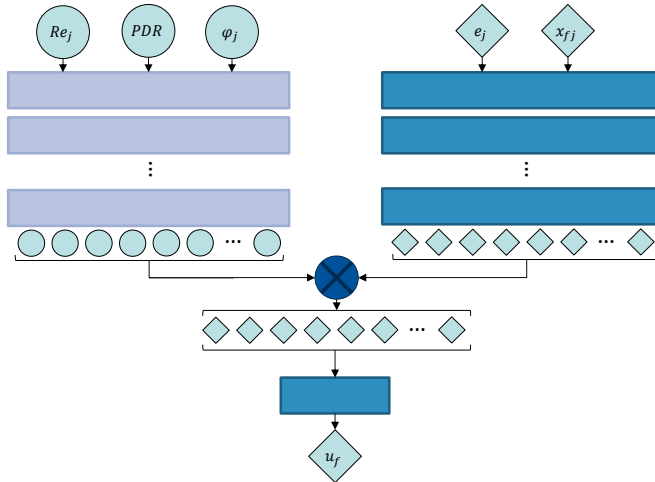
Testing

- Data: $\mathbf{t}_2$
 - Samples are across **the entire simulation**

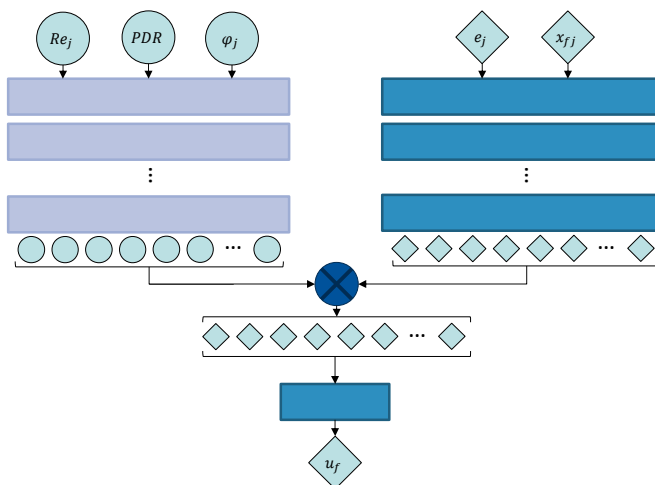


Application of Ensembling UQ to Generative Modeling: UQ Overview*Review from Dr. Lindbeck*

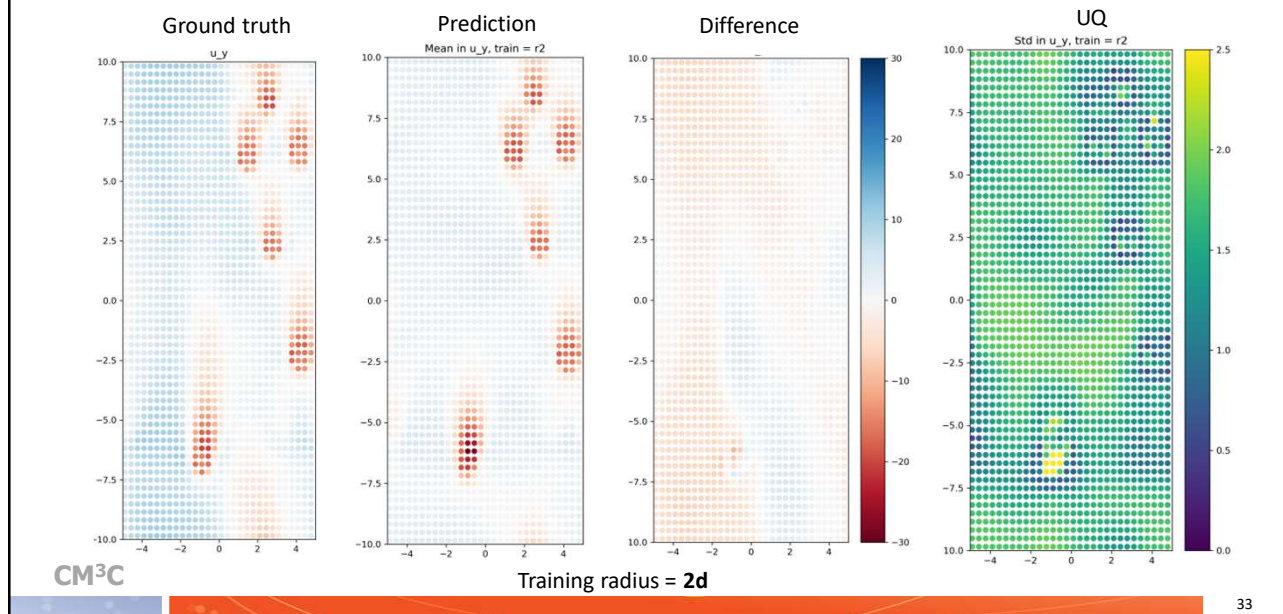
- Training data selection radius
 - 2d, 3d
- Number of layers
 - 3, 5, 7, 9, 11
- Layer size
 - 25, 50, 100, 150, 200, 250

Application of Ensembling UQ to Generative Modeling: UQ Overview*Review from Dr. Lindbeck*

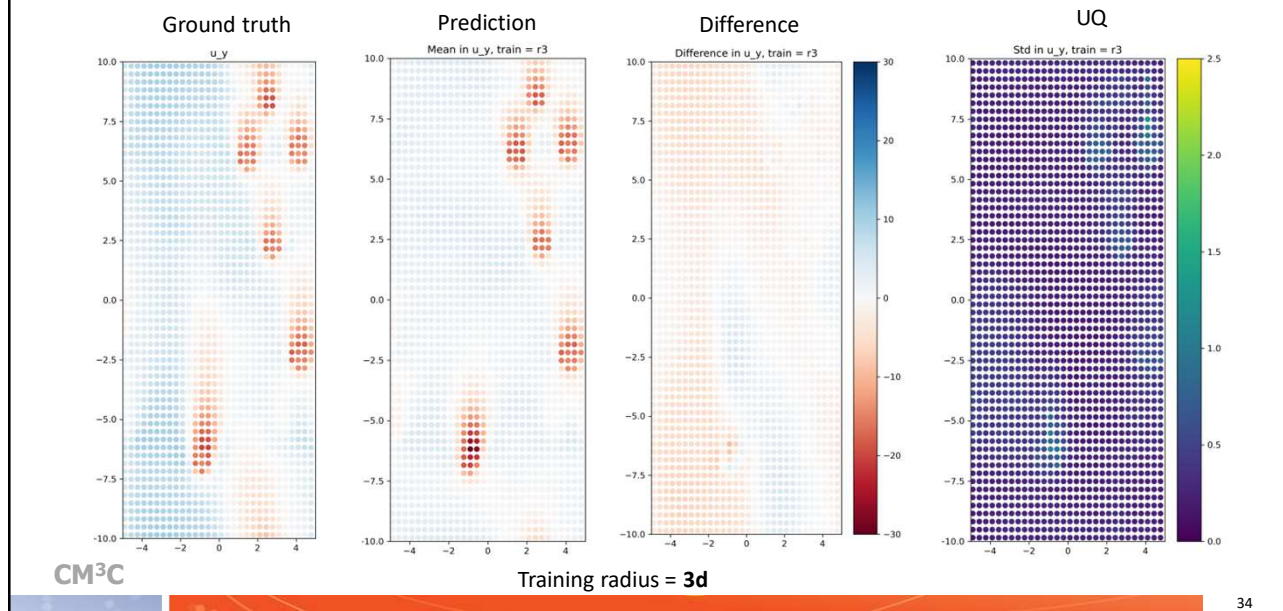
- Training data selection radius
 - 2d, 3d
- Number of layers
 - 3, 5, 7, 9, 11
- Layer size
 - 25, 50, 100, 150, 200, 250
- **Two** ensembles
 - 30 members in each ensemble (num_layers*layer_size)
 - Ensemble_2d (trained with 2d)
 - Ensemble_3d (trained with 3d)



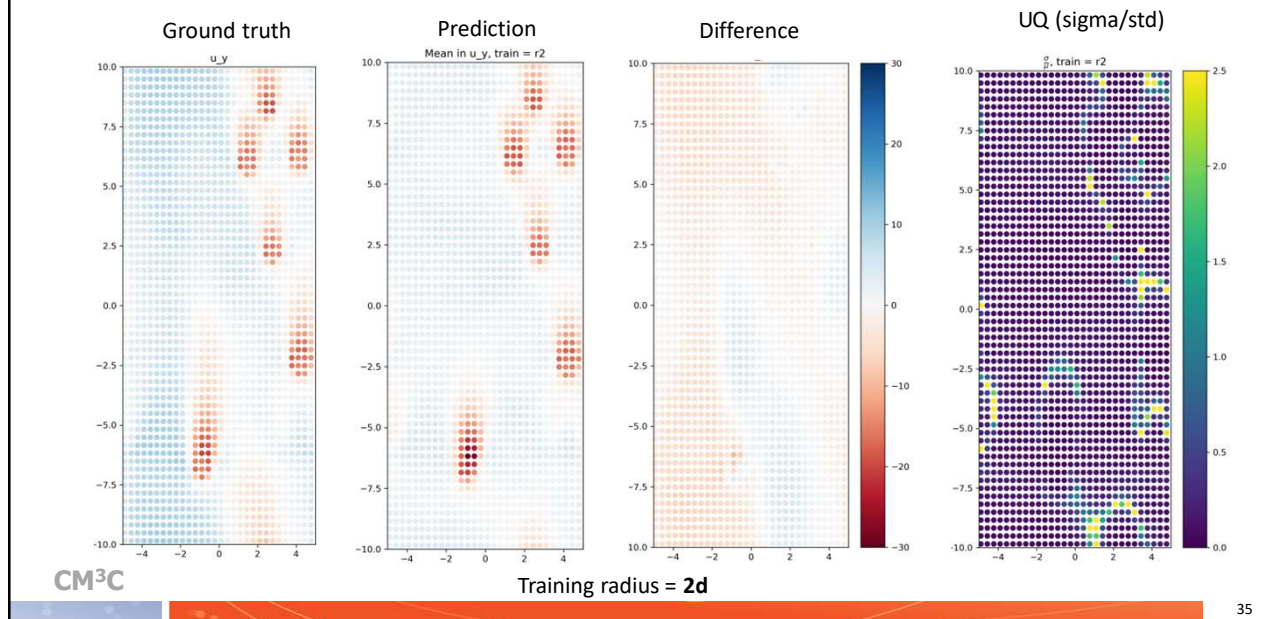
Application of Ensembling UQ to Generative Modeling: Results



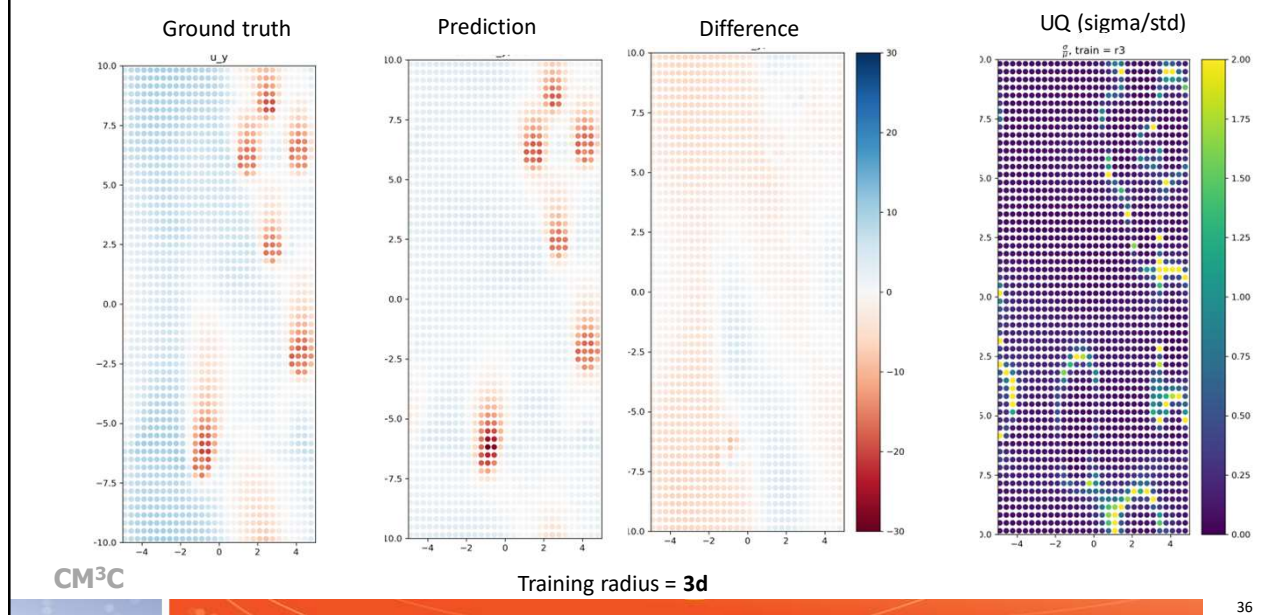
Application of Ensembling UQ to Generative Modeling: Results



Application of Ensembling UQ to Generative Modeling: Results



Application of Ensembling UQ to Generative Modeling: Results



Remaining Questions and Future Work

- Identifying meaningful and interpretable sources of uncertainty for:
 - development and training
 - deployment
- Identifying the impact of uncertain physical parameters and theories on knowledge-guided and physics-informed deep learning models

Questions?

CM³C

CM³C: Center for Multiscale Modeling of Multiphase Combustion

T14: Validation, Uncertainty Quantification, and Uncertainty Budget

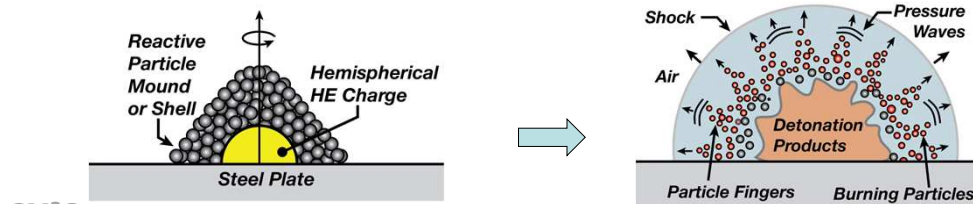
Co-Principal Investigators:
Nam-Ho Kim

Department of Mechanical and Aerospace Engineering



QoIs and Key Sources of Uncertainty

- | | |
|----------------------|--|
| Basic QoIs | <ul style="list-style-type: none"> ■ QoI1: Blast wave locations as a function of time ■ QoI2: Time-dependent pressure and impulse at selected radial distances ■ QoI3: Volume fraction of reactive particulate at selected radial locations |
| Higher-order metrics | <ul style="list-style-type: none"> ■ QoI4: Temperature & heat release rate ■ QoI5: Numbers & amplitude of fingering instabilities ■ QoI6: Time-dependent PDFs of pressure and particulate volume fraction |

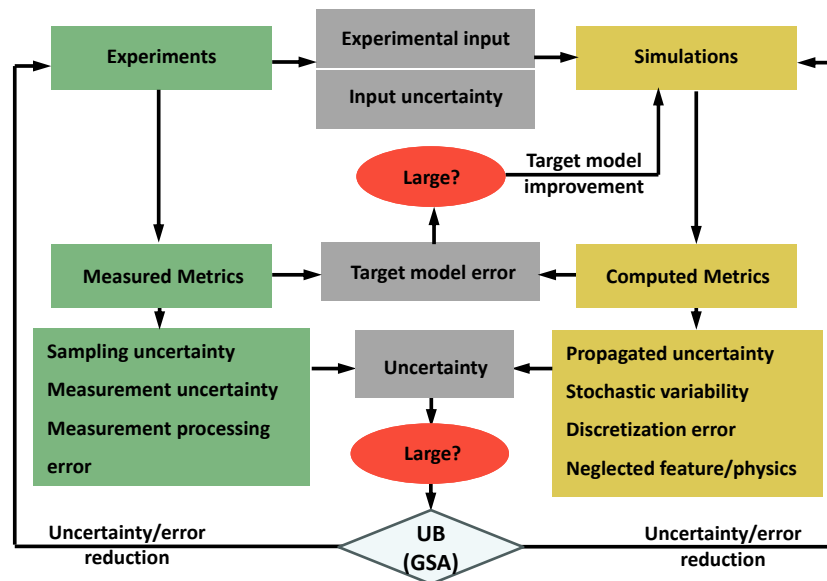


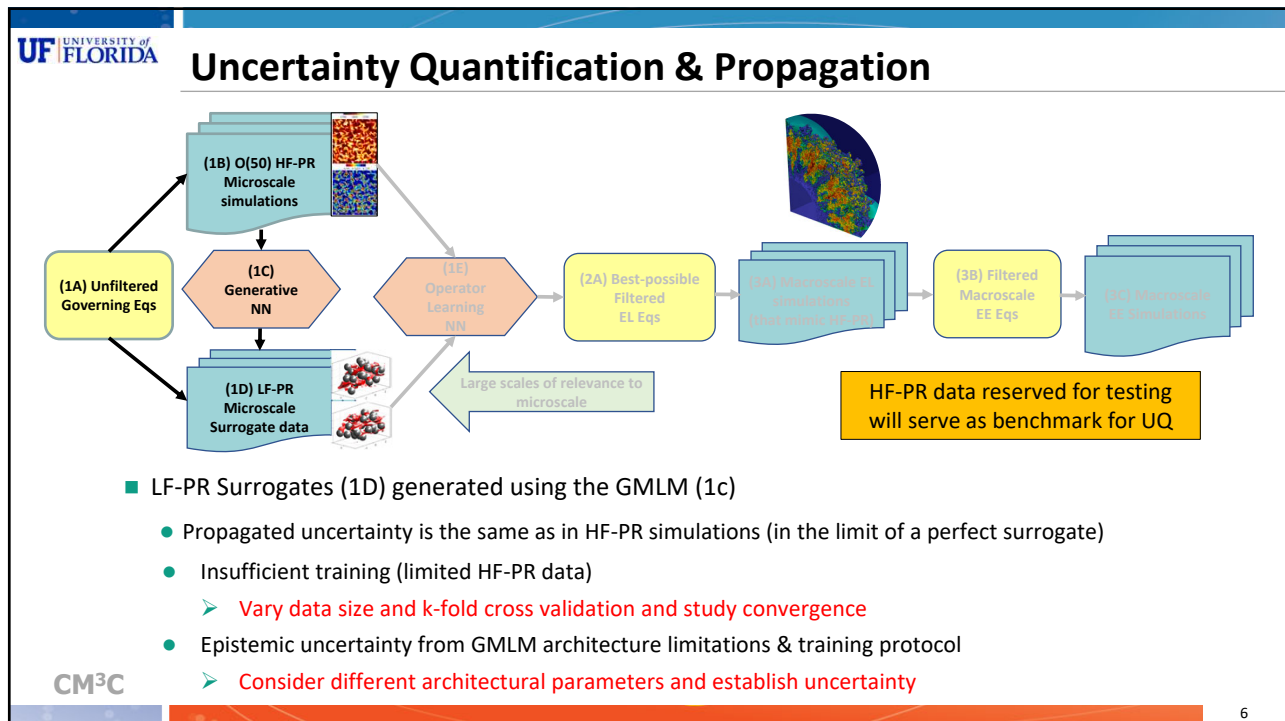
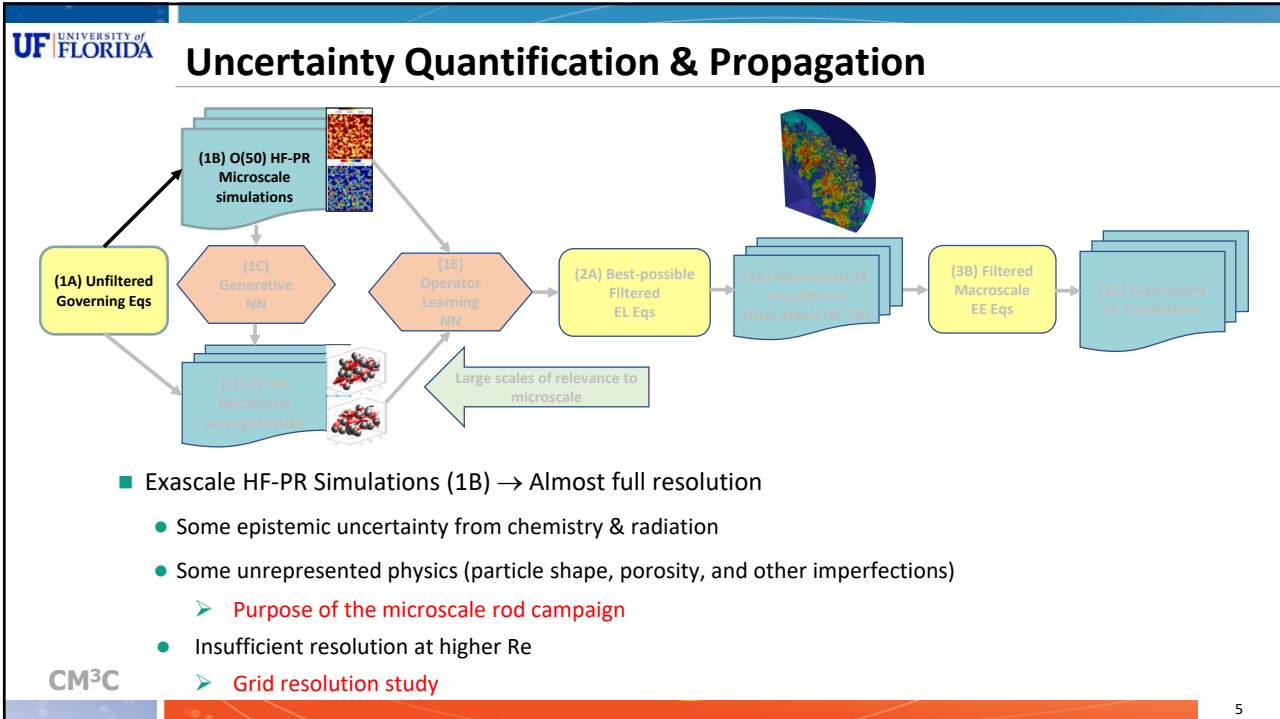
CM³C

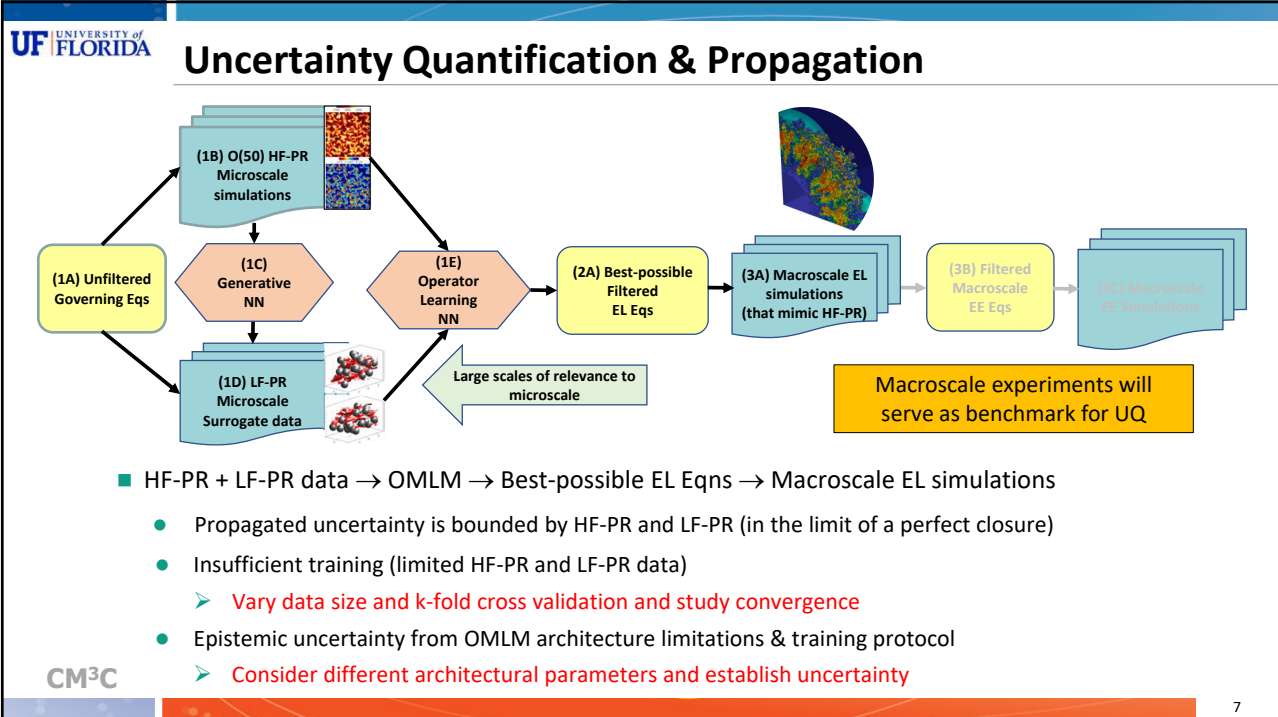
Dominant Sources of Uncertainty

- Stochastic uncertainties (aleatory) and errors
 - Initial random distribution of particle sizes, locations, and density
 - Stochastic detonation characteristics (PETN)
 - Stochastic turbulent mixing
- Experimental measurement uncertainties and errors
 - Shock structures and strength, particle concentrations, pressure history, temperature and species concentration
 - Measurement processing uncertainty and error
- Modeling & numerical uncertainties and errors
 - Spatially filtered gas-phase governing equations
 - Reduced-order surface & gas-phase chemical reactions (C and Mg)
 - GMLM surrogate errors and OMLM uncertainty
 - Radiation and compaction model uncertainty (minimally contributing)

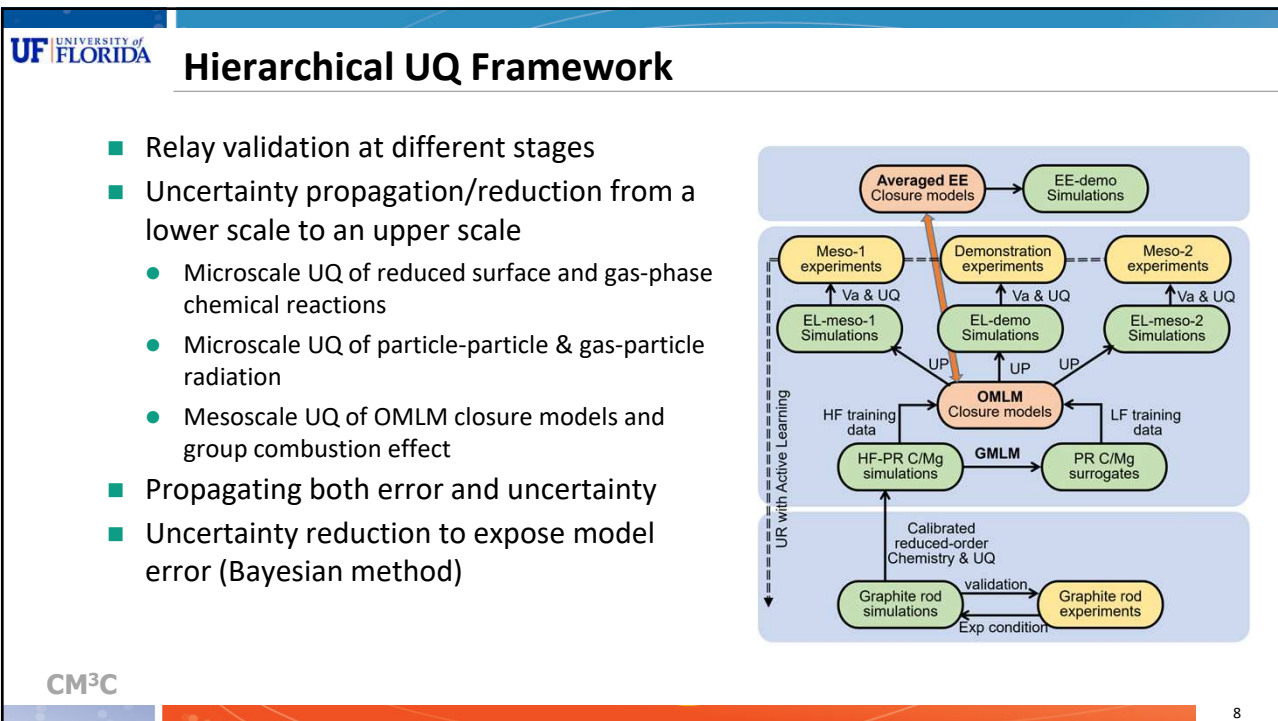
Uncertainty Budget for Uncertainty Reduction (UR)







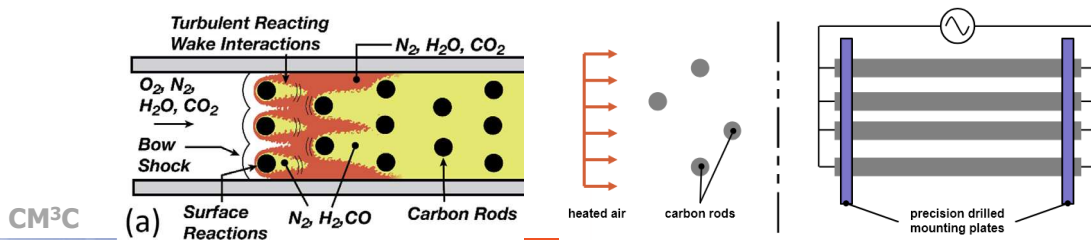
7



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Microscale UQ

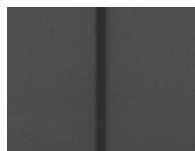
- Experiment: Combustion of carbon rod(s) within a high-speed (up to Ma 1.5) airflow (temp. 500°K)
 - Measurements: surface profiles and overall reaction rates
- Simulation: HF-PR simulation
 - unfiltered gas and particulate phase governing equations in fixed domain
 - heterogeneous surface and gas-phase reactions (optimal reduced-order reactions)
- Objective: UQ of reduced surface and gas-phase chemical reactions
- Initially with a single rod, later multi-rods (**guided by simulations**)



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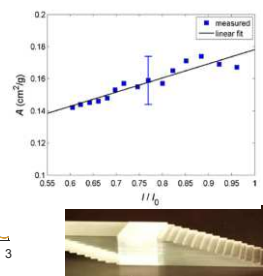
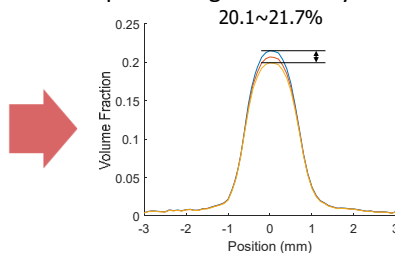
Measurement Processing Uncertainty

- Most experimental UQ focus on estimating variability, but ...
- Measurement processing uncertainty
 - Major source of epistemic uncertainty
 - Commonly happened when QoI cannot be measured directly
 - Rigorous UQ and UR are required to have a meaningful validation
- One-rod HF-PR simulation to UQ in measurements
 - Not only measurement variability but also measurement processing uncertainty

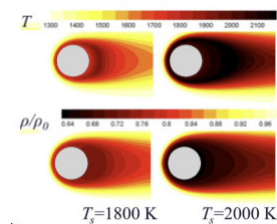


X-ray image

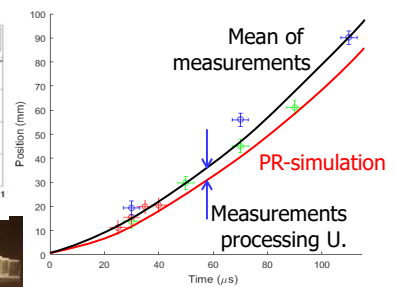
J. Wagner et al. Exp. Fluids, 2012(Sandia)



Calibration exp.



Jiang et al. (2022) Int. Commu. Heat Mass

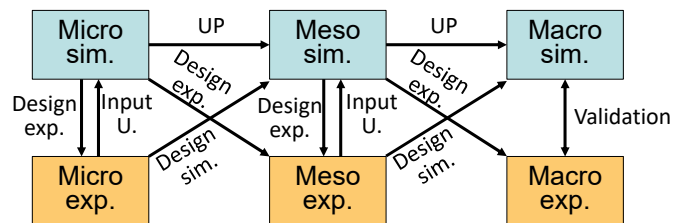


Hughes et al. J. VVUQ (2019)

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Relay Validation: UR cross Simulations and Experiments

- Tradition: Aleatory uncertainty in experiments ↔ Epistemic uncertainty in simulation
- Uncertainty propagation & reduction mostly in simulations
- Our PSAAP-2 experience shows experiments also have major epistemic uncertainty
 - Measurement processing uncertainty
 - Measurement design limitations (for companion simulations)
- Relay validation:
 - Design/reduce epistemic uncertainty in experiments using simulation
 - Where to place the multiple rods to maximize group combustion effect
 - Where to place the measurement probes for maximum sensitivity
 - Lower-scale simulation/experiments are used to quantify input uncertainty in upper-scale

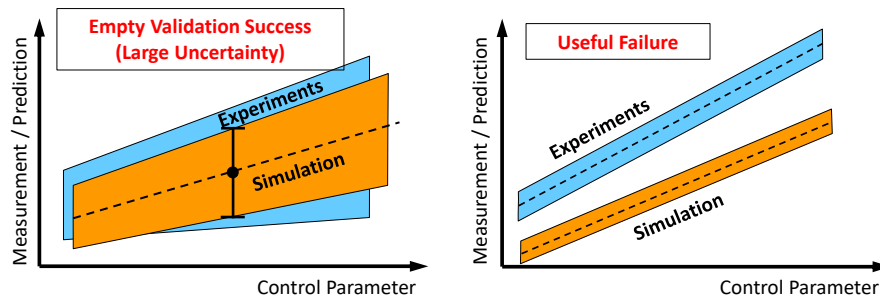


Macroscale UQ

- Experiments: Detonation of HE charge surrounded by reactive particles
 - Scaled versions of proven multiphase blast configuration
 - Measurements: schlieren imaging, pressure probes, tomographic inversion of visible and X-ray extinction, IR pyrometry, LAS within blast-hardened probe
- Simulations
 - EL hero/production simulations with best-possible EL closure models
 - low-fidelity EE simulations with best-possible EE closure models
- Goals:
 - Validation and UQ of blast wave location (QoI1), pressure history (QoI2), and volume fraction (QoI3)
 - Validation and UQ of temperature and emission (QoI4), fingering instabilities (QoI5), PDFs of pressure and volume fraction (QoI6)

Uncertainty Reduction to Expose Model Errors

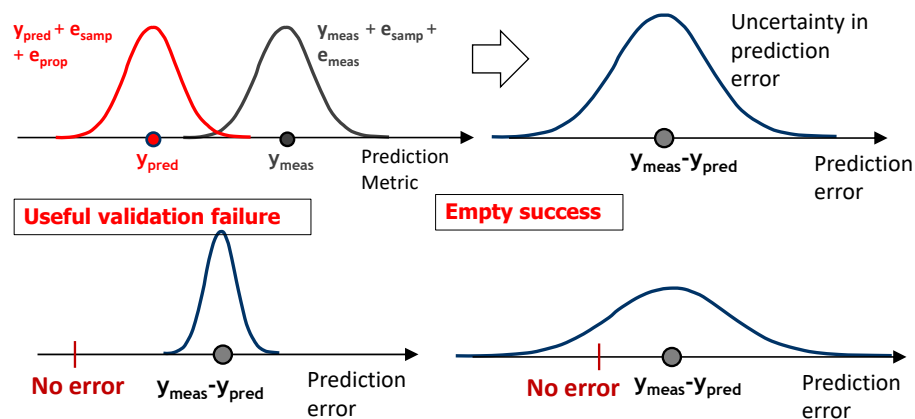
- Separate treatment of error from uncertainty
 - In numerical simulations, quantifying model error is the key issue
 - We reduce uncertainty to expose model error with the goal of improving the model



Prediction Error and Uncertainty

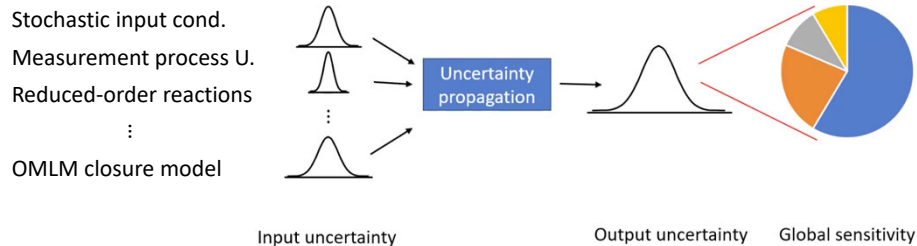
$$e_{\text{pred}} = (y_{\text{meas}} + e_{\text{samp}} + e_{\text{meas}}) - (y_{\text{pred}} + e_{\text{prop}} + e_{\text{samp}})$$

- Prediction error tells us the prediction capability of a simulation
- The corresponding uncertainty tells us the worth of the assessment



Global Sensitivity Analysis for Uncertainty Budget

- How much individual input uncertainties contribute to output uncertainty?

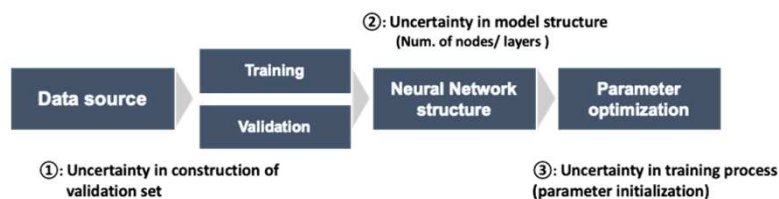


- Modified global sensitivity analysis

- Identifying contributions of different model form uncertainties
- Separating epistemic uncertainty from aleatory uncertainty
- Identifying parameters and configurations that affect output uncertainty the most

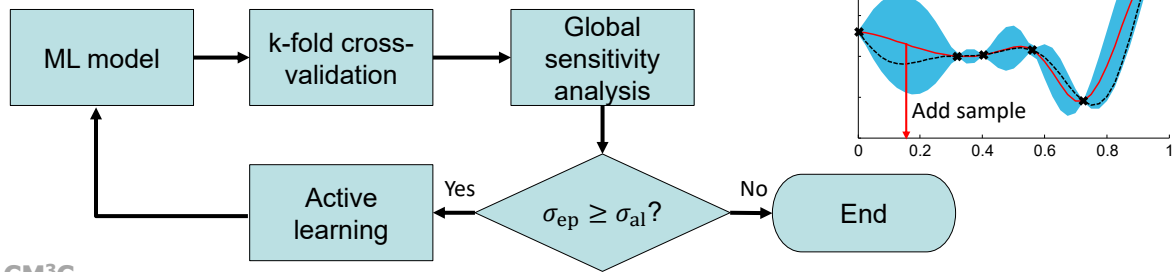
UQ of Machine Learning Model

- Parameter uncertainty (Epistemic uncertainty)
 - Epistemic uncertainty related to weights and biases
- Training uncertainty
 - Uncertainty in construction of training and validation dataset.
 - Initialization of parameter (local minima)
- Data uncertainty (aleatory uncertainty)
 - Test data itself contains the uncertainty (UQ of data will be available)
 - The performance of neural network will be different for which kind of testing set is used for the model evaluation.



UR in Machine Learning Model

- k-fold cross-validation to estimate prediction uncertainty in the ML model
- Modified global sensitivity analysis to separate epistemic uncertainty from aleatory uncertainty
- Active learning strategy to add samples when $\sigma_{\text{epistemic}} \geq \sigma_{\text{aleatory}}$

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Questions?